INDUSTRY STARTER PACKS FOR ENTERPRISE ARCHITECTS

Insurance

What the industry is, how it runs, and how to model it — for architects who have never worked in it

A working guide for an enterprise architect walking into an insurer for the first time. The money and why it behaves the way it does. The operating model end to end. The data, the systems and the names they go by. ACORD, and what a reference architecture is actually for. The regulator, and what it inspects. Then the part that matters most: how to turn all of it into a model somebody in the business would recognise as true.

South African lens. Sourced throughout. October 2026.

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CHAPTER 1

The industry in one chapter

An insurer sells a promise and only finds out years later what it cost. This chapter is the whole book compressed: enough to follow the conversation in an insurer this afternoon, and a map of where the rest of the detail lives.

What insurance actually is

Two transactions, both ordinary. A family in Khayelitsha pays R180 a month so that when the grandmother dies there is R30,000 in the account on the Friday, because the funeral is on the Saturday. A logistics firm in Isando pays R3.4 million a year so that when one of its trucks puts a warehouse on the ground, somebody else pays for the warehouse. Different products, different systems, different licences. The same instrument.

Somebody who stands to lose money hands that exposure to somebody else and pays a fee for the service. On its own that would be a wager. What makes it a business is the second mechanism: the insurer does the same with a hundred thousand households at once, and cannot say which of them will claim this month but can say within a narrow band how many will. Individual uncertainty, aggregate predictability. Every rate table is an application of that one idea, and every catastrophe is a reminder that the idea assumes losses are independent of each other — which a hailstorm is not.

Then the third thing, which shapes your work. The customer pays now and the insurer performs later, if at all, and only if a defined event happens inside a defined window. What is sold is a promise, and the promise is a legal document: the wording, the exclusions, the limits, the excesses and the waiting periods are its working parts.

From which follows the single fact that organises everything else in this book.

> An insurer sells something it cannot cost, and finds out years later what it cost. For a motor book the answer is reasonably clear within eighteen months. For a liability book it may take fifteen years. For a whole-of-life policy sold to a thirty-year-old, the answer arrives when that person dies.

Most of what looks peculiar inside an insurer follows from that. Revenue is recognised now while cost is an estimate made by people who work for the company, so the largest liability on the balance sheet is a judgement and the state supervises solvency rather than only conduct. A promise made in 2019 is honoured on 2019 terms in 2026, so every version of every rule has to be reproducible long after it stopped being used. And because the price was a statistical bet, the only way to learn whether it was a good one is to compare what happened against what was assumed — which means the assumptions have to be data and not a slide.

The seven things an architect must understand

Each of the seven has a chapter of its own later in the book. What is here is the mechanism and the consequence: enough to follow the conversation, ask one useful question, and know which answer should worry you. They build, so read them in order.

The lines of business

An insurer is not one business and insurance is not one product. What an insurer sells is a short list of **lines**, and the line decides nearly everything downstream: the licence it needs, the data it must hold, the system that administers it, the kind of person who underwrites it, and how fast any of it can change. Learn the list first. Every other conversation in this book assumes you already have it.

The first cut is statutory, and the South African words for it are not the global ones. **Short-term** insurance is what most of the world calls non-life, general or P&C. **Long-term** insurance is life. Neither name has anything to do with how long a policy has run, and an insurer needs a separate licence for each. Everything below sits inside one of those two halves, or deliberately outside both.

Short-term business divides into **personal lines** and **commercial lines**. Personal lines is motor, buildings and household contents — the local wording is homeowners and householders — plus all-risks cover on what people carry about with them. Millions of small, similar, modellable risks, priced by algorithm in under a second with nobody in the decision. Commercial lines is the same promise sold to a business: fire and business interruption, liability, engineering, marine, agricultural, trade credit and guarantees. Thousands of large, unlike risks, each priced from a submission, a survey and an argument with a broker over three weeks. One regulator, one balance sheet, almost nothing else in common. Above commercial sits **specialty** — energy, political violence, cyber, satellites, fine art — much of it written in London as a subscription market, where several syndicates each take a share of one risk and 3,015 approved coverholders were binding business on their behalf at the end of 2025 [26]. That market runs on delegated authority and monthly bordereaux rather than on transactions, which is why its data problem is lineage rather than volume.

Life business divides along a different seam. **Individual risk** cover pays on death, disability, critical illness or loss of income. **Funeral cover** is formally a slice of that and in practice its own industry: 6.2 million new individual recurring-premium funeral policies were written in South Africa in 2025, 57% of all new risk policies [25]. Then the savings and **annuity** wrappers, where the insurer is an asset manager and the revenue is basis points on assets rather than an underwriting margin. Then **group** business — group life and group risk cover bought by an employer for its staff as **employee benefits**, sometimes through an **umbrella fund** shared by many employers. There the policyholder is the employer, the lives assured are the employees, one payroll-driven payment covers several thousand people, and the membership changes every month. And **credit life**, sold at the moment a loan is granted, which is why a bank's insurance book is driven by its loan origination system rather than by anything an insurer would recognise as distribution.

Health sits outside all of it here. South African medical schemes fall under the Medical Schemes Act and the Council for Medical Schemes, not the Insurance Act, and the boundary is policed statutorily: the demarcation regulations permit **gap cover** as insurance and prohibit primary healthcare insurance [27].

What breaks is predictable. Insurers organise systems by line, so one human being exists five times over and nobody owns the identity. A composite group buys one platform meant to serve both straight-through personal lines and negotiated specialty underwriting, and it serves neither well. And any number aggregated across lines without a statement of which lines are in it is not a number.

Chapter 2 takes the lines one at a time: what each sells, who buys it, how it is priced and claimed, and what each one does to the architecture.

The money

Four words travel with every premium number and they are not interchangeable. Premium is **written** when the contract is made and **earned** as the exposure elapses [1]; the difference sits on the balance sheet as unearned premium, because it is money the insurer holds and has not yet done the work for. Separately, premium is **gross** before reinsurance and **net** after cession. The two pairs move independently: Santam's gross written premium grew 6.4% in its 2025 financial year while net earned premium grew 14.7% [3]. Anybody reporting "premium growth" without saying which number produced it has already misled the room.

Premium collected now is paid out later, and the money in between is invested for the insurer's own account. That is the float, and it is larger than outsiders expect: South African life insurers held R5.23 trillion of assets at March 2026 against quarterly gross premium of R192.2 billion [2]. A life insurer is an asset manager with an underwriting problem attached.

So there are two results and they must never be summed without both being shown: underwriting is a verdict on risk selection, pricing and expense discipline, investment a verdict on markets. Three ratios carry the underwriting conversation, all against *earned* premium: the loss ratio, the expense ratio, and the combined ratio which adds them together [1]. Below 100% is an underwriting profit; above it is a loss that investment income may still cover [1]. Santam reported 88.7% for 2025 [3].

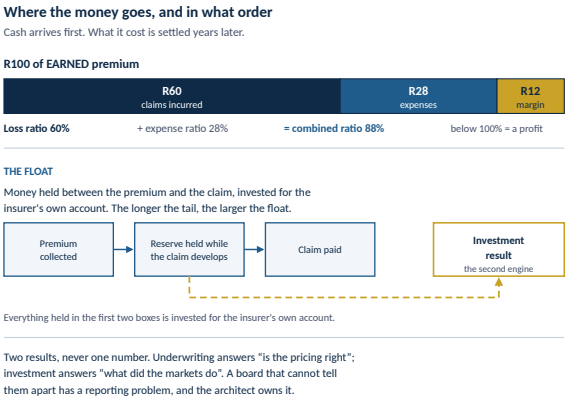


Figure 1.1 The two engines of an insurer, and the float between them. Every figure in this chapter is drawn simpler than its counterpart later in the book; the arithmetic behind this one is Chapter 6. The split of R100 is illustrative, close to a good South African short-term year [1][3].

None of the three has a statutory definition, so two insurers with identical economics can publish combined ratios two points apart depending on whether commission is netted off premium or counted as an expense. Pin the definitions in one governed place.

Inside the loss ratio sits an estimate, and it is the balance sheet's largest number: the money owed on claims that have already happened. Part is countable: case reserves on claims somebody has opened a file for. Most is not. The actuary builds the figure from a loss development triangle, comparing what was known at successive evaluation dates about one accident period [5]. That is a data requirement rather than a report, and most warehouses fail it, because they were built to answer what the position is now instead of what we believed on 31 March 2024. When the warehouse cannot produce the triangle, the actuarial function builds its own estate out of extracts, and within two years that shadow estate is the real system of record for the largest number the insurer publishes.

Last, reinsurance, which architects file under finance and should not, because it decides what the insurer actually owes on every claim it pays. Proportional cover takes the same share of premium and of losses and is computable per policy at bind. Excess of loss pays above an attachment point and is computable only per loss event, after aggregating claims across policies within an hours clause and a geography [4]. That asymmetry is where recovery leakage lives: nobody misses what a system calculates for them.

Chapter 6 does the arithmetic properly: premium states, the float, the ratios with real comparators, the reserving triangle, reinsurance, and the different arithmetic of a life insurer.

The operating model

Ten steps turn a promise into a payment, and they are the same ten in every insurer in the world: define the product, distribute it, quote and bind, underwrite, issue and service the policy, collect the premium, handle the claim, cede and recover under reinsurance, reserve for what is owed, and report to the board and the regulator.

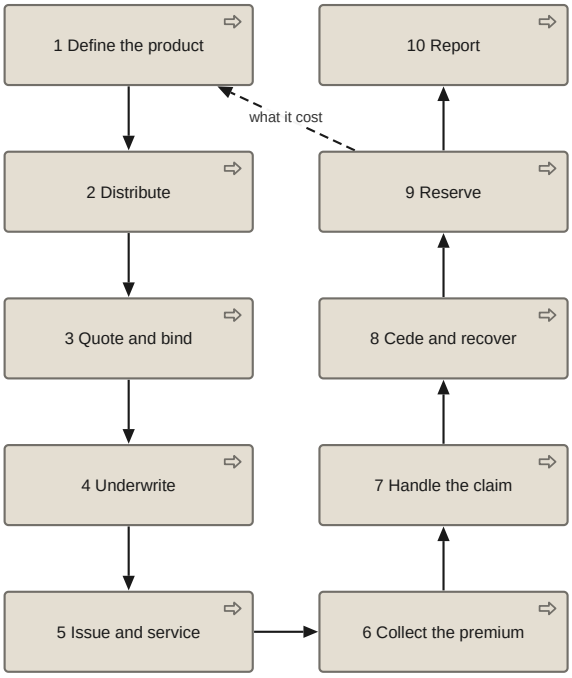


Figure 1.2 The insurance value chain end to end, as ten business processes with the trigger between each. The dashed Flow from reserving back to product is the feedback loop no other industry has: the cost of goods sold arrives after the sale, sometimes decades after.

Walk the chain once and you know where every system in an insurer came from. Walk it again asking of each step who decides and what goes wrong, and you know where the money leaks.

Of the ten, one shapes the architecture more than the other nine together. **Distribution decides where data is created.** An insurer selling direct owns every event: every quote, every abandoned basket, every servicing call, in real time. The same insurer selling the same product through an underwriting management agency under a binder receives a monthly file in somebody else's format, describing business it never touched. Those two insurers need completely different architectures, and no data warehouse closes the gap, because the gap is commercial rather than technical.

Distribution also sets the latency budget — an aggregator wants a bindable price in well under a second, while a commercial broker submission takes three weeks and nobody minds — and it sets who makes the decision the insurer is liable for. A South African UMA holds delegated authority to underwrite, price, issue, service and often settle claims inside a mandate; the insurer remains the principal, carries the risk and must pay valid claims [22], and an insurer outsourcing critical functions remains ultimately responsible for them [7]. The decision is delegated. The liability is not. That turns oversight into a system requirement — entitlements, limits, exception monitoring, audit — rather than a clause in a service level agreement.

Two more local structures belong in your first week's vocabulary. A **binder** is the legal instrument delegating specific activities — entering into, varying or renewing a policy, determining premiums, settling claims — to an intermediary [7]. A **cell captive** is a licensed insurer issuing classes of shares to third parties, so a retailer or a bank can take insurance risk without its own licence; non-life cell captive gross written premium was R9.6 billion in the quarter to March 2026, around a fifth of non-life primary premium [2]. Embedded insurance here usually arrives through a cell.

What breaks is the seams. Quotes produced by one path and binds by another, so prices differ. Entitlements held as convention in a regional manager's head rather than as enforced configuration. Bordereaux that cannot be allocated to policies, so the insurer knows how much money arrived but not whose cover is paid up. None of those is a failure inside a system; every one is a failure between two of them, and the two belong to different people.

Chapter 7 walks all ten steps, with who does it, what gets decided and what goes wrong.

The split that catches everyone

"The insurer's policy system" is almost always the wrong phrase, and asking which one marks you out as somebody who has been here before.

A short-term insurer sells a twelve-month contract, and in South African personal lines often a monthly-renewable one. The risk changes with the car, the driver, the suburb and the weather, and the price is reset at every renewal, so a pricing mistake shows up within a year and is corrected at the next cycle. Claims are investigations: is it covered, who was at fault, what does it cost, and can anything be recovered. A life insurer sells a contract that may run for forty years. The risk is a human being, assessed once, and the price is fixed for the whole term, so a pricing mistake is locked in for a generation. Claims are verifications: somebody died, or did not, and the sum assured is on the schedule.

Everything follows from that. Reserving differs, because one insurer is estimating claims that have happened and the other obligations decades away, where the discount rate matters more than any claim file. Capital differs: catastrophe accumulation on one side, longevity and guarantees on the other. Data differs, because a short-term policy is a thing at a place and a life policy is a person — often several, since extended-family funeral cover is the norm here, and one policy can carry a dozen lives assured.

The systems differ most of all, for a reason worth carrying into every meeting. **A life policy administration system is partly a bank ledger and a short-term one is not.** A life policy can accrue a cash value, pay bonuses, charge a surrender penalty and turn into an annuity, and every one of those values is

contractual: if the system says the surrender value is R184,322.17, the insurer owes R184,322.17. A motor policy accrues nothing. It is a price attached to a risk, re-priced at renewal and pro-rated when a car is added in month seven.

That single difference decides programme strategy. A short-term insurer can write new business on a new platform, stop writing on the old one and run the old book off in two years. A life insurer cannot, because the book will still be paying claims in 2070. It has three options and no fourth: convert the data, keep the legacy behind a façade, or outsource the closed book. Oliver Wyman puts the average United Kingdom life and pensions conversion at 36 to 48 months [14]. Plan against that number, not the one in the proposal.

South Africa licenses the two businesses separately under the Insurance Act 18 of 2017 [6], so a composite group is two regulated entities with at least two cores, two data models and two release cadences. The group architecture problem is not consolidation. It is making two unlike estates answer one question about the same human being.

Chapters 7, 10 and 12 carry this further: each business's operating model, the information model that has to hold both, and the two kinds of core.

The data

Six weeks into any programme somebody will ask you for the conceptual data model on one page and will want *Customer* on it. Do not give them Customer. There is no such thing in insurance.

What there is instead is a **party** — any legal person the insurer deals with — and a **role**, which is something that party does, to something, between two dates. One human being can be, on one Tuesday, the policyholder on their own motor policy, the life assured on a policy their employer owns, a beneficiary on a spouse's policy, a claimant against a stranger's liability cover, and if licensed, an intermediary. Six positions, one party. Flatten them into a customer table and you destroy distinctions the business, the actuary and the regulator all need — among them the one that matters most under privacy law, because a third-party claimant never consented to anything, and their lawful basis, purpose limitation and retention all differ from a policyholder's [12]. ACORD's Information Model names Party as a core concept beside Policy, Product and Claims [9].

The second pattern is as load-bearing. **A product is a definition and a policy is an instance.** The product is the manufactured template — covers, limits, options, rating factors, eligibility rules, wording and pricing algorithm — versioned on its own clock. The policy is one sold instance, with its own. When an insurer tells you a variant takes nine months to launch, the cause is almost always that product rules live inside policy tables and code, so the definition cannot change without touching every instance.

Then the sentence to write on a whiteboard: **a policy number does not identify a policy.** A motor policy with twelve endorsements has one number and thirteen versions, and claims adjudication, cession and accounting measurement all need the policy *as it stood on a particular date* — a version, not a policy.

Three more entities are missing from most outsiders' models, and each absence costs money. The **risk object** — the vehicle, the building, the shipment — has a lifecycle independent of any policy, so making it a child of Policy means you cannot answer "have we insured this vehicle before". The **loss event** is not a claim: one hailstorm yields four thousand claims, and without the event there is no catastrophe aggregation and no excess-of-loss recovery. And a **reserve** is an estimate as at a date, so the claims mart must be a time series of movements rather than a current-state table.

Underneath all of it are two independent time axes. **Valid time** is when a fact was true in the world; **transaction time**, or record time, is when the insurer knew it [10]. A broker advises on 20 June that a vehicle was added on 1 May; in between, the records said there was no cover and the insurer priced, reported and ceded on that basis. Both must survive: the regulator, the auditor and the reinsurer each ask a different one. Most models carry one axis.

Chapter 10 builds the model entity by entity, then tells you where to get one instead of drawing it yourself.

The estate

Ask a chief information officer for the application inventory and you will get a spreadsheet of two hundred to two thousand rows. The part that matters is small and falls into four groups.

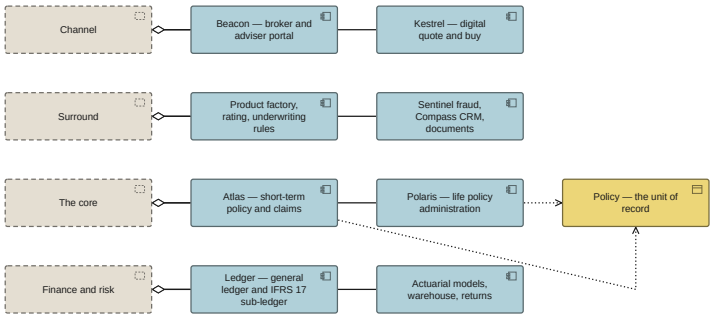


Figure 1.3 The estate at a glance, with Cape Mutual's invented systems in place. Four groupings, and the one passive object both cores hold: the policy, not the customer. Chapter 12 names sixteen categories and what each is authoritative for; this is the shape to carry into a first meeting.

At the centre is the **policy administration system**, which holds the contract. If it says the policy is in force at R2 million sum assured and R450 a month, that is the truth and everything else is a copy. It is not a database of policies but an event processor over time — issue, endorse, alter, renew, lapse, reinstate, cancel, mature, claim, surrender — where each event recalculates premium, re-dates cover, emits documents, triggers commission and writes a new version of the contract with an effective date. Around it is the surround: product configuration, rating, underwriting rules, documents, party master, fraud scoring. Every one of those began life inside the core and was pulled out at a different time, and which ones have been pulled out tells you most of the insurer's technology history. In front is the channel layer, which changes fastest and is architected worst. Behind is the finance and risk stack, whose deadlines are external: a broker portal can slip a quarter, a prudential return cannot.

Why is the core so hard to replace? Four reasons, and only one is age. Product logic is hard-coded, so every product is a code release. Batch sits on the critical path, so the business day is shaped by the overnight run and data is a day old everywhere except the core itself. There is no service boundary, so every downstream system couples to the core's schema, which means the schema can never change, which means the system can never improve. And for life business the values are contractual and must reproduce to the cent, making conversion a correctness problem rather than a data-movement one — and correctness problems do not respond to more contractors.

The market's own timelines are the best calibration available. Santam selected Guidewire for underwriting in December 2012 and for claims in February 2019, and the go-live was announced in June 2026 [19]. Those dates do not measure one implementation and are not a benchmark, but they measure elapsed reality, and they should end any three-year core replacement plan in the room.

The useful move is the opposite of replacement. On the board slide the administration system is one box; inside it are a dozen distinguishable components with different owners, change rates and failure modes — event processing, premium calculation, the policy value engine, billing, documents, commission, posting, batch orchestration. If it is one thing, the only option is replacement. If it is twelve things there are options: externalise rating, lift collections out, put a service in front of the policy read, and leave the value engine alone for a decade. That hour of decomposition is the highest-value hour an architect spends in an insurer.

Chapter 12 names sixteen categories of system and how each goes bad; **Chapter 14** covers who sells them, and how to read a vendor market without producing a shortlist you cannot defend.

The regulator

South Africa has two financial regulators and they want different things. The **Prudential Authority**, inside the Reserve Bank, supervises safety and soundness: can the insurer pay what it has promised. The **Financial Sector Conduct Authority** supervises market conduct: does the insurer treat its customers fairly. Where a matter touches both they issue Joint Standards together [13]. The split is not tidiness — declining valid claims improves solvency, and an insurer can be generous to the point of insolvency, so one supervisor would trade the two off invisibly.

For an architect the consequence is sharp, because the two want data at different grains. The Prudential Authority wants money at a valuation date: balances, provisions, capital, exposures. The FSCA wants process and outcome over a period: who was advised, what was sold, how long it took, what the customer asked for, what they got, who complained. The temptation is to build two reporting estates. Resist it. They are two derivations of the same events — a claim payment is a cash flow for one regulator and a turnaround time with an outcome code for the other — so build the event record once, at the finest grain any consumer needs, and generate both.

Neither regulator will ever ask how your systems are built. They ask questions that only a particular kind of estate can answer, twice, on time, with evidence. Those questions reduce to three properties: **reproducibility**, the same question asked twice over the same period returning the same number, after a restatement or a staff departure; **lineage**, a recorded walk from the published number back to the transaction; and **attribution**, the ability to name who or what made a judgement, on what inputs, under which version of which rule. Section 71 of POPIA makes the last concrete by protecting a person from a decision based solely on automated processing [24]: an insurer must be able to say whether a decline was wholly automated, and reproduce the inputs and rule version behind it.

Four facts belong in your head before the first meeting. Licence class and sub-class under Schedule 2 of the Insurance Act is an attribute of every product, derivable for every contract in force [8], because a product that drifts out of its sub-class is unlicensed business rather than a reporting error. IFRS 17 measures groups of contracts — portfolio, profitability bucket, annual cohort — fixed at initial recognition and never reassessed [11], so the group key belongs on the source event. The POPIA health information regulations took effect on 6 March 2026 and name insurers expressly [12]. And Joint Standard 1 of 2024 requires notification to both regulators at least 30 days before a material outsourcing arrangement [13], so a software shortlist may also be a regulatory filing.

Say this out loud in your first architecture forum, because it is cheap before and ruinous after: **reproducibility and lineage are design requirements, not reporting features.**

Chapter 15 covers Twin Peaks, the solvency regime, Treating Customers Fairly, FAIS, FICA, POPIA, IFRS 17 and the outsourcing standard, and shows how to model a rule so that it traces to the system satisfying it.

One claim, end to end

Everything above is abstract until a car is damaged. Here is one motor claim at Cape Mutual Insurance Group, from the bump in the road to the general ledger, with the systems named.

Cape Mutual is invented — the worked example carried through this book, not a disguised case study. Cape Mutual Life writes risk and funeral business on **Polaris**, a policy administration system in its fortieth year; Cape Mutual Short-Term writes motor, household and small commercial on **Atlas**. **Beacon** is the broker and adviser portal, **Kestrel** the digital quote-and-buy stack, **Sentinel** scores claims for fraud, and **Ledger** is the general ledger and IFRS 17 sub-ledger. The failures below are ordinary; the names are not real.

Monday, 9 March, 06:40, Koeberg Road. A minibus taxi changes lane and clips the front left of a 2019 Polo. Nobody is hurt. The other driver admits nothing, gives a name and a registration number and leaves.

07:20, first notification. The policyholder photographs the damage in the Kestrel app, which opens a claim through the claim service Atlas exposes. She also telephones her broker, who logs the same loss in Beacon forty minutes later. Atlas now holds two claims for one loss and somebody will have to merge them — the first ordinary defect in the thread, and the argument for one intake path behind every channel.

Atlas creates two things, not one. A **loss event**, because one collision can generate several claims — her own damage, the taxi owner's claim against her liability cover if fault goes the other way, an injury claim later — and a **claim** against policy 40188221. An insurer that does not hold the loss event separately can neither aggregate a catastrophe nor compute an excess-of-loss recovery [4], which is where reinsurance money quietly goes missing.

07:22, scoring. Sentinel scores the claim at notification rather than in a monthly batch [23], resolving the vehicle, the bank account and the service providers against prior claims; it comes back below the referral threshold. Scoring before the payment rather than after it is what makes a fraud system a control instead of a report.

17:20, cover confirmed. Eleven hours after notification. The policy was endorsed on 1 March: the policyholder wanted a lower premium and took a higher excess, R6,500 instead of R4,000. The loss is on 9 March, so **version 7** governs and version 7 carries the higher excess. Had the collision been on 25 February, version 6 would have applied and the insurer would have owed R2,500 more. Atlas can say what the policy looks like today in a second; saying what it looked like on a date takes a person and most of a day. That gap is the mechanism behind repudiations that get overturned.

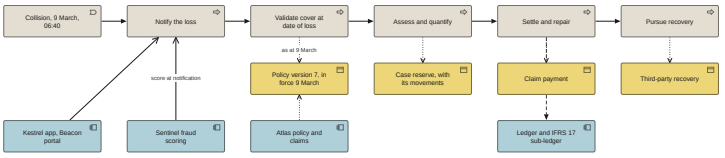


Figure 1.4 One motor claim at the invented Cape Mutual, from the collision to the ledger. The Access edge from cover validation to the policy version — not to the policy — is the single most consequential line on the page.

Cover also depends on the premium being paid. The 1 March collection succeeded under a DebiCheck mandate the policyholder authenticated once at her own bank, the instruments that preceded it having been switched off on 31 October 2021 [15][16]. Had it failed, cover would have been arguable.

Registration, and the first number that is a guess. The handler posts a case reserve of R48,000 from the photographs. It is wrong, which is fine, provided the correction is recorded as a movement rather than written over the top of it: reserve movement history is the actuary's raw material [5].

11 March, assessment. An assessor photographs the car on a tablet at the panel repairer and quantifies the repair at R61,500 against a R280,000 sum insured: damaged, not written off, though vehicles are increasingly written off after accidents that would once have been repaired [17]. The reserve moves, dated.

2 April, settlement. Cape Mutual pays the repairer rather than paying cash, and the choice has tax consequences: a monetary indemnity payment to a vendor insured is a deemed supply by the insured, a trade payment to a supplier who reinstates is not [20]. She is not a vendor, so neither applies — but Atlas could not have told you that, because the attributes deciding it live in the party model and are usually never captured. The R6,500 excess is not income either [20].

Then the part nobody watches. Recoveries. The team pursues the taxi's insurer under subrogation and settles at half four months later; R25,400 arrives in August. That is not a negative payment — different counterparty, different timeline, different accounting, a new legal matter of its own. Book it as a credit against the claim and the insurer can neither age a recoverable nor say whether its recovery function is any good.

Reinsurance barely touches a claim this size: the quota share cession was computed by Atlas at bind, per policy, and the excess-of-loss layer attaches far above R61,500. Had that Monday produced four thousand claims from one hailstorm instead of one from one taxi, the recovery would be computable only at the loss event, after aggregating across policies within an hours clause and a geography — the calculation insurers fail to automate and then fail to claim.

The ledger. Atlas posts the claim incurred, the reserve movements, the payment and the recovery into Ledger, where the IFRS 17 sub-ledger turns cash flows into postings for the general ledger [18]. The claim lands in the liability for incurred claims of the group of contracts policy 40188221 was assigned to at initial recognition — and if that group key was derived at run time rather than stamped on the event, a later change to reference data silently re-groups history [11].

14 August, file closed. One collision, 158 days, five Cape Mutual systems, five parties outside the insurer, nine dates that matter. Not one of the hard parts was inside a system. Every one was at a seam: between the app and the broker portal, between the claim and the policy version, between the claim and the loss event, between operations and finance. That is the work, and it is why this book keeps returning to boundaries rather than boxes.

What this book is, and what it is not

This is a starter pack for an enterprise architect who has never worked in insurance. It assumes you can model, run a programme and read a balance sheet, and that nobody will brief you properly. Its test is narrow: after reading it you should be able to walk into an insurer, follow the conversation, ask the question that reveals the state of the estate, and build a model an insurance person would recognise as true.

It is **not an actuarial text**. The chain ladder and the contractual service margin appear here as things that place demands on data, not as mathematics to be performed. It is **not a product manual**: no insurer's wording, no vendor's configuration guide. And it is **not a vendor shortlist** — Chapter 14 names a great many vendors and recommends none, because a shortlist belongs to one insurer's requirements at one moment, and ownership in that market changes faster than a book can.

Where the South African lens is applied, and why. The mechanisms of insurance are global and are written here as global. The lens goes on wherever the local answer differs materially and an imported assumption would lead you wrong: two regulators rather than one, and Solvency Assessment and Management rather than Solvency II; the binder, UMA and cell captive structures, more prominent here than in most markets; collections, where DebiCheck replaced instruments switched off in 2021 [15] and where failed collections rather than cancellations drive lapse; VAT on short-term insurance, which does not work the way most architects expect [20]; and monthly-renewable personal lines, which removes the annual renewal event an imported design assumes.

When the research was done. October 2026. Every market figure here is a point in time and the point has passed. What dates fastest, in order: vendor ownership and product names, each one acquisition away from wrong; regulatory timetables, with the conduct return, the COFI Bill and the November

2026 outsourcing deadline all in motion [21]; payments governance; market results; and standards versions. What will not date is the mechanisms — product against policy, cover as at date of loss, bitemporality, parties in dated roles, the bordereau boundary, and a batch window that only ever shrinks.

One commitment, because you cannot check what you are told: every claim here either carries a source or is marked as the author's judgement, and where the research could not settle a question — the share of South African short-term premium written through UMAs is the clearest case — the book says so and leaves the number out.

How to read the rest

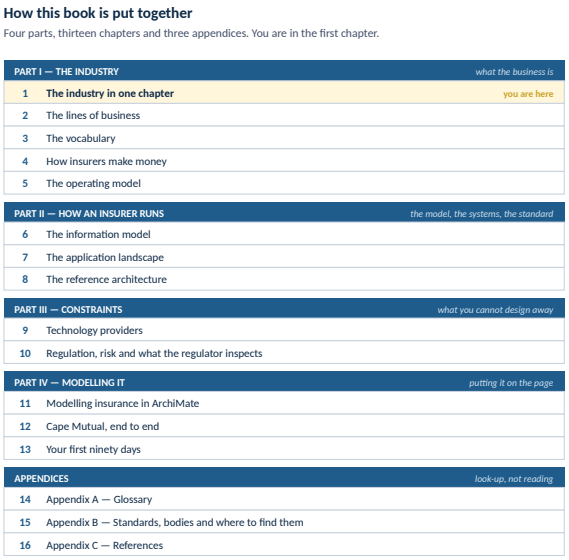


Figure 1.5 The book's structure. Part I is the business, Part II is how an insurer runs, Part III is what you cannot design away, Part IV puts it on the page, and the three appendices are reference rather than reading.

Part I — The industry is the business itself. First the lines of business, one at a time, because what an insurer sells decides what it needs. Then a chapter on the vocabulary, because insurance has borrowed a few hundred ordinary English words and quietly redefined them. Average is not a mean. Excess is not a surplus. Repudiation is not a breach of contract. Then how insurers make money, and the ten-step operating model. Read Part I once, straight through.

Part II — How an insurer runs is the working material: the information model, the sixteen categories of system, and the industry reference architecture — what ACORD actually contains, what it costs, and why a programme whose deliverable is "ACORD adoption" has no deliverable [9].

Part III — Constraints is what you cannot design away: the vendor market, read as capabilities rather than brands, and regulation, read as demands on data.

Part IV — Modelling it is the craft. The mapping from a few hundred insurance concepts onto ArchiMate's 42 elements, with the reasoning kept in so you can handle the concept it missed; a worked model of Cape Mutual, view by view, with the rejected options left in, because those are the ones that teach; and a plan for the first ninety days.

The three appendices are for looking things up rather than reading: a glossary of a little over two hundred terms with the dangerous ones marked, a register of standards and bodies with who owns each one and what it costs to get at, and the references.

Three ways through it, depending on what the week looks like.

The hour before a meeting. Read chapters 1, 2 and 3 and stop. This chapter is the mechanism, Chapter 2 is what the insurer in front of you actually sells, and Chapter 3 is the words they will use as though you already knew them — average, excess, binder, UMA, bordereau, reserve, lapse, treaty. Then ask one question in the meeting rather than three: *can you reconstruct a policy as it stood on an arbitrary past date, and does claims use that?* The answer, and the pause before it, tells you more about the estate than a week of inventory.

The first week in the job. This chapter, then Chapter 2 on the lines of business and Chapter 3 on the vocabulary, then Chapter 18 on the first ninety days, which tells you what to read and whom to ask. Then Chapters 6 and 7 for the money and the operating model. Leave the systems chapters until you have seen three policy documents and three claim files, because the systems make sense only once you know what they are administering.

The architect who must produce a target architecture in a month. You have been given an unreasonable task and this book will not make it reasonable, but it can keep you out of the two traps. Start with Chapter 2, long enough to establish which lines of business you are actually architecting for, because the answer is usually wider than the brief says. Read Chapters 7, 10, 12 and 15 properly — the operating model, the data, the estate and the regulator — and Chapter 17 as a worked example of the sequence of views. Use Chapter 13 to get a capability list you did not have to invent, and present the absences first. Use Chapter 18's six tells to form a view of the estate's health, and write them up as a defect register with named owners, because a defect register survives a budget review and a target state diagram does not. Then decline, in writing, to commit to a core replacement timeline until somebody has answered whether the legacy policy values can be independently reproduced. That one question decides whether a conversion is a project or a fantasy.

A last word before the detail starts. Insurance is not difficult because it is technical. It is difficult because the whole industry rests on a promise whose cost nobody knows at the moment of sale, and every system, every regulation and every argument about a definition is somebody's attempt to manage

that one problem. Hold on to that and the rest of this book is commentary.

THE CHAPTER IN TEN LINES

- Insurance is not one product: the line of business — personal, commercial, specialty, individual risk, funeral, group, credit life — decides the licence, the data, the system and how fast any of it can change.
- An insurer sells something it cannot cost and learns years later what it cost; almost everything peculiar about insurance architecture follows from that one fact.
- Premium is never one number — written or earned, gross or net — and the two pairs move independently, so a growth figure without its basis is not a figure.
- There are two results, underwriting and investment, and they must never be summed without both being shown.
- The balance sheet's largest number is an estimate built from a development triangle, which is a data requirement rather than a report.
- Distribution decides where data is created, and therefore shapes the architecture more than any other single choice.
- Life and short-term insurance are two businesses sharing a word: a life policy administration system is partly a bank ledger and a short-term one is not.
- There is no customer in insurance, only parties in dated roles; a product is a definition, a policy is an instance, and a policy number does not identify a policy state.
- Insurance data is bitemporal, because the regulator, the auditor and the reinsurer each ask a different one of what was true and what was known.
- The policy administration system is never one component, and the hour spent decomposing it decides whether modernisation is affordable at all.
- Two regulators want the same events at two grains, so build the event record once and derive both returns: reproducibility and lineage are design requirements, not reporting features.

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CHAPTER 2

The lines of business

Almost nothing about an insurer's architecture makes sense until you know which lines of business it writes. This chapter is the taxonomy: every box an insurer, a product or a system can go in, and what each box does to the model.

The question that decides everything else

Fatima Patel's third day at Cape Mutual Insurance Group. Pieter van Zyl, the chief information officer, gives her a one-page brief: *rationalise the policy administration estate, target state by March, we think there are four systems but it may be six*.

One question comes before that brief is accepted, and it is not about the systems. **Which lines of business?**

"Policy administration system" is not a category. A platform administering a R180-a-month funeral policy covering eleven family members, and a platform administering a R400 million industrial property risk across two hundred locations with four reinsurers on the slip, are not two implementations of one thing. They share no data model, no release cadence, no claims process, no regulator's attention and no licence. Put them in one programme and it will produce an architecture that suits neither.

> An architect who has not established which lines of business an insurer writes, within the first hour, will build something wrong. Almost every architectural difference between two insurers — the data model, the core system, the change cycle, the capital model, the batch window, the regulator you deal with — traces back to the line of business. Everything else is downstream of it.

This is a reference chapter on purpose. When somebody says "we are mostly commercial, with a bit of engineering through a UMA, and the funeral book sits in the life licence", you need to place every word of that. Cape Mutual is invented: the worked example carried through this book, not a disguised case study.

The first cut: short-term and long-term

South Africa says **short-term insurance** and **long-term insurance**. Those are statutory terms, not descriptions of how long somebody has been a policyholder, and they are the words the business and the regulator use. The rest of the English-speaking world mostly says **general insurance**; North America says **property and casualty**, or **P&C** — which is why vendor documentation, ACORD material and job adverts all use a word your colleagues never say.

The Insurance Act 18 of 2017 commenced on 1 July 2018 with 42 Prudential Standards, and it licenses the two businesses separately [3]. Schedule 2 sets out the classes and sub-classes in two tables: Table 1 for life business, nine classes, and Table 2 for non-life, seventeen [1][2]. A licence authorises named classes and sub-classes. A product that drifts out of its sub-class is not a reporting error. It is unlicensed business.



Figure 2.1 Schedule 2 of the Insurance Act 18 of 2017: nine life classes, seventeen non-life classes, and a separate licence for each side [1][2]. Ask for this schedule in week one, then ask whether every in-force contract can be mapped to a class from data.

That licence structure is the most useful reference data an architect can get hold of in week one. Every product maps to a class and sub-class, so every contract in force has one, derivable, and every prudential return is cut by it. If the insurer cannot produce that mapping from data — and many cannot, because it lives in a compliance spreadsheet — the product model is not finished, whatever the product team says.

Why they are two different businesses

	Short-term (non-life)	Long-term (life)
Contract length	Twelve months, or monthly-renewable	Ten to sixty years, or whole of life
What is insured	A thing at a place, an activity, a legal exposure	A person, and often several people
Price	Reset at every renewal	Fixed at inception, for the term
Reserving	Mostly claims that have already happened	Mostly obligations decades away
What consumes capital	Catastrophe accumulation, claims inflation, reserve error	Longevity, old guarantees, new business strain
A claim is	An investigation: covered? who is liable? what does it cost? what can be recovered?	A verification: did the event happen, is the person on the policy, pay the sum assured
The core system is	A transaction processor over a risk	Partly a bank ledger: it accrues contractual values
Change cycle	Fast: short contracts, short feedback loops	Slow: closed books still administered in 2070, and many lives per contract

Hold on to the row about the ledger, because it decides programme strategy. A life policy accrues contractual values: if the system says the surrender value is R184,322.17, the insurer owes R184,322.17, and a migration reproducing R184,322.09 has failed. A motor policy accrues nothing. Chapter 6 does the arithmetic of both; Chapter 12 does the two kinds of core.

At March 2026 the Prudential Authority counted 159 registered insurance entities: 59 life primary insurers, 56 non-life primary, four life cell captives, six non-life cell captives, six non-life captives, 14 life microinsurers, three composite microinsurers, nine professional reinsurers, four composite reinsurers, one non-life reinsurer and four branches of foreign reinsurers [4]. That is a list of *licences*, not companies. One group can hold several.

> **TWO LICENCES MEANS TWO OF EVERYTHING**
>
> A composite group is two regulated entities: two sets of prudential returns, two Heads of Actuarial Function, two solvency calculations, at least two policy administration systems, two data models, two release cadences. The group architecture problem is not consolidation, because the two estates were never meant to be one. It is making two unlike estates answer one question about the same human being. An architect who proposes a single core for a composite will spend a year discovering the mistake.

Personal lines and commercial lines

Inside short-term insurance the second cut is **personal lines** against **commercial lines**. It looks like a customer segmentation. It is a different manufacturing process, and the statute knows it: most non-life classes in Table 2 divide literally into "personal lines" and "commercial lines" [2].

	Personal lines	Commercial lines
Who buys	A household	A business, from a corner café to a mine
Sold how	Direct, aggregator, bank, tied adviser, retailer, broker	Almost always a broker; the larger the risk, the more certainly
Underwriting effort	Seconds, by rules	Days to six weeks, with surveys and negotiation
Rating	Many factors, one algorithm, at volume	Few factors, much judgement, a rated schedule per location
Policy structure	One or two risk objects, three or four covers	Sections, schedules, hundreds of risk objects, bespoke wording
Premium	Monthly, debit order	Annual, often financed; adjustable on declaration

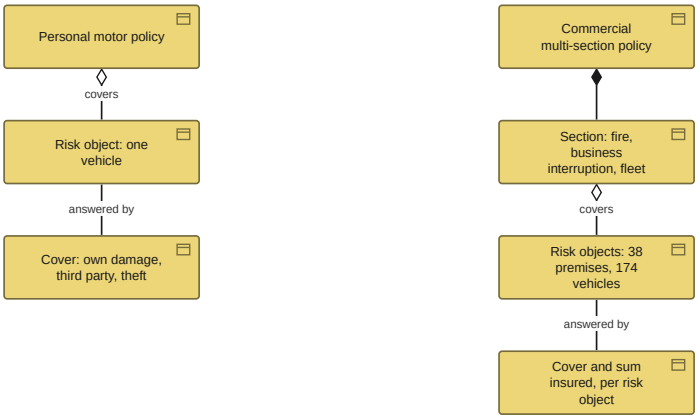


Figure 2.2 Personal and commercial lines are not two sizes of the same policy. Both chains end the same way: cover hangs off the risk object, not off the policy. The counts are illustrative; the structural difference is the point.

Four consequences.

Commercial policies are schedule-bearing and personal policies are not. A household policy has a building and some contents. A commercial property policy has a schedule of locations, each with its own address, construction, occupancy, sum insured, business-interruption figure and sometimes its own excess. A fleet policy has a schedule of vehicles whose membership changes daily. A model holding covers against a *policy* rather than against risk objects beneath it cannot represent commercial business at all. Chapter 10 builds the risk object, the entity outsiders most often leave out.

Rating complexity sits in different places. Personal lines complexity is in the *algorithm*: dozens of factors, interaction terms, geocoded peril scores, telematics, a retrained model. Commercial lines complexity is in the *structure*: a rate per location per section, loadings negotiated line by line, a minimum and deposit premium, and an adjustment at period end against declared turnover or payroll. Buy a rating engine designed for one, apply it to the other, and you discover the difference in user acceptance testing.

The broker is a system, not a channel. In commercial lines the broker holds the relationship, prepares the submission, runs the market exercise and often administers the policy under a binder. So the line of business decides how much of the value chain the insurer actually operates. Chapter 7 has the detail.

Adjustable premium breaks period accounting. A commercial premium finalised four months after expiry against declared exposure needs a deposit premium, an adjustment transaction, a re-cession and a restatement of earned premium for a closed period. Personal lines never does this, so in a personal-lines core it ends up in a spreadsheet.

Inside short-term insurance

Seventeen statutory classes, perhaps eight words people actually use. The statutory class on the left, the market vocabulary beside it [2].

Statutory class	What the market calls it
Motor	Motor, fleet, motor trade
Property	Household, homeowners, HOC, commercial property, assets all risks
Agriculture	Crop, hail, livestock, agri assets
Engineering	Contract works, CAR, machinery breakdown, electronic equipment
Marine	Cargo, hull, goods in transit
Aviation	Hull, passenger and third-party liability
Transport	Goods in transit, carriers' liability
Rail	Rolling stock, rail liability
Legal expense	Legal cover
Liability	Public, product, professional indemnity, directors and officers, employers
Consumer credit	Credit protection, retrenchment cover
Trade credit	Credit insurance
Guarantee	Bonds, construction and rehabilitation guarantees
Accident and health	Personal accident, hospital cash plan, gap cover
Travel	Travel insurance
Miscellaneous	Whatever the above does not fit
Reinsurance	Inwards reinsurance

Motor is the volume business and the one newcomers assume is simple. It is two businesses in one cover. Own damage settles in weeks against a repair quotation, through a panel of repairers, with salvage and a subrogation recovery against the other driver's insurer. Third-party bodily injury settles in years, against a legal claim by somebody who never contracted with the insurer: a long tail hiding inside a short-tailed class. Repair cost inflation is the live pressure, with vehicles increasingly written off after accidents once repaired [20].

Household and homeowners are two covers people conflate. Homeowners — "HOC" — insures the building, is usually required by the mortgage lender and is very often sold by the bank at the point of the bond. Householders insures the contents. Different risk objects, different perils, frequently different insurers on one property, which is where the doctrine of contribution starts to matter.

Commercial property is where the architecture earns or loses money. A schedule of locations, a sum insured each, construction and occupancy driving the rate, geocoded accumulation, a condition of average on every item, and a reinsurance placement that depends on the largest single location rather than the total. Weather catastrophe was the largest single force on South African short-term claims costs in the first half of 2026, and Santam's weather catastrophe losses have roughly doubled over ten years [20]. Accumulation modelling on live exposure, not a quarterly extract, is a design requirement here.

Business interruption attaches to property and behaves nothing like it. It insures a *flow* — the gross profit the business would have earned had the loss not happened. Two parameters do the work. The **indemnity period** is the maximum time the insurer will pay, running from the event until the business is expected to reach the turnover it would have achieved anyway, and twelve months is the usual floor [22]. The **gross profit** figure comes from the last full financial year as a rate, applied to the coming period's forecast [22]. Get either wrong and average cuts the claim: cover of R100,000 against an actual gross profit of R200,000 pays half [22]. Financial statements exclude VAT, so leaving it out creates about 15% under-insurance before anything happens [22]. This is the clearest case of a cover whose *sum insured is a derived financial quantity rather than a value*, and it needs its own fields, indemnity period, declaration cycle and average test. Insurers hold it as one more amount on the property section, and then cannot explain a settlement.

Liability has no risk object. Nothing to geocode, nothing to value. Exposure is measured by turnover, payroll, headcount, professional discipline and jurisdiction, and what triggers cover is a *finding against the insured*, which may arrive fifteen years after the advice that caused it. Two trigger bases exist and conflating them is a real error: an **occurrence** policy responds to events happening during the period whenever they are reported; a **claims-made** policy responds to claims notified during the period whenever the event happened. Professional indemnity is usually claims-made. So a liability book needs the dates of the act, of manifestation, of first awareness and of notification, and a claims model with one date of loss cannot adjudicate it.

Engineering is a family, not a class: contract works and contractors' all risks, machinery breakdown on named plant, electronic equipment, deterioration of stock in cold storage. Its risk object has a *project* lifecycle — a contract period, a maintenance period after it, an escalating declared contract value — which no annual-policy data model represents without work.

Marine splits into cargo and hull, and cargo breaks systems. An open cover under which the insured declares shipments as they happen, each with its own voyage, conveyance, values and currency, generates transactions the way a bank account does. Cover on goods in international transit may also be zero-rated for VAT while the rest of the same policy is standard-rated, so the premium must be apportioned between tax treatments — a requirement Chapter 10 shows most policy systems cannot meet.

Agriculture is seasonal, highly correlated and parametric in places. Santam writes crop cover under four headings: hail, fire, multi-peril and sundry [19]. Multi-peril brings the correlation problem with it: one bad season hits every policy in a district at once. The demand is geospatial — field boundaries, planted hectares by crop and date, and a loss quantified as a percentage yield reduction rather than a repair cost.

Specialist and niche is a commercial reality rather than a statutory class: cyber, political violence, kidnap and ransom, fine art, event cancellation, mining rehabilitation guarantees, renewable energy. In South Africa these are usually written by a UMA under an insurer's licence, because the expertise is too thin to sit inside a composite.

Sasria, and why every South African policy system has a field for it

Sasria SOC Ltd is the state-owned special risks insurer, created in 1979 after the 1976 Soweto Uprising, covering loss from strike, riot and public disorder. The July 2021 unrest caused damages exceeding R31 billion and reshaped the global political-violence market; Sasria's funds have since risen to R18.6 billion against a target of R30 billion by 2029, and its Wrap Cover product for large corporates relaunched in 2026 as excess cover above a R500 million primary coupon, with a R500 million limit [18].

Sasria does not sell directly. Its cover attaches to an underlying policy written by another insurer. So every South African non-life policy administration system needs a construct for a cover it did not underwrite, a premium collected on somebody else's behalf, a separate remittance and accounting flow, and a claims path routing to a different carrier. That attachment mechanism is my description of standard market practice; the figures are sourced [18].

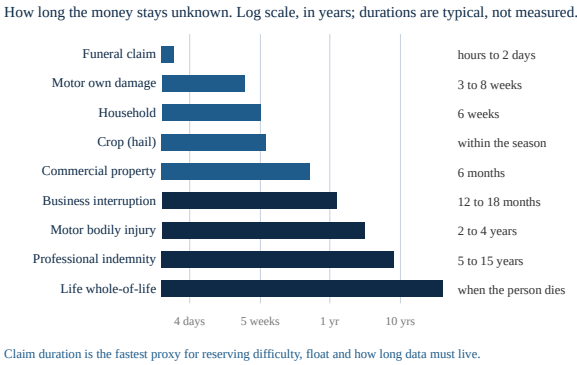


Figure 2.3 Claim duration by line of business, log scale. Typical practice rather than measured data. The motor bar hides two businesses: own damage settles in weeks, bodily injury in years.

Inside life insurance

Life business divides twice, and both cuts matter more than the product names.

Risk business and investment business

Schedule 2's life table draws the first cut cleanly. Classes 1 to 4 are risk: Risk (individual and group death, health and disability), Fund Risk, Credit Life and Funeral. Classes 5 to 8 are money: Life Annuities, Individual Investment, Fund Investment, Income Drawdown. Class 9 is reinsurance of any of them [1].

	Risk business	Investment business
What the customer buys	A payout on death, disability or illness	A wrapper around their savings
Where margin comes from	Mortality and morbidity, expenses	Basis points on assets under management
What the system holds	Lives, benefits, waiting periods, exclusions	Units, funds, switches, fees, guarantees
What breaks	Underwriting and persistency	Unit pricing, and a guarantee written in 1994

The mix explains numbers that otherwise look like incompetence. Momentum's value of new business was R491 million on R93.8 billion of present value of new business premiums — 0.5% — while OUTsurance Life reported 25% [5]. That is product mix, not a ranking of skill. Chapter 6 works it through.

The practical consequence: a "life policy administration system" may be two systems with one name — a protection engine and a unitised investment ledger, with different batch cycles, reconciliation obligations and auditors' attention. Ask which one you are being shown.

Individual and group — the cut that changes the party model

The second cut is **individual** against **group**, and it catches architects, because group business has a *completely different party model*. On an individual policy the person who pays is usually the person insured. On group business they are different parties, and the person insured usually has no contract with the insurer at all.

- **Group life** and **group risk** mean an insured arrangement covering a defined body of people, usually the employees of one employer, for death, disability, critical illness or funeral benefits. Table 1 carries Group Death, Group Health and Group Disability as named sub-classes, and Funeral splits into individual and group [1].
- **Fund risk** is the statutory class for risk benefits provided through a retirement fund rather than to an employer directly [1]. The distinction is who holds the policy, and it is not cosmetic.
- **Employee benefits** is the market term for the whole bundle: the fund, the group risk cover, and the administration of both.
- An **umbrella fund** is one registered retirement fund with many participating employers inside it, each as its own sub-fund. The Sanlam Umbrella Fund is a public worked example: each subsidiary in a corporate group must be a separate sub-fund; a new employer needs at least 20 members or R300,000 of annual contributions; the board has six trustees, three appointed by the sponsor and three elected; group life and lump-sum disability

can be structured through the fund or through the employer's own policy, while income disability, critical illness, accident, spouse's cover and funeral go only through the employer's policy; and the document states plainly that the participating employer "remains the contracting party" [11].

Read that last clause again, because it is the architectural point. The member is not the policyholder. Depending on structure the policyholder is the employer or the fund's trustees. The member is a life assured — a party in a dated role, identified on a monthly membership schedule submitted by somebody else.

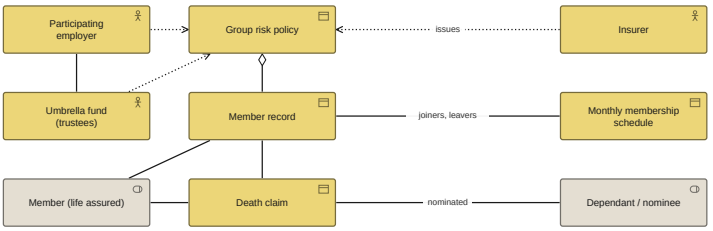


Figure 2.4 The group scheme party model. The member is a life assured in a dated role, not the policyholder: the employer — or the fund — is the contracting party [11], and membership arrives as a monthly schedule rather than as transactions.

Five consequences, each of them a thing that goes wrong.

1. **There is no policy per person.** One policy, thousands of lives. A model with `policy.life_assured_id` cannot hold group business, and cannot hold multi-life funeral cover either.
2. **Membership arrives as a schedule, not as transactions.** A monthly file of joiners, leavers and salary changes, from which premium is recalculated, so the insurer's view of who is covered is only as current as the last file.
3. **Cover is defined by a formula, not an amount.** "Three times annual pensionable salary, to a free cover limit of R4 million." Every member's sum assured is derived and moves when the salary does. A system storing an amount has lost the rule that produced it.
4. **The claim comes from a party the insurer has never met.** A group death claim arrives through the employer or the fund, with a beneficiary nomination held by trustees who have a statutory discretion over distribution. The insurer's beneficiary model does not govern.
5. **Administration is frequently somebody else's.** The fund administrator holds the member record, the insurer holds the risk, both believe they have the membership data, and reconciling them is a standing cost.

Credit life, funeral, annuities

Credit life pays off a debt if the borrower dies, becomes permanently disabled or loses their job. It is its own class in Table 1 [1], sold at the point of credit rather than the point of advice, and price-regulated: the credit life regulations in force from 9 August 2017 cap the cost at R4.50 per R1,000 of the deferred amount for most credit agreements and R2 per R1,000 for mortgages, with retrenchment cover paying instalments for up to twelve months and not sellable to people who are not employed [10]. Three things follow. The policy is triggered by a loan origination event in a *banking* system, so the integration is into credit origination. The sum assured amortises with the loan balance, making it a function of another system's data, recalculated monthly. And a capped price makes margin a function of expenses and claims only, which puts straight-through processing on the critical path of the business case.

Funeral cover is the dominant retail insurance relationship in South Africa and the product most imported designs cannot hold. In 2025, 6.2 million new individual recurring-premium funeral policies were written — 57% of the 10.8 million new risk policies that year, against 1.9 million credit life and 2.7 million for all other protection combined [5]. About 60% of South Africans report having an insurance product; excluding funeral cover that falls to 19% [24]. Its shape is the opposite of a commercial property risk: high volume, low premium, high lapse, **multi-life** — extended-family cover is the norm, so one contract carries a dozen lives assured with different ages, waiting periods and benefit amounts — collected by debit order from constrained incomes, and claimed fast, because the funeral is on Saturday. Underwriting is close to nothing: the waiting period does the work a medical examination would do on a larger sum assured. The architecture that follows is minimal cost to serve, mobile and retail onboarding, collections-retry intelligence as a core capability, phone-based document capture at claim, and a data model that handles households rather than individuals.

Annuities convert capital into income, and the two South African kinds put the risk in opposite places. A **guaranteed life annuity** pays for life: the insurer carries longevity and investment risk, and the capital is gone. A **living annuity** keeps the capital invested in the retiree's name and they draw within a regulated band, carrying both risks — the Actuarial Society's worked example is blunt, with a guaranteed R7,117 a month from R1 million needing an 8.54% drawdown to match, and the 17.5% cap leaving about R7,006 a month at age 85 [12]. Schedule 2 separates them: Life Annuities is class 5, Income Drawdown class 8 [1]. An annuity book is a **payment engine with a mortality dependency**. No renewal, no endorsement, no claim in the ordinary sense: a monthly payroll run, an escalation rule, a guarantee period, a reversion to a spouse, and a proof-of-life process whose failure mode is paying somebody who has died. Protection platforms are usually poor at it, which is why annuity administration so often sits on its own system.

Health: a different regulatory universe

A **medical scheme** is a not-for-profit mutual fund registered under the Medical Schemes Act 131 of 1998 and supervised by the **Council for Medical Schemes** — not the Prudential Authority and not the Financial Sector Conduct Authority. Its members pay **contributions**, not premiums. It must practise community rating and open enrolment and provide prescribed minimum benefits, and may not rate for health status. **Health insurance** is an insurance product under the Insurance Act, priced on risk, sold by an insurer.

The boundary is statutorily policed. From 1 April 2017 the definition of "the business of a medical scheme" was amended so health insurance products are prohibited unless the regulations explicitly permit them [7]:

Product	Status	What it does
Gap cover	Permitted as insurance	Pays the difference between what the scheme pays and what the provider charges, under strict underwriting and marketing conditions
Hospital cash plan	Permitted as insurance	Pays a fixed amount per day in hospital, for incidental costs and lost income — not medical cost
Primary healthcare insurance	Not permitted as insurance	GP visits, medication, dentistry, optometry: must be provided under the Medical Schemes Act

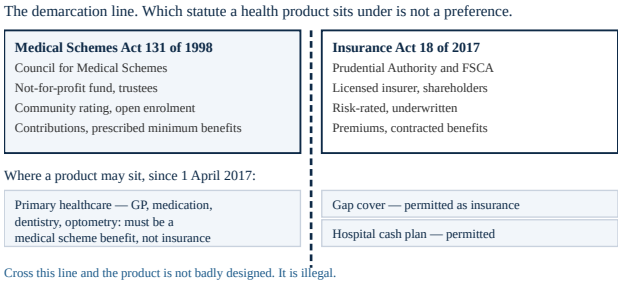


Figure 2.5 The health boundary. Two statutes, two regulators, two vocabularies, and a statutory list of which products may sit on which side [7].

This is the **demarcation** question, and the one product boundary where a team can build something that is not a bad product but an illegal one. Transition ran from 1 April 2017 for new policies, on variation or renewal for existing long-term health policies, and by 1 January 2018 for existing short-term accident-and-health policies [7].

Size the scheme world, because architects are asked to work across this boundary inside one group. At 2024 there were 70 registered medical schemes, down from 131 in 2005, covering about 9.04 million beneficiaries and 4.11 million principal members; open schemes held 56.1% of principal members, down from 67.5%; and two schemes — Discovery Health Medical Scheme at 33.1% and GEMS at 21.4% — account for more than half of principal membership [8].

Four architectural differences from insurance, and they are large.

The scheme is not the administrator. A medical scheme is a fund with trustees; the systems are usually run by a **third-party administrator** that must be accredited by the Council for Medical Schemes under section 58 of the Medical Schemes Act read with Regulation 17, against standards covering operational structure, member record management, contribution management, claims management, financial management, information management and data control, broker remuneration management and customer services [9]. That standard is a surprisingly good functional decomposition of a health administration platform, and free to read.

The claim is an adjudication against a tariff, not an investigation of a loss. Volume is enormous, each claim small, the claimant usually a provider rather than a member, and the decision turns on diagnosis and procedure coding, benefit limits, sub-limits, formularies and managed-care protocols. Nothing in a short-term claims platform resembles it.

The vocabulary shifts under you. Contribution for premium, option or plan for product, benefit for cover, co-payment and deductible for excess. Chapter 3 maps the three dialects onto one structure — and a group running insurance and medical-scheme administration side by side cannot have one premium measure spanning both. That is not a simplification; it is a reporting defect with a regulator at each end.

The live uncertainty is National Health Insurance, before the Constitutional Court in 2026 on challenges to the public participation process behind the NHI Act, with no ruling reported as at mid-2026 [21]. Do not plan around an assumed outcome.

Reinsurance as a line of business

Reinsurance is insurance bought by insurers, and a line of business with its own licences. The Prudential Authority counted nine professional reinsurers, four composite reinsurers, one non-life reinsurer and four branches of foreign reinsurers at March 2026 [4], and Schedule 2 carries reinsurance as the final class in both tables [1][2].

Chapter 6 covers what reinsurance *is* — treaty against facultative, proportional against excess of loss — and Chapter 7 the operational cycle. The point here is routinely missed: **a reinsurer's architecture is nothing like a primary insurer's.**

	Primary insurer	Reinsurer
Contract	A policy with a policyholder	A treaty or facultative slip with a cedant
What it insures	A risk object or a life	Another insurer's book, or one large risk
How business arrives	Channels, quotes, applications	A broker, a submission, a slip, a signed line
Unit of record	Policy, policy version, risk object	Contract, section, layer, cession
Risk data	Captured at source, in full	Received as a bordereau: aggregated, lagged, in the cedant's coding
Premium	Collected from the policyholder	An estimated premium income, adjusted later against actuals
Claims	Notified, investigated, quantified, settled	Advised by the cedant, checked against terms, settled on a technical account
Accounting	Policy and claim transactions	Statements of account, technical accounts, funds withheld, portfolio transfers

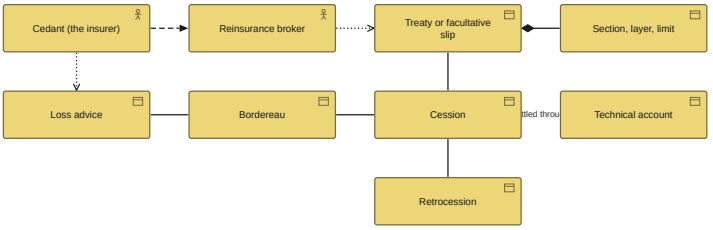


Figure 2.6 A reinsurer's core objects. No policy, no policyholder, no quote and no collections: contract, section, layer, cession, bordereau, technical account — and retrocession, which has no primary-insurer equivalent.

A reinsurance administration platform has a different functional shape: underwriting and administration of treaty and facultative business, assumed and ceded and retroceded, reinsurance accounting and cash management to audit and statutory requirements, and calculation of ceded paid, reserve and incurred metrics for financial reporting [14]. Notice what is absent — no quote, no bind, no policyholder servicing, no collections retry strategy — and what is present with no primary-insurer equivalent: retrocession, reinsurance of reinsurance, which creates a chain of authority and accounting several hops deep.

This matters to every insurer. Most primary insurers also write **inwards** reinsurance, often as a small line inside a larger entity — Santam Re is about 80% of Santam's non-South African specialist gross written premium [6]. Inwards reinsurance is not a product on the policy system. It needs the contract-section-layer-cession model above, and insurers that bolt it onto the direct book cannot reconcile inwards and outwards views.

Specialty and delegated business

The last structural family is not about what is insured but about **who holds the pen**: arrangements under which somebody other than the insurer underwrites, prices, issues, services or settles. Chapter 7 treats these as distribution. Here they are market structures, because they change what an insurer *is*.

Who holds the pen, and what the insurer sees as a result.

Holds the pen	May decide	Insurer's data position
The insurer	Underwrites, prices, issues, services, settles	Every event, in real time
Binder holder (broker)	Issues, services, settles within limits, on the insurer's paper	A monthly bordereau
UMA / MGA / coverholder	Underwrites and prices too, inside a mandate	A bordereau in its own format
Cell owner	Takes the underwriting result; the licensed insurer keeps the licence	Cell accounts, not transactions
Administrator	Administers and pays, takes no underwriting authority	Whatever the contract says
Lloyd's chain	Managing agent → coverholder → service company, each able to sub-delegate	Bordereaux, several hops away

The further down the list, the more the architecture is ingestion and oversight rather than transaction.

Figure 2.7 Delegated business arranged by who makes the decision the insurer is liable for. Authority is delegated; liability is not [17][23].

The subscription market. At Lloyd's several syndicates each take a share of one risk, negotiated through brokers. At the end of 2025 the market had 103 syndicates, 57 managing agents, 401 registered brokers, 3,015 approved coverholders and 409 service companies [13]. A managing agent can authorise a coverholder to bind on its behalf, and a service company can sub-delegate further [13] — so authority arrives in multiple hops and data arrives as bordereaux rather than transactions. A South African insurer meets this world when it places a large or unusual risk abroad, with the same data consequence the local UMA model produces: the insurer's share of a risk is described in somebody else's document and arrives as a line on a statement.

Binders. A binder is the written mandate delegating specific insurer activities — entering into, varying or renewing a policy, determining wordings or premiums, or settling claims — to an intermediary. The 2017 and 2018 conduct reforms limit who may hold one, which activities may be outsourced under one and what remuneration may be paid, to manage conflicts of interest, and an insurer outsourcing critical functions remains ultimately responsible for it [17].

UMAs and MGAs. An underwriting management agency — MGA in most other markets, coverholder at Lloyd's — holds delegated authority to underwrite, price, issue, service and often settle claims within a mandate, and must be an authorised financial services provider in its own right. The insurer is the principal: it carries the risk, must pay valid claims, and must enforce conduct standards [23]. South Africa's UMA sector is unusually large for a market this size, and specialist lines are frequently written this way.

> **A NUMBER NOT TO PUBLISH**
>
> You will hear confident figures for the UMA share of South African short-term premium. None of them traces to a primary source. Describe the model, which is real and important. Do not publish a share figure, because somebody will build a business case on it.

Cell captives. A licensed insurer issues classes of shares — cells — to third parties; the cell owner participates in the underwriting result of its own ring-fenced book while the licensed insurer holds the licence and carries ultimate liability. It is the standard South African route for a retailer, a bank or an affinity group to take insurance risk without obtaining a licence, and the country's native vehicle for embedded insurance. Non-life cell captive gross premium was R9.6 billion in the March 2026 quarter, roughly a fifth of non-life primary premium, at a 38.6% claims ratio but a median solvency cover of only 1.2 times [4]. Guardrisk lists cell captives, UMA, microinsurance and embedded insurance as named business lines [16].

The administrator model. Distinct from a UMA: a firm that administers policies, collects premium and pays claims for an insurer or a fund without taking underwriting authority. In the medical scheme world it is a formal, accredited role under section 58 of the Medical Schemes Act [9]; in insurance it arrives through binders and outsourcing [17]. Either way the system of record for the contract may belong to a third party, and the insurer's estate becomes an ingestion, oversight and reconciliation estate.

Microinsurance and inclusive insurance

The Insurance Act created a distinct, lighter **microinsurance** licence class for financial inclusion, and a microinsurance licence can include both life and non-life products [15] — the only licence in the South African framework that crosses the first cut in this chapter. At March 2026 the Prudential Authority counted 14 life microinsurers and three composite microinsurers [4], and Guardrisk Microinsurance is licensed as a composite [16].

I could not source the benefit limits, policy-term limits or the reduced capital and governance requirements attached to the class. The topics are named in the published summaries but the detail sits in the Act and the Prudential Standards [15]. Confirm them there before designing to them: they are product constraints, not compliance review items.

What the class means architecturally is a cost ceiling. These products are small, simple, high-volume and sold through channels that are not insurance channels — retailers, mobile networks, burial societies, employers. Per-policy cost to serve has to be close to zero: no manual underwriting, no paper, no call centre as the default, and collections that work against irregular income [21].

The table to keep

One page, the whole chapter. Durations are typical rather than measured; the regime column is South African, where "life" and "non-life" mean a class under Schedule 2 of the Insurance Act [1][2].

Line of business	Policy lifetime	Rating complexity	Claim duration	Data volume	Regime	Typical core
Personal motor	1–12 months	Very high: models, telematics	Weeks, or years for injury	High	Non-life	Short-term policy and claims platform
Household and homeowners	12 months	High	Weeks	High	Non-life	Short-term platform, often bank-distributed
Commercial property	12 months, adjustable	Simple algorithm, complex structure	Months	Heavy schedules	Non-life	Commercial platform with sections
Business interruption	12 months	Derived sum insured	6–18 months	Low	Non-life	Attached to property; needs its own fields
Liability and professional indemnity	12 months, claims-made or occurrence	Judgement, not algorithm	2–15 years	Low volume, long history	Non-life	Commercial or specialty, often a UMA's
Engineering	Project plus maintenance	Moderate	Months to years	Low	Non-life	Specialty platform
Marine cargo	Open cover, declared voyages	Low each, high volume	Weeks	High transaction	Non-life, VAT apportionment	Specialty platform
Agriculture	Season	Geospatial, parametric	Weeks, seasonal peak	Moderate	Non-life	Specialist agri platform
Individual life risk	10–60 years	Medical underwriting	Days once verified	Very long histories	Life	Life policy administration system
Group and fund risk	Annual renewable, perpetual scheme	Scheme-level formula	Days to weeks	Membership schedules	Life	Employee benefits platform
Credit life	Term of the loan	Capped price, no rating	Days to weeks	Very high	Life, credit life regulations	Bank-integrated platform
Funeral	Monthly, perpetual	Waiting periods instead	Hours to two days	Enormous, multi-life	Life or microinsurance	Mass-market administration platform
Annuities	For life	Mortality and yield curves	No claim; a payroll	Long-lived, low churn	Life	Annuity payment platform
Investment and savings	Decades	Fees, not risk rating	Withdrawals, maturities	Unit-level, very high	Life	Unitised investment ledger
Medical scheme	Annual, community-rated	None on health status	Days, at huge volume	Enormous: provider claims	Medical Schemes Act, CMS	Accredited health administration platform
Reinsurance assumed	Treaty year	Exposure and experience rating	Years	Low records, heavy accounting	Reinsurance class	Reinsurance administration platform
Microinsurance	Monthly	Minimal	Days	High volume	Microinsurance licence	Low-cost mass platform

What to do in the first hour

Four questions, in this order, before you accept any brief.

1. **Which licences does this entity hold, and which classes and sub-classes?** Ask for the licence schedule, not a product list. Then ask whether every in-force contract can be mapped to a class from data. The pause before the answer tells you whether the product model exists.
2. **Which of these businesses is writing new business, and which is in run-off?** A closed book still needs administering, still generates claims and still appears in the returns, so a decommissioning plan built on new business volumes is wrong.
3. **Where is the pen held?** Per line: is the insurer underwriting, or a UMA, a binder holder, a cell owner, an administrator? That answer decides whether your architecture is a transaction architecture or an ingestion and oversight one.
4. **How long does a claim take to settle?** The fastest proxy for everything else: reserving difficulty, float, data retention, bitemporal depth, and how long the system has to survive.

Thandi Nkosi, Cape Mutual's chief operating officer, puts it more bluntly than I would. There is no such thing as an insurer. There are seventeen non-life classes and nine life classes, a licence for each, and a company is whichever subset it has chosen to write. Ask which subset first and the rest of the conversation makes sense. Ask in month three and you will have designed for an insurer that does not exist.

The next chapter gives you the words. Chapter 6 does the money, and Chapter 7 walks the operating model all of these lines run through.

THE CHAPTER IN TEN LINES

- Ask which lines of business an insurer writes before anything else: almost every architectural difference between two insurers traces back to that answer.
- South Africa licenses short-term and long-term insurance separately under Schedule 2 of the Insurance Act — nine life classes, seventeen non-life — so a composite group is two regulated entities with two of everything.
- A life core accrues contractual values and a short-term core does not, which is why one book can be run off in two years and the other cannot.
- Personal and commercial lines are different manufacturing processes: commercial policies are schedule-bearing, so covers must hang off risk objects rather than off the policy.
- Within short-term insurance the class decides the tail, and the tail decides reserving difficulty, float, data retention and how long the system must survive.
- Group business has a completely different party model: the member is a life assured in a dated role, the employer or the fund is the contracting party, and membership arrives as a monthly schedule.
- Funeral, credit life and microinsurance are mass-market, multi-life, collections-driven businesses whose only viable architecture is near-zero cost to serve.
- Medical schemes are a different regulatory universe — Medical Schemes Act, Council for Medical Schemes, contributions, community rating — and the demarcation line decides whether a health product is legal.
- A reinsurer's architecture has no policy, no policyholder and no collections: contract, section, layer, cession, bordereau and technical account, plus retrocession.
- Delegated structures — binders, UMAs, cell captives, administrators, the Lloyd's chain — move the pen outside the insurer and turn its estate into an ingestion, oversight and reconciliation estate.

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CHAPTER 3

The vocabulary

Insurance runs on a few hundred ordinary English words used with precise and different meanings. This chapter defines them — the money words with the arithmetic shown, then the contract, the people, the traps and the acronyms — in the order that stops you nodding along.

Three sentences, no words you can trust

Fatima Patel's second day at Cape Mutual Insurance Group, in a Tuesday morning risk forum. Riaan Botha, the Chief Risk Officer, says this:

> *"The treaty renews on 1 January and we are carrying too much average on the commercial book, so the excess structure has to move. Nomsa, where are we on the repudiation reason codes — the Authority asked again. And please stop describing the run-off as a product."*

Every word in that paragraph is English. None means what it looks like. *Treaty* is not a diplomatic instrument. *Average* is not an arithmetic mean — it is a penalty for under-insurance, and the most misunderstood clause in short-term insurance. *Excess* is not a surplus. *Repudiation* is not a breach of contract but the ordinary, legitimate act of declining a claim, and it now carries a reason code because a conduct regulator collects them. *Run-off* is a book nobody sells any more but everybody still owes money on. And *product* means something so specific that using it loosely in front of an actuary marks you as an outsider for the rest of the programme.

This chapter is the decoder, and it runs in five parts, in the order that builds.

1. **The money words**, defined arithmetically, on one worked set of numbers, so you can see how they chain.
2. **The contract words**: what is in a policy and what happens to it over time.
3. **The people words**: who everybody is, and which of them has a contract with the insurer.
4. **The words that have been redefined** — the ordinary English insurance has borrowed and quietly repurposed, and the pairs that look like synonyms and are not.
5. **The acronyms of the first week**, as a table, with the South African ones marked.

Parts one to three give you the plain meaning. Part four gives you the trap. The full alphabetical glossary is Appendix A; this is narrower and ordered to be read straight through.

Cape Mutual is fictional — a composite carried through this book so the examples have somewhere to live. Nothing in it is a case study.

> The test of whether you have learned insurance vocabulary is not whether you can define a word. It is whether you notice when somebody else uses it loosely. Most data defects in an insurer began as two people using one word for two things, and neither of them checking.

Part one: the money words, with the arithmetic

A definition of *earned premium* is forgettable. The arithmetic is not. So here is one set of numbers — Cape Mutual Short-Term's year to 31 December 2026, entirely invented — and every money word in the chapter defined against it. Figures are in millions of rand.

Premium

Premium is the price of cover. On its own the word names nothing useful, because it exists in several states and at two scopes, and the states and the scopes are independent.

Gross written premium, GWP, is the premium contracted in the period, before any reinsurance and before any earning. It is the industry's top line and the number in the headline. Cape Mutual wrote **R1,000.0** of it.

Reinsurance premium ceded is what the insurer paid its reinsurers for cover on that book: **R220.0**.

Net written premium is gross written premium less reinsurance ceded. It is what the insurer retained the risk for.

> $1{,}000.0 - 220.0 = \mathbf{780.0}$

Earned premium is the part of the premium relating to cover the insurer has already provided, recognised proportionally as exposure elapses [6]. It is calculated from exposure dates, never from billing dates, and the gap between written and earned sits on the balance sheet as the **unearned premium reserve** — money the insurer holds and has not yet done the work for. Cape Mutual's unearned premium reserve was R410.0 at the start of the year and R450.0 at the end, so the reserve grew by R40.0.

Gross earned premium is gross written premium less the increase in the unearned premium reserve.

> $1{,}000.0 - (450.0 - 410.0) = \mathbf{960.0}$

Reinsurers hold their own share of that unearned premium — R92.0 at the start, R100.0 at the end — so **ceded earned premium** is 220.0 − 8.0 = 212.0, and **net earned premium**, NEP, is gross earned premium less ceded earned premium.

> $960.0 - 212.0 = \mathbf{748.0}$

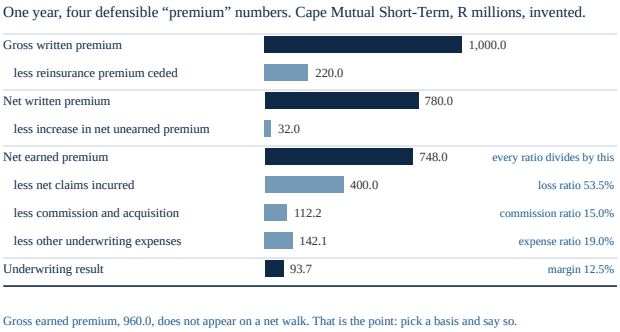


Figure 3.1 The premium words on one invented year, net of reinsurance. Four numbers all fairly called “premium”, and the only one the ratios use is net earned premium.

So one year produced four defensible "premium" numbers: 1,000.0, 780.0, 960.0 and 748.0. Growth rates on them move independently — Santam's gross written premium grew 6.4% in its 2025 financial year while net earned premium grew 14.7% [9] — and a growth figure without its basis is not a figure.

Premium income is the loosest word in the set and you will hear it constantly. It usually means premium actually received in the period, which is a cash measure and matches none of the four above. Treat it as a question, not a number.

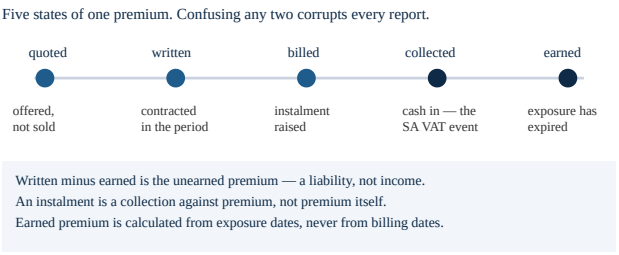


Figure 3.2 Premium exists in five states, and the word "premium" on its own names none of them. In South Africa the VAT event is collection, not billing or inception [1].

Claims

Four words, and the arithmetic that joins them.

Claims paid is cash out: **R430.0** gross in the year.

Outstanding claims are the insurer's estimate of what it still owes on claims it knows about — the sum of the handlers' **case reserves**. R260.0 at the start of the year, R295.0 at the end.

IBNR — incurred but not reported — is the actuarial estimate of claims that have happened but of which the insurer is still ignorant [5]. The collision happened; the call has not come. R120.0 at the start, R141.0 at the end.

IBNER — incurred but not *enough* reported — is the expected further development on claims already known, because first estimates are usually too low [5]. It is normally a component of the IBNR figure rather than a separate line: of Cape Mutual's R21.0 IBNR movement, R13.0 was genuinely unreported losses and R8.0 was strengthening on claims already open.

Claims incurred is what the year actually cost: cash paid, plus the closing case reserves and IBNR, less the opening case reserves and IBNR.

> $430.0 + (295.0 + 141.0) - (260.0 + 120.0) = \mathbf{486.0\ gross\ claims\ incurred}$

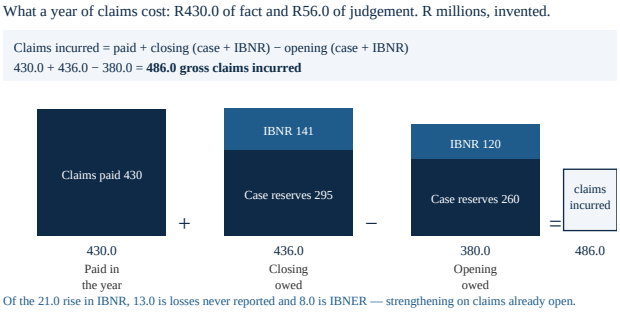


Figure 3.3 Claims incurred is cash paid plus the movement in what is still owed. The IBNR bands are an estimate made by people who work for the company, which is why the reserving actuary is a statutory control role.

Net of reinsurance recoveries and the movement in reinsurers' share of those reserves — R86.0 in total — **net claims incurred** is 486.0 − 86.0 = **400.0**.

Notice what has just happened. R486.0 of "claims" contains R430.0 of fact and R56.0 of judgement, and the judgement was made by people who work for the company. That is why the actuarial function is a statutory control role and why Chapter 6 spends so long on the development triangle.

The ratios

All three of the headline ratios are expressed against *earned* premium, and each one has to be stated gross or net [6]. Cape Mutual's are net.

Ratio	Numerator	Denominator	Cape Mutual
Loss ratio (claims ratio)	Incurred claims plus loss adjustment expenses	Earned premium	400.0 / 748.0 = **53.5%**
Commission ratio	Commission and acquisition costs	Earned premium	112.2 / 748.0 = **15.0%**
Expense ratio	Other underwriting and management expenses	Earned premium	142.1 / 748.0 = **19.0%**
Combined ratio	Claims plus all underwriting expenses	Earned premium	654.3 / 748.0 = **87.5%**

Below 100% the underwriting made money; above it, the underwriting lost money and investment income may still rescue the year [6]. Cape Mutual's underwriting result is 748.0 − 654.3 = R93.7, an underwriting margin of 12.5%. Santam reported a combined ratio of 88.7% for 2025 [9]; the South African non-life primary claims ratio was 52.1% in the quarter to March 2026 [10].

Solvency cover is a different kind of ratio and the one the Prudential Authority watches. It is eligible own funds divided by the **solvency capital requirement** — how many times over the insurer can meet the capital it is required to hold. Cape Mutual holds R493.0 of eligible own funds against an SCR of R290.0, a cover of **1.7 times**. Median cover at March 2026 was 1.8 times for life primary insurers, 1.7 for non-life primary, and only 1.2 for non-life cell captives [10]. There is a lower **minimum capital requirement** underneath, with its own cover ratio, and breaching it is a different order of event.

> THREE DENOMINATORS THAT WILL EMBARRASS YOU
>
> **The same ratio name, two denominators.** The Prudential Authority's March 2026 data reports a life individual lapse ratio of 68.7% as a percentage of new policies issued in the period, and 6.9% on an in-force basis, in one table [10]. Both are correct and they answer different questions.
>
> **Life and non-life claims ratios are not comparable.** Life "claims" include maturities, annuity payments and surrenders, which are repayments of the policyholder's own money rather than losses. The same table shows a life primary claims ratio of 90.8% beside a non-life primary ratio of 52.1% [10]. The life insurer is not nine times worse at underwriting.
>
> **None of these ratios has a statutory definition.** Whether commission is netted off premium or counted as an expense, how loss adjustment expense is split, and how head-office cost is allocated all vary by insurer. Two carriers with identical economics can publish combined ratios two points apart. Pin the derivations in one governed place and expose them.

Part two: the contract words

What a policy is made of, and what happens to it over time.

Word	What it means	What to watch
Policy	One sold, in-force instance of an insurance contract, with a number, parties, risk objects, covers and a premium	A policy number identifies the contract, not its state
Product †	The reusable *definition* policies are made from: covers, limits, options, rating factors, eligibility rules, wording, pricing algorithm	Versioned on its own clock, independently of any sale
Cover	A promise to pay in defined circumstances, with its own limit, excess and conditions	One cover answers many perils; the relationship is many-to-many
Benefit	The payout. In short-term insurance usually an add-on to a cover, such as car hire. In life insurance it does the work *cover* does	A dialect difference, not a structural one
Sum insured	The declared value of the insured thing and the ceiling on an indemnity payment	Tested against actual value by the condition of average
Sum assured †	The agreed amount a life policy pays on the insured event, regardless of measurable loss	Not the same calculation as a sum insured, and usually the same column
Limit	The ceiling on what the insurer pays: per claim, per event, per year, or as a sub-limit inside a larger one	A small structure, not an amount; some limits reinstate after a loss
Excess †	The first slice of every claim the insured carries themselves	Compulsory, voluntary, fixed or percentage, and often several stacked
Deductible	The American and health-insurance word for an excess	Same mechanism, different market vocabulary
Endorsement †	A formal mid-term change — add a vehicle, raise a sum insured, change an excess — and the document effecting it	Produces a new policy version and usually a premium adjustment
Schedule †	The personalised pages saying who and what is covered, for how much, at what premium	Not a timetable. It *is* the data, rendered
Inception	The date and time cover starts	Not the date the policy was captured or the premium received
Renewal	Offering and accepting a further period of cover on a policy that has reached the end of its term	Much South African personal lines business is monthly and never renews
Anniversary	The yearly date on a policy that does not expire — used for escalations, reviews and premium increases	A life policy has anniversaries, not renewals
Lapse †	The policy ending because the premiums stopped	Not a cancellation and not an expiry. In mass-market books the usual cause is a failed collection
Surrender †	Cashing in a savings-type life policy for its surrender value	A contractual amount the system must compute to the cent
Paid-up	A life policy where premiums have stopped but reduced cover continues from the value already built up	A third state beside in-force and lapsed, and routinely missing from status models
Cancellation	Either party ending the contract deliberately, with notice and usually a pro-rata refund	A decision, where a lapse is a consequence
Run-off †	A book the insurer no longer writes but still owes claims on	Not closed, and not a product

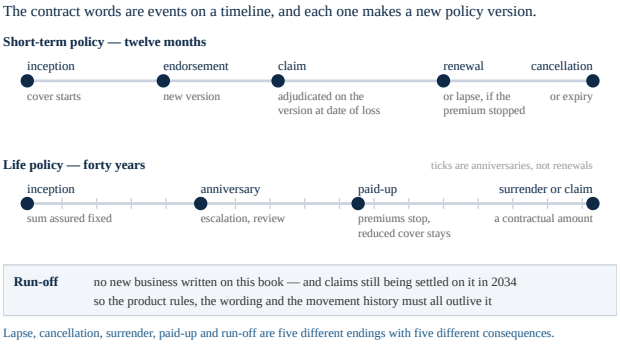


Figure 3.4 The contract vocabulary laid on two timelines. A short-term policy renews; a life policy has anniversaries and may go paid-up. Both end in several different ways, and the difference between those endings is reportable.

Two notes an architect needs from that table before anything else.

Every one of those events creates a version, not an update. A motor policy with twelve endorsements has one number and thirteen versions, and claims adjudication, cession and accounting measurement all need the policy as it stood on a particular date. Chapter 10 builds that properly.

Lapse, cancellation, surrender, paid-up and run-off are five different terminations with different causes, different accounting and different conduct consequences. Insurers that hold one `status` column with the value `INACTIVE` cannot report any of them, and the regulator asks about several.

Part three: the people words

Insurance has more party roles than almost any other industry, and the first mistake an outsider makes is assuming that two of them are the same person.

Word	Who they are	Where it diverges
Insurer	The licensed entity carrying the risk	A composite group holds more than one licence (Chapter 2)
Underwriter †	Whoever decides to accept a risk and at what price	Not an administrator. In a UMA or at Lloyd's they do not work for the insurer
Broker	An intermediary who represents the *client* and places business with several insurers	In South Africa may also issue, service and settle under a binder
Agent	An intermediary appointed by, and representing, one insurer	Tied. Whose mandate they hold is the whole difference
Adviser	Somebody giving regulated advice before a sale	The advice record is retained data, linked to policy and adviser
Intermediary	The umbrella term for broker, agent, adviser and UMA	Not a channel code: a party with a licence, a mandate and a remuneration arrangement
FSP	South Africa: a Financial Services Provider, licensed under FAIS to advise or render intermediary services	Every intermediary party needs a licence number, category and dated authorisation status
Policyholder	Contracts with the insurer, owes the premium, may cancel, amend, assign or surrender	The employer on a group scheme, the company on a keyman policy, the bank on credit life
Insured	The party whose interest the policy protects	Commercial policies carry additional insureds and interested parties
Life assured †	The person on whose death, disability or illness the benefit rests	Joint-life policies have two; funeral cover has a dozen; group schemes have thousands
Member	On a group scheme or a medical scheme, the covered individual	Usually not the policyholder and usually not a contracting party
Beneficiary	Nominated to receive the proceeds	A dated set summing to 100%, with classes, not a foreign key
Claimant	Whoever asserts a claim	In liability business, a stranger who has no relationship with the insurer at all
Cedant †	An insurer when it is buying reinsurance; the reinsurer's "customer"	Not a person
Reinsurer	The party accepting risk from a cedant	Buys its own cover in turn, which is retrocession
UMA	An underwriting management agency holding delegated authority to underwrite, price, issue, service and often settle claims [13]	MGA elsewhere, coverholder at Lloyd's [21]. The insurer stays liable
Administrator	A firm that administers policies, collects premium and pays claims without underwriting authority	In medical schemes a formally accredited role under section 58 of the Medical Schemes Act [25]
Assessor	Appointed to inspect and quantify damage	Usually a specialist in one thing: vehicles, buildings, crops
Loss adjuster	An independent specialist appointed to investigate a larger or disputed claim and recommend a settlement	Works on the insurer's instruction but owes a professional duty of independence

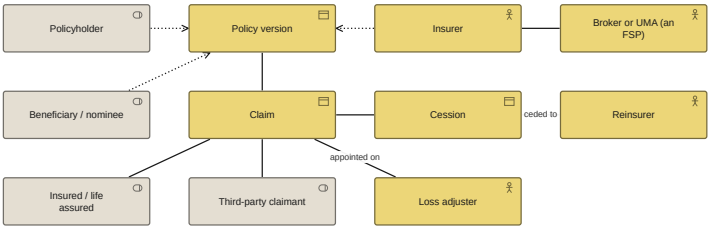


Figure 3.5 One short-term claim, with every party named. Only the policyholder, the insurer and the intermediary have a contract; the third-party claimant has no relationship with the insurer at all and never consented to anything.

The lesson in that table is one sentence long and it is the most expensive sentence in insurance data modelling. **There is no customer.** There are parties, and dated roles those parties play against a contract, a claim or a risk object. One human being can be policyholder on their own motor policy, life assured on a policy their employer owns, beneficiary on a spouse's, third-party claimant against a stranger's liability cover, and a licensed intermediary — at once, in one insurer. Chapter 10 builds the pattern.

Part four: the words that have been redefined

Eleven ordinary English words, each with a harmless everyday meaning, a precise insurance meaning and a consequence for the model. The consequence is why they are here.

One structure, three dialects. The same shape is described with different words.

Short-term (SA)	Life	Health
product	plan	option / plan
cover	benefit	benefit
peril	insured event	event
sum insured / limit	sum assured	annual limit
excess	(none)	deductible / co-payment
premium	premium	contribution

Figure 3.6 Short-term, life and health describe one structure in three vocabularies. A canonical model that treats the dialects as different entity types has guaranteed that no group-level report will reconcile.

Average

In ordinary English, a mean. In short-term insurance, **the condition of average** reduces a claim payment in proportion to under-insurance. Insure a R1 million building for R600,000 and you are insured to 60% of value; suffer a R200,000 loss and the insurer pays 60% of it, because you only ever bought 60% of the cover. Without the clause everyone would insure every building for its largest plausible partial loss and pay a fraction of the premium.

Element	Amount
Replacement value	R1,000,000
Sum insured on the schedule	R600,000
Degree of insurance	60%
Loss suffered	R200,000
Payable before excess	R120,000
Less excess	R5,000
Settlement	R115,000

For the model. Average cannot be computed unless the system holds, per risk object, both the sum insured *and* a value to test it against — two numbers that bad warehouses store in one column. It is also where the conduct exposure sits: the policyholder who expected R600,000 and received R115,000 is the complaint that lands at the Ombud, and an insurer that cannot reproduce the value basis loses it.

Excess

Not a surplus. The excess is the Anglo-South African word for what American and health insurers call a *deductible* [2], and it comes in four forms that behave differently: compulsory, voluntary (chosen in exchange for a lower premium), fixed and percentage. Many motor policies carry two or three, stacked.

For the model. The excess is an attribute of the cover on the risk object, not of the policy, and it is effective-dated — raising an excess mid-term is an endorsement with a premium consequence. There is a tax trap too: in South Africa the excess is not consideration for a taxable supply, so neither party accounts for VAT on it [1]. Insurers that book the excess as income get the revenue line and the VAT line wrong together.

Treaty

A **treaty** is a standing reinsurance agreement under which the insurer *must* cede and the reinsurer *must* accept every risk inside defined terms, including policies not yet written [3]. Its opposite is **facultative** reinsurance, negotiated risk by risk, which the reinsurer may decline [3]. A treaty is an obligation engine, not a contract you consult when convenient.

For the model. Treaty terms are machine-readable rules or they are nothing. In most insurers they live in a spreadsheet and a PDF, cessions are calculated by hand each month, and the recovery on a large loss is found late or not at all. Treaty terms as modelled data is among the highest-return modelling work in a short-term insurer, and the least often done.

Repudiation

In general contract law, repudiation is a refusal to perform. In insurance it is **the decision to decline a claim** — normal, frequent and often entirely correct. Nobody at an insurer flinches at it. What matters is that it is a *structured* outcome: a repudiation carries a reason code, and reason codes are now conduct-reporting data, because the regulator compares declines by cause across the sector [7].

For the model. Repudiation is an outcome entity with a reason taxonomy, not a status value, and the taxonomy must be identical across the life platform, the short-term platform and every binder holder's platform. Expect four taxonomies and a free-text field.

Run-off

A **run-off** book is one the insurer no longer writes but still owes claims on. A liability book closed in 2014 can still be paying in 2034. It is not "closed", and it is not a product.

For the model. Run-off is why you cannot decommission a system by pointing at new business volumes. The data must outlive the book, the product rules must remain reconstructable to adjudicate a claim on 2014 terms, and the reserving actuary needs the full movement history of a book nobody sells. Every life insurer carries closed blocks, which is why a short-term insurer can migrate in eighteen to twenty-four months and a life insurer cannot [15].

Producer

Nothing to do with manufacturing. **Producer** is the North American word for whoever sold the policy — agent, broker, anyone. You will meet it not because South Africans say it but because the systems do: packages built for the United States market carry `PRODUCER_CODE` fields throughout, and the older exchange formats are built around agency-channel concepts.

For the model. Map `producer` to your intermediary party and its role once, at the integration boundary. Let it into the internal model as a second concept beside *intermediary* and you will spend a year reconciling two commission hierarchies.

Contribution

Two meanings, unrelated. In general insurance, **contribution** is the doctrine that where two policies cover the same loss, each insurer pays a proportionate share. In South African health, **contribution** is the word for the money a medical scheme member pays every month — what an insurer would call premium. Medical schemes sit under the Medical Schemes Act and the Council for Medical Schemes, not the Insurance Act, and they use a different vocabulary for the same mechanics (Chapter 2).

For the model. The doctrine of contribution needs other insurers' policies represented against the risk object — which almost no model does, and why double-insurance recoveries are routinely missed.

Cession

In reinsurance a **cession** is the act and the record of passing a share of a risk to a reinsurer — the join between a direct policy and a reinsurance contract. In South African retail life the same word means **ceding a policy** to a bank as security for a loan, so the bank is paid first out of the proceeds. One word, two unrelated structures, both live in a composite insurer.

For the model. Reinsurance cession is a first-class entity at policy-version and claim level, reported periodically on a **bordereau** [22]. A retail cession is a dated party role against the policy, with its own notification and release process. Naming them both `cession` in one warehouse is a fault that surfaces at a death claim.

Reserve

The word that costs programmes the most. A **reserve** is not money set aside. It is an **estimate of an amount the insurer expects to pay** — an accounting liability. Part one did the arithmetic: case reserves, outstanding claims, IBNR and IBNER [5].

For the model. A reserve is meaningless without an evaluation date. "The reserve on claim 4471" is not a number; "the reserve on claim 4471 as at 31 December 2025" is. A claims mart that holds the current reserve and overwrites it on revision has destroyed the actuary's raw material, which is the time series of movements, not the latest value.

Development

Not building something. **Development** is how an estimate changes over successive valuation dates. A 2023 accident year looked like R180 million at the end of 2023, R214 million a year later, R208 million the year after. Arrange those by accident period down and valuation period across and you have a **development triangle** — the fundamental artefact of reserving, and why the previous paragraph matters.

For the model. Development is a *comparison across as-at dates*, and cannot be produced from a current-state table at any cost. This is the first hard requirement for the bitemporal data in Chapter 10 [23], and the commonest reason an actuarial team builds a shadow data estate and quietly becomes the system of record.

Recovery

A **recovery** is money flowing *back in*, reducing the net cost of a claim: salvage proceeds, a payment from whoever caused the loss, a share from another insurer, a reinsurance recovery. It is not a negative payment. Different counterparty, timeline and accounting treatment, and in South Africa a different VAT position — a subrogation recovery carries no output tax [1].

For the model. Recoveries are their own entity with their own parties and dates. Insurers that store them as negative payment rows cannot age a reinsurance recoverable, cannot report gross and net claims cost consistently, and cannot tell a R2 million salvage from a R2 million clawback.

The pairs that are not synonyms

The eleven words above make you sound like an outsider. These pairs make you build the wrong model.

Policy and product. A product is a definition; a policy is an instance. The product is the manufactured template — covers, limits, options, rating factors, eligibility rules, wording, pricing algorithm. The policy is one sold, priced, in-force occurrence of it. This is the most consequential distinction in the book and Chapter 10 spends several pages on it. The short version: when an insurer says a product variant takes nine months to launch, product rules have been built into policy tables, so changing the definition means changing every instance. When a vendor says its product model is "a blueprint" from which a configured instance is produced [16], it is selling you the separation.

Cover and benefit. This is a dialect difference, not a structural one, and that is the trap. Two business units using different words for one structure is survivable. A canonical model that treats life benefits and short-term covers as unrelated entity types *because the words differed* is not: no group-level product or claims report will ever reconcile.

Claimant and beneficiary. A beneficiary is nominated on a policy, has a share, a class and sometimes an irrevocable nomination. A claimant is whoever asserts a claim, and in liability business has no relationship with the insurer at all. Beneficiary nominations are a dated *set* summing to 100%, not a foreign key, because the superseded nomination must stay queryable when the family disputes the payout. The third-party claimant is the hardest party in the model: the insurer holds detailed personal and often health data about somebody who never contracted with it and never consented, so their lawful basis, purpose limitation and retention under POPIA all differ from a policyholder's.

Broker and agent. Both sell insurance; the difference is whose mandate they hold. In South Africa both must be authorised financial services providers under FAIS, and an intermediary may additionally hold a **binder**: a written mandate to perform insurer functions such as issuing policies, servicing them or settling claims. Who may hold one, which functions it may cover and what remuneration may be paid are all regulated, to manage conflicts of interest [8]. Model intermediary type as a channel code and you cannot answer the first question a conduct supervisor asks: what was this intermediary authorised to do, when, and what were they paid for it [19].

Premium and contribution. Covered above, and it belongs here for an architectural reason: a South African composite group commonly runs insurance and medical-scheme administration side by side, under two statutes with two regulators. One "premium" measure spanning both is not a simplification; it is a reporting defect with a regulator at each end.

Insured, policyholder and life assured. Three parties, routinely assumed to be one, and almost never one in the business that matters. The costliest outsider mistake in insurance data modelling is building `Customer` as an entity, and this trio is where the mistake usually starts.

> THREE MORE PAIRS, FOR WEEK TWO
>
> **Sum insured and sum assured.** Sum insured is an indemnity ceiling: the insurer pays the lesser of the loss and the sum insured. Sum assured is an agreed value, paid in full on the insured event regardless of economic loss. Usually the same column in a bad warehouse.
>
> **Written and earned.** Written premium is what was contracted in the period; earned premium is the portion relating to cover already provided, calculated from exposure dates. The two move independently — Santam's 2025 gross written premium rose 6.4% while net earned premium rose 14.7% [9].
>
> **Gross and net.** Gross is before reinsurance, net after. Read every insurance number with three qualifiers attached: gross or net, written or earned, and on what basis (statutory, IFRS 17 or management "normalised"). Architects who drop them ship reconciliation defects.

Part five: the acronyms of the first week

The South African ones are marked, and there are more of those than in most industries, because South Africa regulates insurance under two supervisors and a stack of local statutes.

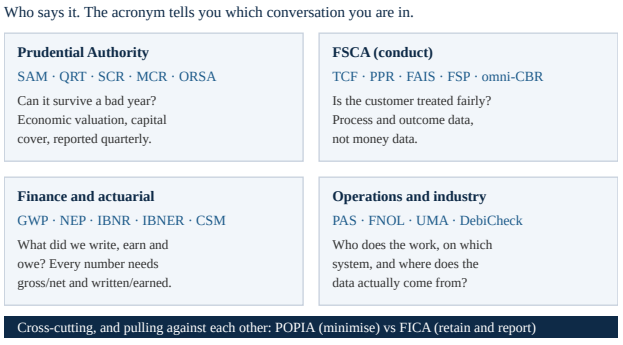


Figure 3.7 The acronyms sort themselves by who is asking. Hearing SAM and QRT means a prudential conversation; TCF and PPR mean a conduct one; and the two ask for different data about the same policy.

Acronym	Stands for	Plain English	What it means for a model
GWP	Gross Written Premium	Premium contracted in the period, before reinsurance and before earning	A written-premium measure needs contract dates, not billing dates
NEP	Net Earned Premium	Premium after reinsurance ceded and after the earning calculation	Derived from exposure dates and cession records: two calculations, two input sets
UPR	Unearned Premium Reserve	Written premium not yet earned. A liability, not income	Calculated from exposure dates per risk object, not pro-rata on the policy
IBNR	Incurred But Not Reported	Estimate of claims that have happened but the insurer does not yet know about [5]	Needs accident-period and development-period granularity, gross and net, with claim counts
IBNER	Incurred But Not Enough Reported	Expected further development on claims already open [5]	A component of IBNR, and only visible if reserve movements were never overwritten
ALAE / ULAE	Allocated / Unallocated Loss Adjustment Expense	The cost of handling claims, attributable to one claim or not	Two different allocations; the loss ratio depends on which is included
XoL	Excess of Loss	Non-proportional reinsurance paying above an attachment point, in layers [3]	Computable only per loss event, after aggregating claims across policies
FNOL	First Notification of Loss	The moment a claim is first reported, on any channel	Attributes analysed *at notification* flag severity and fraud before money moves [18]
PAS	Policy Administration System	The system of record for the contract; usually the oldest and most expensive thing in the estate	Count them. PAS instance count is the best single proxy for architectural debt
EV / VNB	Embedded Value / Value of New Business	The life valuation measures: present value of future profit on the book, and on the year's new business	Needs every in-force contract's full attribute set projected to run-off
SCR / MCR	Solvency Capital Requirement / Minimum Capital Requirement	The capital an insurer must hold, and the floor beneath it	Cover ratios for both are reported to the Prudential Authority [10]
ORSA	Own Risk and Solvency Assessment	The insurer's own forward-looking view of its risks and capital	An annual cycle with a data dependency, not a document
IFRS 17	International Financial Reporting Standard 17, Insurance Contracts	The accounting standard that measures groups of contracts rather than policies	Groups are fixed at initial recognition and never reassessed [20], so the group key belongs on the source event
CSM	Contractual Service Margin	Unearned profit on a group of contracts, released as service is delivered	A movement analysis, not a balance: every component must be traceable
LRC / LIC	Liability for Remaining Coverage / Liability for Incurred Claims	IFRS 17's successors to unearned premium and outstanding claims	Two liabilities with different drivers, from one set of events
PAA / VFA	Premium Allocation Approach / Variable Fee Approach	The simplified IFRS 17 measurement for short contracts, and the model for contracts with direct participation	Eligibility must be tested against the full model, so PAA is no free pass
FSP	Financial Services Provider (SA)	A firm or person licensed under FAIS to give advice or render intermediary services	Every intermediary party needs a licence number, category and dated authorisation status
UMA	Underwriting Management Agency (SA; MGA elsewhere)	An intermediary delegated authority to underwrite, price, issue, service and often settle claims [13]	The insurer's policy data arrives as a periodic schedule in someone else's format, not as live records [19]
TPA	Third Party Administrator	A firm administering policies or a scheme without underwriting authority	In medical schemes, an accredited role with published standards [25]
TCF	Treating Customers Fairly	The outcomes-based conduct philosophy the FSCA supervises against: suitability, clear information, fair advice, performance meeting expectation, no unreasonable post-sale barriers [7]	Conduct outcomes are process data, not money data. The hardest thing to retrofit
PPR	Policyholder Protection Rules (SA)	Conduct rules effective 1 January 2018: premium review principles, minimum data governance, claims-management minimums, complaints reporting, binder limits [8]	"Minimum data governance requirements" is a rule, not an aspiration
COFI	Conduct of Financial Institutions Bill (SA)	The intended successor conduct framework, outcomes-based and cross-sector	Still pending at the FSCA's July 2026 regulation plan [24]; plan for it, do not design to it
CBR	Conduct of Business Return (SA, the "omni-CBR")	The periodic conduct return, with indicators by product, channel and outcome	Pushes granularity down to cover and intermediary level [7]
SAM	Solvency Assessment and Management (SA)	South Africa's risk-based prudential regime, Solvency II-aligned, delivered through the Insurance Act 18 of 2017 and its Prudential Standards [11]	Economic valuation of assets and liabilities, reported to the Prudential Authority
QRT	Quantitative Reporting Template	The structured prudential return filed with the Prudential Authority, solo and group	Designed under IFRS 4 and not aligned to IFRS 17; premium terms keep their pre-IFRS 17 meaning in the return [12]
FAIS	Financial Advisory and Intermediary Services Act 37 of 2002 (SA)	Licenses advisers and intermediaries: fit and proper, competence, advice records, disclosure [13]	The advice record and needs analysis are retained data, linked to policy and adviser
FICA	Financial Intelligence Centre Act (SA)	Anti-money-laundering: risk-based due diligence, record keeping, reportable transactions	Pulls against POPIA — retain and report versus minimise. Resolve it explicitly, per data element

POPIA	Protection of Personal Information Act 4 of 2013 (SA)	Data protection: lawful basis, purpose limitation, data-subject rights, operator obligations, cross-border limits	Insurance runs on special personal information, because life and health underwriting *is* health data
DebiCheck	(SA, not an acronym)	Authenticated debit order: the customer confirms the arrangement with their own bank once, after which the bank permits only collections within the approved terms [4]	The mandate is a bank-held, long-lived object with its own lifecycle. Something must master it
ASISA / SAIA	Association for Savings and Investment South Africa / South African Insurance Association	The life and non-life industry bodies; both publish sector statistics and definitions [14]	Where comparable industry numbers and definitions come from
ACORD	Association for Cooperative Operations Research and Development	Insurance's standards body: messaging standards and a seven-facet Reference Architecture [17]	Its Business Glossary carries over 6,000 terms and is subscription material: treat it as procurement
AL3 / NGDS	ACORD Level 3 / Next Generation Digital Standards	The old one-way batch exchange format, and the modern fine-grained one	Interchange formats. Translate at the boundary; never restructure internal data to match a message schema

Two notes on that table.

AEDO and NAEDO are gone. The older authenticated and non-authenticated early debit order instruments were switched off on 31 October 2021; from 1 November 2021 those instructions were not permitted [4]. DebiCheck replaced them. AEDO logic in a collections system is dead code with a live-looking name.

SAM and IFRS 17 do not agree, and both are compulsory. The Prudential Authority issued interim guidance because its templates predate the accounting standard: premium terms keep their pre-IFRS 17 meaning, gross technical provisions are reported as one total without breakdown, and the first post-implementation submission reports the current period on IFRS 17 with the prior period left on IFRS 4 [12]. One cash flow therefore supports an IFRS 17 measurement, a pre-IFRS 17 premium definition for the return, and a SAM economic valuation. Three bases over one event store — build the event store first and derive all three.

What to do with this

Four things, in order.

1. **Build the glossary before you build the model.** Not a document — a governed artefact with an owner, versioned, each definition traced to a source. ACORD's Business Glossary carries over 6,000 insurance terms and is the obvious starting point [17]; it is subscription material, so treat it as procurement. The alternative is years of argument about whether a cancelled-and-rewritten policy counts as a renewal.
2. **Pin every measure to a derivation.** Numerator basis, denominator basis, period, scope, gross or net. Part one of this chapter is the template: if a number cannot be written as an arithmetic line on named inputs, it is not yet a measure.
3. **Ask what a word means every time, for the first month.** The worst outcome of this chapter would be that you now recognise the words well enough to stop asking.
4. **Write down the disagreements.** When two people in one insurer define *endorsement*, *lapse* or *recovery* differently, that is the location of a future data defect — and worth more than any diagram you will draw in the first month.

Chapter 6 takes the money words and shows where the money actually comes from.

THE CHAPTER IN TEN LINES

- Premium is never one number: gross or net, written or earned, and the four answers for one year can differ by a quarter.
- Every headline ratio divides by earned premium, and none of loss, expense, commission or combined ratio has a statutory definition — so pin the derivations or two identical insurers will publish different numbers.
- Claims incurred is paid plus the movement in case reserves and IBNR, which means the loss ratio always contains an estimate made inside the company.
- A ratio without its denominator is not a definition: one Prudential Authority table carries a lapse ratio of 68.7% and 6.9% for the same quarter.
- A policy number identifies a contract, not its state, and lapse, cancellation, surrender, paid-up and run-off are five different endings with five different consequences.
- There is no customer in insurance — only parties and the dated roles they play, which is why policyholder, insured, life assured, member and claimant must be modelled separately.
- Average is not a mean, excess is not a surplus, a treaty is an obligation engine, repudiation is routine, and a reserve is an estimate rather than cash.
- Development is a comparison across as-at dates and cannot be produced from a current-state table at any price.
- South Africa adds a local acronym stack — FSP, UMA, PPR, SAM, QRT, FAIS, FICA, POPIA, DebiCheck — and POPIA and FICA pull in opposite directions by design.
- Build a governed, sourced glossary before you build the model, and record every definitional disagreement: each one marks a future data defect.

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CHAPTER 4

How the business model works

An architect who cannot draw the business model on one page has no way of telling a system that is merely old from a system that is load-bearing. Two instruments fix that: the nine-block canvas, and the customer journey with the systems named and the failures marked.

The question you will be asked in week three

Somebody will put a system in front of you and ask whether it matters. Usually it is phrased as a budget question. Do we replace Polaris. Do we keep paying maintenance on the broker portal. Do we let the funeral business buy its own thing.

You cannot answer that from a technical assessment. A system matters in proportion to the part of the business model it carries, and to the part of the customer's experience it decides. An architect who cannot draw the business model on one page has no way to tell a system that is merely old from a system that is load-bearing. Polaris is forty years old and runs the life book: old and load-bearing. The reporting database that three analysts query is also old. One of those is an architecture problem and the other is a licence renewal.

This chapter gives you two one-page instruments. The first is the business model on nine blocks. The second is the customer journey, done properly, with the systems named and the failures marked. Between them they let you say, in a sentence a chief financial officer will accept: this system sits on the revenue stream for 60% of our policies and it is the only place the cover decision is made, so it is not a licence renewal.

Neither instrument is insurance-specific. The nine-block canvas comes from Osterwalder and Pigneur [1] and journey mapping comes from service design. What makes them useful here is that insurance punishes the generic version of both. A canvas for an insurer that says "customers" in the customer segment box and "peace of mind" in the value proposition box is worse than no canvas, because it will be believed. Insurance has at least four customer segments that need four different operating models, and the value proposition is not peace of mind — it is money, in a particular amount, at a particular speed, with a particular amount of argument.

> Fill the canvas with the invented names, the invented systems and the real numbers. A canvas you cannot argue with is a canvas nobody read.

The canvas for a composite insurer

Here is the canvas filled in for Cape Mutual Insurance Group: a life licence and a short-term licence under one holding company, a large independent broker force, a funeral book sold through agents and kiosks, a group risk business, and one underwriting management agency holding the pen on specialist liability. Cape Mutual is invented. Nothing in it is a case study of any South African insurer. The *shape* is typical of a mid-sized South African composite, and the shape is the point.

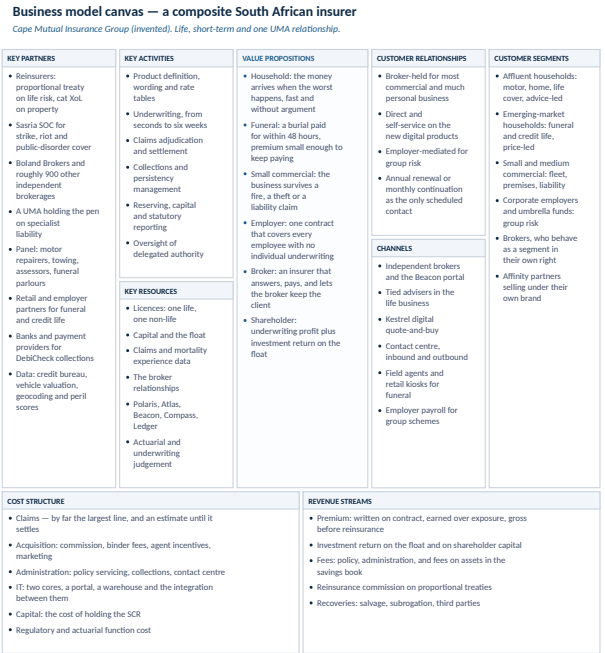


Figure 4.1 The business model canvas for an invented composite South African insurer, Cape Mutual Insurance Group. Not a case study of any real insurer; the shape is typical of a mid-sized composite with brokers, a funeral book, group risk and one UMA.

Now walk the blocks. Each one implies a capability, and most of them imply a system you will have to find, name and decide about.

Customer segments

An insurer that writes affluent household business, emerging-market funeral cover, small commercial and group risk has four businesses, not one market. The segments differ in premium size by three orders of magnitude, in the advice they need, in how they pay, and in what "good service" means. South African funeral cover is the extreme: 6.2 million new individual recurring-premium funeral policies were written in 2025, which is 57% of all new risk policies [2]. Against that, the same industry writes R6 million property risks that take three weeks to quote.

There is a fifth segment on the canvas that outsiders always miss: **the broker is a customer**. The broker chooses where to place the business, and the insurer competes for the broker's attention with service levels, binder terms and commission. If your canvas has brokers only under "channels" you will under-build everything the broker touches, because channels get integration budget and customers get experience budget.

Architecture implication. Segments that differ this much cannot share one cost-to-serve. The funeral business needs a per-policy administration cost measured in rands per year; the commercial business can afford a human underwriter for six hours. One policy administration system configured to serve both ends up too expensive for the one and too rigid for the other. This is the single most common cause of the two-core estate you will walk into, and it is often the right answer rather than an accident.

Value propositions

Write these as the customer would say them after the event, not as marketing says them before it. For a household: *the money arrived in four days and nobody argued.* For a funeral policyholder: *they paid within 48 hours and the premium is small enough that I kept paying it.* For an employer: *every employee is covered and I did not have to underwrite anybody.* For a broker: *they answer the phone, they pay claims, and they do not try to take my client.*

Notice that three of the four are about the claim and one is about the relationship. The product is a promise, and the promise is tested exactly once. That is why claims capability dominates the architecture of a retail insurer, and why a programme that modernises quote-and-buy and leaves claims alone is improving the part the customer does not judge you on.

Architecture implication. If the value proposition is speed and the absence of argument, then the architecture must answer two questions fast: is this person covered, and was the cover in force on the date of the event. Both are as-at questions. Both are the questions legacy policy systems answer worst, because they store current state.

Channels

Cape Mutual sells through independent brokers, tied life advisers, a digital quote-and-buy stack, a contact centre, field agents and retail kiosks, and employer payroll. Six channels is normal. Santam reported direct channels at 22% of business against a target above 30% [3], which tells you that even the insurers with the strongest direct brands run mixed distribution and intend to keep doing so.

Architecture implication. Channels set the latency budget and they set where data is created. An aggregator expects a priced, bindable quote in close to real time, and pays per lead, per policy or by commission [4]; a commercial submission can take three weeks. The honest conclusion is one rating and binding service behind many front ends, with what each channel may do — quote, bind, issue, endorse, settle — held as enforced configuration rather than as code in each front end. The dishonest conclusion, which you will find already built, is one stack per channel, each with its own copy of the product rules.

Customer relationships

In most of this book's market the insurer does not hold the relationship. The broker does. For funeral business the agent does, and when the agent moves the book often moves with them. For group risk the employer and the benefit consultant do, and the member — the person actually at risk — has no relationship with the insurer at all until there is a death.

Architecture implication. An insurer that does not hold the relationship cannot assume it holds the data. The customer record of record may be in a brokerage's administration system. That single fact decides whether a customer data platform is a sensible investment or a monument. Decide it deliberately, and write the decision down.

Revenue streams

Premium, investment return on the float, fees, reinsurance commission, recoveries. Only the first is obvious and even that one is four numbers: written or earned, gross or net [5]. The second is large and volatile. The fifth — salvage, subrogation and third-party recoveries — is the one nobody puts on a canvas, and it is the one that quietly disappears when a claims system is replaced without anybody owning recoveries.

Architecture implication. Each revenue stream needs a derivation that can be reproduced. If the finance team cannot show how the premium number on the board pack was built from policy transactions, the board pack is an opinion. Chapter 6 does the arithmetic; the canvas is where you notice there are five streams rather than one.

Key activities

Define the product. Underwrite. Administer. Collect. Pay claims. Reserve, report and hold capital. Oversee everybody to whom you have delegated authority.

The last one is the South African addition and it is not optional. Under a binder or a UMA arrangement the insurer remains liable for every policy issued in its name and must pay valid claims [6], while the decisions are made in somebody else's system. Oversight is therefore an activity of the insurer, with its own data requirement, not an audit event.

Architecture implication. Rank these activities by how often they change. Product definition changes monthly; underwriting rules weekly in personal lines; reserving methodology annually; the general ledger chart of accounts almost never. An architecture that puts a monthly-changing thing inside an annually-changing system has made a decision, usually without noticing.

Key resources

Two licences. Capital and the float. Claims and mortality experience data. The broker relationships. The systems. Actuarial and underwriting judgement.

Put the data on the resource list in writing, because it is the resource most often treated as a by-product. An insurer's pricing advantage is its own claims experience. A core replacement that migrates balances but not history destroys the resource and nobody notices for two years, when the pricing actuary cannot fit a model.

Key partners

Reinsurers, who are partners and not suppliers: on a proportional treaty the reinsurer takes a share of every risk and pays a ceding commission, so the treaty is part of the product economics. Sasria, the state-owned special-risks insurer that covers strike, riot and public-disorder loss [14], which every South African short-term policy system has to accommodate — Chapter 2 explains how. Brokers and the UMA. The repair, towing, assessing and funeral-parlour panels. Retail and employer partners for mass-market products. Banks and payment providers for collections. Data providers for credit, vehicle values, geocoding and peril scores — and South African insurers have rebuilt pricing around spatial analytics and geocoded climate risk rather than historical trends [7].

Architecture implication. Every partner on that list is an integration with a contract behind it, and most of them are bidirectional. The panel repairer needs the claim; the claim needs the repair cost. The reinsurer needs the bordereau; the recovery needs the treaty terms. Count the partners on your canvas, then count the integrations in your estate. If the second number is smaller, something is being done by email.

Cost structure

Claims first, by a distance, and claims are an estimate until they settle. Acquisition second: commission, binder fees, agent incentives, marketing. Then administration, then technology, then the cost of holding regulatory capital.

Architecture implication. When non-life profitability is forecast to fall — global non-life return on equity is projected to drop from 14% in 2025 to 7.7% by 2028 [8] — cost-to-serve becomes the primary measure and claims leakage becomes the primary target. That changes which projects get funded. Straight-through processing rates and expense ratios outrank new features, and an architecture roadmap that cannot be expressed as a movement in the combined ratio will not survive a planning round.

What the canvas leaves out

Be clear about the instrument's limits before you take it into a room with an actuary.

The canvas has no time axis, and insurance is a business about time. A life policy sold at age 32 has its price fixed for forty years; a motor policy is re-priced annually. Nothing on the nine blocks shows that difference, and it is the difference that decides whether a pricing mistake is correctable.

The canvas has no risk. It shows that premium comes in and claims go out, but not that the cost of goods sold is unknown at the point of sale, that the biggest number on the balance sheet is an estimate, or that the business can be destroyed by one accumulation nobody measured. Add a note on the canvas naming the two or three risks that would end this business model. For a South African short-term insurer in 2026, weather catastrophe is one of them [7].

The canvas has no capital. An insurer must hold capital against the risks it writes, and the cost of that capital is a real cost that the canvas records only as a line in the cost block. Two business models with identical canvases can have very different capital intensity, and the capital-light one wins.

The canvas has no regulator, and in financial services the regulator is a party to the business model, not an externality. Two supervisors here — one prudential, one conduct — each with its own data demands [11].

None of this invalidates the instrument. It means the canvas is the first page, not the only page. Chapters 6 and 15 supply what it omits.

The canvas is different in every line of business

Here is where the instrument earns its place. Take the same nine headings to a direct personal-lines motor insurer and to a group risk business, and you get two enterprises with almost nothing in common.

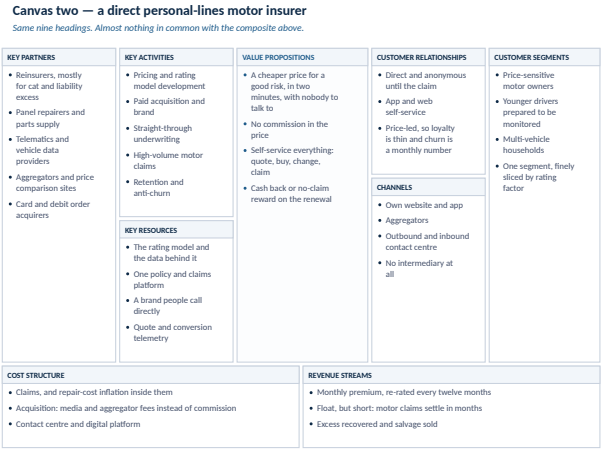


Figure 4.2 The same nine headings for a direct personal-lines motor insurer. Compare the channels, the customer relationships and the acquisition cost with the composite above.

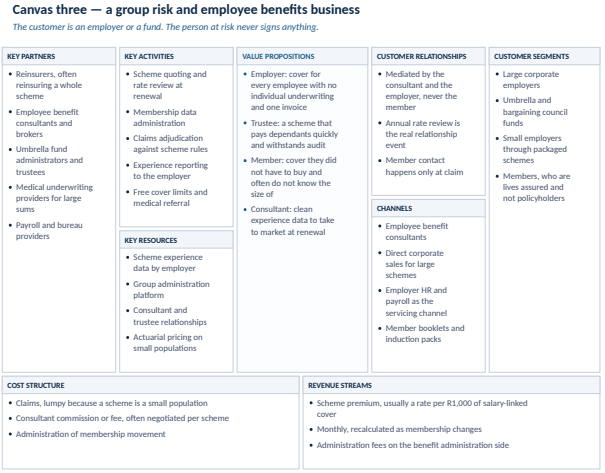


Figure 4.3 And for a group risk and employee benefits business. The customer is an employer or a fund; the person at risk is a member, not a policyholder, which is why a retail party model cannot hold this business.

Read the three canvases side by side.

Block	Composite	Direct motor	Group risk
Who the customer is	Households, small commercial, employers, and brokers	One segment sliced finely by rating factor	An employer or a fund; the insured person is not the customer
Who holds the relationship	Mostly the broker	The insurer, thinly	The consultant and the employer
What is actually sold	A promise mediated by advice	A price, in two minutes	A scheme, renegotiated annually
The decisive activity	Underwriting across five orders of magnitude of effort	Rating model development	Scheme pricing on a small population, and membership administration
Where the data is created	Partly outside the insurer	Entirely inside	In a payroll system the insurer does not own
Acquisition cost	Commission and binder fees	Media and aggregator fees	Consultant fee, negotiated per scheme
The renewal event	One at a time, broker-driven	Monthly continuation, churn-driven	One annual rate review per scheme, consultant-driven
What breaks the model	Losing a brokerage	A rating model that drifts	A membership schedule that is wrong

Three consequences follow, and they are the reason to do this exercise before any platform decision.

One. "A policy administration system" is not one requirement. The direct insurer needs sub-second rating, a thin party model and a high-volume motor claims process. The group business needs no rating engine of that kind at all; it needs to ingest a monthly membership schedule, compute salary-linked cover, apply free cover limits, and produce scheme-level experience. Putting both on one platform is possible, but it is a decision with a price, and the price should be stated before the contract is signed rather than discovered in year two.

Two. The canvas tells you which capabilities are shared and which only look shared. Payments are genuinely shared. Document generation is genuinely shared. Party is shared only if you model it properly — the group business has members who are lives assured and not policyholders, and a

party model built for retail will not hold them, as Chapter 2 sets out. Rating is not shared at all.

Three. The cost structures point in different directions. The direct insurer spends on acquisition and technology and almost nothing on intermediaries. The group business spends on administration and consultant remuneration. A shared-services programme that averages across them will overcharge the business with the thin cost base, which will then leave and buy its own system. This is how estates fragment, and it is a modelling failure before it is a governance failure.

> Do not draw one canvas for a group and call it the business model. A composite insurer's canvas is a superimposition of three or four canvases, and the blocks that conflict are exactly where the architecture arguments will happen.

Customer journeys, done honestly

A customer journey map is the second instrument. Done badly it is a row of smiling faces above a row of frowning ones. Done properly it carries four rows and earns its place in a technical review:

1. **What the customer does** — in their words, with the inconvenience left in.
2. **What the insurer does** — including "nothing", which is often the true answer and always the interesting one.
3. **Which systems are touched** — named, including the ones outside your organisation.
4. **Where it breaks** — the drop-outs, the dropped threads, the data that does not survive the hand-off.

The fourth row is the one that makes the map useful to an architect. Everything else is description. Four maps follow: buying short-term cover through a broker, buying funeral cover at a kiosk, making a motor claim, and a scheme member's family claiming a death benefit. They are drawn from how these processes typically run in the South African market; the systems are Cape Mutual's invented ones.

Buying short-term cover through a broker

Journey one — buying short-term cover through a broker

A small commercial client, R6m of property and a fleet of nine vehicles.

Stage	What the customer does	What the insurer does	Systems	Where it breaks
Trigger	Buys a building, loses a tender for want of liability cover, or gets a renewal increase he will not accept.	Nothing. The insurer does not know this is happening.	None	The insurer has no signal and no chance to retain.
Needs analysis	Spends an hour with the broker listing vehicles, premises, stock values and contracts.	Nothing yet. The data is being created in the broker's system.	Broker system	Sums insured guessed, not surveyed. Underinsurance is created here and discovered at claim.
Market the risk	Waits. Asks twice. Compares two or three quotes on price and excess.	Receives a submission by email or portal. Underwriter prices, may ask for a survey, may decline.	Beacon, Atlas	Three weeks of silence. Re-keying from email. Each insurer asks for the same data in a different shape.
Advice and bind	Takes the broker's recommendation, signs, pays the first premium.	Accepts the risk, or the broker binds it under a binder within mandate.	Beacon, Atlas	Record of advice lives only with the broker. Cover incepts before the insurer has the data.
Documents	Receives a schedule. Reads it only after a loss.	Issues schedule and wording; files the policy version.	Atlas	Document and system disagree. The broker's own schedule differs from the insurer's.
Mid-term change	Buys a tenth vehicle and phones the broker.	Endorses the policy, pro-rates the premium, re-rates or does not.	Beacon, Atlas	Cancel-and-rewrite destroys continuous cover history. Endorsement never reaches the insurer.
Renewal	Sees an increase, asks the broker to re-market.	Issues renewal terms; may lose the account without ever speaking to the client.	Atlas, Beacon	The insurer learns it has lost the client from a cancellation notice.

▲ make-or-break moment Systems named in blue are invented Cape Mutual systems.

Figure 4.4 Buying short-term cover through a broker. Note the first row: for the first third of its own sales process the insurer does not know the customer exists.

The first thing to notice is the first row. The customer's journey starts with a trigger — a new building, a lost tender, a renewal increase — and the insurer does not know it is happening. For broker-placed business the insurer is usually absent from the first third of its own sales process. Everything it will later know about this risk is being created in a brokerage's system during an hour-long conversation it does not attend.

The second thing is the length of the middle. Marketing a commercial risk takes weeks, most of it silence. The broker chases, the underwriter asks for a survey, the client asks twice. There is no status anybody can see. When an insurer wins the business it often wins it on turnaround rather than price, which is a statement about workflow rather than about underwriting.

Where it breaks.

- *Sums insured are guessed.* Nobody surveys a R6 million building to sell a policy. Underinsurance is created at this point and discovered at claim, when average applies and the client learns that a 40% shortfall in sum insured means a 40% reduction in the settlement. The customer experiences this as a repudiation. It is actually a data defect two years old.
- *Submissions arrive as email and get re-keyed.* Each insurer asks for the same data in a different shape, so the broker maintains the risk in their system and re-enters it per insurer. Every re-key is a divergence between the broker's record and yours.
- *The record of advice lives with the broker.* Under South African conduct rules the advice process is documented and regulated [9], and the document sits in a brokerage. When a complaint arrives at the ombud, the insurer must ask a third party for the evidence of its own sale.
- *Cover incepts before the insurer has the data.* Under a binder, the broker binds; the entry in your system follows. In the gap, your exposure reporting is wrong and you do not know by how much.

- *Endorsements evaporate.* The tenth vehicle is added by telephone. Whether it reaches the insurer's system, and with what effective date, decides whether a claim three months later is paid.
- *You learn you have lost the client from a cancellation notice.* There is no earlier signal, because every earlier signal belongs to the broker.

Buying funeral cover at a kiosk or from a field agent



Journey two — buying funeral cover at a kiosk or from a field agent

Mass market, multi-life, cash-constrained. The volume product of South African insurance.

Stage	What the customer does	What the insurer does	Systems	Where it breaks
Trigger	A relative dies and the family cannot find R20,000, or an agent arrives at the taxi rank.	Runs a campaign, pays the agent per policy.	None	Volume-paid agents sell to people who cannot sustain the premium.
Sale	Gives an ID number, names six family members, signs on a tablet in under fifteen minutes.	Light or no underwriting. Waiting period instead of a medical.	Kestrel, Polaris	Relationships to the lives covered captured as free text. Duplicate policies on the same family, sold twice.
Mandate	Gives bank details or agrees to a payroll or retail deduction.	Sets up a DebiCheck mandate and a collection day.	Collections	Mandate fails authentication. Collection day falls after the money is gone.
Premium	Pays, misses, pays again. Rarely cancels on purpose.	Retries, dunning, grace period, then lapse.	Collections, Polaris	Most lapses here are failed collections, not decisions. Cover status at a given date becomes ambiguous.
Claim	Phones or walks in within a day of the death, needs money for the funeral within 48 hours.	Verifies identity, death certificate, waiting period, premium history.	Polaris, Compass	Premium history cannot prove cover was in force. Claim declined for arrears the customer did not know about.
Payment	Receives cash into an account, or the parlour is paid directly.	Pays the nominated beneficiary or the service provider.	Polaris, Ledger	Beneficiary details years out of date. Payment to the wrong person is almost impossible to reverse.
After	Tells the family whether the insurer paid. That is the entire brand.	Usually does nothing. The surviving policies on the same family are not reviewed.	Compass	No household view, so no cross-sell and no retention of a family that just saw the product work.

▲ make-or-break moment Systems named in blue are invented Cape Mutual systems.

Figure 4.5 Buying funeral cover at a kiosk or from a field agent — the highest-volume customer journey in South African insurance, and the one least like the textbook.

This is the highest-volume customer journey in South African insurance and the least like the textbook. Fifteen minutes, a tablet, an identity number, six family members named as covered lives, a waiting period instead of underwriting, and a debit order against an income that will not always be there. Then 8.7 million risk policies lapsed across the industry in 2025, against 10.8 million new ones written [2].

Those two numbers side by side are the business model. The product is sold in bulk and lost in bulk, and the economics depend on how much of the loss is avoidable.

Where it breaks.

- *Volume-paid distribution sells to people who cannot sustain the premium.* Commission or incentive paid per policy, with a clawback window shorter than the lapse curve, produces exactly this. It is a remuneration design that shows up as an architecture requirement: you will be asked to build persistency-adjusted agent remuneration, and you will need lapse attribution by agent to do it.
- *The family is captured as free text.* Six covered lives with relationships typed into a notes field cannot be used for a claim, for duplicate detection or for cross-sell. Funeral cover is multi-life by nature, and modelling it as one policyholder with a comment is the most common data modelling error in the South African mass market.
- *The mandate fails.* Debit order authentication fails, or the collection day falls after the money has gone. In low-income markets collection *failure* rather than deliberate cancellation is the leading cause of lapse, which means retry-day strategy is a retention capability and belongs in the architecture as one.
- *Cover status at a given date is ambiguous.* Two missed premiums, one retry, a grace period and a reinstatement produce a policy whose in-force status on the day of death is a matter of interpretation. If the system cannot derive that status deterministically from the premium ledger, the claims handler will decide it, and the ombud will later disagree.
- *The collection instrument itself changed under the industry's feet.* The early debit order streams, AEDO and NAEDO, were switched off on 31 October 2021 [12]. DebiCheck replaced them: the customer authenticates the arrangement once with their own bank, and the bank then permits only collections within the approved terms [13]. That last clause is an architecture requirement hiding in a payments scheme. If the premium changes — an annual escalation, an added life, a benefit increase — the mandate may have to be re-authenticated, and a mandate that cannot be amended silently stops collecting.
- *The beneficiary record is years old.* Paying the wrong person is close to irreversible, and in this market the claim arrives within a day of the death with a family standing in the branch.
- *Nothing happens after the claim is paid.* The insurer has just done the only thing the customer ever wanted from it, in front of an extended family, and has no household view with which to do anything about it.

> Penetration figures for funeral cover are hard to source. The best available study, from 2022, found that about 60% of South Africans report holding an insurance product but only 19% once funeral cover is excluded [10]. Treat current penetration as unknown rather than quoting a newer number you cannot source.

Making a motor claim

Journey three — making a motor claim

The claim is the product as the customer experiences it. Everything before was paperwork.

Stage	What the customer does	What the insurer does	Systems	Where it breaks
The loss	Collides at 06:40 on a Monday. Shaken, at the roadside, on a phone with low battery.	Nothing until notified.	None	No towing or help at the moment of maximum need, so the tow is arranged by a stranger with an incentive.
Notification	Phones the broker, the insurer or the app, whichever number he can find.	Registers the claim, captures date, time, place, parties, damage.	Atlas, Kestrel, Beacon	Three notification channels, three data shapes. Duplicate claims. Loss date captured as the reporting date.
Cover check	Waits to be told whether he is covered.	Validates cover against the policy version in force at the date of loss, not today's version.	Atlas	Current-state policy tables cannot answer the as-at question, so cover is confirmed by hand.
Assessment	Surrenders the car, chases for news, borrows a car he is not insured to drive.	Assessor inspects, quotes repair, decides repair or write-off, posts a case reserve.	Atlas, Sentinel	Assessor and repairer data arrive as PDFs. Fraud screening happens after payment instead of at registration.
Decision	Learns the excess, the betterment, and that the salvage value is deducted.	Authorises, declines, or repudiates with reasons.	Atlas	Repudiation wording that an ombud will not accept. No audit trail of why the decision was made.
Repair or settle	Collects a repaired car, or receives a payment less than the car was worth to him.	Pays the repairer or the policyholder; closes the file.	Atlas, Ledger	Payment made before salvage and third-party recovery are set up, and then nobody owns them.
Recovery	Hears nothing, and is surprised by the excess refund if it ever comes.	Pursues the third party, sells the salvage, claims on reinsurance if the event qualifies.	Atlas, Ledger	Recoveries are the quietest leak in the business. Nobody complains when they do not happen.

△ make-or-break moment

Systems named in blue are invented Cape Mutual systems.

Figure 4.6 A motor claim, from the roadside to the recovery nobody sees. More systems touch this transaction in a shorter time than any other in a short-term insurer.

This is the customer journey that the customer judges you on, and the one where the most systems touch one transaction in the shortest time.

Start at 06:40 on a Monday. The policyholder is at the roadside and has to decide who to phone. If the insurer is not the easiest number to find, the tow is arranged by whoever arrives first, and the vehicle enters a supply chain the insurer does not control at a price it did not agree.

Then the cover check, which is the architectural heart of the whole chapter. The question is not "is this person covered" but "was this person covered at 06:40 on that Monday, under the terms in force at that moment". A system that stores current state can answer the first question and only guesses at the second. Chapter 7 calls this the as-at problem; here it is the reason a claim sits for two days while somebody reconstructs a policy history from endorsement letters.

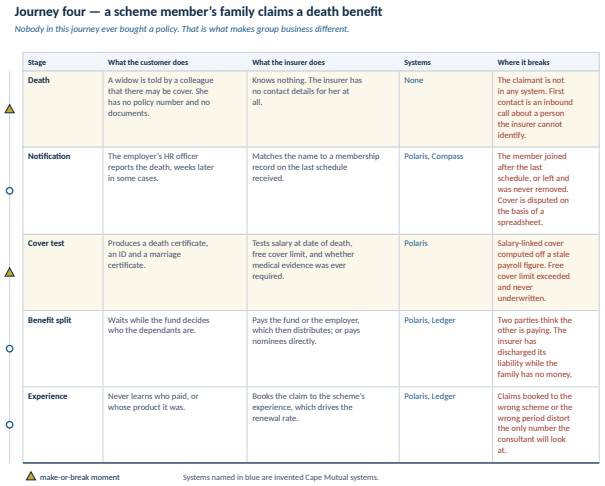
Then assessment, where the data arrives as PDFs from assessors and repairers, and fraud screening competes with cycle time. Screen at registration and you catch things before money moves; screen after payment and you are running a recoveries exercise against someone who has already gone.

Then the decision, where the customer discovers the excess, the betterment and the salvage deduction. Most complaints that reach an ombud are about this moment, and most of them are about being told badly rather than about being told no.

Then recovery, which the customer never sees. Salvage, subrogation, the third party's insurer, and the reinsurance recovery if the event qualifies. Excess-of-loss recovery can only be computed per loss event, after aggregating claims across policies within an hours clause and a geography — a different data problem from the per-policy cession, and the place most reinsurance leakage happens.

Where it breaks. Three notification channels producing three record shapes; loss date captured as reporting date; cover validated by hand; assessor data outside the system; fraud scoring after payment; and recoveries that nobody owns because the claim file is closed. Weather catastrophes were the largest force on South African short-term claims costs in the first half of 2026 [7], which adds one more: surge. Ten to fifty times normal volume for a week. A claims process that works at normal volume and collapses at surge volume has not been designed, it has been observed.

A scheme member's family claims a death benefit



Journey four — a scheme member's family claims a death benefit

Nobody in this journey ever bought a policy. That is what makes group business different.

Stage	What the customer does	What the insurer does	Systems	Where it breaks
Death	A widow is told by a colleague that there may be cover. She has no policy number and no documents.	Knows nothing. The insurer has no contact details for her at all.	None	The claimant is not in any system. First contact is an inbound call about a person the insurer cannot identify.
Notification	The employer's HR officer reports the death, weeks later in some cases.	Matches the name to a membership record on the last schedule received.	Polaris, Compass	The member joined after the last schedule, or left and was never removed. Cover is disputed on the basis of a spreadsheet.
Cover test	Produces a death certificate, an ID and a marriage certificate.	Tests salary at date of death, free cover limit, and whether medical evidence was ever required.	Polaris	Salary-linked cover computed off a stale payroll figure. Free cover limit exceeded and never underwritten.
Benefit split	Waits while the fund decides who the dependants are.	Pays the fund or the employer, which then distributes; or pays nominees directly.	Polaris, Ledger	Two parties think the other is paying. The insurer has discharged its liability while the family has no money.
Experience	Never learns who paid, or whose product it was.	Books the claim to the scheme's experience, which drives the renewal rate.	Polaris, Ledger	Claims booked to the wrong scheme or the wrong period distort the only number the consultant will look at.

▲ make-or-break moment

Systems named in blue are invented Cape Mutual systems.

Figure 4.7 A scheme member's family claiming a death benefit. The claimant has never been a record in any of the insurer's systems, and the only evidence of cover is a monthly schedule.

This is the short one, and it is here because it breaks the model that the other three share.

Nobody in this customer journey ever bought a policy. The member was covered because they were employed. The widow calling the insurer has no policy number, no documents and no record in any of its systems. The contracting party is the employer or the fund; the member is a life assured in a dated role; membership arrives as a monthly schedule, and that schedule is the only evidence of who was covered when.

So the failures are different in kind. The member joined after the last schedule was sent, or left and was never removed, and cover is argued over a spreadsheet. Salary-linked cover is computed off a stale payroll figure. The free cover limit — the amount above which the member should have been medically underwritten — was exceeded, and nobody checked at the time. The benefit is paid to the fund, which then has to identify dependants, and the family waits while two organisations each believe the other is paying.

For an architect the lesson is blunt: a party model and a policy model built for retail insurance will not hold group business. Member, employer, fund, scheme, benefit category and dependant are first-class objects. If your conceptual model has "customer" and "policy" you have already excluded a third of a composite insurer's book.

The moments that decide the relationship

Every customer journey has a handful of moments that carry the whole relationship, and they are not evenly spread. They are marked on the four figures. Collected, they look like this.

Customer journey	The moment	What the customer is deciding	What the architecture must do at that moment
Broker-placed commercial	The three weeks of marketing the risk	Whether this broker, and by extension this insurer, is competent	Make status visible to the broker without a telephone call; turn a submission into a priced answer without re-keying
Broker-placed commercial	Inception and the first schedule	Whether what was promised is what was bought	Produce a schedule the insurer can reproduce from its own system, on the day cover starts
Funeral	The fifteen minutes of the sale	Whether this is worth R180 a month out of a tight income	Capture the family properly, once, with relationships as data; set a collection day against the salary date
Funeral	The 48 hours after a death	Everything. This is the only test the product ever faces	Prove cover was in force without a human reconstructing a premium history
Motor	The roadside, in the first ten minutes	Who to call, and whether to accept the tow in front of them	Be the easiest number to find; register a claim from any channel into one record
Motor	Being told the excess, the betterment and the salvage deduction	Whether they were treated fairly, which is not the same as whether they were paid	Produce a decision with its reasons attached, in language an ombud will accept
Group death benefit	The first call from a person you have no record of	Whether the employer's benefit is real	Identify a claimant from a name and an employer, from the schedule rather than from a policy

Two patterns are worth noticing. The moments cluster at the beginning and at the claim, and the middle of a policy's life — the part where almost all the servicing cost sits — contains almost none of them. And the make-or-break moments are disproportionately about *information*, not money: silence during marketing, a decision explained badly, a family that cannot find out whether there is cover. Those are the cheapest failures to fix and they are usually nobody's project.

> Resist the temptation to score these moments out of ten and plot a satisfaction curve. The curve will be invented. What is defensible is the list of moments, the decision the customer is making at each, and what the systems must be able to do. Keep the map to things you could defend in front of the head of claims.

The same five failures, every time

The same five failures, in all four journeys

Different industries, different products, one list.

	Broker	Funeral	Motor claim	Group death
Identity: the person is not one record	The client belongs to the broker. You hold a policy, not a person.	Six lives on one policy, related by free text.	Three notification channels create three records of one person.	The claimant has never been a record at all.
As-at: the state at a past date cannot be derived	Which schedule was in force when the tenth vehicle was added?	Was cover in force on the day of death, given the arrears?	Which policy version applied at 06:40 on the Monday?	Which membership schedule was current at date of death?
The document and the system disagree	The broker's schedule and the insurer's differ, and the client holds the broker's.	Nothing was ever issued that the family can find.	The repudiation letter cites a clause the system never applied.	The member booklet promises what the scheme rules do not.
Hand-off: the thread is dropped between parties	Submission by email, re-keyed, and nobody owns the gap.	Agent, kiosk, collections and admin never meet.	Assessor, repairer, salvage and recoveries each in their own system.	Insurer, fund and employer each think the other is paying.
Status: nobody can say what happens next, or when	Three weeks of silence during marketing.	Lapsed by failed collection, and the customer is not told.	The single most common complaint: no one told me anything.	Weeks pass while dependants are identified.

Author's assessment from South African engagements, not measured data.

Figure 4.8 Four products, four customers, five failure classes. The author's assessment from South African engagements rather than measured data; the pattern is the point.

Four customer journeys, four products, four very different customers, and the failures sort into five classes. That is the finding worth taking out of this chapter.

Identity. The person is not one record. They are a broker's client, three channel records, six names in a free-text field and a membership line. Every cross-sell business case, every fraud model, every single-customer-view programme and every fair-outcomes report depends on resolving this, and it is almost never solved inside the policy administration systems.

As-at. The state of a contract at a past date cannot be derived. This is the same defect in all four, and it is the one with legal consequences, because every claim is adjudicated against the terms in force at the date of loss.

The document and the system disagree. The policy wording is the product. When the schedule the client holds is not reconstructable from the system that administers the cover, you have two products and one of them is enforceable against you.

The hand-off. Broker to insurer, agent to administration, assessor to repairer to recoveries, insurer to fund to family. Each hand-off is a point where the thread is dropped and nobody is accountable for the gap.

Status. Nobody can say what happens next, or when. This is the cheapest of the five to fix and the one most reliably ignored, because it produces no reconciliation break — only complaints.

From two pages to an architecture

The canvas and the customer journeys are not decoration for a strategy pack. They produce three concrete things.

First, a capability list with a reason attached. Capabilities derived from the canvas inherit a justification you can defend, which is the difference between a capability model that is used and one that is admired.

Capability	Where it comes from	What it needs to be able to do
Know the party and the household	Four segments, multi-life funeral, group members	Resolve one person across channels, policies and roles; hold households and schemes
Define product, wording and rates	Product is the key activity; wording is the product	Effective-dated wording, rates and rules; reconstruct any policy's terms as at any date
Quote, rate and bind	Six channels, one rating truth	One service, many front ends, channel entitlements as configuration
Administer a policy as at a date	The as-at failure in all four customer journeys	Event-sourced, effective-dated policy history, not mutable current state
Collect premium and derive status	Mass-market lapse economics	Retry strategy as a decision; in-force status derived from the premium ledger
Adjudicate and settle claims	The value proposition is the claim	Cover validation against the as-at policy, supply chain, fraud at registration
Recover and cede	Revenue streams nobody names	Per-claim recovery and per-event reinsurance recovery, derived rather than reconstructed
Oversee delegated authority	UMA and binder arrangements	Entitlements, limits, bordereau ingestion, aggregate exposure, exception monitoring

Second, a map from those capabilities to the estate. This is the figure to draw second, after the canvas and before anybody says the word "roadmap".

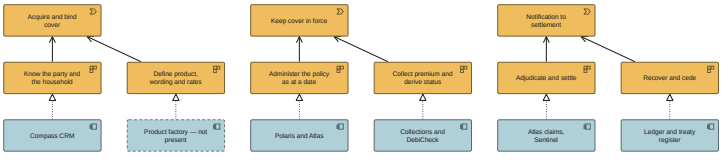


Figure 4.9 From value stream to capability to application, with the gap drawn honestly: Cape Mutual has no product factory, so the capability that changes most often is realised twice, in two cores, with the wording somewhere else.

Draw it with the gaps honest. At Cape Mutual there is no product factory: product is configured in two places, once in Polaris and once in Atlas, with the wording in a document management system that neither of them reads. Drawing the empty box is the most valuable thing an architect does in the first month, because an empty box on a capability map is a budget conversation and a paragraph in a report is not.

The same two pages also tell you which systems you must find, and in which order to go looking. Work from the customer journeys, because they name the systems that are actually touched rather than the ones on the architecture poster.

What the two pages show	The system category it implies	What to ask for in week one
Four segments with incompatible cost-to-serve	Policy administration, usually more than one	Which book runs on which, and what each costs per policy per year
Six channels, one rating truth	Quote and rating service, portals, partner APIs	Where rates are held, and how many copies exist
The broker holds the relationship	Intermediary management, commission, binder feeds	Who masters the client record, and what arrives as a bordereau
The claim is the product	Claims, assessing, fraud, supplier network, payments	How cover is validated at the date of loss, and by whom
Mass-market lapse economics	Collections, mandate management, dunning	Share of collections by instrument, and the retry rules
Delegated authority	Bordereau ingestion, exposure aggregation, oversight reporting	How much of the book the insurer does not transact itself
Reserving and reporting on all of it	Data warehouse, actuarial platform, ledger and sub-ledger	Whether a loss development triangle can be produced without a spreadsheet

Two cautions. A system category is not a product: "collections" may be a module of the policy system, a bank's service and three spreadsheets. And the systems that appear in the customer journeys but not on the architecture poster — the assessor's mobile application, the brokerage's own administration platform, the agent's tablet — are part of your architecture whether or not you own them.

Third, an order of work. You cannot model everything. Rank by two things only: how much of the canvas and the customer journeys a concept touches, and how much the business currently disagrees about it.

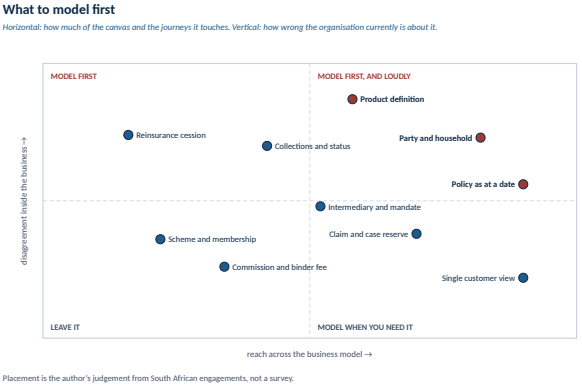


Figure 4.10 What to model first. Reach across the business model on one axis, how much the business disagrees with itself on the other. Placement is the author's judgement.

That is a judgement, not a measurement, and it is drawn from South African engagements rather than from a survey. The top right is where to start: party and household, product definition, and the policy as at a date. These three touch every block of the canvas and every stage of every customer journey, and in most insurers three people will give you three different definitions of each.

> Everything in Part II and Part III of this book — the processes, the data model, the application model, the reference architecture — is an elaboration of these two pages. If the canvas and the customer journeys are wrong, the elaboration is wrong, and it will be wrong at a scale that takes two years to discover.

A last practical note. Do both instruments with the business in the room, not at your desk. The canvas takes ninety minutes with the right six people and will produce at least one argument you did not expect — usually about who the customer is, occasionally about whether the broker is a channel or a customer, and frequently about whether a line of business is actually profitable. Write the arguments down. They are the real output. A canvas everybody agreed with immediately is a canvas that was filled in from the annual report.

THE CHAPTER IN TEN LINES

- A system matters in proportion to the part of the business model it carries and the part of the customer's experience it decides, which is why the canvas comes before the estate review.
- Fill the canvas with named segments, named partners and named systems: a canvas that says 'customers' and 'peace of mind' is worse than none, because it will be believed.
- A composite insurer has four or five customer segments whose cost-to-serve differs by orders of magnitude, and the broker is one of them.
- The value proposition is tested exactly once, at the claim, so claims capability outranks quote-and-buy in any retail insurer's architecture.
- The same nine headings produce three different enterprises for a composite, a direct motor insurer and a group risk business — which is why 'a policy administration system' is not one requirement.
- The canvas has no time, no risk, no capital and no regulator in it; add those as notes or it will mislead an actuary.
- A customer journey map earns its place only when it names the systems touched and the point at which the data breaks.
- Across buying through a broker, buying funeral cover, claiming on motor and claiming a group death benefit, the failures sort into five classes: identity, as-at state, document versus system, the hand-off, and status.
- The make-or-break moments cluster at the sale and at the claim, and most of them are failures of information rather than of money.
- Model party and household, product definition and the policy as at a date first: they touch every block of the canvas and every stage of every customer journey, and nobody agrees on what they mean.

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CHAPTER 5

Who does what

Every role in an insurer, what the person actually does all day, the decision they own, the data they need and what they will complain about. This is the chapter that lets you run a discovery session without being found out.

The org chart will not help you

You will be given an organisation chart in your first week. It will show divisions, headcounts and reporting lines, and it will tell you almost nothing, because insurance work is organised around decisions rather than around departments. The person who decides whether a R40 million risk is accepted may be three levels below the person whose name is on the division. The person who decides whether a claim is paid is a clerk with a limit. The person who can stop your programme is a statutory control function with a dotted line to the board and no staff at all.

What you need instead is a map of who owns which decision, what data they need to make it, and where they currently get that data from. That map is what this chapter is. Work through it once and you can run a discovery workshop without being found out — which matters, because the fastest way to lose an insurance audience is to use a role name wrongly in the first ten minutes.

Three warnings before the roles themselves.

Titles vary and the job does not. One insurer's "claims consultant" is another's "claims handler" and a third's "claims technician". One insurer's "underwriter" prices risk; another's checks documents against a checklist and refers everything else. Never take the title. Ask what they decide on their own, and what they have to refer.

The most important people are often outside the company. In a market where brokers hold the client and underwriting management agencies hold the pen, a large part of the operating model is staffed by people who do not appear on any chart you will be shown.

Everyone will tell you their system is the problem. They are usually half right, and the half they are right about is almost never the half they name. Listen for the workaround, not the complaint. When a claims handler says the system is slow, watch what they have open beside it. That spreadsheet is your architecture.

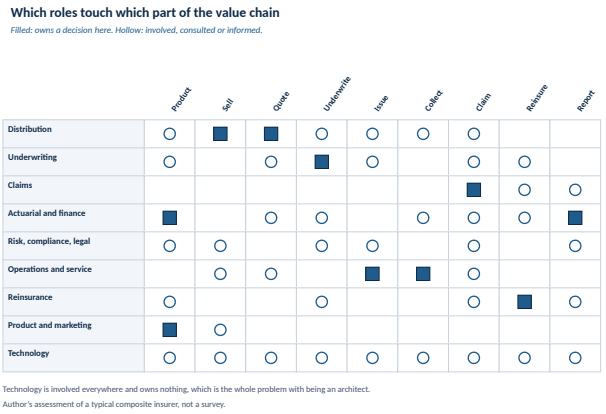


Figure 5.1 Which role groups own a decision at which step of the value chain, and which are merely in the room. The author's assessment of a typical composite insurer rather than a survey.

Distribution

This is the group most outsiders get wrong, and it is also the group that shapes the architecture most, because distribution decides where data is created and who owns the client. Chapter 7 covers the distribution *models*; here we are concerned with the people.

Role	What they do all day	The decision they own	Data they need	Where they live	What they complain about
Tied agent / adviser	Sees clients, does needs analysis, sells one insurer's products	Which product to recommend to this client	Client history, product rules, illustration values	Adviser portal, CRM, illustration tool	Commission statements they cannot reconcile; clawbacks they did not expect
Independent broker	Advises clients, markets risks to several insurers, services the book, often handles claims	Where to place the risk	Risk data, market appetite, claims status	Their own brokerage system, plus your portal	Being made to re-key; no status visibility; slow quotes
Brokerage principal	Runs the firm, negotiates binders and commission, carries the licence	Which insurers the firm supports	Profitability by insurer, retention, compliance exposure	Management reporting, binder agreements	Binder fee caps; data demands; insurers courting their staff
UMA / underwriting manager	Underwrites and administers a book under a mandate from an insurer	Accept, price, and often settle, within mandate	Full policy and claims data for their book	Their own system; the insurer sees a bordereau	Insurer oversight requests; system change they cannot get
Bancassurance desk	Sells cover at the point of credit or in-branch	Which product attaches to which banking event	Customer banking data, credit event triggers	The bank's systems, not the insurer's	Legacy bank platforms; slow product change
Direct contact centre agent	Quotes and sells inbound and outbound, all day, measured by the minute	Nothing. The rules decide; they execute	Quote in one screen, scripts, objection handling	One quote-and-buy stack	Screen count, handover time, repeating what the customer typed online
Aggregator	Collects consumer details once and sends them to several insurers	Which insurers see the lead, in which order	Rateable quote in under a second	Their platform, your API	Insurers who cannot return a bindable price in real time
Field agent (funeral)	Signs up families at taxi ranks, workplaces, kiosks and in homes	Who to approach and what to sell	Identity verification, a working tablet, airtime	A mobile application, often offline	Devices, signal, commission timing
Employee benefit consultant	Advises an employer or fund, runs the annual market review	Which insurer the scheme goes to	Scheme experience data, membership, rates	Their own spreadsheets and the insurer's reports	Experience data that arrives late and does not reconcile

Four of those rows need a note before we get to the broker.

UMA, MGA and coverholder are the same idea in three markets. An underwriting management agency is the South African name; a managing general agent is the American and British one; at Lloyd's the equivalent is a coverholder, authorised by a managing agent to bind business on its behalf. The scale is worth knowing, because it tells you this is a structural feature of the industry rather than a local quirk: Lloyd's reported 3,015 approved coverholders at the end of 2025, and a service company can sub-delegate authority further still [14]. Multi-hop authority chains, data arriving as bordereaux rather than as transactions, and lineage as the dominant problem — that is delegated business everywhere, not just here.

The bancassurance desk does not work for you. It works for the bank, inside the bank's systems, against the bank's targets, and it sells at the moment of a credit event. The insurer's product has to fit into a loan origination flow it does not control. Older bank platforms are called out specifically as a constraint on scaling digital insurance servicing in South Africa [7].

The aggregator is a channel that sets your clock. It does not care about your product; it cares whether you can return a bindable price in real time, every time. Insurers typically pay per lead, per policy or by commission [15]. Once a consumer can see five prices side by side, rating sophistication becomes the only durable defence, which pushes investment into pricing data and the rating engine rather than into the front end.

The contact centre agent owns no decisions at all, which is why their complaints are the most useful in the building. They are the only role that experiences your integration gaps at conversational speed, with a customer listening.

The broker, properly

If you learn one role in this chapter, learn this one. Almost everything outsiders believe about brokers is slightly wrong in a way that produces bad architecture.

What a broker actually does. A broker is not a salesperson for an insurer. A broker acts for the client. The work is: find out what the client has to lose; decide what cover is needed; approach several insurers; compare terms that are not directly comparable; recommend one; place the risk; issue or arrange the documents; service the policy through the year; argue with the insurer when a claim is disputed; and re-market the account at renewal. In commercial lines the broker is the risk manager of a business too small to employ one. That is the value, and it is why the client's loyalty attaches to the broker and not to you.

Who pays them, and how. The client pays a premium. The broker remits it to the insurer net of commission. So the broker is paid by the insurer out of the client's money, which is the structural conflict the whole South African conduct regime is built to manage [1].

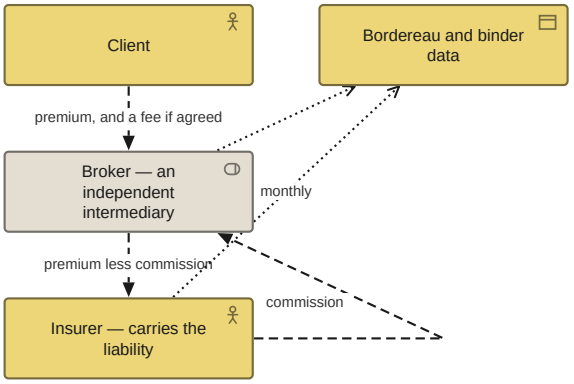


Figure 5.2 How a South African short-term broker is paid, and what travels back the other way, drawn as Flow because what moves is money rather than control. Commission caps are regulated; binder fees must be commensurate with actual cost; profit share is negotiated. Note the object on the right: under a binder the insurer's record of its own book is a monthly file.

Four remuneration forms, which you must be able to tell apart because they land in different systems:

- **Commission.** A percentage of premium, paid by the insurer, capped in regulation. In short-term business the caps sit in regulations made under the Short-Term Insurance Act: 12.5% of the premium on a motor policy and 20% on other short-term policies, with a sliding scale on certain accident and health business [2]. Note the direction of that difference, because most people guess it the wrong way round. Confirm the current numbers before you rely on them: intermediary remuneration is a live item in the conduct reform programme [1].
- **Fees.** Agreed with the client, disclosed, and paid by the client rather than the insurer. Common in large commercial and in employee benefits, rare in personal lines. A fee-based broker is harder for an insurer to influence, which is the point of it.
- **Binder fees.** Paid for performing one of the insurer's own functions. More on this below.
- **Profit share or contingent commission.** A share of the underwriting result on the book the broker places. Negotiated rather than regulated, and the subject of long-standing argument about whether it aligns a broker with the insurer or with the client.

> Commission is paid on premium. If your system pays commission on premium *due* rather than premium *received*, every lapse creates a clawback, and clawback is the single most reliable source of argument between an insurer and its brokers.

Binder versus non-binder. This is the distinction that changes the architecture, and it is specific enough that getting it right will earn you credibility fast.

A **binder** is an agreement under which the insurer outsources one or more of its own functions to an intermediary. The functions named in the legislation are: entering into, varying or renewing a policy; determining the wording; determining premiums; determining the value of policy benefits; and settling claims [3]. The middle three are the underwriting functions. An intermediary with a binder is, for those functions, acting as the insurer.

Remuneration for binder functions must be "reasonable and commensurate with the actual cost of performing the binder function", and for non-mandated intermediary advisers the regulations set caps: 3.5% for entering into policies, 5% where that is combined with one or more of the underwriting functions, 0% for underwriting functions alone, and 4% for settling claims [3]. The zero is deliberate: it stops an adviser being paid to underwrite without also administering, which is where the conflict is sharpest.

A **non-binder broker** advises and places, and the insurer does everything else. The policy is issued by the insurer, the claim is handled by the insurer, and the insurer's systems hold the transaction as it happens.

The architectural consequence is not subtle:

	Non-binder	Binder
Who transacts	The insurer	The intermediary
When the insurer learns	At the time	On the bordereau, weeks later
Where the client record lives	Both, with yours authoritative	The brokerage, with yours derived
Exposure reporting	From your own transactions	From files you ingest and normalise
Claims authority	Yours	Theirs, within limits you must monitor
What the insurer must build	Portals and quote services	Ingestion, reconciliation, oversight and exception monitoring

The vocabulary, and why it matters. South African regulation uses a precise set of words for intermediation, and the everyday words people use in workshops are not those words.



What people say, and what it means in South African law

The everyday words on the left are not licence categories. The words on the right are.

IN THE WORKSHOP	IN THE REGULATIONS
"Broker"	An independent intermediary, usually a licensed FSP in its own name, acting for the client. May or may not hold a binder.
"Agent"	Usually a representative of one insurer, rendering financial services for that insurer and under its licence.
"Adviser"	Not a licence category. It describes the act — giving advice — which is what FAIS actually regulates.
"Consultant"	In employee benefits, the intermediary between an employer or fund and the insurer. Often remunerated by fee.
"UMA" or "MGA"	An underwriting manager: holds a mandate from the insurer to underwrite, price, issue and often settle claims.
"Intermediary"	The umbrella word. Split in the regulations into mandated intermediaries, non-mandated intermediaries and underwriting managers.
"The FSP"	The licensed entity. It must have at least one key individual, who manages or oversees the rendering of financial services.
"The rep"	A representative renders financial services for or on behalf of an FSP, under supervision until competent.

Ask which one the person in front of you is. The answer decides who is liable for the advice.

Figure 5.3 The words used in a workshop, and the words used in the regulations. Only the right-hand column decides who is liable for the advice.

A few points to hold on to. A **financial services provider** is a licensed entity under FAIS; brokers and UMAs are authorised FSPs [4]. Every FSP must have at least one **key individual**, defined as the natural person responsible for managing or overseeing the activities of the entity relating to the rendering of financial services [5]. A **representative** renders financial services for or on behalf of an FSP. An **independent intermediary** acts for the client; a **representative** acts for the insurer. The regulations further split intermediaries into **mandated intermediaries**, **non-mandated intermediaries** and **underwriting managers**, with a non-mandated intermediary defined as a representative or independent intermediary that is neither of the other two [6].

Why an architect should care: these are not synonyms, they carry different liability, and the entitlements in your systems should follow them. "Broker" is not a system role. "Non-mandated intermediary with a claims-settling binder up to R250,000" is.

Why the broker holds the client, and what that does to your data. The client met the broker. The client phones the broker. When the client moves house, the broker finds out; you find out when the policy lapses. The broker has every commercial reason to keep it that way, and in a consolidating market — South African broker consolidation is described as accelerating, strengthening procurement power and influencing pricing [7] — the larger brokerages have the standing to insist on it.

Three consequences you will meet:

1. **Your customer record is a shadow.** The master of name, contact details, household composition and risk history is a brokerage administration system you have no access to. Any single-customer-view business case must say honestly which part of the customer it can actually see.
2. **Your data arrives late and in somebody else's shape.** Under a binder you receive a bordereau. Dozens of partners, dozens of formats, monthly. Normalising them is not an interface project; it is a product with a roadmap.
3. **You are liable for decisions you did not make.** The insurer remains liable for every policy issued in its name and must pay valid claims, even where a UMA or binder holder made the decision [4], and an insurer outsourcing critical functions remains ultimately responsible [1]. Oversight must therefore be designed in: entitlements, limits, exception reports, aggregate exposure, and evidence that somebody looked.

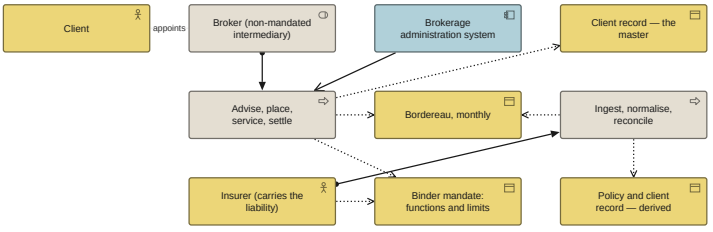


Figure 5.4 What it means when the broker is the system of record. The master client record sits in the brokerage; the insurer's record is derived from a monthly file, while the insurer carries the liability for every policy in it.

> The most expensive architecture mistake in this market is designing a customer-centric estate for a business whose customers belong to somebody else. Decide, explicitly and in writing, whether you are building to own the client relationship or to serve the intermediary who owns it. Both are defensible. Drifting between them is not.

Underwriting

Underwriting is risk selection and pricing. The word covers everything from a rules engine deciding in 40 milliseconds to a team spending six weeks on a petrochemical plant.

Role	What they do all day	The decision they own	Data they need	Where they live	What they complain about
Underwriter (personal lines)	Work the referral queue; the engine handles the rest	Accept, decline, load, exclude, raise the excess	Rating factors, claims history, third-party enrichment	Underwriting workbench or the policy system	Referral rules that send them everything; no history on screen
Underwriter (commercial)	Read submissions, order surveys, negotiate with brokers, draft terms	Terms, limits, deductibles, retention	Survey reports, loss history, exposure data, treaty terms	Email, spreadsheets, a workbench if they are lucky	Submissions arriving as PDFs; no view of accumulation
Chief underwriter	Set appetite, authority limits and guidelines; sign the big ones	What the insurer will and will not write	Portfolio performance by segment, accumulation, treaty capacity	Portfolio reporting, usually rebuilt by hand	Reporting that is a month old; rules in people's heads
Pricing actuary	Build and monitor rating models; measure rate adequacy	The technical price	Exposure and claims at policy level, with rating factors	Modelling tools, a data extract	Data quality; rating factors captured as free text
Referral desk	Clear exceptions against guidelines, fast	Whether an exception is within appetite	The case, the guideline, the precedent	Workflow queue	Volume; cases arriving without the information needed
Medical underwriter (life)	Read medical evidence, apply ratings, order tests	Standard rates, loading, exclusion or decline	Application disclosures, medical reports, reinsurer manuals	Life underwriting system, reinsurer rating manual	Paper; illegible reports; chasing evidence

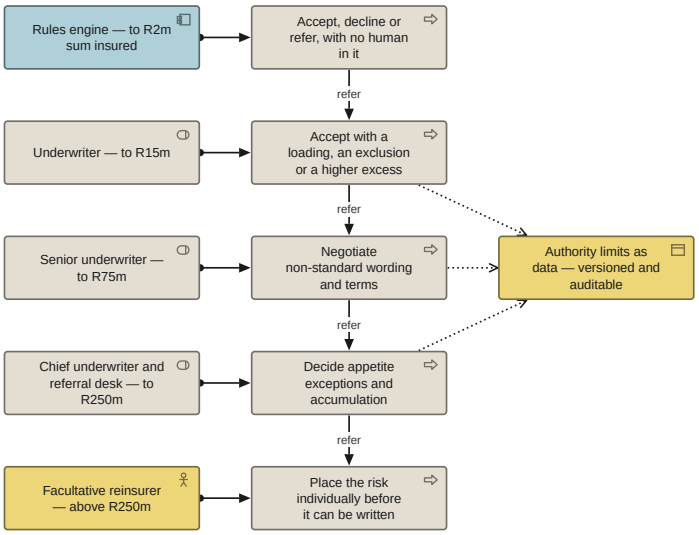


Figure 5.5 The underwriting authority ladder: each rung an active structure element assigned to the decision it may take, and a referral trigger to the rung above. Limits are invented but the shape is typical. Note the object on the right — every limit is data a system must read, version and be audited against.

Two things to take from the ladder. First, **authority is data**: every limit is a rule that must live somewhere a system can read, be versioned, and be audited after the fact. In most insurers it lives in a document, and the system enforces an older version of it.

Second, **the straight-through rate is an architecture metric with a business name.** Loosen referral rules and cost-to-serve falls while risk quality drifts; tighten them and the referral desk drowns. The dial between those two is usually nobody's explicit decision, which means it is set by whoever last edited the rules.

The medical underwriter in life insurance deserves a separate note, because that role carries the industry's most sensitive data. Health information, family history, test results — special personal information under South African data protection law, processed because the business cannot function without it. Every architecture decision about the life underwriting estate is also a privacy decision, and the reinsurer is usually in the data flow too, because large sums assured are rated against the reinsurer's manual.

Claims

Claims is where the product is delivered and where the money leaves. It is also the function with the most roles per transaction.

Role	What they do all day	The decision they own	Data they need	Where they live	What they complain about
Claims handler	Work a queue of 40 to 120 open files; phone people; chase documents	Cover, quantum and the case reserve, within authority	Policy as at the loss date, cover terms, documents, supplier quotes	Claims system plus email plus a spreadsheet	Switching between systems; policy history they cannot see
Assessor (motor)	Inspect vehicles at a repairer or roadside; agree the repair method and cost	Repair or write off; the agreed cost	Vehicle data, parts pricing, images, the schedule	A mobile application, often a different vendor's	Connectivity; re-entering the same assessment twice
Loss adjuster (commercial)	Investigate larger or disputed losses independently; report to the insurer	What happened, what is covered and what it is worth	Policy wording, accounts, site evidence	Their own firm's systems; reports arrive as documents	Being appointed late, after the scene has been cleared
Forensic investigator	Pursue suspected fraud; interview; build the file	Whether to repudiate and whether to refer for prosecution	Claim history across the industry, networks, scoring output	Fraud platform, external data	Being called after payment; scores with no explanation
Recoveries clerk	Chase third parties, salvage buyers and other insurers	How hard to pursue, and when to write off	Liability evidence, the paid amount, third-party details	A spreadsheet, usually	Claims closed before recovery is set up; no prompts
Claims manager	Run the queue, the service levels, the leakage and the panel	Authority limits, supplier appointments, escalations	Cycle time, leakage, reopen rates, supplier performance	Operational reporting	Reporting that measures volume rather than leakage

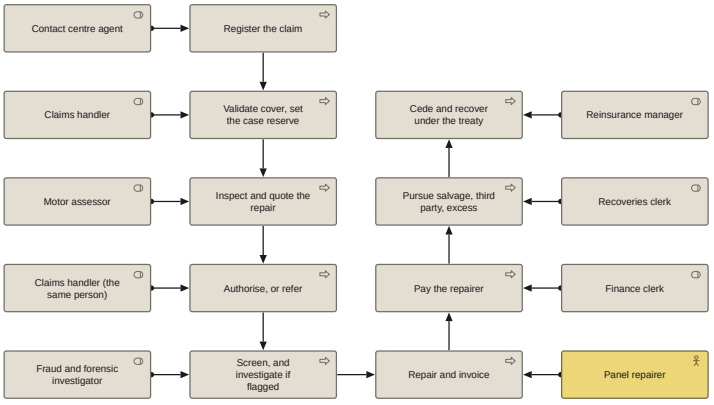


Figure 5.6 One motor claim and the nine pairs of hands that touch it: each role assigned to the step it performs, each step triggering the next. Two of the nine are the same person, and nobody owns the gaps between the rest.

Three distinctions that will mark you out as someone who has been here before.

Assessor and loss adjuster are not the same job. An assessor looks at the damage and agrees what it costs to put right — high volume, mostly motor, usually working to the insurer's panel rates. A loss adjuster is appointed on larger or contested losses to establish independently what happened, whether it is covered and what it is worth, and reports in writing. Different volumes, different turnaround, different data, different systems.

Claims leakage is not fraud. Leakage is paying more than the contract required: cover validated loosely, a settlement agreed above the policy limit, betterment not applied, an excess not deducted, a salvage not recovered, a reinsurance recovery not claimed. It is mundane, entirely internal, and larger than most fraud exposure. The architecture that reduces it is unglamorous — automated cover validation against the as-at policy, supplier integration, recovery triggers at closure rather than at month end.

Recoveries is the role nobody protects. When a claims system is replaced, recoveries is the piece most likely to be deferred to a later phase, because no customer complains about it. It is also pure margin: money the insurer has already paid out and is entitled to get back.

Actuarial and finance

These are the people who will ask you for data the operational systems were never designed to produce. Understanding what they are actually doing is the difference between being asked for an extract and being asked to design the foundation.

Role	What they do all day	The decision they own	Data they need	Where they live	What they complain about
Pricing actuary	Fit models, monitor rate adequacy, sign off product pricing	The technical price and the assumption set	Policy-level exposure and claims with rating factors, over years	Modelling tools fed by extracts	Extracts that change shape; no history after a migration
Reserving actuary	Build loss development triangles; estimate IBNR; explain the movement	The booked reserve, within governance	Paid and incurred by accident and development period, by class, gross and net	Actuarial platform and a warehouse they do not trust	Triangles that cannot be reproduced; reopened claims that vanish
Head of Actuarial Function	Statutory control function: opinion on technical provisions, pricing and reinsurance adequacy	Whether to sign	Everything the reserving actuary has, plus governance evidence	Board and committee papers	Being asked to sign off numbers whose lineage nobody can show
Capital actuary	Run the capital model and the ORSA; test scenarios	The own view of required capital	Exposure, correlations, stress scenarios, reinsurance structure	Capital modelling platform	Reinsurance terms held in PDFs rather than as data
Chief financial officer	Close the books, report to the board and the market, hold the plan	What the numbers are	A reconciled ledger and a consistent management view	Ledger, consolidation, board pack	Two systems disagreeing about the same month
Financial accountant	Reconcile premium, claims, commission and suspense	Which entries are right	Transaction-level detail that ties to cash	Ledger, bank statements, suspense accounts	Suspense accounts cleared by hand every month
IFRS 17 team	Group contracts into cohorts, measure, release the margin, explain the movement	How contracts are grouped and measured	Contract-level cash flows, discount curves, movement analysis	A separate actuarial-finance data layer	A sub-ledger reconciled to the general ledger manually

The two actuarial roles are not interchangeable, and confusing them in a workshop is a tell. **Pricing** looks forward and asks what to charge. **Reserving** looks back and asks what is already owed. They use different data from the same events, and in many insurers they maintain different versions of it.

What the reserving actuary needs is worth stating precisely, because it is the hardest data requirement in the enterprise: paid and incurred amounts and claim counts, by class of business, by accident or underwriting period down the rows and development period across the columns, gross and net of reinsurance, reproducible as at any past valuation date. That is a loss development triangle, and the standard reserving methods — chain ladder, Bornhuetter-Ferguson, Benktander, Cape Cod — all consume it [8]. If the warehouse cannot produce it reliably, the actuarial function builds its own data estate, and that estate becomes the de facto system of record for the largest number on the balance sheet.

The IFRS 17 team is usually a programme that became a department. Contract-level measurement, grouped into cohorts by profitability and issue year, with full movement analysis — none of which legacy policy systems were built to produce. The evidence that it is live in South African reporting is in the disclosures: Momentum reported an onerous contract value of R349 million for its 2026 financial year, down 39% [9]. Expect to find a separate actuarial-finance data layer, and expect its reconciliation to the general ledger to involve a spreadsheet.

Risk, compliance and legal

Insurance is supervised by two authorities — a prudential one for safety and soundness, and a conduct one for how customers are treated — with overlapping data demands and standards issued jointly [10]. That produces a set of roles with real authority and small teams.

Role	What they do all day	The decision they own	Data they need	Where they live	What they complain about
Chief Risk Officer	Maintain the risk framework, run the ORSA, report to the risk committee	Whether an exposure is inside appetite	Risk register, incidents, capital model output, exposure	GRC platform and a lot of documents	Risk data assembled by hand each quarter
Compliance officer	Monitor against FAIS, the conduct rules and the Policyholder Protection Rules; report breaches	Whether a practice is permitted	Advice records, disclosures, complaints, marketing material	GRC platform, complaints system	Evidence held by third parties; marketing that ships without review
Key individual (FAIS)	Manage or oversee the rendering of financial services in a licensed entity	Whether a representative may act, and under what supervision	Representative register, competence records, supervision evidence	The FSP's own records	Representative registers kept in spreadsheets
Information officer (POPIA)	Encourage and ensure compliance, deal with data subject requests, assist the Regulator	How personal information may be used	Processing register, consents, purposes, retention rules	Privacy tooling, the PAIA manual	Nobody can say where all the copies of the data are
Ombud liaison	Respond to complaints escalated outside the company, within deadlines	What the insurer's formal answer is	The complete file, including the record of advice	Complaints system plus whatever the broker sends	Files reconstructed from email after the fact
Legal counsel	Wordings, contracts, litigation, regulatory correspondence	What the contract actually says	Policy wording versions, binder and outsourcing agreements	Document management	Wording versions that cannot be tied to the policies sold on them

Three of these deserve a closer look.

The key individual under FAIS. Not an honorific. The FAIS Act defines a key individual as the natural person responsible for managing or overseeing the activities of the entity relating to the rendering of financial services [5]. Every licensed FSP must have one, including the brokerages and UMAs you deal with. When a key individual is unavailable or departs, the entity's ability to render financial services is affected — which makes it an operational dependency in the distribution network, not just a compliance detail.

The information officer under POPIA. The default information officer of a private body is the head of the organisation — in practice the chief executive — who may authorise another person in writing, at executive level, while retaining accountability. The information officer must register with the Information Regulator before performing the role's duties. Those duties include encouraging compliance, dealing with requests, assisting the Regulator with investigations, and under the regulations developing and maintaining a compliance framework, conducting personal information impact assessments and maintaining the PAIA manual. Deputy information officers may be designated from among employees, and delegation does not move the accountability [11].

For an architect, two things follow. First, a personal information impact assessment is a requirement with a named owner, so build the record of processing purposes as data rather than as a document. Second, insurance is unusually exposed, because the business requires special personal information: health data for life and medical underwriting, biometric and criminal data for fraud. Minimisation and retention pull against each other, since financial-crime obligations require you to keep and report what privacy law asks you to minimise.

The ombud liaison. When a complaint is not resolved internally it goes outside. The arrangements consolidated recently: the National Financial Ombud Scheme South Africa began operating on 1 March 2024, formed by combining four previous schemes — banking, credit, long-term insurance and short-term insurance — and runs divisions for non-life insurance, life insurance, and banking and credit [12]. Complaints about advice rendered by an FSP go to the FAIS Ombud. The conduct rules require complaints to be monitored, analysed and reported to the regulator and the ombuds, and require insurers to display ombud information [1].

The architectural demand is specific and routinely missed: on request you must assemble the complete file — the record of advice, the quote, the schedule, the wording in force, the premium history, every contact and the claim decision with its reasons — for a policy sold five years ago, possibly by a broker. If that takes three weeks of email, it will take three weeks every time.

All of this sits on a statutory frame. The Insurance Act 18 of 2017 commenced on 1 July 2018 with 42 Prudential Standards covering governance, risk management, internal controls and financial soundness, applied proportionally to the nature, scale and complexity of each insurer's risks [16]. The control functions those standards require are the people in the figure below, and between them they hold a veto over anything that touches the numbers or the customer.

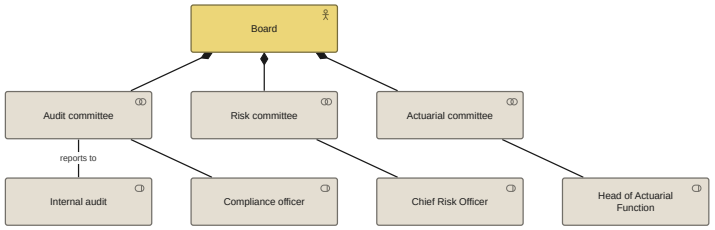


Figure 5.7 The statutory control functions of a licensed South African insurer, with the board committees drawn as collaborations and the reporting line as an association — ArchiMate has no element for a dotted line to a chief executive, so read every association here as the solid line to the board. Each of the four will ask for data the operational systems were not built to produce.

Operations and service

The unglamorous group where most of the staff and most of the IT cost sit.

Role	What they do all day	The decision they own	Data they need	Where they live	What they complain about
Policy servicing clerk	Process endorsements, address and bank-detail changes, cancellations, reinstatements	Whether a change is in order, and from what date	The policy as it stands, the request, the rules	Policy administration system	Changes that force a cancel-and-rewrite
Contact centre agent	Answer, identify, resolve or escalate, measured on handling time	What to do within the script and their mandate	One screen with the whole customer on it	Three screens with parts of the customer on them	Having to ask the customer what they already typed online
Collections clerk	Chase failed premium, manage mandates and retries, work the unpaid report	When to retry, when to lapse, when to reinstate	Unpaid reason codes, salary dates, mandate state	Collections platform, bank files	Reason codes nobody has mapped to an action
Administrator / TPA	Run the whole back office of a book for an insurer or a scheme	How the book is administered day to day	Everything about that book	Their own platform	Insurers wanting data in a shape nobody agreed up front
Underwriting assistant	Chase documents, prepare files, issue schedules	Nothing, and everything moves through them	Checklists, document status	Workflow and email	Being the integration layer between two systems

Two observations worth carrying into a design review. First, servicing is where continuity lives: an endorsement that cancels and rewrites a policy destroys the continuous cover history that claims, reserving and renewal pricing all depend on. Second, collections in this market is a risk function wearing a back-office costume. Most lapses in mass-market business are failed collections rather than decisions, so retry strategy, mandate management and salary-date alignment are retention capabilities. Treat the collections clerk as a decision-maker and give them reason codes mapped to actions.

Reinsurance

Small teams, large numbers, and almost always under-served by the estate.

Role	What they do all day	The decision they own	Data they need	Where they live	What they complain about
Reinsurance manager	Operate the programme: cessions, bordereaux, technical accounts, recoveries	Which losses are notified and recovered	Treaty terms as data; claims by event; cession records	A spreadsheet beside the core, usually	Treaty terms in PDFs; recoveries computed by hand
Treaty negotiator / head of reinsurance	Design and place the programme; negotiate retentions and layers at renewal	How much risk to keep	Loss history by event, exposure, capital impact	Actuarial and capital output	Exposure data the reinsurer asks for and nobody has
Reinsurance broker	Place the cedant's programme in the market; advise on structure	Which markets to approach and with what submission	A clean submission: exposure, history, structure	Their own systems	Cedant data arriving late and inconsistent
Reinsurer's underwriter	Price and accept treaties and facultative risks across many cedants	Accept, price, decline	The cedant's submission and their own experience	The reinsurer's systems	Cedants who cannot explain their own data

The architectural point sits in one distinction. On a proportional treaty, cession is computable per policy at the moment of binding. On an excess-of-loss treaty, recovery can be computed only per loss event, after aggregating claims across policies within an hours clause and a geography [13]. Those are different data problems, and the second one is where recovery leakage lives. If the reinsurance manager runs the programme from a spreadsheet — and in most mid-sized insurers they do — the recoveries are as good as that spreadsheet.

Product and marketing

Role	What they do all day	The decision they own	Data they need	Where they live	What they complain about
Product manager	Own a product line: features, wording, performance, competitive position	What the product is and when it changes	Sales, persistency, claims experience, competitor terms	Product documents, performance reports	Time to market measured in quarters
Product actuary	Price the product and set the assumptions; test profitability	The pricing basis and the profit target	Experience data, expense allocation, assumptions	Modelling tools	Expense allocations they cannot defend
Marketing	Campaigns, brand, acquisition cost per channel	Where acquisition spend goes	Conversion by channel, cost per policy, lapse by cohort	Campaign tooling, analytics	Attribution; lead data that does not survive the hand-off

The product manager is the role most affected by whether product is data or code. Where wording, rates and rules are configured and effective-dated, a product change is a release of configuration. Where product is hard-coded across two cores and a document system, the same change is a programme. This is the clearest case in the book of an architectural choice determining an operating-model outcome, and the product manager is the person who feels it.

Technology, and where you sit

Role	What they do all day	The decision they own	What they need	Where they live	What they complain about
CIO	Keep it running, land the portfolio, answer for cost	Where the money goes	Portfolio status, run cost, risk exposure	Steering committees	Business cases built on hope
Enterprise architect	Model the business, govern the target, make the trade-offs visible	Nothing directly; everything by influence	A truthful picture of capability, systems and data	Everywhere, owning nothing	Being invited after the decision
Solution architect	Design a change inside a programme	How this change is built	Standards, current state, constraints	A delivery team	A target architecture that does not survive contact
Head of data	Warehouse, reporting, governance, increasingly the actuarial foundation	What the authoritative source is	Lineage, definitions, quality measures	Platform and governance forums	Being asked for a number without its definition
Vendor account manager	Sell you the next module	Nothing of yours; a great deal of your roadmap	Your roadmap	Your meetings	Nothing, out loud

Your real counterparties are not in this table. An enterprise architect in an insurer gets things done through six or seven people who own decisions: the chief operating officer, who owns cost-to-serve; the head of claims, who owns the number everyone watches; the chief underwriter, who owns appetite; the Head of Actuarial Function, who owns whether the numbers can be signed; the chief risk officer and the compliance officer, who can stop things; the head of distribution, who speaks for brokers who are not in the building; and the chief financial officer, who decides whether any of it is funded.

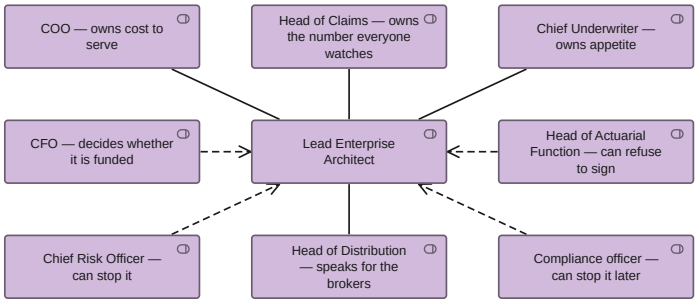


Figure 5.8 The architect's real counterparties. Three of the eight can stop a programme and none of them report to technology, which is why influence rather than authority is the job.

The practical consequence: an architecture artefact is only useful if one of those people would recognise it as true and use it in their own argument. A capability model the chief operating officer uses to explain cost-to-serve is worth ten that nobody opens.

One process, every role

Put the roles against a single end-to-end process and the hand-offs become visible.

One end-to-end process, every role that touches it

A broker-placed motor policy from quote to reporting. R responsible, A accountable, C consulted, I informed.

	Broker	Under-writer	Claims handler	Assessor	Pricing actuary	Reserving actuary	Compliance	Finance
Quote the risk	R	C			A		I	
Accept and bind	R	A			C		I	
Issue the schedule	C	A					I	I
Collect the premium	C	I						A
Register the claim	C		A			I		
Validate cover at loss date	I	C	A				C	
Assess the damage	I		A	R				
Set the case reserve			A	C		C		I
Settle or repudiate	I	C	A	C			C	I
Pursue recoveries			C			I		A
Reserve and report		C	C		C	A	C	R

Read the column, not the row. The claims handler is accountable for six steps and owns none of the data created in the first four. That gap is the architecture.

Figure 5.9 A broker-placed motor policy from quote to reporting, with every role that touches it. Read down the columns: the claims handler is accountable for six steps and owns none of the data created in the first four.

Read down the columns rather than across the rows. Three things fall out of this one.

The **claims handler** is accountable for six of the eleven steps and owns none of the data created in the first four. Everything they need — the sum insured, the rating factors, the wording version, the premium history — was captured by somebody else, in another system, sometimes in another company. That gap is the architecture problem in one picture.

The **broker** is responsible at the start and merely informed later, which is exactly the opposite of how the client sees it. The client's expectation is that the broker remains their representative throughout the claim. Build the broker a view of claim status or accept that your contact centre will carry those calls.

Nobody is accountable for recoveries except finance, late and at a distance. That is the honest version of what most RACI charts in this industry would show if they were drawn truthfully.

Running the discovery session

A closing practicality, because the point of knowing the roles is to be able to sit in a room with them.

Four questions work on anybody in this chapter, regardless of function:

1. **What do you decide on your own, and what do you have to refer?** This gets you the authority model, which is usually undocumented.
2. **What do you need to know to make that decision, and where do you get it today?** This gets you the real integration map, including the email and the spreadsheets.
3. **What do you do when the system will not let you do the right thing?** This gets you the workaround, which is the requirement nobody wrote down.
4. **What would you have to explain to the regulator or the ombud about this, and could you?** This gets you the evidence and lineage requirements while the person is still being honest.

Two rules of conduct. Never correct an insurance person on insurance in their own workshop, even when you are right; take it offline and check first, because in this industry the exception you have not heard of is usually real. And never leave a session without the artefact in front of them — the value

chain, the canvas, the RACI — marked up in their handwriting. A model that somebody in the business has corrected is a model they will defend. Everything else is your opinion, drawn well.

THE CHAPTER IN TEN LINES

- Insurance work is organised around decisions, not departments: ask what a person decides alone and what they must refer, and ignore the title.
- A large part of the operating model is staffed by people outside the company — brokers, underwriting managers, administrators, assessors — and none of them appear on the chart you will be shown.
- A broker acts for the client, is paid by the insurer out of the client's premium, and that structural conflict is what the South African conduct regime exists to manage.
- Commission on short-term business is capped in regulation at 12.5% of premium on motor and 20% on other policies, which is the opposite of what most people guess.
- A binder outsources the insurer's own functions — entering into, wording, premiums, benefit values, settling claims — and turns the insurer's estate into an ingestion, reconciliation and oversight estate.
- Broker, agent, adviser and intermediary are not synonyms in South African law, and the system entitlement should follow the regulatory category rather than the everyday word.
- Underwriting authority is data: every limit is a rule that must be versioned, enforced and auditable, and the straight-through rate is an architecture metric with a business name.
- Assessor and loss adjuster are different jobs, claims leakage is not fraud, and recoveries is the role most often deferred to a later phase and never built.
- The reserving actuary needs a reproducible loss development triangle, and when the warehouse cannot produce one the actuarial function becomes the de facto system of record.
- The statutory control functions — Head of Actuarial Function, Chief Risk Officer, compliance officer, internal audit — have small teams, a line to the board and the standing to stop a release.
- Your real counterparties are the six or seven people who own decisions, and an artefact is only useful if one of them would use it in their own argument.

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CHAPTER 6

How insurers make money

An insurer sells something it cannot cost, collects the money first, and finds out years later whether the price was right. Every strange thing about insurance architecture grows out of that one fact.

The promise you are actually selling

Nobuhle pays Cape Mutual Life R210 a month. If her mother dies, Cape Mutual pays R40,000, and the family expects the money within about two days, because the funeral is on Saturday. A logistics firm in Isando pays Cape Mutual Short-Term R3.4 million a year. If one of its trucks puts a warehouse on the ground, Cape Mutual pays for the warehouse.

Cape Mutual Insurance Group is invented. It is the worked example that runs through this book and nothing about it is a case study of a real insurer. The two transactions, though, are entirely ordinary, and they are the same instrument.

Somebody who stands to lose money hands that exposure to somebody else, and pays for the service. Five things have to be present for it to work:

- **An insurable interest.** The buyer must actually stand to lose something. You cannot insure your neighbour's car, and the rule exists so that insurance does not become a wager on other people's misfortune.
- **A premium.** Certain, comparatively small, paid now.
- **A defined contingent event.** Death, fire, a motor collision, a finding of legal liability. Defined narrowly, because the definition *is* the edge of the promise.
- **A defined response.** Cash, a repair, a replacement, or a service.
- **A term.** The window in which the event counts.

There is a sixth thing that is not a mechanism but a duty. Both sides must deal in good faith, and the buyer must disclose what a prudent insurer would want to know. Non-disclosure of a material fact is the lever an insurer pulls to avoid a claim, and it is the single largest source of complaints to the ombud schemes.

Two families of contract come out of that, and the difference between them explains more about insurance systems than any other single distinction.

	Indemnity contract	Benefit contract
Promise	Put you back where you were, no better	Pay an agreed sum on a defined event
Typical products	Motor, property, liability, engineering	Life, funeral, critical illness, disability lump sum
Amount known when?	Only after the loss is investigated and measured	At inception, written on the schedule
Timing known when?	Within months of the event	Unknown, possibly decades away
What the claims function does	Assess, quantify, negotiate, repair, recover	Verify identity, verify the event, pay
What this costs to run	High touch per claim, supply chain, assessors	Low touch per claim, high volume, document capture

An indemnity claim is an investigation. A benefit claim is a verification. Build one claims platform for both and you will serve neither well. This is the first of many places in the book where the words "insurance company" cover two unlike businesses.

Three consequences you cannot design around

Everything peculiar about insurance architecture follows from three facts about the contract.

You do not know what the product cost until long after you sold it. There is no bill of materials. Price comes from a statistical model applied to a population, not from a cost sheet applied to a unit. For a motor book the true margin is reasonably clear within eighteen months. For a liability book it may take fifteen years. For a whole-of-life policy sold to a thirty-year-old, the answer arrives when that person dies.

Cash arrives before cash leaves. Premiums come in monthly or annually; claims go out later. The accumulated money in between is invested for the insurer's own account. It is called the float, and for long-tailed business it is enormous.

The product is a legal document. Not described by a document — *is* a document. The policy wording is what the insurer owes. Rating tables, exclusions, limits, excesses and waiting periods are the product's working parts. This is why effective-dated versioning of wording and rules is a first-class architectural concern and not metadata housekeeping. A claim notified in 2026 on a policy incepted in 2019 must be adjudicated on the 2019 terms, and somebody has to be able to produce them.



Figure 6.1 Premium is certain and early; cost is an estimate that settles late. Every other peculiarity of insurance architecture follows from this shape.

A fourth consequence follows from the first three and it is the one that trips up newcomers hardest. Because cost is unknown and cash arrives early, an insurer can look magnificently profitable for several years while quietly writing business that will ruin it. The premium is real and recognised now. The claims are an estimate and the estimate is made by people who work for the company. Every regulatory regime in the world exists largely to constrain that temptation, which is why insurance is supervised on solvency rather than only on conduct, and why the reserving actuary is a statutory role rather than a job title.

> Every other business you have modelled knows its cost of goods sold at the point of sale. An insurer does not. Treat that as the root fact and most of the strange things in the architecture stop being strange.

Premium: four words that must travel together

Ask an insurance person what premium was last year and you will get a number. Ask which of four numbers it is and you will usually get a pause.

Premium is **written** when the contract is made. It is **earned** as exposure elapses, recognised proportionally as time passes over the life of the policy [1]. The gap between the two sits on the balance sheet as the unearned premium reserve, because it is money the insurer has but has not yet done the work for. Separately, premium is **gross** before reinsurance and **net** after cession. Gross premium measures market presence; net premium measures retained risk.

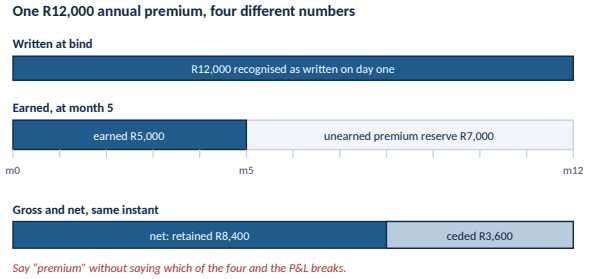


Figure 6.2 Written, earned, unearned, gross and net are five views of one R12,000 premium. Illustrative figures.

The two pairs move independently, and the difference is not a rounding error. Santam's gross written premium grew 6.4% in FY2025 while its net earned premium grew 14.7%, because of new partnership business written through its reinsurance arm [2]. Anyone reporting "premium growth" from that set without saying which number produced it has already misled their audience.

For an architect the practical rule is blunt. In the information model, premium is never a single attribute. It is at minimum an amount with a basis (written or earned), a scope (gross, ceded, net), a period, and a currency. Any reporting layer that lets a user drag a field called "Premium" onto a chart without choosing those four things is a defect factory. The reconciliation tickets will arrive within the quarter.

There is a second rule that follows. Growth figures in insurance are nearly meaningless without the basis attached. A book that grows 20% in written premium may be growing exposure, or it may be re-rating an existing book upward in a hard market, or it may be writing one large account that will not repeat. Earned premium lags written premium by roughly half the average policy term, so in a growing book earned always looks worse and in a shrinking book it always looks better. The board sees one number; the actuary sees three.

The float, and why it changes the answer

Premiums are held before they are paid out. Reserves for claims not yet settled are held for longer. All of that money is invested. South African life insurers held R5.23 trillion of assets at March 2026 against quarterly gross premium of R192.2 billion [3]; ASISA puts life assets under management at R5.2 trillion at the end of 2025 [4]. The asset book, not the premium book, dominates a life insurer's balance sheet.

How long the float is held depends entirely on how long claims take to settle. A motor claim settles in weeks. A liability claim arising from professional advice given in 2026 might settle in 2041. Long-tailed classes therefore generate much more investment income per rand of premium than short-tailed ones, which is why the economics of a liability book and a household book are not comparable even at identical loss ratios.

This is also why an insurer can lose money on insurance and still make money. Investment income can make a company with a combined ratio above 100% profitable overall [1]. That single sentence explains a great deal of pricing behaviour.

It cuts the other way too. Investment results are volatile and they are not underwriting skill. Santam booked roughly R1 billion of foreign-currency investment losses in FY2025 [2]; OUTsurance's normalised investment income rose 27.7% in the half to December 2025 on equity markets [12]. Neither number says anything about whether the underwriting was any good.

Two results, not one

An insurer has two profit engines and they must be read separately.

- The **underwriting result** is what is left of earned premium after claims and the cost of running the business. It is a verdict on risk selection, pricing and expense discipline.
- The **investment result** is the return on the float and on shareholders' capital. It is a verdict on asset allocation and on markets.

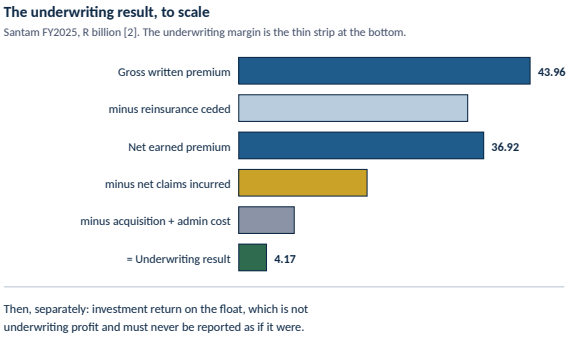


Figure 6.3 From gross written premium to the underwriting result, drawn to scale from Santam's FY2025 figures [2]. Investment return sits outside this picture entirely.

A board that lets a good investment year cover a bad underwriting year is storing up trouble, because markets mean-revert and bad pricing does not fix itself. A board that cannot tell which engine produced the profit has a reporting problem, and the architect usually owns it.

Running an underwriting loss on purpose

Some insurers plan to lose money on underwriting. It is not always folly. Three reasons are defensible.

The first is float arithmetic. If a class holds premium for eight years and the insurer can earn a real return on it, a combined ratio of 104% can still clear the cost of capital. This logic belongs to long-tailed classes and nowhere else. Applied to motor it destroys capital quickly, because there is no float to speak of.

The second is deliberate purchase of scale. A new entrant underprices to build a book, intending to re-rate at renewal. In short-term insurance this is survivable, because contracts are twelve months and mispricing is correctable within a year. In life it is close to suicidal, because you are locked into the price for decades.

The third is strategic cross-subsidy. A bank sells credit life at thin margin because it protects the lending book. A composite group accepts a weak funeral margin because it is the entry point to a lifetime customer relationship.

What makes all three dangerous is that they are indistinguishable, in the management accounts, from being bad at underwriting. The architectural answer is that the plan has to be in the data. Target loss ratio by product and cohort, expected float duration, and expected investment return have to be recorded at the point the product is approved, and actual experience has to be comparable against them. If the only place the plan exists is a slide, nobody will ever be able to say whether it worked.

Reading the ratios

Three ratios carry most of the conversation in a short-term insurer. All three are expressed against *earned* premium.

Ratio	How it is built [1]	What it is telling you
Loss ratio (claims ratio)	Incurred losses plus loss adjustment expenses, divided by earned premium	Quality of risk selection and adequacy of rates
Expense ratio	Underwriting expenses divided by earned premium	Cost of distribution and cost of operating
Combined ratio	Incurred losses plus loss adjustment expenses plus other underwriting expenses, divided by earned premium	Underwriting profitability. Below 100% is a profit, above 100% is a loss, before any investment income

Note that "incurred" is not "paid". Incurred losses include the change in reserves for claims that have happened but not yet been settled, and for claims that have happened but not yet been reported at all. The loss ratio therefore contains an actuarial estimate. More on that shortly.

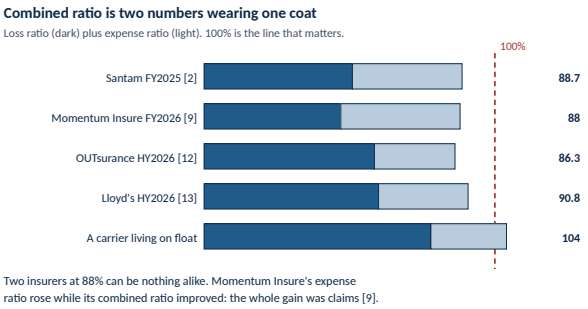


Figure 6.4 Published combined ratios decomposed into claims and expenses. The last row is illustrative: a long-tailed carrier running an underwriting loss and living on investment return.

Comparators are worth having in your head, because they calibrate what "good" means.

Insurer and period	Claims ratio	Expense ratio	Combined
Santam, FY2025 [2]	—	—	88.7%, underwriting margin 11.3%
Momentum Insure, FY2026 [9]	47% (from 51%)	41% (from 39%)	88% (from 90%)
OUTsurance Group, HY2026 [12]	58.6% short-term	—	about 86%
Lloyd's, HY2026 [13]	—	—	90.8%
SA non-life primary, March 2026 quarter [3]	52.1%	—	not reported on this basis
SA non-life cell captives, March 2026 quarter [3]	38.6%	—	not reported on this basis

Look at Momentum Insure. The combined ratio improved by two points while the expense ratio got two points *worse*. Every bit of the gain and more came from claims. A dashboard showing only the combined ratio would have reported an efficiency story that did not happen.

One more piece of calibration, because it tells you where the money for your programme will come from. Global non-life premium was about USD 4.6 trillion in 2024 against a life pool of about USD 3.0 trillion, with the United States roughly 45% of the world total [15]. Swiss Re's mid-2026 view had total real premium growth slowing to 1.3% in 2026, non-life growing just 0.6% against a long-term trend of 3.6%, and non-life return on equity falling from 14% in 2025 to a projected 7.7% by 2028 [5]. Read that as a cost story. When growth stalls and margin is forecast to halve, boards fund expense-ratio reduction, claims automation and straight-through underwriting rates ahead of growth features. Cost to serve becomes the primary measure, and the architecture portfolio follows the measure.

> TWO TRAPS IN THE RATIO TRIANGLE
>
> **Life and non-life claims ratios are not comparable.** Life "claims" include maturities, annuity payments and surrenders. Those are repayments of the policyholder's own saved money, not losses. The Prudential Authority's March 2026 table shows a life primary claims ratio of 90.8% next to a non-life primary ratio of 52.1% [3]. The life insurer is not nine times worse at underwriting. It is a different number.
>
> **None of these ratios has a single statutory definition.** They are management and trade measures. Whether commission is netted off premium or counted as an expense, how loss adjustment expense is split, and how head-office cost is allocated all vary by insurer. Two carriers with identical economics can publish combined ratios two points apart. Pin the definitions in the semantic layer and expose the derivation, or the first external benchmarking exercise will produce nonsense that somebody acts on.

The biggest number on the balance sheet is an estimate

An insurer's largest liability is the money it expects to pay on claims that have already happened. Part of that is countable. Most of it is not.

- **Case reserves** are handler estimates on claims the insurer knows about. Somebody has opened a file and written down a number.
- **IBNR** — incurred but not reported — is the actuarial estimate of claims that have occurred but of which the insurer is still ignorant. The collision happened; the call has not come.
- **IBNER** — incurred but not enough reported — is the expected development on claims already known, because first estimates are usually too low.

The working tool is the loss development triangle: cumulative incurred and paid amounts, with accident period down the rows and development period across the columns, split by class, gross and net of reinsurance, with claim counts alongside.

A loss development triangle

Cumulative incurred, R million, motor own damage. Rows: accident year. Columns: months of development.

dev →	12	24	36	48	60
2021	62	108	131	140	143
2022	58	101	127	136	
2023	71	126	151		
2024	66	118			
2025	74				

everything below the diagonal is IBNR

The shaded cells have not happened yet, in the sense that nobody has told the insurer about them. The actuary estimates them. They go on the balance sheet as a liability, and they are usually the largest one.

Figure 6.5 A loss development triangle. Illustrative figures. The blank cells below the diagonal are the claims nobody has told the insurer about yet; the actuary estimates them and they become the largest liability on the balance sheet.

From the triangle the actuary projects to ultimate. Four methods account for most of what you will see [7].

Method	Mechanism	Where it bites
Chain ladder	Latest reported multiplied by a cumulative development factor. The past is taken as indicative of the future	Needs no prior view, but is volatile where the development factor is large: one big claim moves the whole ultimate
Bornhuetter-Ferguson	Ultimate equals losses reported so far plus an a priori expected loss amount scaled by the share still to develop	Leans on the prior for immature years and on experience as years mature
Benktander	An iterated Bornhuetter-Ferguson. At one extreme it is the expected-loss method, at the other it converges on chain ladder	A tunable blend, which also means a tunable answer
Cape Cod	Bornhuetter-Ferguson structure, but the a priori loss ratio is derived from the triangle itself	Avoids an arbitrary prior, at the price of over-smoothing

Two cautions come from the same source and both matter to anyone building the reporting around this. Bornhuetter-Ferguson and Cape Cod smoothing can make historical loss ratios look artificially smooth, understating future volatility. And for high excess layers, slow development patterns often produce projected ultimates that are either zero or very large by year, which is unstable by construction [7].

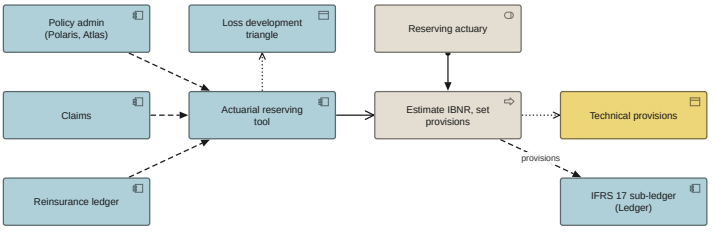


Figure 6.6 The reserving data foundation. When the flows on the left cannot be produced reliably, the actuarial team builds a shadow data estate and it becomes the de facto system of record.

Tail length drives everything downstream. Long-tailed classes — liability, workers' compensation, the old asbestos books — develop over decades, carry large IBNR relative to case reserves, have high reserve uncertainty and throw off a large float. Short-tailed classes are the mirror image.

Here is the architecture point, and it is the one that costs programmes money. The data requirement above is not a report. It is a contract-level, movement-complete, effective-dated foundation that can be regenerated identically on demand. Most warehouses cannot do it, because they were built to answer "what is the position now" rather than "what did we believe on 31 March 2024, and why". When the warehouse cannot produce the triangle reliably, the actuarial function builds its own data estate out of extracts and spreadsheets. Within two years that shadow estate is the de facto system of record for the largest number on the balance sheet, it is maintained by three people, and it is not in anyone's architecture repository.

Reinsurance is an architecture concern

Reinsurance is insurance for insurers. Architects tend to file it under finance and leave it alone. That is a mistake, because reinsurance determines what the insurer actually owes on every claim it pays, and the calculation has to happen inside the transaction systems.

Two questions, asked independently, give you the whole taxonomy.

What is covered — treaty or facultative? Facultative reinsurance covers a single risk or a defined package; the reinsurer underwrites each one individually and may decline it, and a cedant may need several reinsurers to complete one large risk. Treaty reinsurance covers a whole book or class; the reinsurer must accept all business falling in the agreed class, including policies not yet written, and therefore never sees the individual risks and relies on the cedant's underwriting instead [6].

How losses are shared — proportional or excess of loss? Under a proportional treaty the reinsurer takes the same share of premium and of losses. Quota share takes a fixed percentage of every risk; surplus share lets the cedant keep a line and cede the surplus above it, so the cession varies with risk

size. Proportional cover is easy to administer and protects against frequency as well as severity, and the reinsurer usually pays a ceding commission to offset the cedant's policy-issue costs. Under excess of loss the cedant keeps losses up to a retention and buys cover above it, in layers, up to a limit — "R50 million excess of R50 million". It is cheaper in premium and administration but protects only against severity, not against a bad year of small claims [6].

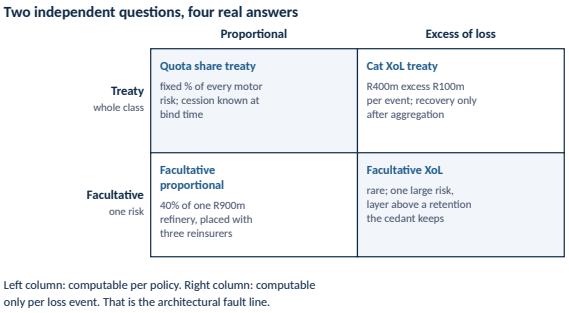


Figure 6.7 Treaty or facultative answers 'what is covered'. Proportional or excess of loss answers 'how is the loss shared'. The two questions are independent [6].

Reinsurance also sells two different things, and people conflate them constantly. One is risk transfer: genuine loss is moved and volatility falls. The other is capital relief: the regulatory capital requirement falls, so the insurer can write more business on the same equity. Proportional treaties are often bought mainly for the second.

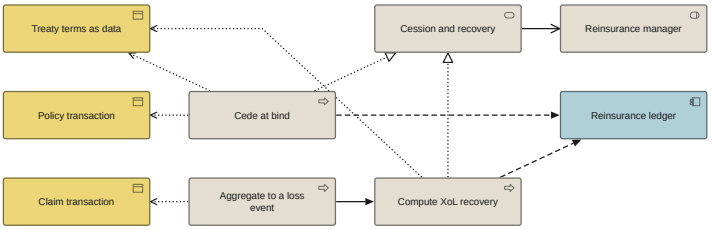


Figure 6.8 Cession and recovery as a modelled capability. Proportional cession resolves against one policy transaction; excess-of-loss recovery resolves only after claims are aggregated into a loss event.

Now the architecture. Proportional cession is computable per policy at bind time. It is a function of the treaty terms and the risk, both known the moment the policy is issued, and it belongs in the policy transaction. Excess of loss recovery is computable only per loss event, after aggregating claims across many policies within an hours clause and a defined geography, and sometimes across classes. That is a wholly different data problem, performed over data that arrives late and from several systems.

This is why most reinsurance recovery leakage happens on excess of loss. The recoveries that get missed are the ones nobody can compute automatically. The common pattern is treaty terms living in PDFs and spreadsheets rather than as machine-readable rules, cessions calculated by hand in a quarterly exercise, and inwards and outwards views that cannot be reconciled. In a hardening market, where insurers retain a larger share of each loss [14][16], the cost of getting this wrong rises every year.

Life insurance has different arithmetic

A life insurer is not a short-term insurer with longer contracts. The profit arises in a different way, over a different horizon, and it is measured with different instruments.

Where the margin comes from

Short-term insurance has one underwriting margin: premium minus claims minus expenses, settled within the year. Life insurance has several, and they are additive.

Source of margin	What it is	What moves it
Mortality and morbidity margin	Premium charged for risk, less claims actually incurred	Underwriting quality, selection effects, medical advances, pandemics
Expense margin	Expense loading charged, less expenses actually incurred	Scale, automation, cost-to-serve per policy
Investment spread	Return earned on assets backing the liability, less the return credited or guaranteed to the policyholder	Asset allocation, rate movements, guarantees written years ago
Fee income on assets	Basis points on assets under management in savings and annuity wrappers	Market levels and net client cash flow, not underwriting at all
Lapse margin, positive or negative	The effect of policies ending earlier than priced for	Persistency, and whether the product was priced on an assumption that lapses help or hurt

That last row is where most of the damage happens, so it gets its own section.

New business strain

Writing a life policy costs money upfront. Commission, medical underwriting, system set-up and the initial reserve are all incurred now; the profit emerges over the following twenty or forty years. Under any accounting basis that recognises costs as incurred, a life insurer writing a great deal of profitable new business reports *worse* current earnings than one writing none at all.

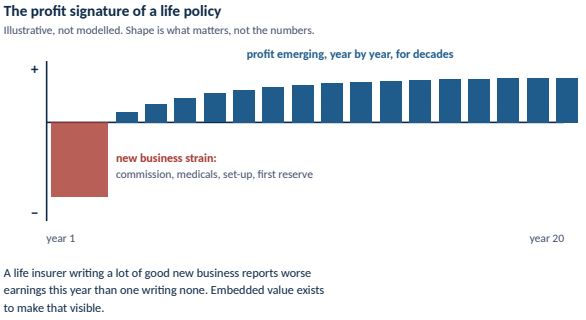


Figure 6.9 New business strain and profit emergence on a life policy. Shape, not scale: the figures are illustrative.

This is called new business strain. It is standard actuarial usage rather than a term with a convenient citable definition, so take the name on my authority and the mechanism on the arithmetic. The visible, sourced consequence in current South African reporting is the treatment of contracts expected to lose money: Momentum disclosed an onerous contract value of R349 million for FY2026, down 39%, representing expected losses recognised immediately rather than as they occur [9].

Strain is also the reason embedded value exists. Embedded value is net asset value plus the present value of future profits on the book already written, less the cost of holding capital against it. Discovery reported embedded value of R143 billion and a return on embedded value of 14.1% for the year to 30 June 2026 [8]. **Value of new business** is the same idea applied only to business written in the period, and the **new business margin** is that value divided by the present value of the new business premiums.

Those margins look alarming until you understand what they measure.

Group, 2026 reporting	Value of new business	Margin
Momentum Group [9]	R491 million on PVNBP of R93.8 billion	0.5%
Old Mutual [10]	R569 million, up 32%	1.4% against a 2–3% target
Sanlam, Q1 [11]	Down 22%	1.5%
OUTsurance Life [12]	R285 million, up 61.9%	25%

A spread from 0.5% to 25% is not a ranking of competence. It is product mix. Capital-light savings business produces huge premium volumes at razor-thin margin; pure protection business produces small volumes at fat margin. Sanlam's value of new business fell precisely *because* volumes shifted towards savings [11]. If your management information cannot decompose margin by product line, somebody senior will eventually read the group number as a performance signal and act on it.

A last word on embedded value, because architects meet it as a data demand rather than a concept. Producing it requires projecting every in-force contract's cash flows to run-off under a set of assumptions, then discounting them. That means the policy administration system must be able to emit, per contract, the full set of attributes the projection model needs — sum assured, premium, term, age, smoker status, product variant, option history, and the current state of every guarantee. Legacy life platforms frequently cannot, because attributes were overwritten rather than versioned when policies were altered. The actuarial team then reconstructs them from paper files and historic extracts, once a year, under time pressure. That is a data-lineage problem wearing a valuation costume.

Persistency and lapse

A lapse is a policy terminating because the premiums stopped. A surrender is a policyholder cashing in a savings policy. Persistency is the complement: the proportion still in force after some number of months. Thirteen-month and twenty-five-month persistency are the measures usually quoted, though that convention is market practice rather than something I could source.

The volumes in South Africa are extraordinary. In 2025, 8.7 million risk policies lapsed, against 10.8 million new individual recurring-premium risk policies written in the same year [4]. The book grew, but barely more than it churned.

Because acquisition cost is spent upfront, a policy that lapses before that cost is recovered destroys value outright. It would have been better never to have sold it. That is why Old Mutual described itself as "scaling back pockets of lower quality sales to improve persistency" [10], and why Santam reported improved persistency in FY2025 as a result in its own right [2].

THE LAPSE-RATIO DENOMINATOR TRAP

In the Prudential Authority's data for the quarter to March 2026, the individual lapse ratio for life primary insurers is **68.7%**, up from 58.8%. In the same table, the individual lapse ratio is **6.9%**, up from 6.1% [3].

Both are correct. The first is expressed as a percentage of new policies issued in the period. The second is expressed against policies in force. One is a comment on sales quality this quarter; the other is a comment on the stability of the whole book. They differ by an order of magnitude and they answer different questions.

This is the canonical example of why a definition without a denominator is not a definition. If the two had been loaded into the same warehouse column called `lapse_ratio` — and that happens — the resulting dashboard would be indefensible. Every ratio in your information model needs its numerator basis, its denominator basis and its period stated as structure, not as a footnote in a data dictionary that nobody opens.

What all of this means for systems and data

Pull the economics back into architecture and nine demands fall out. They are not a wish list; they are the things that are hard to retrofit.

1. **Premium is a measure with a basis, not a number.** Written, earned, gross, ceded, net, by period. Make it structural in the model and make the reporting layer force the choice.
2. **Product is data, not code.** Wording, rating tables, exclusions, limits and rules, all effective-dated, all reconstructable as at any past date. Without it you cannot adjudicate an old claim correctly and you cannot explain a historic price.
3. **The transaction log is the system of record, not the current state.** Reserving, IFRS 17 and conduct supervision all ask what you believed at a past date. Mutable rows cannot answer that.
4. **Reserving needs a contract-level, movement-complete foundation, on demand.** Triangles by class, accident period and development period, gross and net, with counts, regenerable and identical each time. If finance and actuarial cannot get it from the warehouse, they will build a shadow one, and you will inherit it.
5. **Treaty terms belong in the model as machine-readable rules.** Cession computed per policy at bind; recovery computed per event with hours clauses and geography. Not a quarterly spreadsheet exercise.
6. **The float must be visible by liability duration.** Asset-liability matching is only as good as the liability cash-flow projection the policy systems can produce.
7. **Separate the two results everywhere.** Underwriting result and investment result should never be summed in a management report without both components shown.
8. **Ratio definitions live in one governed place.** Numerator basis, denominator basis, period, scope. Every ratio, every time.
9. **Life needs profit-signature reporting, not period profit.** New business strain, value of new business by product line, and persistency cohorts, or the business will misread its own numbers.

At Cape Mutual, all nine land on the same two or three awkward places. Polaris, the forty-year-old life administration platform, holds policy state but not a reliable movement history, which is why the actuarial team keeps its own extracts. Atlas, on the short-term side, computes proportional cession but not excess-of-loss recovery, which the reinsurance manager reconstructs quarterly from a claims extract and a treaty PDF. Ledger, the general ledger and IFRS 17 sub-ledger, receives summarised postings and cannot be traced back to contracts. None of that is unusual. All of it is expensive.

Fatima Patel, the group's lead enterprise architect, puts the point this way when she is briefing a new programme: in most industries the finance function reports on what the business did, and in insurance the finance and actuarial functions are *part of* what the business does. Price, reserve and capital are operational decisions, taken continuously, on data the operational systems produce. An architecture that treats them as downstream reporting has already failed, and the failure will surface as a reconciliation project that never ends.

The next chapter walks the operating model that produces all of this data, step by step, and asks of each step who does it, what they decide, what they need, and what goes wrong.

THE CHAPTER IN TEN LINES

- An insurance contract transfers a defined contingent loss for a certain premium, and the wording is the product.
- Indemnity contracts pay what the loss turns out to be; benefit contracts pay an agreed sum, and the two need different claims machinery.
- Premium is never one number: written or earned, gross or net, and the pairs move independently.
- The float is money held between premium and claim, and it is why a combined ratio above 100% can still be profitable.
- Loss ratio, expense ratio and combined ratio are management measures with no statutory definition, so pin them in one governed place.
- Life and non-life claims ratios are not comparable, because life claims include repayments of the policyholder's own money.
- The largest liability on the balance sheet is an estimate built from a loss development triangle, and most warehouses cannot produce one reliably.
- Proportional reinsurance cedes per policy at bind; excess of loss recovers per loss event, which is where the leakage lives.
- Life profit is strained upfront and emerges over decades, which is why embedded value and value of new business exist.
- A lapse ratio without its denominator is not a number: the same South African table reports 68.7% and 6.9% for the same thing.

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CHAPTER 7

The operating model

Ten steps turn a promise into a payment. Walk them once properly and you will know where every system in an insurer came from, and which seams between them are quietly costing money.

Two businesses that happen to share a word

Start here, because everything else in the chapter depends on it.

A short-term insurer sells a twelve-month contract. In South African personal lines it is often a monthly-renewable contract, which is shorter still. The risk changes with the car, the driver, the suburb and the weather. The price is reset at every renewal. A pricing mistake shows up in the loss ratio within a year and is corrected at the next renewal cycle. Claims are investigations: somebody has to decide whether the event is covered, who is liable, what it costs, whether the quote from the panel beater is honest, and whether anything can be recovered from a third party.

A life insurer sells a contract that may run for forty years. The risk is a human being, assessed once, at the outset. The price is fixed at that point, usually for the whole term. A pricing mistake is locked in for a generation. Claims are verifications: somebody died, or did not; the sum assured is on the schedule.



Figure 7.1 A short-term contract and a life contract on the same time axis. Almost every difference between the two estates follows from this picture.

The consequences run all the way down.

- **Reserving.** A short-term insurer reserves mostly for claims that have happened and not yet settled. A life insurer reserves mostly for obligations decades away, which means discount rates and mortality assumptions matter more than claim files.
- **Capital.** Short-term capital is consumed by catastrophe accumulation. Life capital is consumed by longevity, by guarantees written years ago, and by new business strain.
- **Data.** A short-term policy is a thing-at-a-place: a vehicle, a building, a liability exposure. A life policy is a person, and often several people, because in South Africa extended-family funeral cover is the norm rather than an edge case.
- **Change cycles.** Short contracts mean short feedback loops, which is the structural reason short-term insurers change systems more often than life insurers do.

> When somebody says "the insurer's policy system", ask which one. A composite group like Cape Mutual runs at least two, they have different data models, different release cadences and different regulators' attention, and nobody has ever successfully merged them. The group-level architecture problem is usually not consolidation. It is making two unlike estates answer one question about the same customer.

With that established, walk the chain.

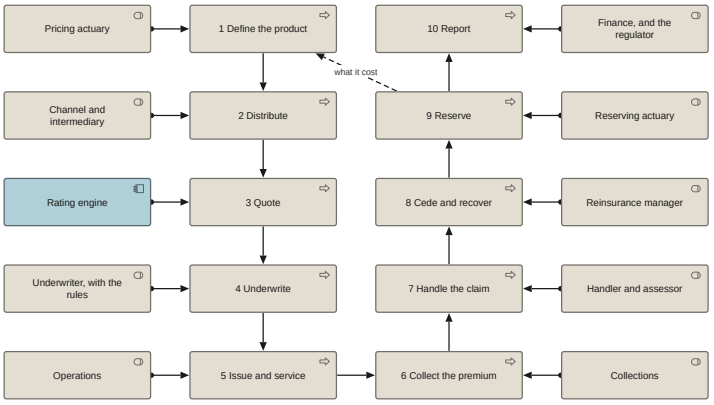


Figure 7.2 The insurance value chain end to end, with the role or system assigned to each step. Step three is assigned to an application component rather than a person, which is the whole of the straight-through argument. The dashed Flow from reserving back to product is the loop that makes insurance different: what the product actually cost is known only once the claims have developed.

For each step the same four questions: who does it, what gets decided, what data it needs, and what goes wrong. The last one is the most useful, because it is what you will walk into.

1. Product development and filing

What happens. Somebody decides what the cover is. Not in the marketing sense — in the legal sense. The team defines the insured events, the exclusions, the limits, the excesses or deductibles, the waiting periods and the conditions. A pricing actuary builds the rating algorithm and the rate tables, priced to a target loss ratio. Legal and compliance draft and review the wording. The reinsurer is frequently involved, because for anything novel the reinsurer co-designs and prices the cover. The product is then configured in the policy administration system, and the disclosure documents are produced.

Who. Product manager, pricing actuary, legal and compliance, reinsurer, and distribution, which turns up demanding features and a commission rate.

What gets decided. Target segment, price level, rating factors, underwriting appetite, reinsurance structure, commission, and which channels may sell it.

On filing. What "filing" means depends entirely on where you are. In some jurisdictions rates and forms must be approved by a regulator before use, which puts a statutory gate on the product cycle. South Africa does not work that way for most business. The Insurance Act 18 of 2017 commenced on 1 July 2018 with 42 Prudential Standards covering governance, risk management, internal controls and financial soundness, applied proportionally to the nature, scale and complexity of each insurer's risks [19]. The conduct constraint comes through the Policyholder Protection Rules and the broader conduct regulations, effective 1 January 2018, which set principles for premiums and premium reviews, minimum data governance requirements, and rules on how products may be marketed [1]. The practical effect for an architect is that product approval in South Africa is largely an internal governance gate with external consequences, rather than an external gate. It is still a gate, and it still needs evidence.

What goes wrong.

- Time to market measured in quarters, because the product is coded rather than configured. Every new variant is a release.
- The wording and the system configuration drift apart. The document promises something the policy system cannot administer and the claims system will not pay. This is a conduct finding waiting to happen.
- Rating factors that are impermissible or unfair, which is direct exposure under the conduct regime [1].
- No version history, so a claim on a 2019 policy cannot be adjudicated on 2019 terms.

A worked example of why this is hard. Cape Mutual wants to add a R5,000 "repatriation benefit" to its funeral range: if the deceased dies more than 200 kilometres from the place of burial, the policy pays to move the body. It sounds like a line item. It is not. The wording needs an exclusion for travel outside South Africa and a definition of "place of burial" that will survive a dispute. The rating needs a loading, which needs an assumption about how often this happens, which nobody has data on. The claims process needs a new proof requirement. The policy schedule template changes. The claims system needs a new benefit type with its own limit. Reinsurance needs to know whether the benefit falls inside the existing treaty. And every one of the 180,000 in-force funeral policies must be marked as being on the old version or the new one, for ever. An insurer whose product is configuration does this in a sprint. An insurer whose product is code does it in a release train, and in the meantime the brokers sell it anyway on a promise.

Two South African product boundaries worth knowing. The first is the demarcation line between insurance and medical schemes. From 1 April 2017 the definition of "the business of a medical scheme" was amended so that health insurance products are prohibited unless the regulations explicitly permit them. Gap cover and hospital cash plans are permitted as insurance; primary healthcare insurance covering general practitioners, medication, dentistry and optometry is not, and must be provided under the Medical Schemes Act [15]. A product team that drifts across that line has not built a bad product; it has built an illegal one. The second is the microinsurance licence, a distinct lighter licence class created by the Insurance Act for financial inclusion, which can cover both life and non-life products [16]. The Prudential Authority counted 14 life microinsurers and three composite microinsurers at March 2026 [4]. If a product is being built for that licence, the benefit limits and governance requirements are part of the product definition, not a compliance review at the end.

Architecture. The product must be data. A product factory holding effective-dated wording, rate tables and rules, able to reconstruct any policy's complete terms as at any date, is the capability with the largest payoff in an insurer. Most insurers know this. Few have it, because retrofitting it means admitting that the current product definitions exist in six places: the policy system, the rating engine, the document templates, a spreadsheet the actuary maintains, the broker portal, and a Word file in legal.

2. Distribution, and why it shapes the architecture more than anything else

This is the longest section in the chapter, on purpose. If you remember one thing from this book about why two insurers selling the same product have completely different system estates, it is distribution.

The models

Model	Who owns the customer	How they are paid	What the channel may do	What it demands of the architecture
Tied agents, own advisers	The insurer	Salary plus commission	Quote, advise, submit	Adviser portal, needs analysis, hierarchy and commission engine
Independent brokers	The broker	Commission, sometimes binder fees and profit share	Quote across several insurers; in South Africa often issue, service and settle claims under a binder	Broker portal and API, quote services, bordereaux exchange, delegated-authority controls
Bancassurance	The bank	Intra-group transfer or commission	Sell at the point of credit, in branch or in the app	Integration into banking channels, credit life triggered off loan origination, shared identity
Direct to consumer	The insurer	No commission; marketing spend instead	Everything	Lowest-latency quoting, straight-through underwriting, self-service, high automation
Aggregators and comparison sites	The platform	Per lead, per policy, or commission [12]	Quote, sometimes fulfil	A real-time rateable quote API; price transparency makes rating sophistication the only defence
Affinity and embedded	The partner brand	Commission or profit share	Sell insurance as a feature of a non-insurance purchase	Embeddable API, partner-specific product variants, white-label documents, per-partner settlement
UMA or MGA	The UMA	Fee plus profit share	Underwrite, price, issue, service, often settle claims under delegated authority	The insurer consumes rather than generates data: bordereaux ingestion, aggregate monitoring, oversight reporting

Six reasons distribution dominates

One: it sets the latency budget. An aggregator or an embedded checkout needs a priced, bindable quote in well under a second, because the customer is mid-transaction and will not wait. A commercial broker submission can take three weeks and nobody minds. Those are not two settings of the same system.

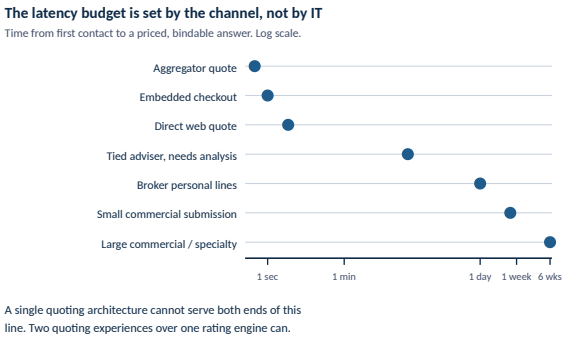


Figure 7.3 Time from first contact to a priced, bindable answer, by channel. Indicative durations, log scale.

An insurer serving both ends of that line needs two quoting experiences over one rating engine. The common failure is to build the slow one first, because that is where the money was, and then bolt a cache in front of it when the aggregator deal is signed. The cache goes stale, the bound price differs from the quoted price, and the complaints start.

Two: it sets where data is created. This is the one architects underestimate. If an insurer sells direct, it owns every event: every quote, every abandoned basket, every servicing call. If it sells through a UMA under a binder, it receives *periodic summaries, in somebody else's format, after the fact*. Its data model is downstream of another firm's operational system, with all the lag and reconciliation that implies. Those two insurers can sell identical products and need entirely different architectures.

Put that concretely. The same motor policy, sold five ways, gives the insurer five different data positions.

Sold through	What the insurer sees at the moment of sale	What it sees afterwards	Latency of the truth
Direct web	Every keystroke, every abandoned quote	Every servicing event, every login	Real time
Tied adviser	The application and the needs analysis	Most servicing, some adviser-held detail	Minutes to days
Independent broker, no binder	The application as the broker keyed it	Servicing requests, not broker-side context	Hours to days
Broker under binder	A policy record created by the broker's system	A monthly bordereau	Up to a month
UMA	Nothing at the time	A bordereau in the UMA's own format	Up to a month, plus normalisation

The insurer in the bottom two rows cannot do same-day accumulation monitoring, cannot detect a fraud ring across the book in time to matter, and cannot run a pricing feedback loop faster than monthly. None of that is fixable with a better data warehouse. It is a consequence of the commercial model, and the architecture has to be honest about it.

Three: it sets who makes the decision the insurer is liable for. A South African insurer remains liable for every policy a UMA issues in its name and must pay valid claims on them [2]; an insurer outsourcing critical functions remains ultimately responsible for them [1]. The decision is delegated. The liability is not. That turns oversight into a system requirement: entitlements, limits, exception monitoring and audit have to be built as product features, not promised in a service-level agreement.

Four: it sets the economics the systems must support. Commission, binder fees, lead fees, profit shares and ceding commissions are different remuneration models with different regulatory constraints. South Africa's 2017 and 2018 reforms specifically limit who may hold a binder, which activities may be outsourced under one, and what remuneration may be paid, to manage conflicts of interest [1]. A commission engine is therefore not a calculation; it is a regulated calculation with an audit trail.

Five: it sets the identity problem. A multi-channel insurer holds the same human being as several customers: once in the life system from a tied adviser, once in the short-term system from a broker, once in the digital stack from a quote they abandoned in 2021. Cross-sell, fraud detection, single customer view and fair-outcomes testing all depend on resolving that. It is almost never solved inside the policy administration systems, because they were never asked to.

Six: channel mix is a strategic target, so the architecture has to flex. Santam's direct channels were 22% of business in FY2025 against a stated target above 30% [6]. Discovery Bank drew 70% of its new business from outside the group [17]. South African broker consolidation is accelerating, strengthening procurement power and influencing pricing, while banks expand their insurance ecosystems and retailers and mobile networks extend into funeral, device and micro short-term cover [7]. None of that is a five-year plan. It is this year's board paper, and the architecture has to absorb it.

UMAs, binders and cell captives

Three South African structures deserve their own treatment because they change what an insurer actually is.

A **UMA** — underwriting management agency, called an MGA in most other markets — holds delegated authority from a licensed insurer to underwrite, price, issue, service and often settle claims within an agreed mandate. UMAs must be authorised financial services providers in their own right. The insurer is the principal: it carries the risk, it must pay valid claims, and it must manage the relationship and enforce conduct standards [2]. South Africa's UMA sector is unusually large relative to comparable markets, and insurers court it. Guardrisk, for instance, lists UMA as one of its named business lines alongside cell captives, microinsurance and embedded insurance [3].

> **A NUMBER NOT TO QUOTE**
>
> You will hear confident figures for the UMA share of South African short-term premium. I could not trace any of them to a primary source, and neither could anyone I asked. Describe the model, which is real and important. Do not publish a share figure you cannot source, because somebody will build a business case on it.

A **binder** is the legal instrument that delegates specific activities — entering into, varying or renewing a policy, determining wordings or premiums, or settling claims — to an intermediary. Binders are ordinary in South African short-term insurance: a large independent broker will commonly issue the policy, service it, and settle claims up to a limit, all on the insurer's paper. The 2017 reforms restricted who may hold one and what may be paid for it [1].

A **cell captive** is a different thing again. The licensed insurer issues classes of shares — cells — to third parties. The cell owner participates in the underwriting result of its own ring-fenced book while the licensed insurer holds the licence and carries ultimate liability. It is the standard South African route for a retailer, a bank or an affinity group to take insurance risk without obtaining its own licence, and it is the country's native vehicle for embedded insurance. The Prudential Authority reports cell captives as separate categories and they are material: non-life cell captive gross premium was R9.6 billion in the March 2026 quarter, around a fifth of non-life primary premium, at a 38.6% claims ratio but a median solvency cover of only 1.2 times [4]. Capitec ran its credit life and funeral books through two cell captive agreements before obtaining its own licence, effective 3 October 2022 [14].

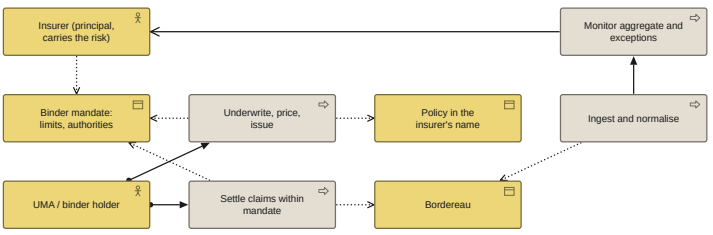


Figure 7.4 Delegated authority. The UMA or binder holder makes the decisions; the insurer carries the liability and sees the business as a bordereau after the fact [2].

What goes wrong. Mis-selling and poor advice, which the conduct reforms target explicitly [1]. Quote-to-bind leakage nobody can measure because the quote happened in a partner's system. Channel conflict: the same risk priced differently direct and through a broker, discovered by a customer. Commission structures that reward volume over quality and produce early lapse. And point-of-sale data that is unusable downstream because every channel invented its own schema.

Architecture. In a binder-heavy market the insurer's "policy administration system" is frequently not where policies are administered. The real architecture becomes delegated-authority management, bordereaux ingestion and normalisation across dozens of partner formats, aggregate exposure monitoring over transactions the insurer did not create, claims authority limits and oversight, and reinsurance cession computed on somebody else's

data. It is a data integration and governance architecture wearing a transaction-processing costume, and it is the most distinctive feature of South African non-life IT.

3. Quote and bind

What happens. A risk is described, priced, offered, accepted, and a contract comes into existence. In personal lines this is seconds. In commercial lines it is a negotiation with documents.

Who. The customer, an adviser or a broker, a rating engine, and in referral cases an underwriter.

What gets decided. The price, the terms offered, and whether the channel is allowed to bind at all without referral.

What it needs. The risk attributes, the product definition as at today, the rate tables, enrichment data, and the channel's entitlements.

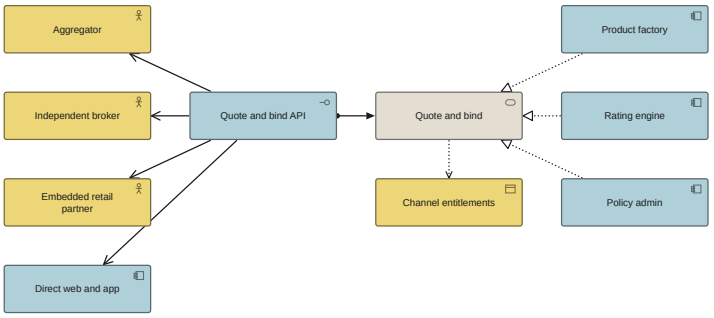


Figure 7.5 Many front ends over one quote-and-bind service, with channel entitlements as enforced configuration rather than per-channel code.

What goes wrong. Three things, repeatedly.

The first is quote-and-bind divergence. The quote is produced by one path and the bind by another, usually because the digital stack was built later and separately. Prices differ. Nobody notices until a customer screenshots both.

The second is entitlements held as convention. "Boland Brokers can bind up to R5 million on commercial property" lives in an email and in the head of the regional manager, not as enforced configuration. When the manager leaves, the limit leaves with him.

The third is that quotes are not kept. A quote is a priced offer made by the insurer. It is evidence of what was offered, on what information, under what rules. Insurers that discard unconverted quotes lose their conversion analytics, their pricing feedback loop and their defence in a conduct complaint, all at once.

Architecture. Many front ends over one quote-and-bind service, with channel entitlements — quote, bind, issue, endorse, settle — as explicit, policy-enforced configuration rather than code branches per channel. Quotes stored as first-class objects with the rule version that produced them.

4. Underwriting

Underwriting is risk selection and pricing for a specific case. The word covers an enormous range.

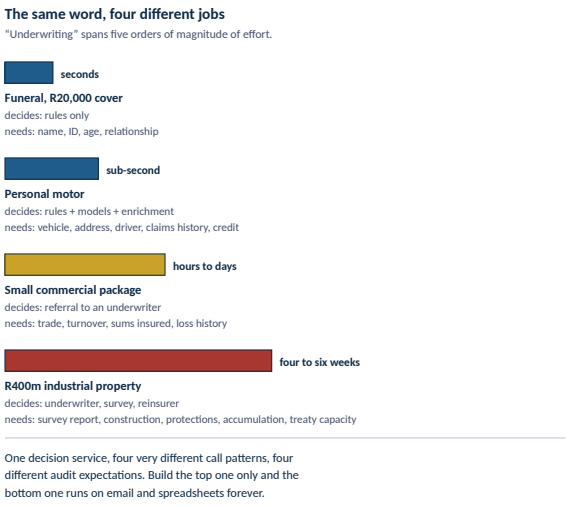


Figure 7.6 The range the word 'underwriting' covers. Durations are indicative of normal practice, not measured.

There are three modes and their architectures are not related.

Straight-through. Personal lines, funeral, simple credit life. Rules and models decide, no human is involved, and the answer comes back in well under a second. A funeral policy for R20,000 may ask for a name, an identity number, ages and relationships, and nothing else. The underwriting is the product design: the waiting period does the work that a medical examination would do on a larger sum assured.

Referral. Rules route the exceptions to an underwriter's workbench. The underwriter sees the case, the reason it referred, and the supporting data, and decides.

Manual and negotiated. Commercial and specialty. A submission arrives from a broker. There may be a survey of the premises by a risk engineer, five years of loss history, a schedule of two hundred locations, and a negotiation over wording. For a R400 million industrial property risk, four to six weeks is normal, and the insurer will be talking to its reinsurers before it can quote at all.

At the extreme end of that range sits the subscription market. At Lloyd's, several syndicates each take a share of one risk, negotiated face to face through brokers; the market reported 103 syndicates and 3,015 approved coverholders at the end of 2025 [11]. A South African insurer meets this world when it places a large or unusual risk abroad, and the data consequence is the same one the UMA model produces at home: the insurer's share of a risk is described in somebody else's document, and arrives as a line on a statement rather than as a transaction.

Who. A rules engine; an underwriter; for life, a medical underwriter and often the reinsurer on large sums assured; for commercial property, a surveyor or risk engineer.

What gets decided. Accept, decline, or accept with a loading, an exclusion or a higher excess. Sums insured and limits. How much to retain and how much to cede. Whether to require a survey or a medical examination.

What goes wrong.

- **Anti-selection.** Price too low for bad risks and the bad risks find you. They are better at this than you are, because they know things about themselves that you do not.
- **Accumulation blindness.** Four hundred policies in one flood plain, or on one city block, with no aggregate view. This is no longer theoretical in South Africa: weather catastrophes were the largest single force on short-term claims costs in the first half of 2026 [8], and Santam's weather catastrophe losses have roughly doubled in size over the past ten years [8].
- **Referral rules so loose that straight-through rates collapse.** Every percentage point of referral is cost-to-serve. Underwriters end up rubber-stamping cases the rules should have taken.
- **Model drift**, and rating factors captured as free text so that nobody can analyse them later.

Architecture. A decision service with versioned rules and an audit trail of *why* each decision was made, which is increasingly required where models assist the decision. Geospatial accumulation on live exposure, not on a quarterly extract. Third-party enrichment: credit data, telematics, geocoded peril scores, vehicle data. South African insurers have rebuilt pricing models around spatial analytics and geocoded climate risk rather than relying mainly on historical trends [8], and that is an integration programme before it is an analytics one.

5. Policy issue and servicing

This is where most of an insurer's IT cost and most of its customer dissatisfaction live, and it is routinely the least interesting part of the programme to everyone senior.

What happens. The schedule and documents are issued. Then the contract starts changing. A vehicle is added. An address changes. Cover is increased. A beneficiary is nominated. Bank details change. The policy renews, or in a monthly-renewable book it continues. It lapses, and it may be reinstated. Somebody complains.

Who. The service centre, a self-service portal, the broker, or a binder holder, which in South Africa is extremely common.

What gets decided. Whether an endorsement triggers a re-rate. What terms to offer at renewal. Whether to reinstate a lapsed policy, and on what basis.

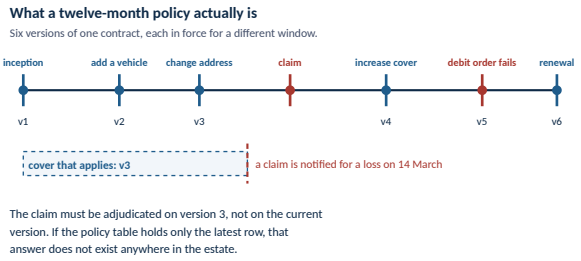


Figure 7.7 A twelve-month policy is a sequence of versions. A claim is adjudicated against the version in force at the date of loss.

On renewals, and the South African wrinkle. In most markets a short-term policy runs for twelve months and renews on an anniversary, which gives the insurer one annual opportunity to re-rate, one renewal invitation to produce, and one retention conversation. Much South African personal lines business is written on a monthly basis instead, continuing month to month until cancelled. That sounds like a minor contractual difference. It changes the systems substantially: there is no annual renewal event to hang a re-rate on, so rate changes are applied as mid-term adjustments with notice periods; retention is a continuous activity rather than an annual campaign; and the natural unit of exposure measurement becomes the month, not the policy year. An architect arriving from a market where annual renewal is assumed will look for a renewal batch and not find one.

What goes wrong. This list is long because the function is large.

- **Endorsements implemented as cancel-and-rewrite.** The old policy is cancelled, a new one issued. Continuous cover history is destroyed, which matters for no-claim bonuses, for claims history, and for conduct.
- **Mid-term premium not pro-rated correctly**, so the customer is charged for cover they did not have or given cover they did not pay for.
- **The system of record disagreeing with the document that was sent.** The document is the contract. If they differ, the insurer loses.
- **Several inherited policy administration systems**, each with its own customer identifier, so there is no single view of the customer. Legacy bancassurance platforms are called out specifically in South Africa as a constraint on scaling digital servicing [7].
- **No ability to answer the as-at question.** "What did this policy look like on 14 March 2023, and who changed it?" If the answer requires a database restore, the insurer cannot defend a claims decision or a conduct complaint.

Architecture. The policy lifecycle is an event-sourced, effective-dated transaction log, not a set of mutable rows. Every version of the contract is retained with the period it was in force, the change that produced it, and the person or process that made it. This is the single most valuable thing a core replacement can buy, and it is the first thing cut when the programme runs late.

6. Premium collection, which is harder here

In most mature markets collection is a back-office function. In South Africa it is a risk function, and the difference is worth understanding properly.

What happens. Premium is collected by debit order, card, payroll deduction, cash at a retailer, or premium finance for commercial clients. Failed collections are retried according to a strategy. There are grace periods. Everything has to be reconciled, including premium collected by intermediaries and held in trust before remittance.

Who. Collections operations, the bank or payment provider, the broker or UMA collecting on the insurer's behalf, the employer for group schemes, and the retailer for in-store products.

What gets decided. The collection day. The retry strategy. When a missed premium becomes a lapse. And the question that causes the most trouble: is cover in force during the grace period?

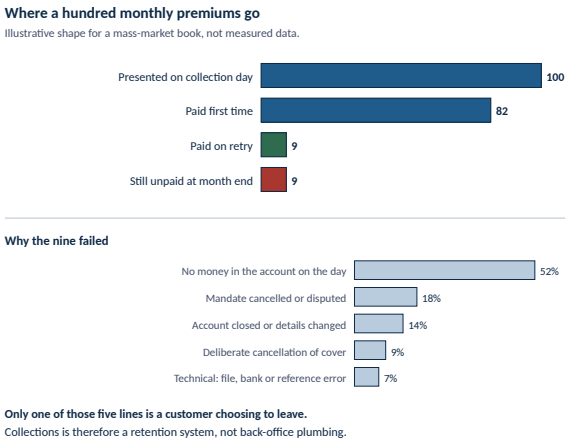


Figure 7.8 A mass-market collections funnel and the reasons for failure. Illustrative shape, not measured data: the point is the proportions, not the digits.

Why it is harder in South Africa. Several reasons stack.

The first is the income profile of the mass market. For a household collecting an income on the 25th, a debit order on the 1st and a debit order on the 28th have radically different success rates. Retry-day optimisation is not a marginal gain; it moves lapse rates measurably.

The second is the payment infrastructure itself. The two early debit order instruments that insurers relied on for years, AEDO and NAEDO, have been retired in favour of DebiCheck, which requires the customer to authenticate the mandate electronically with their bank. The replacement was instructed by the payments association and ran as a phased programme with deadlines that moved more than once; the published plan in early 2020 described the phase-out ending that October [21], and the instruments were finally switched off on 31 October 2021. DebiCheck is better for the customer and worse for a collections operation that was built around the old behaviour, because an unauthenticated mandate is now a dead mandate rather than a retryable one.

The third is the sheer scale of churn. In 2025, 8.7 million risk policies lapsed in South Africa [5]. In a mass-market book the dominant cause of lapse is not a customer deciding to cancel. It is a collection failing. That distinction is the whole argument for treating collections as a retention capability.

The fourth is intermediated collection. When a broker or UMA collects, the insurer receives an aggregate remittance and a bordereau. If the bordereau cannot be allocated to policies, the insurer knows how much money arrived but not whose cover is paid up. At claim time, cover-in-force status is then genuinely ambiguous, and the ambiguity is resolved in the customer's favour or in the insurer's, depending on who is in the room.

What goes wrong. Unreconciled intermediary bordereaux. Policy status derived differently by three systems. Grace-period logic implemented in code rather than configured, so a change in product terms needs a release. And cover-in-force at loss date computed at claim time by a human reading a ledger.

Architecture. Policy status and cover-in-force must be derived deterministically and reproducibly from the premium ledger, as at any date, by one service that everything else consults. Retry strategy must be configurable and measurable per product and per segment, because it is a lever with real money

attached.

7. Claims, from first notification to closure

Claims is the product as the customer experiences it. Everything before this is paperwork.

What happens. The loss is notified. The claim is registered. Cover is validated against the policy as it stood at the date of loss. Liability is assessed. The amount is quantified. The claim is screened for fraud. A case reserve is posted. The claim is settled — in cash, or by repair, replacement or a service. Recoveries are pursued. The reinsurance recovery is calculated. The file is closed, and it may be reopened.

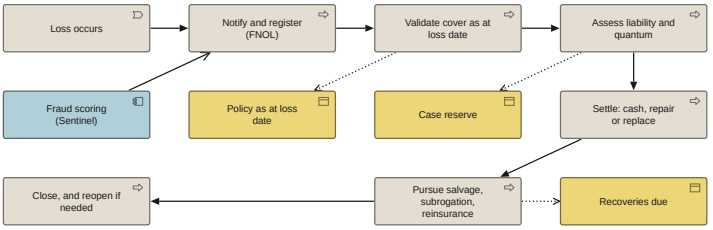


Figure 7.9 Claims from first notification to closure, with recoveries treated as part of the chain rather than as an afterthought.

Who. A claims handler. A loss adjuster or assessor for anything substantial. A panel repairer, a medical provider or a funeral parlour. A special investigations unit for suspected fraud. A recoveries team. The reinsurer. And in South Africa, very often a claims binder holder doing most of the above on the insurer's behalf.

What gets decided. Is it covered? Who is liable, and for how much? Repair or write off? Settle or litigate? What case reserve to post? Is it fraudulent?

Recoveries deserve their own paragraph, because they are where money quietly disappears. **Salvage** is the residual value of the damaged property the insurer now owns, having paid for it: the written-off vehicle sold at auction. **Subrogation** is the insurer standing in the shoes of its own insured to pursue whoever caused the loss, usually that person's insurer. **Third-party recovery** and **reinsurance recovery** complete the set. All four are computed after the claim is paid, by different people, often in different systems, and each one is a leak if it is missed.

What goes wrong.

- **Claims leakage**: paying more than the contract requires. Weak cover validation, over-settlement, missed recoveries, missed reinsurance recoveries. It is rarely one big error; it is a thousand small ones.
- **Under-reserved case reserves**, which distort the development triangle and therefore the whole reserving exercise.
- **Catastrophe surge.** Ten to fifty times normal volume, arriving in three days, with no elastic capacity. Weather catastrophes were the largest driver of South African short-term claims costs in the first half of 2026 [8]; Old Mutual Insure's May floods cost R376 million after reinsurance [20]; Santam's early-2026 northern flooding and Western Cape wildfires cost about R300 million in two months [6]. Surge is a design requirement now, not a contingency.
- **Repair cost inflation.** Vehicles are increasingly written off after accidents that would once have been repaired [8], driven by materials, labour and imported parts costs [7].
- **Fraud detected after payment**, which is to say not detected.

Architecture. First notification on any channel. Automated cover validation against the policy as at the loss date — which is only possible if section five was done properly. Supply chain integration with panel repairers, medical providers and assessors. Fraud analytics at registration, not in a monthly batch. Reinsurance recovery derived per claim rather than reconstructed quarterly. And elastic capacity for surge, which in practice means the claims intake path must be able to run without the core system in the loop for a few days.

	Motor own damage claim	Funeral claim
Notified by	Customer, broker, or a tow operator at the scene	Family member, often by phone, often within hours
First question	Is it covered, and who was at fault	Who died, and are they on the policy
Evidence	Accident report, photographs, assessor's report, quotations	Death certificate, identity documents, proof of relationship
External parties	Assessor, panel repairer, salvage yard, third-party insurer	Funeral parlour, sometimes none
Typical cycle time	Weeks	Hours to two days
Cost to handle	Hundreds of rands, sometimes thousands	As close to zero as possible
What the architecture optimises	Accuracy, recovery, leakage control	Speed, document capture, payment

Both of those are "claims". Cape Mutual runs both, and the market expectation of a near-immediate funeral payout is a practitioner norm rather than a regulated service level — but an insurer that misses it loses the family, and the family is the distribution channel.

8. Reinsurance operations

Chapter 6 covered what reinsurance is. This is the operational cycle.

What happens. The programme is designed, usually annually. It is placed, normally through a reinsurance broker. Business is ceded according to treaty terms. Bordereaux and technical accounts are produced for the reinsurers. Large losses and catastrophes are notified. Recoveries are claimed. Reinstatements of exhausted cover are managed. The programme renews — 1 January and 1 July are the main dates in the global market.

Who. The reinsurance manager, the cedant's actuarial function, a reinsurance broker, the reinsurers, and their own retrocessionaires.

What gets decided. The retention level. Proportional or excess of loss [9]. Limits and reinstatements. Which reinsurers, which is a credit decision as much as a pricing one. And how much capital relief to buy as against how much risk to transfer.

What goes wrong. Recoveries that are not automatically derived, and are therefore missed or claimed late — pure cash leakage. Treaty terms held in spreadsheets and PDFs rather than as machine-readable rules, so cession is a manual quarterly exercise. Inwards and outwards views that cannot be reconciled. And a hardening market, which raises net retentions and earnings volatility directly [7][8] and makes every modelling weakness more expensive.

One South African structure that lands in the policy system. Sasria, the state-owned special risks insurer created in 1979 after the 1976 Soweto Uprising, covers strike, riot and public-disorder loss. The July 2021 unrest caused damages exceeding R31 billion and reshaped the global political-violence market; Sasria's funds have since risen to R18.6 billion against a target of R30 billion of capital reserves by 2029, and its Wrap Cover product for large corporates relaunched in 2026 as excess cover above a R500 million primary coupon, with a R500 million limit [18]. The architectural consequence is that Sasria cover is attached to an underlying policy written by another insurer, which means every South African non-life policy administration system needs a construct for it plus a separate remittance and accounting flow. That mechanism is standard market practice rather than something stated in the source above, so take the attachment mechanic as my description and the figures as sourced.

Architecture. Treaty terms modelled as data; cession and recovery computed per transaction and reconciled continuously. This is one of the most commonly under-architected domains in an insurer, and the reason is organisational: reinsurance sits in finance, the data sits in operations, and no one owns the join.

9. Reserving and the actuarial cycle

What happens. Every quarter, and more intensively at year end, the actuarial function estimates what the insurer owes. For short-term that is case reserves plus IBNR, built from development triangles [10]. For life it is the present value of future obligations under a set of assumptions. The solvency capital requirement and minimum capital requirement are computed. IFRS 17 measurement is produced. Regulatory returns are filed. Embedded value is disclosed.

Who. The reserving actuary, the Head of Actuarial Function as a statutory control role, finance, the external auditor, and the Prudential Authority.

What gets decided. Method and assumptions. The margin for prudence. Discount rates. Whether to change an assumption, and if so how to explain the effect.

What goes wrong. Reserve data unavailable at the granularity required — by class, by accident period, by development period, gross and net of reinsurance — so the work is done on manual extracts. Reserve releases or strengthening that nobody in the business can explain. IFRS 17 and statutory bases reconciled by hand. Adverse development discovered years late.

On assumption changes, because they confuse newcomers. When an actuary changes a mortality or lapse assumption, the reported result moves, sometimes by a lot, with nothing having happened in the real world that quarter. The movement is real in accounting terms and meaningless in operational terms. Management reporting therefore has to decompose the result into experience variance (what actually happened against what was expected) and assumption change (what we now think will happen). If the data foundation cannot produce that split, the business will argue about the number instead of about the business, every quarter, for ever.

Architecture. Actuarial and finance need a contract-level, movement-complete, effective-dated data foundation, available on demand rather than as a quarterly extract. It is usually the hardest data problem in the enterprise, and it is where most insurers' data programmes actually get funded, whatever the business case says on the cover.

10. The reporting calendar

An insurer's year has a shape, and the shape is non-negotiable.

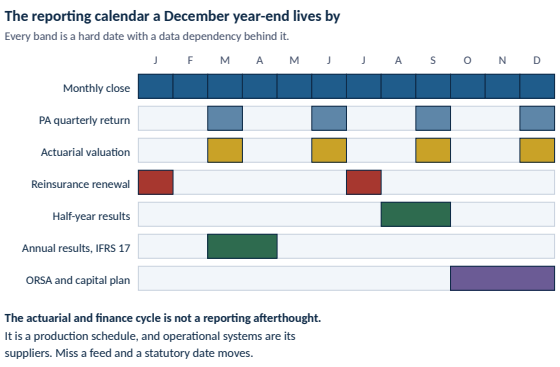


Figure 7.10 The reporting calendar for a December year-end. Each band is a statutory or contractual date with an operational data dependency behind it.

Monthly close. Quarterly returns to the Prudential Authority, which publishes sector data from them [4]. Quarterly actuarial valuation. Reinsurance renewal at 1 January and often 1 July. Half-year and annual results, with IFRS 17 disclosures. An own risk and solvency assessment and capital plan on an annual cycle. For life insurers, embedded value disclosure alongside results [17].

There is a regulatory programme calendar sitting on top of all that. The Financial Sector Conduct Authority's plan for 2026 to 2029 puts a Joint Standard on Governance as its top priority, with themed frameworks ahead of the Conduct of Financial Institutions Bill, cross-sector licensing, Joint Standards on third-party service provision, and draft Joint Standards on operational risk and resilience and on cloud computing and offshoring of data [13]. Those last two are the items that constrain technology strategy directly: cloud region, data residency, exit planning and provider concentration stop being procurement preferences and become regulated matters.

What goes wrong. A missed data feed moves a statutory date. That sounds dramatic until you see it happen, at which point it becomes a very short conversation with a regulator. The architectural implication is unglamorous and important: the data pipelines that feed the actuarial and finance cycle are production systems with service levels, not analytics plumbing, and they should be operated as such.

Walking it at Cape Mutual

Put the whole chain against the fictional group and the pattern that every architect meets appears immediately.

Step	Where it happens at Cape Mutual	The problem
Product	Polaris for life, Atlas for short-term, plus a spreadsheet of rates	Three definitions of the same product; wording lives in legal's file share
Distribution	Beacon for brokers and advisers, Kestrel for digital, bordereaux from the liability UMA	Three customer identities for one person; UMA data arrives monthly in CSV
Quote and bind	Kestrel quotes; Atlas binds	Prices can differ; unconverted quotes are discarded after 30 days
Underwriting	Rules in Atlas, referrals by email	No audit trail of why a referral was accepted
Servicing	Atlas and Polaris; brokers endorse under binder	Endorsements in Polaris are cancel-and-rewrite
Collection	One collections module per platform	Cover-in-force derived two different ways
Claims	Atlas claims, Sentinel for fraud scoring, manual funeral claims	Reinsurance recovery reconstructed quarterly from an extract
Reinsurance	A spreadsheet maintained by one person	No machine-readable treaty terms
Reserving	Extracts into the actuarial tool	The shadow data estate is the real system of record
Reporting	Ledger receives summarised postings	Contract-level traceability does not exist

Cape Mutual is invented and nothing here describes a real company. The pattern, though, is close to universal, and it is worth naming what it is. None of these are failures of any individual system. Every one of them is a failure at a seam — between two platforms, between an insurer and an intermediary, between operations and finance. That is the useful thing to know walking in. The architecture work in an insurer is almost never inside a box. It is between the boxes, and the boxes belong to different people.

What to take from this chapter

Thandi Nkosi, Cape Mutual's chief operating officer, has a line she uses on new architects: *every one of these ten steps thinks it is the main one, and every one of them is right*. Product thinks the chain starts with it. Claims thinks everything before it is administration. Actuarial thinks the rest of the business exists to generate its data. All three are defensible positions, and an architecture that privileges one of them produces a business that is excellent at one step and incoherent across the chain.

The counter to that is boring and effective. Model the chain end to end before you model any part of it in detail. Establish for each step who decides, what they decide, what data they need and what they produce. Then look at the seams, because that is where the work is, and that is where the money goes.

The next chapters take the same ground from the other side: the sixteen categories of system that implement this chain, who sells them, and the data model underneath.

THE CHAPTER IN TEN LINES

- Life and short-term insurance are different businesses that share a word, and no architecture should pretend otherwise.
- Product must be data: effective-dated wording, rates and rules, reconstructable as at any past date.
- Distribution shapes the architecture more than any other factor, because it sets latency, data ownership, liability, economics and identity.
- Under a binder or a UMA the insurer carries the liability but sees the business only as a bordereau, after the fact.
- Quote and bind belong in one service behind many front ends, with channel entitlements as enforced configuration.
- Underwriting spans five orders of magnitude of effort, from a seconds-long funeral decision to a six-week commercial risk.
- Servicing is where most IT cost lives, and the as-at question is the one legacy systems cannot answer.
- In South Africa most lapses are failed collections rather than cancellations, which makes collections a retention capability.
- Claims is the product as experienced, and recoveries — salvage, subrogation, third party and reinsurance — are where money quietly leaks.
- The actuarial and reporting calendar is a production schedule, and the operational systems are its suppliers.

SOURCES FOR THIS CHAPTER

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CHAPTER 8

The process model

A process architecture is the inventory of what an insurer does; a process model is the inside of one of them. Here are both, at level one and level two, drawn in BPMN with the roles in lanes — because the question that matters is always who is holding the work right now.

The 412 pages

In her second week at Cape Mutual, Fatima Patel asked for the process model. She was sent a folder containing 412 pages of diagrams, last edited in 2023, and a spreadsheet index. Most of it described motor claims at a level of detail that included which tab to open in Atlas. None of it described how a claim gets from a broker's email to an assessor's phone, which is the step that takes four days.

That is the normal starting position, and it is a specific kind of failure. The organisation had documented procedures and called it a process architecture. Nobody had ever written down the list of its processes.

> A **process architecture** is the inventory: which end-to-end processes exist, what triggers each one, what outcome ends it, who owns it, and how they fit together. A **process model** is the inside of one of them: steps, roles, decisions, data. The first is a map of the country. The second is a street plan of one town. You can have a thousand street plans and still not know how many towns there are.

An architect needs both, for four reasons that have nothing to do with documentation.

A process is where a capability becomes work with a cost. "Claims management" is a capability; it has no cycle time and no unit cost. "Handle a motor claim" is a process; it takes eleven days and R1,340 per claim at Cape Mutual, and both numbers are negotiable. Capabilities are for deciding what matters. Processes are for deciding what to change.

Processes are the unit the business argues about. No operations director has ever disputed a capability map, because nothing is at stake in one. Put a swimlane diagram of claims registration on the wall and the claims manager will dispute it within ninety seconds, which is the point. A model that provokes no argument has told you nothing.

A process model is the only architecture artefact that forces you to name who decides. Application landscapes hide this. Data models hide it completely. The moment you draw lanes you have to answer: who repudiates a claim, who may bind R5 million of commercial property, who decides a premium has failed for the last time. In an insurer those answers are frequently not written down anywhere, and the gap between the authority people have and the authority the systems enforce is a standing audit finding.

Systems are bought and replaced along process boundaries. When an insurer buys a claims platform it is buying a span of process. If the inventory says claims ends at payment, recovery and reinsurance recovery fall outside the procurement, which is how insurers end up with recoveries in a spreadsheet for the next decade. A wrong boundary at L1 becomes a wrong contract at signature.

Why this chapter is drawn in BPMN

ArchiMate is the right notation for relating a process to a capability, an actor, a business object and an application service, and Chapter 16 does that properly. It is a poor notation for sequence. It has no gateway, its triggering relationship carries no condition, and it cannot show a handover between two performers without a convention you have to teach your audience.

BPMN 2.0 [1] is a sequence notation with a lane per performer. It answers the operations manager's question — who is holding this work right now, and what are they waiting for — and it answers the architect's question — where does the work cross an organisational boundary, because that is where it stops. Use both notations. Use BPMN when the roles are the point, which in this chapter they are.

This chapter is the one place in the book that leaves ArchiMate, and it is worth being plain about why. ArchiMate models the process as an element in an architecture: one box, named for what it achieves, with a role assigned to it, a capability it realises, the objects it accesses and the applications that serve it. BPMN models what happens inside that box — the order, the conditions, the waiting, and the hand-off from one pair of hands to the next. The two are complementary, and the join between them is mechanical. Each pool below is one of the ArchiMate Process elements from the L1 map: the quote-and-bind pool is process 3, the two claim pools are process 8, the cession and recovery pools are process 11. Each lane is a Role assigned to that Process in the ArchiMate model, and each system lane is an application component serving it. Keep the names identical in both notations and the two models stay reconcilable. Let them drift and you have two process architectures, and a meeting about which one is true.

The roles in the lanes are the jobs from Chapter 5. A short reminder of the ones that appear in this chapter, because an architect who cannot name the jobs cannot run the workshop:

Role in the lanes	What the person actually does all day
Broker	Holds the client relationship, shops the risk, and under a binder issues and services the policy on the insurer's paper
Underwriter	Decides whether to accept a risk, on what terms, and at what price, for the cases the rules would not take
Policy servicing clerk	Takes change requests on the phone and in writing, keys them, and decides from when they apply
Claims handler	Works a queue of forty to a hundred open files, decides cover and quantum up to an authority limit, and talks to upset people
Claims manager	Owns the authority limits, the panel and the escalations, and signs the claims above the handler's limit
Assessor or loss adjuster	Goes and looks at the damage, verifies the quote, and writes a report the insurer pays attention to
Collections clerk	Builds the collection files, reads the unpaid reason codes, and decides when to retry and when to lapse
Recoveries clerk	Chases salvage proceeds and other insurers, for years, on claims everyone else has closed
Reinsurance manager	Owns the treaty programme, computes cessions, and claims recoveries; frequently one person with a spreadsheet

L0, L1, L2, L3 — and the warning

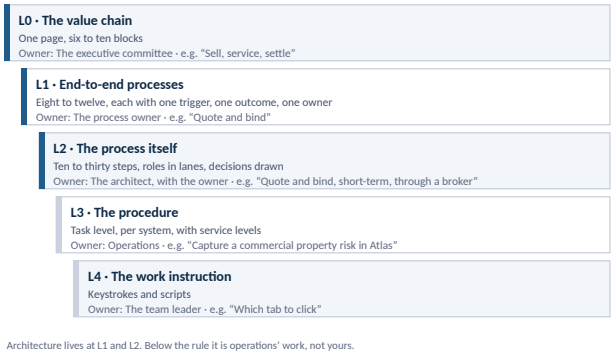


Figure 8.1 The level scheme this chapter uses. The labels are conventions, not standards, and other frameworks number them differently — settle yours in writing before the first workshop.

Those are the definitions this chapter uses. They are not the only ones in circulation and they are not authoritative.

APQC's cross-industry process framework numbers its levels category, process group, process, activity and task, which makes "level 2" a process *group* [2]. ACORD's Reference Architecture takes a different tack again: its Capability Model is a functional decomposition published deliberately *without* a predefined workflow, and the Process Model is a separate facet of workflow-oriented reference processes aligned to it [3]. In many insurers "L1" means the one-page value chain, which this chapter calls L0, and everything shifts down a number.

> **SETTLE THE LEVELS IN WRITING BEFORE THE FIRST WORKSHOP**
>
> This sounds pedantic. It is the single most expensive piece of pedantry you will skip. A programme with two definitions of "level 2" produces two populations of models at different grain, discovers the mismatch at integration, and reconciles by hand. Write the definitions on one page, get the process owners to initial it, and put it in the modelling conventions. Half an hour, week one.

Two more points about levels before the map.

The levels are not a decomposition of equal depth. It is normal and correct for an insurer to have twelve L1 processes, six of which are modelled at L2 because they carry the volume, two at L3 because they are being automated, and four not modelled below L1 at all because nothing is changing there. A programme that insists on uniform depth is producing documents, not architecture.

Below L2, stop. L3 and L4 belong to operations. They change weekly, they are tied to a screen layout, and an architecture repository that holds them is a repository that is wrong. Model L1 and L2, point at the operational procedures, and do not take ownership of them.

The L1 process map

Here is a defensible L1 inventory for a composite insurer. Twelve processes, grouped four ways.

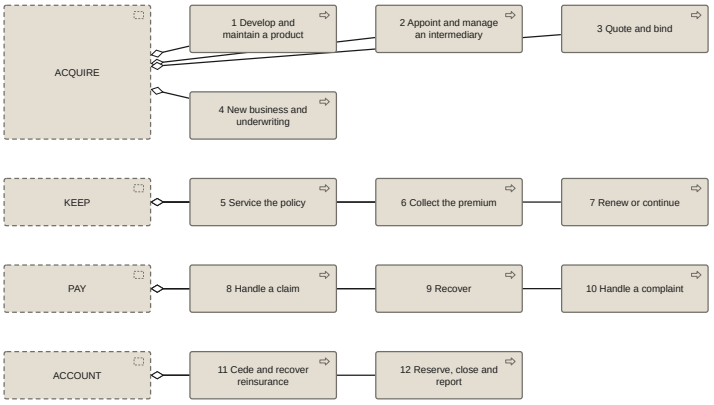


Figure 8.2 An L1 process inventory for a composite insurer: twelve ArchiMate Process elements aggregated into four groupings by what they are for rather than by which department performs them. The order inside a grouping carries no meaning — that is what the L2 models are for.

And here is the part that makes it an architecture rather than a list: each one has a trigger, an outcome and an owner, and the honest note at the end of the row.

#	Process	Trigger	Outcome	Honest about it
1	Develop and maintain a product	A market, pricing or regulatory decision	A product version live and consistent in every consuming system	A governed **lifecycle**, not a triggered process. Model it as a value chain with gates
2	Appoint and manage an intermediary	An application to transact	An accredited intermediary with a mandate, a licence check and a remuneration agreement	A real process. Almost never modelled. Almost always broken
3	Quote and bind	A request for a price	A priced offer, accepted or lapsed	The most forked process in an insurer: one variant per channel
4	New business and underwriting	A submitted application or an accepted offer	A policy in force, or a decline, with reasons	Seconds in funeral, six weeks in commercial property
5	Service the policy	A request to change something	A new policy version and the document that proves it	Where most of the IT cost and most of the dissatisfaction live
6	Collect the premium	A billing date	Cash allocated and cover status derived	In South Africa a risk process, not back-office plumbing
7	Renew or continue	A date	A contract continuing on current or revised terms	Has no trigger event at all in a monthly-renewable book
8	Handle a claim	A notification of loss	A settled or repudiated claim, closed	The product as the customer experiences it
9	Recover	A paid claim with a recovery prospect	Cash in, and a final net claim cost	Starts after the claim closes and runs for years
10	Handle a complaint	An expression of dissatisfaction	A resolved complaint with a recorded root cause	A first-class regulated process since the conduct reforms [4]
11	Cede and recover reinsurance	A bound risk; separately, a large loss	A recorded cession; a collected recovery	Two processes sharing one name. Split them
12	Reserve, close and report	A calendar date	A filed return and a signed set of accounts	A production schedule with statutory deadlines

Four rules make an L1 inventory usable.

One trigger, one outcome, one owner. If a candidate process has two triggers it is two processes. "Cede and recover reinsurance" fails this test, which is why the table says so out loud; most insurers keep them together because one person does both, and that is an organisational fact rather than a process fact.

Nothing between the trigger and the outcome appears on the list. "Validate cover" is a step inside process 8, not a thirteenth process. This rule is what keeps the inventory at twelve items rather than forty.

Cross-cutting work is a service, not a process. Fraud scoring, document production, payment execution, party matching and rating are consumed by several processes and owned by none of them. Put them on the application and service side of the model. A process architecture that lists "document generation" as an L1 process has confused a shared capability with an end-to-end flow, and will then be unable to say which process owns the outcome.

Be honest about which items are not processes. Three of the twelve above are cycles or lifecycles rather than triggered flows. Leave them in the inventory, because excluding them makes the business think you have missed the work, but mark them, because trying to draw a swimlane diagram of "reserve, close and report" produces a Gantt chart with lanes.

Where L1 goes wrong, in order of frequency. Processes named after departments, so the inventory is the org chart and every handover is invisible by construction. Processes named after systems — "Atlas processing" — which guarantees that replacing Atlas changes the process architecture. One name meaning several things: "onboarding" meaning intermediary appointment to distribution, application capture to new business, and party creation to the

data team. And the four missing processes, which are almost always the same four: intermediary appointment, complaints, recovery, and reinsurance, because each of them is owned by somebody with no seat at the architecture table.

1. Quote and bind: short-term commercial, through a broker

A broker has a client with a R40 million warehouse. The broker wants a price today and a bindable answer this week.

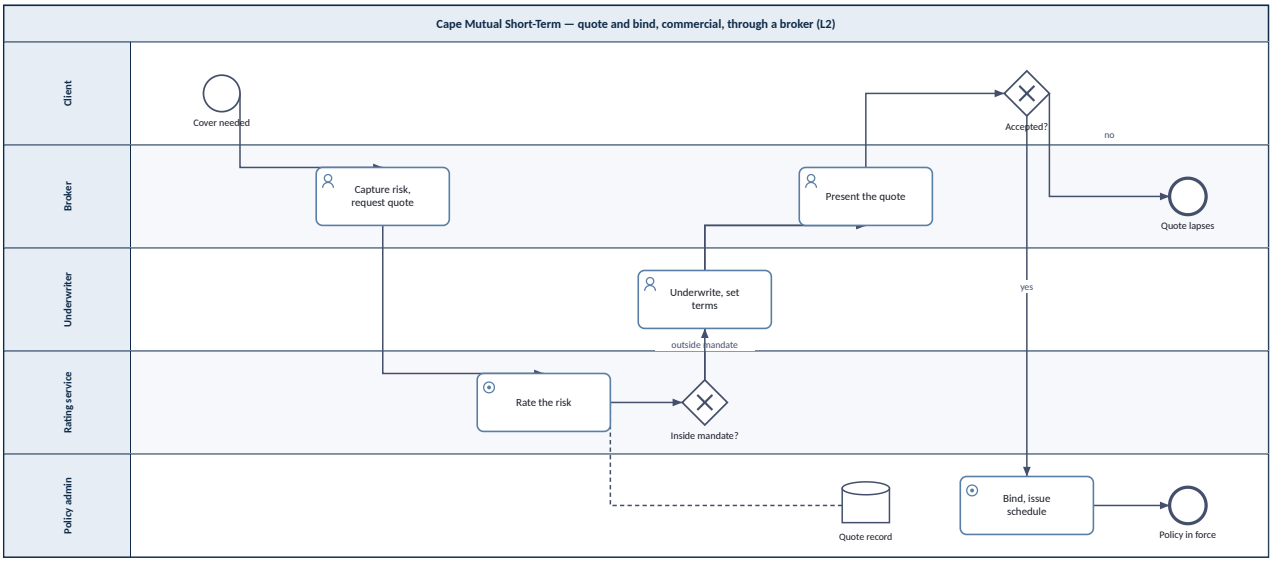


Figure 8.3 Quote and bind for a commercial risk through a broker. The load-bearing element is the mandate gateway: in most insurers it is a human decision, not the configured one drawn here.

The decision points. There are only two that matter, and the first one is the architectural one. *Is this inside the broker's mandate?* — a question about delegated authority, evaluated against the broker's binder scope, the class of business, the sum insured and sometimes the postcode. *Does the client accept?* — a question about price and terms that the insurer does not control. Everything else in the figure is work.

The data created or changed. A **risk description** (the warehouse: construction, occupancy, location, fire protection, sums insured by section). A **quote**, which is a priced offer with the rate version that produced it. On acceptance, a **policy**, its first **version**, a **premium** broken into its components, a **schedule** document, and a **cession** if the risk exceeds the treaty line. The quote is the one people forget: it is evidence of what was offered, on what information, under which rules, and insurers that discard unconverted quotes lose their conversion analytics, their pricing feedback loop and their defence in a conduct complaint at the same moment.

The systems. The broker portal (Beacon), the rating service, the underwriting workbench for referrals, the policy administration system (Atlas), document composition, and the reinsurance system for the cession. Note that the rating service appears in a lane of its own. That is deliberate: in a modern estate rating is externalised so that actuaries can deploy a rate change without a core release [5], which makes it a participant in the process rather than a subroutine inside the core.

The measures that matter. Quote turnaround, from submission to priced answer, split by referred and not referred. Straight-through rate — the proportion of quotes priced and bindable without an underwriter touching them. Quote-to-bind conversion, by broker. Referral rate and referral reason distribution, which is the only honest measure of whether the underwriting rules express the appetite. And the one nobody has: the proportion of bound policies whose bound price differs from the quoted price.

Where it goes wrong. Three failures, every time.

Quote and bind are two code paths. The digital stack was built later and separately, so a quote produced on one path is bound on another and the prices diverge. Nobody notices until a customer screenshots both.

Mandates are held as convention. "Boland Brokers can bind up to R5 million on commercial property" lives in an email and in the regional manager's head, not as enforced configuration. When the manager leaves, the limit leaves with him. The gateway in the figure is drawn as a system decision; in most insurers it is a human one, and the model should say so rather than flatter the estate.

The variants multiply unmanaged. One quote-and-bind process per channel, each with its own validation, its own question set and its own idea of what a complete risk description is. The target architecture is one service behind many front ends with channel entitlements as configuration, and the reason it is hard is not technical: it is that each channel's process was built by a different programme with a different sponsor.

2. New business and underwriting: life, with the medical path

Short-term underwriting fits in the quote-and-bind figure. Life does not, because a life case is a case-management problem with external parties and a cycle time measured in weeks. Two figures, split at the underwriting decision.

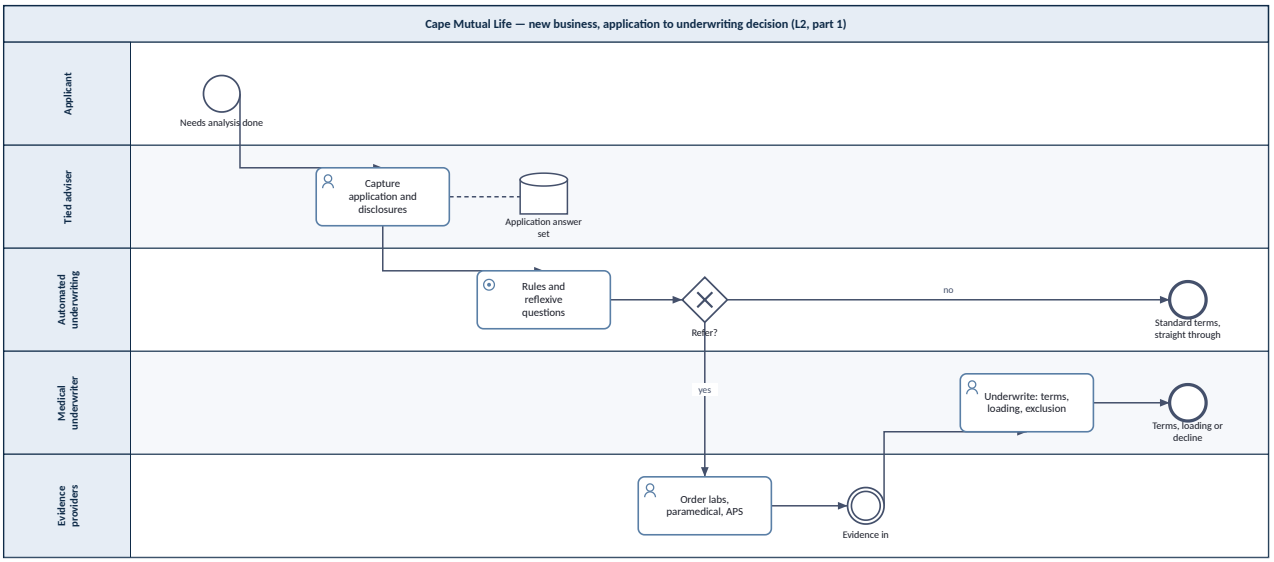


Figure 8.4 Life new business, application to decision. The referral gateway honestly has three exits, not two — decide, refer, and cannot decide.

The decision points, part one. *Can the rules decide this?* The honest version of that gateway has three exits, not two. The Society of Actuaries' survey of life automated underwriting decomposes the outcome as a final decision taken without review, a recommendation referred for review, and — the one architects miss — **no decision at all**, arising from data errors or complexity [6]. Ask what happens in the third case, and whether the system still manages the workflow when it cannot decide. That is where straight-through processing quietly leaks into manual process. *Then, if referred: accept, decline, or accept with a loading or an exclusion?*

The data created or changed. The **application answer set**, which is the evidential basis of every future non-disclosure dispute and must survive the policy term plus prescription — decades. **Requirements** and their status. **Evidence**: bloods, a paramedical report, an attending physician's statement, which is the main source of case delay [6]. The **underwriting decision** with its reasons, loadings and exclusions. And **consent**, which in life is consent to process health data and now runs into regulations in force since 6 March 2026 that apply expressly to insurance companies [7].

The systems. The adviser portal, the new business and onboarding stack, the automated underwriting rules engine, the underwriter's workbench, pathology and paramedical provider integrations, and Polaris at the end. Note that the medical evidence set an architect may have read about — the Medical Information Bureau, motor vehicle records, pharmacy prescription profiles — is North American infrastructure. South African insurers work from the application, tele-underwriting, pathology and paramedical providers and industry data sharing. I could not source a definitive description of the South African evidence landscape, so treat that as market practice rather than a sourced fact.

The measures. Straight-through underwriting rate, and the referral reason mix. Cycle time from application to decision, and the distribution rather than the mean, because the mean is dominated by the cases waiting on a doctor. Requirement ageing: which evidence type most often delays a case, measured, not guessed. Not-taken-up rate by channel and reason. And decline rate with reasons, which the conduct regulator will eventually ask for.

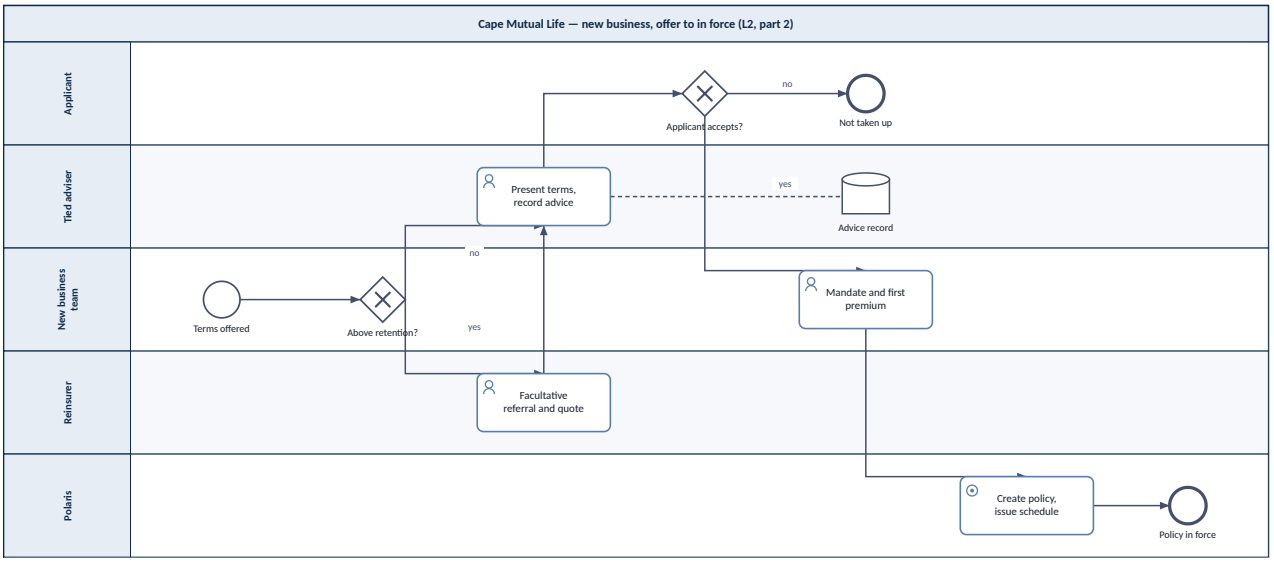


Figure 8.5 Life new business, offer to policy in force, with the facultative referral path for sums assured above the retention.

The decision points, part two. *Is the sum assured above the retention?* — if so the case goes facultatively to a reinsurer, who underwrites it independently and may take weeks. *Does the applicant accept the terms?* A counter-offer with a 50% loading has a materially lower acceptance rate than standard terms,

and the gap between terms offered and policies issued is where life insurers lose new business value.

Where it goes wrong. The front end is rebuilt every three years while the application data model never changes, so each generation of portal adds fields nowhere downstream can store. Three channels run three onboarding stacks writing into one policy system with different validation, so the same product has three definitions of a complete application. Evidence arrives as PDFs and is never structured, which means the underwriting data an actuary needs to analyse mortality experience does not exist. Not-taken-up cases are not tracked, so conversion leakage is invisible. And consent is captured per channel and never reaches the party master, which makes a subject access request a manual search of six systems.

3. Service the policy, and the endorsement

This is the least glamorous process in an insurer and it consumes the most IT money.

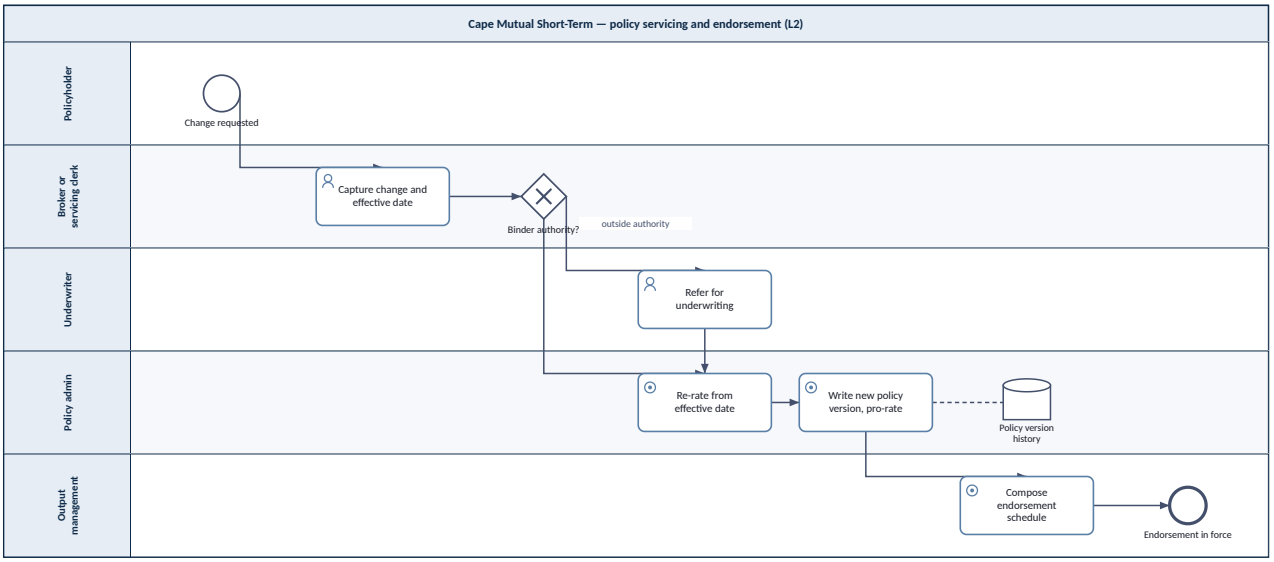


Figure 8.6 Policy servicing and endorsement. The back-dated endorsement, which invalidates premium, cession and commission for a closed period, would not fit in seven columns — which is itself the finding.

The decision points. *Is the change inside the binder holder's authority?* In South African short-term insurance a large independent broker commonly issues, services and settles on the insurer's paper, so this gateway is real and is evaluated dozens of times a day. *Does the change require a re-rate?* Adding a vehicle does; changing a postal address does not; changing a residential address does, because it is a rating factor. Insurers get this wrong in both directions and the second direction is worse: a change that should have re-rated and did not is an unpriced exposure.

The data created or changed. A new **policy version**, dated from the effective date of the change rather than the date it was keyed. A **premium adjustment**, pro-rated across the remaining term. An **endorsement** document. Possibly a change to a **risk object**, a **party role** or a **beneficiary nomination**. And, where the effective date is in the past, a recalculation of everything downstream that has already been reported: earned premium, cession, commission.

The systems. Policy administration, rating, document composition and output management, the collections module if the premium changed, and the mandate management layer if the collection amount changed, because a DebiCheck mandate authorises collections only within its approved terms [8].

The measures. Endorsement volume per policy per year, which is the honest driver of servicing cost. First-contact resolution. The proportion of endorsements requiring manual intervention. The proportion applied with a retrospective effective date, which is the leading indicator of reconciliation pain. And the one that should be on the operations dashboard and never is: the proportion of endorsements where the document sent to the customer does not match the system of record.

Where it goes wrong. Endorsements implemented as cancel-and-rewrite, which destroys continuous cover history and with it no-claim bonuses, claims history and the conduct answer to "how long have you been our customer". Premium not pro-rated correctly. No ability to answer the as-at question — "what did this policy look like on 14 March 2023, and who changed it?" — which Chapter 10 treats properly, because it is a data problem dressed as a process problem. And the back-dated endorsement, which is a single business event that invalidates premium, exposure, cession and commission for a period already closed. The gateway for it is not in my figure because it would not fit in seven columns, and that is itself a finding: the exception that breaks the most downstream systems is the one nobody draws.

4. Collect the premium, and the retry cycle

In most mature markets collection is a back-office function. In South Africa it is a retention function, and the difference is worth drawing.

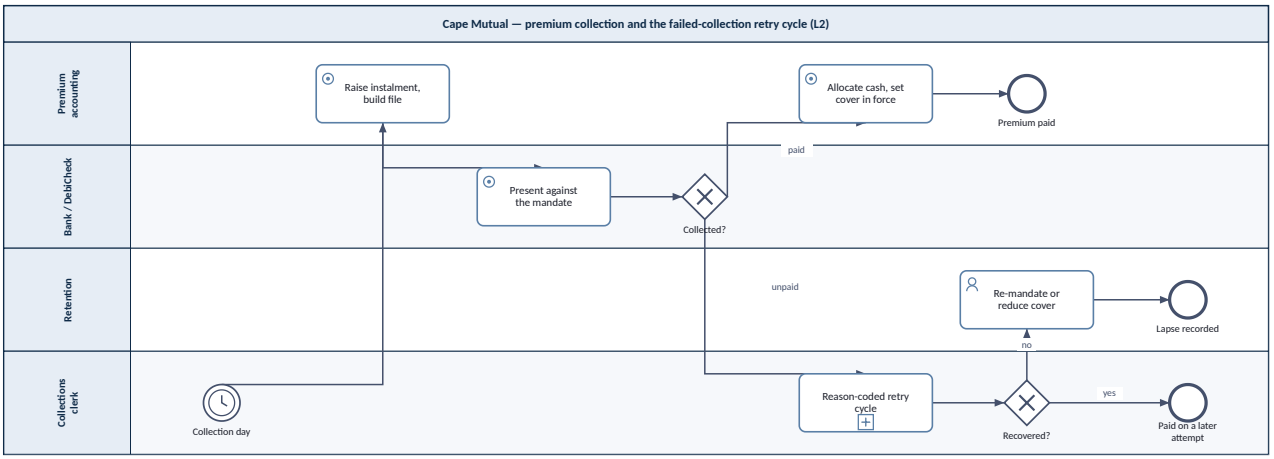


Figure 8.7 Premium collection and the failed-collection retry cycle under DebiCheck. The retry subprocess is drawn as one box because its internals are a reason-code decision table, not a flow; a collection recovered on a later attempt rejoins the allocation step.

The decision points. *Did the collection succeed? What does the unpaid reason code mean, and does it justify a retry?* — the whole value of the process is in that second question. *Are the retries exhausted, and is cover now lapsed?* And the question the systems answer inconsistently: *is cover in force during the grace period?*

The data created or changed. An **instalment** raised against the premium, which is not itself premium. A **collection attempt** with an outcome and a reason code. **Cash**, allocated to a policy or sitting in suspense. **Cover-in-force status**, derived. And a **mandate**, which since the retirement of the old early-debit-order instruments is a long-lived object held at the debtor's bank rather than a line in the insurer's billing table. AEDO and NAEDO were switched off on 31 October 2021 [9]; DebiCheck replaced them, with the mandate registered by the creditor through its bank, stored at the debtor's bank, authenticated once by the debtor, and then validated against its approved terms on every collection [8]. The architectural consequence is concrete: something in the estate has to master the mandate, and it is usually a mandate management layer between billing and the banks [8].

The systems. Billing and premium accounting, the mandate management layer, the bank's authenticated collection service, the policy system for status, and the retention or campaign tooling at the end.

The measures. First-presentation success rate. Recovery rate by retry attempt and by reason code — the central measure, because a uniform retry strategy is by definition the wrong one. Collection success by presentation day, which in a mass-market book moves lapse rates measurably: a household paid on the 25th behaves entirely differently to a debit order on the 1st and one on the 28th. Suspense balance and its age. And lapse decomposed into *failed collection* and *customer cancellation*, which most insurers cannot do and which is the whole argument for treating this as a risk process. In 2025, 8.7 million risk policies lapsed in South Africa [10]; in a mass-market book the dominant cause is a collection failing, not a decision to stop.

Where it goes wrong. Unpaid reason codes not mapped to an action, so the retry strategy is uniform. Mandate data diverging from policy data after an alteration, so collections fail silently. Premium due reconciling to premium received only in aggregate, so suspense grows and is cleared by hand. Commission paid on premium due rather than premium received, creating clawback exposure. Intermediated collection, where a broker or UMA collects and remits an aggregate with a bordereau the insurer cannot allocate to policies — so the insurer knows how much money arrived but not whose cover is paid up, and resolves the ambiguity at claim time, under pressure, in whichever direction the person in the room prefers.

One vendor figure worth knowing and treating as a vendor figure: Altron FinTech reports that 35% to 40% of monthly TT1 mandate requests fail because customers miss the notification or cannot reach their banking app in time, with a merchant fee charged per failed attempt, and claims above 99.5% success for the in-person TT3 variant where a card is inserted and a PIN entered at the point of sale — explicitly including field sales of funeral cover [11]. If those numbers are even approximately right, mandate acquisition is a process in its own right and belongs in the L1 inventory under number 4.

5. Handle a claim: notification to the cover decision

Claims splits naturally at the cover decision, and the split is where the architecture changes too: everything before it is intake and adjudication, everything after it is supply chain and money.

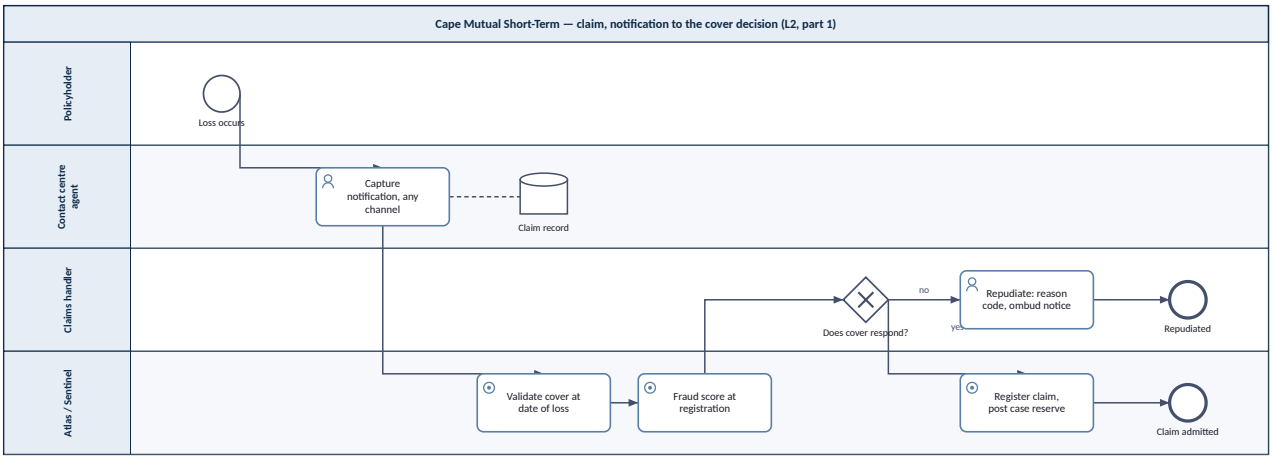


Figure 8.8 Claim notification to the cover decision. Cover is validated against the policy as it stood at the date of loss, and the repudiation path is drawn as a first-class outcome rather than a note.

The decision points. *Does the cover respond?* — evaluated against the policy as it stood at the date of loss, not as it stands today, which is the single most consequential sentence in insurance claims architecture. *Is this claim genuine?* — a fraud decision, which in a well-built estate happens at registration rather than in a monthly batch. And beneath them the quiet one: *is this one claim or several?* One collision yields an own-damage claim, a third-party property claim and possibly three bodily-injury claims, all from one loss event.

The data created or changed. A **loss event**, which is not the same thing as a claim. A **claim**, with its four dates — loss, notification, registration, settlement — which drive different things and are routinely collapsed into one. A **case reserve**, which is an estimate as at a date and the actuary's primary raw material. A **fraud score** with its contributing factors. And, if the claim is declined, a **repudiation** with a reason code, which is now conduct-reporting data rather than an internal note [12].

The systems. Claims management, the policy system's point-in-time reconstruction, fraud scoring (Sentinel at Cape Mutual), document capture, and the party master for entity resolution — because link analysis across claims, addresses, bank accounts and service providers only works if somebody has solved identity first.

The measures. Notification-to-registration time. Registration-to-cover-decision time. Repudiation rate with reason mix, and repudiation rate overturned on complaint, which is the measure of whether the first decision was right. Fraud detection rate before payment versus after payment — the second number being the count of failures. And cover validation accuracy, which almost nobody measures and which the point-in-time question makes measurable.

Where it goes wrong. The claim links to the *current* policy version rather than the one in force at the date of loss, producing wrong cover decisions that are discovered years later. Cause-of-loss coding is free text, or a 200-value list used inconsistently, which destroys claims analytics and with it peril-level pricing. Reserve movements are overwritten rather than versioned. The fraud score arrives after payment. And the repudiation path is modelled as an afterthought — a note on the happy path saying "or decline" — when repudiation is a regulated outcome with disclosure requirements, an internal appeal, an ombud route and a reason taxonomy that has to be consistent across platforms [4][12].

6. Handle a claim: quantum to settlement

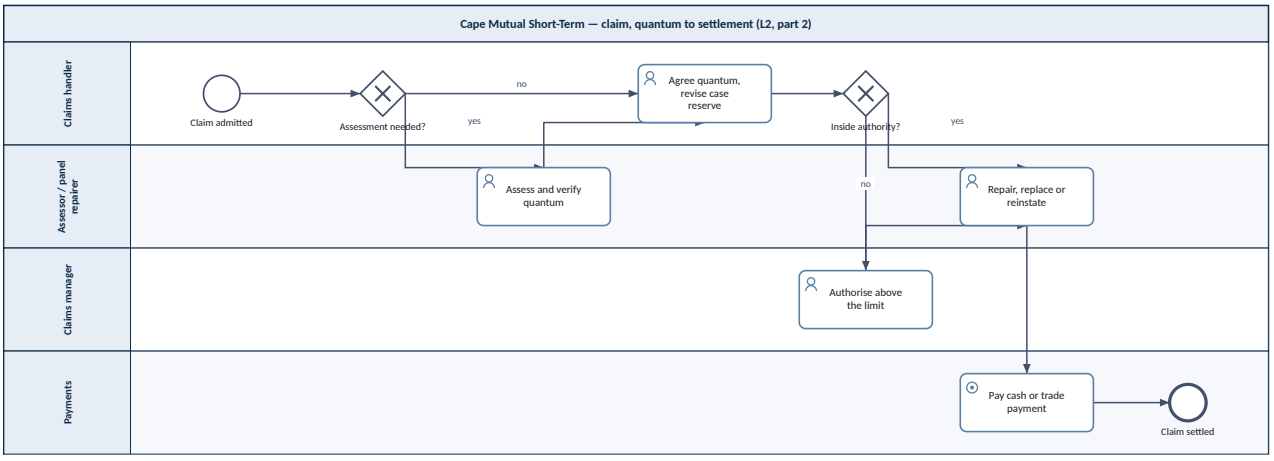


Figure 8.9 Claim quantum to settlement. The cash-versus-trade-payment choice at the end is a tax decision as well as an operational one.

The decision points. *Does this need an assessor?* — a cost decision: an assessor costs money and adds days, and under-assessment is the largest single component of claims leakage. *Is the quantum inside the handler's authority? Repair or write off?* — increasingly answered "write off", because repair cost inflation has pushed more vehicles over the economic threshold than a decade ago [13]. *Cash settlement or payment to a supplier?* That last one looks like

an operational preference. It is a tax decision. In South Africa a monetary indemnity payment to a vendor insured for an enterprise loss is a deemed taxable supply by the insured, whereas a trade payment to a third party who reinstates is not, and the two are accounted for differently [14]. Chapter 10 works through why the claims model usually lacks the attributes needed to tell them apart.

The data created or changed. A revised **case reserve**, as a movement rather than an overwrite. **Payments**, each with a payee, a date, a head of damage and a tax treatment. **Supplier instructions** to a panel repairer, a medical provider or a funeral parlour. **Salvage title**, if the insurer takes the wreck. And the **claim outcome** with its final incurred cost — which is not final, because recoveries are still to come.

The systems. Claims management, the assessor's mobile tooling, supplier and panel integrations, payments and banking, and the general ledger. Payments deserves its own lane because outbound payment is a controls problem as much as an integration one; bank-detail change is one of the highest-risk events in an insurer.

The measures. Cycle time from admitted to settled. Average claim cost, split by settled-in-cash and settled-through-supplier. Reserve accuracy — initial case reserve against final incurred — which is the measure the reserving actuary actually cares about. Leakage, sampled by file review rather than computed, because a system cannot detect the money it should not have paid. Re-opened claim rate. And authority breaches, which should be zero and are not.

Where it goes wrong. Authority limits enforced in the user interface rather than in the service, so a direct integration bypasses them. Part-payments and recovery offsets handled as adjustments to a single amount, so the payment history cannot be reconstructed. Supplier performance invisible because the insurer buys repairs through a panel it does not measure. And catastrophe surge: ten to fifty times normal volume arriving in three days with no elastic capacity. That is no longer a contingency in South Africa. Weather catastrophes were the largest single force on short-term claims costs in the first half of 2026 [13], and the process consequence is that the intake path has to be able to run for a few days without the core system in the loop.

7. Recover, and subrogate

The process that starts after everyone else has closed the file.

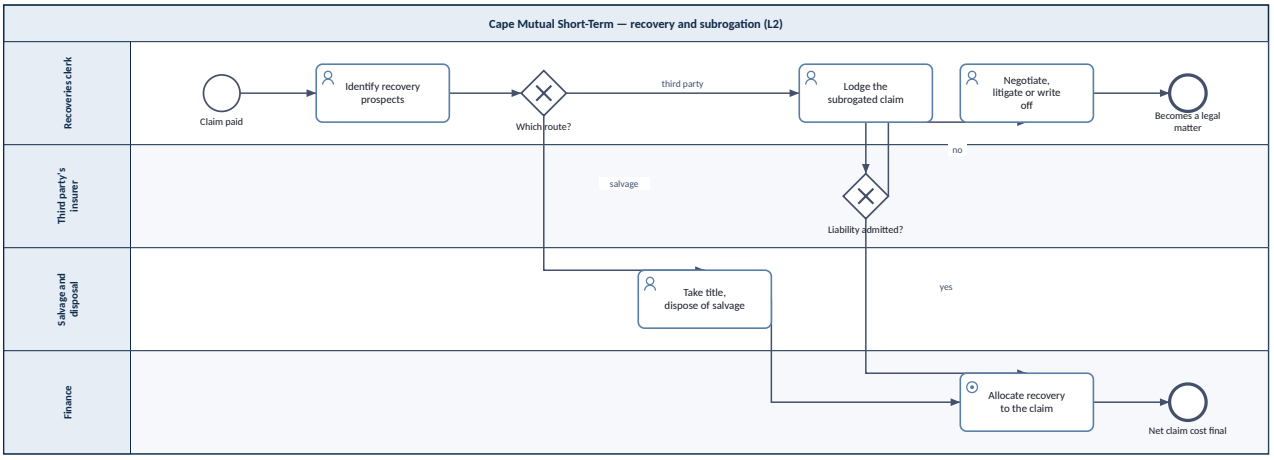


Figure 8.10 Recovery and subrogation, the process that begins after every other file is closed. A denied subrogated claim becomes a legal matter with its own parties and timeline, which is why it leaves this process rather than looping inside it.

The decision points. *Which recovery route, if any?* Salvage, subrogation against the party who caused the loss, contribution from another insurer covering the same loss, or nothing. *Did the other side admit liability? Is it worth pursuing?* — a commercial decision with a threshold, and the threshold is usually a number in somebody's head rather than a configured rule.

The data created or changed. A **recovery**, which is emphatically not a negative payment: different counterparty, different timeline, different accounting, and if you model it as a negative payment you cannot age a recoverable. A **transfer of title** on salvage. A **legal matter**, with its own parties and dates, if it goes to attorneys. And a revised **net claim cost**, which is the number the loss ratio is built from.

The systems. Claims management, a recoveries module if one exists, salvage disposal platforms, the ledger. In most insurers the honest answer is "a spreadsheet", and that is worth writing on the diagram rather than drawing an aspirational system lane.

The measures. Recovery rate as a percentage of recoverable claims, by route. Recovery cycle time, which runs to years in subrogation. Recoverable ageing. And identification rate — the proportion of claims where a recovery opportunity was spotted at all, which is the measure that exposes the real loss. Claims platforms increasingly identify subrogation opportunities from claim data and intervene early to create and preserve them [15]; an insurer doing that manually at file-close finds a fraction of them.

Where it goes wrong. Recoveries tracked outside the claims system, so leakage is unmeasurable by construction. The opportunity identified after the file is closed and the evidence is gone. Salvage proceeds not allocated back to the claim, so the net claim cost is overstated and the loss ratio with it. And subrogation treated as a legal matter rather than a process, so nobody owns the funnel and nobody reports on it.

8. Cede and recover reinsurance

The L1 table called this two processes sharing one name, so draw it as two. They share a system, a contract and usually a single person, and they have almost nothing else in common.

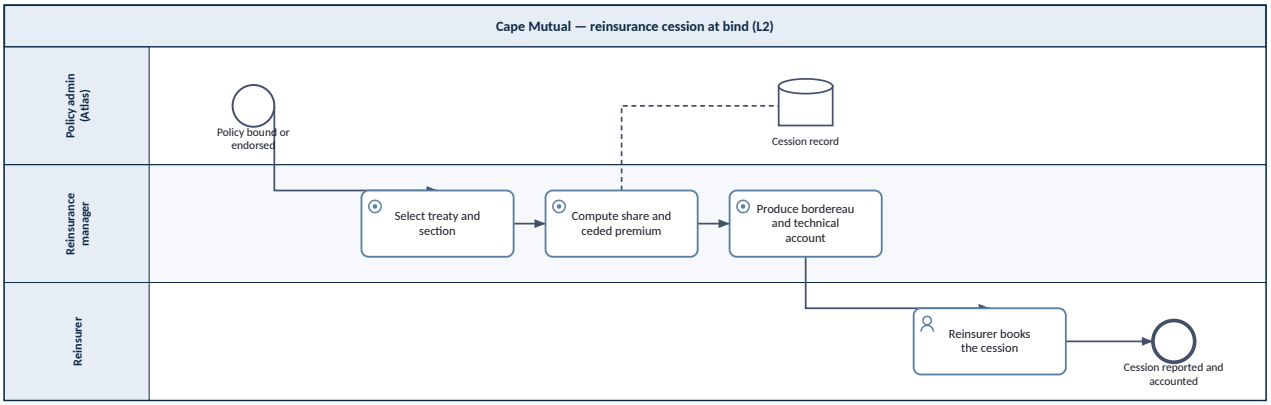


Figure 8.11 Cession at bind. Proportional cession is computable per policy at the moment the risk is written, which is why this flow is short and mostly automatable.

Cession is the easy half. A risk is written, the treaty terms say what share goes where, and the computation can be done at bind time, per policy, automatically. An insurer whose cession is a manual quarterly exercise has not got a hard problem; it has an unmodelled treaty.

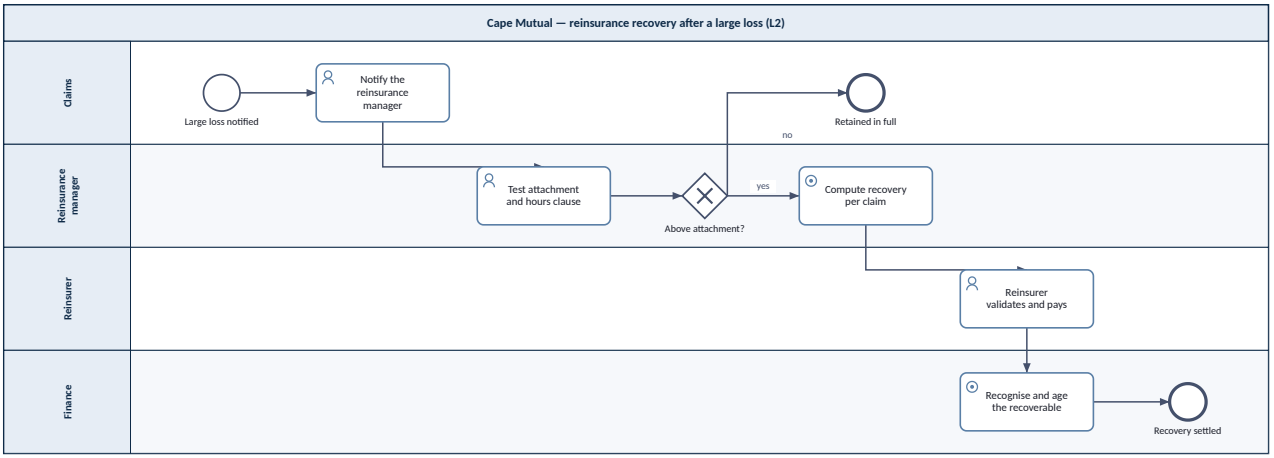


Figure 8.12 Recovery after a large loss. Excess-of-loss recovery is computable only per loss event, after aggregating claims across policies inside an hours clause — which is why this is the flow that leaks.

Recovery is the hard half, and the asymmetry is the point of showing them side by side.

The decision points. On cession: *which treaty, which section, and what share?* On recovery: *does this loss reach the attachment point, aggregated how, over what period?* Excess-of-loss recovery is computable only per loss event, after aggregating claims across policies within an hours clause and a geography — a fundamentally different data problem, and the place most reinsurance recovery leakage lives.

The data created or changed. A **cession** record per risk, which is the join between a direct policy and a reinsurance contract. **Ceded premium**. A **premium bordereau** and a technical account. A **recovery** on the asset side with its own reserving — reinsurance IBNR, bad-debt provisioning and reinsurer credit exposure are all separate estimates. And **aggregate erosion and reinstatements**, which are the running state of the programme and are usually tracked by hand.

The systems. Policy administration for the risk and premium, a reinsurance management system for terms and cessions, claims for the loss notification, the ledger for the recoverable. Reinsurance platforms published capability lists cover treaty and facultative terms, quota share and excess of loss per risk, per occurrence and aggregate, risks-attaching versus losses-occurring bases, ceding and profit commission, reinstatement premium, and bordereau production with statements and supporting detail [16]. If you are assessing an insurer, read that list and ask which of it is done in a system and which in Excel.

The measures. Cession accuracy at policy level. Recovery identification rate per large loss. Recoverable ageing by reinsurer, which is a credit measure, not an operations measure. Time from large loss to recovery claimed. And the reconciliation of the inwards view to the outwards view, which should balance and frequently does not.

Where it goes wrong. The programme lives in a spreadsheet maintained by two people, which is routinely the largest single-point-of-knowledge risk in an insurer. Cessions computed in aggregate rather than at risk level, so a specific recovery cannot be evidenced to a specific treaty clause. A mid-year treaty restructure that the system cannot represent. Inuring order — which contract applies before which — existing only in someone's head. And, structurally, the fact that reinsurance sits in finance, the data sits in operations and nobody owns the join.

9. Renew, or continue

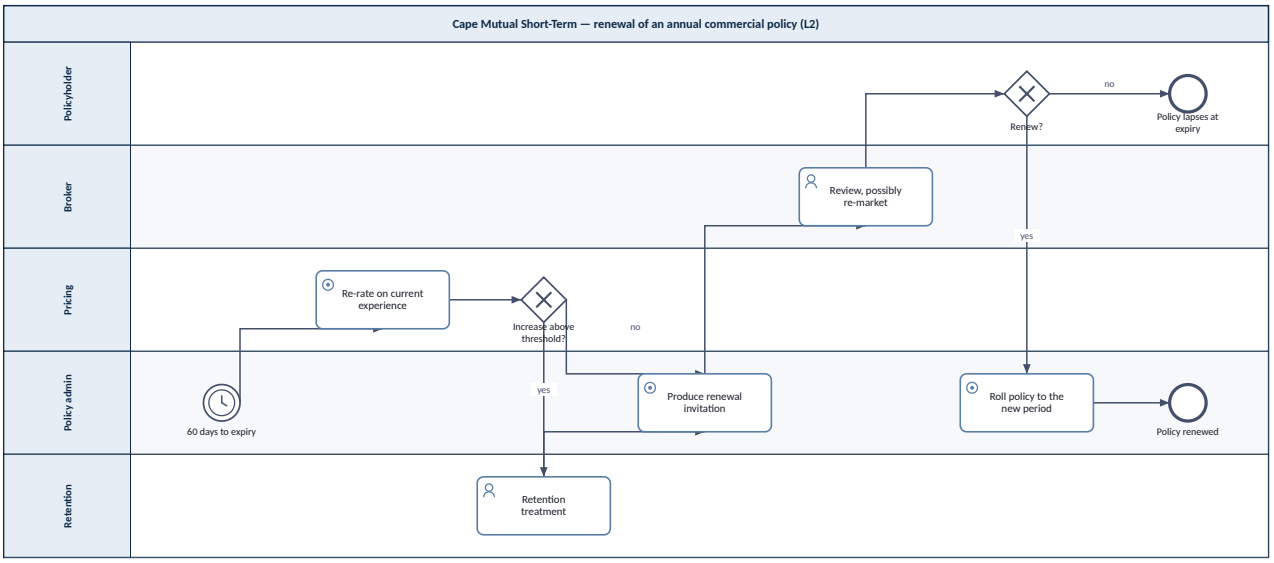


Figure 8.13 Renewal of an annual commercial policy. A monthly-renewable personal lines book has no equivalent trigger, so modelling it this way describes something that does not happen.

The decision points. *What terms to offer? Is the increase above the threshold that triggers a retention intervention? Does the broker re-market the risk?* — a decision the insurer does not make and cannot see. *Does the client renew?*

The data created or changed. A **re-rate** on current experience and current rates. A **renewal invitation**, which in most jurisdictions carries statutory notice requirements. A new **policy period** or, in a monthly-renewable book, nothing at all. And a **lapse** if nothing happens, which is the quiet outcome: non-renewal is usually silence rather than a decision.

The systems. Pricing and rating, the policy system's renewal batch, document composition, the broker portal, retention campaign tooling.

The measures. Retention rate by segment, by broker and by rate-increase band — the third of those being the one that tells you what your pricing is costing you. Renewal invitation lead time. Re-market rate, where you can see it. Rate achieved against rate quoted. And persistency, which is the life-side name for the same idea.

Where it goes wrong, and the South African wrinkle. Much South African personal lines business is written monthly rather than annually, continuing month to month until cancelled. An architect arriving from a market where annual renewal is assumed will look for the renewal batch and not find one, because there is no renewal event to hang a re-rate on. Rate changes are applied as mid-term adjustments with notice periods; retention is a continuous activity rather than an annual campaign; and the natural unit of exposure measurement becomes the month, not the policy year. The process in my figure is the annual commercial one. The monthly personal lines "process" is a rate review and a notification, and modelling it as a renewal produces a diagram of something that does not happen.

The other failure is more ordinary. Renewal is run as a batch that produces invitations, and the retention treatment is bolted on afterwards from a report. By the time the retention team sees the 28% increase, the invitation has already been posted.

From here to a target process architecture

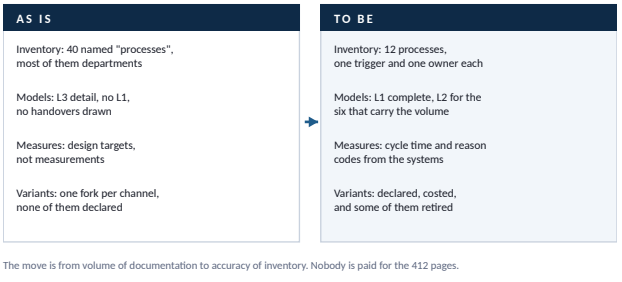


Figure 8.14 What moving to a target process architecture actually changes. The gain is in the accuracy of the inventory, not the volume of the documentation.

Four moves, in order. They are boring and they are the work.

Fix the inventory before you model anything. Get to twelve processes with one trigger, one outcome and one named owner each. This takes two workshops and produces more argument than any other artefact in the programme, because naming an owner for complaints or recoveries is a political act. Do it anyway. Everything downstream is unsound without it.

Measure the current process before you redesign it. Not with a survey. From the systems: event timestamps rather than status flags, reason-code distributions rather than averages, and a random sample of thirty closed cases traced end to end by hand. The measurement is what makes the redesign arguable; without it the target process is a preference.

Decide which variants you keep. An insurer does not have one quote-and-bind process; it has five, one per channel, and the useful question is not how to unify them but which forks are justified. A commercial broker submission and an embedded checkout genuinely need different processes, because one needs three weeks and the other needs 400 milliseconds. A tied-adviser path and a call-centre path probably do not. Declare the variants, cost them, and retire the ones nobody can justify. A target architecture that pretends there is one process is a target architecture that will be ignored.

Attach each process to a capability and each step to a service. This is where BPMN hands off to ArchiMate. The L2 model says who does what in what order; the ArchiMate model says which capability the process realises, which application service each step consumes, and which business object each step touches. Do that and the target becomes testable: you can ask whether the application estate can actually support the process you have drawn, and you will usually find two steps where it cannot. Those two steps are the programme.

One honest note on sequencing. Do not model all twelve at L2. Model the six that carry volume or pain — in a short-term insurer that is quote and bind, servicing, collections, both halves of claims, and renewal — and leave the rest at L1 until something changes. Twelve L2 models is a documentation project. Six is an architecture.

The trap: modelling the process you were told about

Every process model you are handed describes an idealisation, and the idealisation is not a lie. It is what the process owner believes, and they believe it because they designed it and nobody has told them otherwise.

The signs are consistent and easy to spot:

- **No rework loops.** Real processes are mostly rework. A model with no loop back for incomplete information is describing a world where forms arrive complete.
- **Every gateway resolves one way.** The model has a "cover responds?" gateway with a "no" branch that leads to a single box called "decline", and no further detail, because the person who drew it has never handled a repudiation.
- **Cycle times that are service levels.** "Two days" in the model, eleven days in the data. The model is quoting the target.
- **One path where the systems have forty reason codes.** Count the exits from the gateways in the model. Count the distinct reason codes the system records at that decision. If the model has two and the system has forty, the model is a summary of somebody's intention.
- **A step with no owner in any lane.** Usually the handover. In the 412 pages at Cape Mutual, the broker-email-to-assessor step was not missing because someone forgot. It was missing because nobody owns it, and therefore nobody described it.

Four things defeat it, in increasing order of cost.

Sit with the work for a day. Not an interview — a day at the desk, watching, counting the times the handler opens a second system or picks up the phone. One day produces more correction than a week of workshops.

Pull thirty closed cases at random and trace each one. Not thirty cases the operation selects; thirty from a random sample, including the ones that went wrong. The variance will be larger than anyone told you, and the distribution of paths through the process is the actual process.

Count the reason codes, and read the free text. The configured list of reason codes is the organisation's considered view of what can happen. The free-text notes are what actually happens. Where a reason code called "other" accounts for a third of cases, the process has a branch nobody has modelled.

Find the spreadsheet. There is always a spreadsheet, and it sits exactly where the process crosses an organisational boundary that no system spans: between claims and recoveries, between operations and reinsurance, between the broker's bordereau and the premium ledger. The spreadsheet is not a workaround to be eliminated. It is a precise map of the missing part of the architecture, maintained for free by the person who needs it most, and the first thing to ask for.

Cape Mutual is invented and nothing in this chapter describes a real insurer. The pattern is not invented. Every L2 figure above has a gateway whose "no" branch is where the money goes, and in most insurers that branch is the one not drawn.

THE CHAPTER IN TEN LINES

- A process architecture is the inventory of end-to-end processes; a process model is the inside of one of them, and a programme that commissions the second without the first cannot tell what is missing.
- Define L0 to L3 in writing and have the process owners initial it before the first workshop, because APQC, ACORD and your own insurer all number the levels differently.
- A good L1 inventory has eight to twelve processes, each with exactly one trigger, one outcome and one named owner, and nothing between the trigger and the outcome on the list.
- Fraud scoring, document production, payments, party matching and rating are shared services, not processes; listing them at L1 destroys the inventory.
- Three of the twelve — product development, reinsurance, and the reporting close — are cycles rather than triggered processes; keep them in the inventory and mark them.
- The four processes missing from almost every insurer's map are intermediary appointment, complaints, recovery and reinsurance, because each is owned by someone with no seat at the architecture table.
- Draw the delegated-authority gateway honestly: in most short-term insurers the question of whether a change is inside a binder holder's mandate is answered by a person, not by configuration.
- Cover validation against the policy as it stood at the date of loss is the single most consequential step in claims, and it is only possible if servicing was built on versions.
- Proportional cession is computable per policy at bind; excess-of-loss recovery is computable only per loss event, which is where reinsurance recovery leakage lives.
- Every model you are handed describes an intention: look for missing rework loops, gateways with one live exit, cycle times that are service levels, and the spreadsheet at the organisational boundary — which is a free map of the missing architecture.

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CHAPTER 9

The conceptual data model

Fifteen boxes, no attributes, no keys and no system names. This is the model an architect draws on a whiteboard in the first week, and it is still on the wall when the database underneath it has been replaced twice.

The model you draw in the first week

Fatima Patel drew Cape Mutual's conceptual model on a whiteboard on her fourth day. It took about ninety minutes, with the chief operating officer, a claims manager and a pricing actuary in the room. Fifteen boxes. Somebody photographed it.

Three years later that photograph is still the first thing a new architect is shown. The logical model underneath it has been rewritten twice and the physical schema three times. The photograph has not changed, because the business has not changed.

That is what a conceptual model is for. An insurer in 1975 had parties, policies, covers, claims and reserves. So does an insurer in 2026, and so will one in 2050, whatever is in the database by then. The conceptual model is the layer with a ten-year half-life, and it is cheap to build and almost impossible to buy back once a programme has started without one.

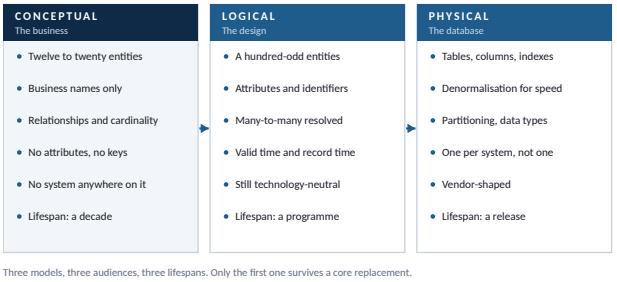


Figure 9.1 Conceptual, logical, physical. The prohibitions in the first column are what give it the long lifespan.

What it is not

Five prohibitions. Each one is load-bearing, and each one gets argued about in the first hour.

No attributes. Not even the obvious ones. No `inception_date` on Policy, no `amount` on Claim.

No keys. Not natural, not surrogate. "Policy number" is not on the model, which is just as well, because Chapter 10 spends a page explaining why a policy number does not identify a policy state.

No tables. No join entities, no resolution of many-to-many relationships, no history tables. A many-to-many relationship stays many-to-many here; resolving it is a design decision and this is not a design.

No systems. The words Atlas, Polaris and Kestrel do not appear. If the model changes when a system is replaced, it was not a conceptual model.

No regulatory or technical structure. No IFRS 17 cohorts, no POPIA consent flags, no valid-from and valid-to. Those are real and they are mandatory and they belong one layer down.

The point of the prohibitions is not minimalism. It is audience. The moment attributes appear, the meeting becomes a negotiation about attributes, and the three people who can actually tell you whether the model is *true* — the claims manager, the actuary, the head of distribution — stop being able to read it. A conceptual model that a business person cannot argue with has failed, whatever else is right about it.

Watch it happen once and you will enforce the rule for ever. Somebody adds `sum_insured` to Cover, because obviously a cover has a sum insured. Within ten minutes an underwriter is explaining that commercial property has a sum insured per location and an aggregate across the schedule, somebody else points out that life has a sum *assured* and it behaves differently, and a third person asks whether the figure is indexed. All three are right, all three matter, and none of them belongs in this meeting. The model that was going to take ninety minutes now takes three sessions and comes out as a logical model nobody owns.

What it is actually used for

Three things, and if a model is not being used for at least one of them it is decoration.

Scoping a programme. Draw a boundary round the boxes a core replacement will touch and you have the most honest one-page scope statement anyone will produce. It survives contact with procurement because it is in business language: "this programme touches policy, cover, risk object and premium; it does not touch claim, reserve, payment or recovery" is a sentence an executive can hold a supplier to.

Settling vocabulary. Most of the arguments in an insurer's data programme are definitional rather than technical. Is a cancelled-and-rewritten policy a renewal? Is a lapse a cancellation? Is a reopened claim a new claim? Those questions are answered on this page or they are answered inconsistently in nine systems.

Testing a package. When a vendor demonstrates a product, take the fifteen boxes into the demonstration and ask where each one lives. The silences are informative. In my experience the boxes that produce the longest pause are Loss Event, Risk Object and Recovery, in that order — though that is a pattern I have noticed rather than one I can source.

Three tests for whether something is an entity

1. **Can you define it in one sentence using words the business already uses**, without saying "a record of" or "a table holding"?
2. **Does the business count it, name it or argue about it?** "How many policies are in force" is a real question that people fight over. "How many policy-cover junction rows" is not.
3. **Would it still exist if you replaced every system tomorrow?** A Claim would. A "Claim Header" would not.

The third test is the sharpest and it kills most candidates.

How many

Twelve to twenty. Below twelve you have a diagram rather than a model, and you have almost certainly collapsed something that matters. Above twenty-five you have started the logical model and lost the room. Fifteen is a good number for an insurer and it is the number this chapter uses.

> The conceptual model is the only artefact in the whole data architecture that the business owns rather than tolerates. Treat it as theirs. Draw it in their words, keep it on one page, and let them change it. Everything below it is yours.

The spine: a party, a contract, a thing

Six entities carry most of the weight, and if you get these right the rest follows.

> **Why these are in crow's foot and not in ArchiMate.** The rest of this book models in ArchiMate 4, and the second half of this chapter draws the same entities as Business Objects. The figures that carry cardinality are drawn in crow's-foot notation on purpose. A conceptual data model argues about multiplicity — one party or many, one cover per risk object or several — and crow's foot states that on the line, where the business can read it off the page and disagree with it. ArchiMate has no notation for multiplicity on a relationship, so the argument would have to go in a note. Draw the argument in crow's foot; hold the model in ArchiMate, where it connects to the processes, the applications and the data objects that realise it.

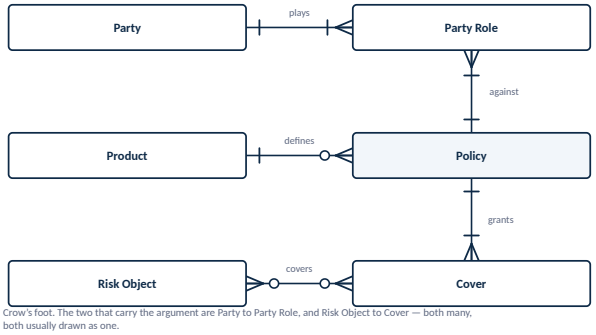


Figure 9.2 The spine of the model in crow's-foot notation: a party, in a role, against a contract instantiated from a product, granting covers over things.

Party. Any legal person the insurer deals with, natural or juristic. Party holds identity and nothing else: who this is, and whether they are the same as that. It is at the conceptual level because the business genuinely talks about people and companies, and because every relationship in the model eventually lands on one. *The question it answers:* who are we dealing with, and have we dealt with them before? ACORD's Information Model names Party as one of its four core concepts alongside Policy, Product and Claims, which is a useful thing to be able to say in a room where somebody is insisting on Customer [1].

Party Role. Something a party does, to something, between two dates. Policyholder, insured, life assured, beneficiary, claimant, intermediary, cedant. This is the classic party pattern from the data-modelling literature, where Party is supertyped and every relationship is carried by a dated role [4], and the health standards make the same move explicitly: HL7's vocabulary carries a named-insured *role* class rather than a named-insured person type [5]. It is at the conceptual level, despite looking like a design pattern, because the business distinguishes these roles constantly and will tell you so. *The question:* in what capacity is this party involved, and for what period?

Product. The manufactured definition: what is covered, what is excluded, what the limits are, how it is priced. Vendors describe it the same way — a blueprint from which a quote and then a configured instance are produced [19]. *The question:* what do we sell?

Policy. One sold, in-force instance of a product. The agreement. *The question:* what have we promised, to whom, for how long? Note that Policy and Product are separate boxes on a conceptual model and not on most insurers' actual databases, and that single separation is the most valuable line on the page.

Cover. A specific promise with its own limit, excess and conditions. A motor policy grants own-damage cover, third-party liability cover and perhaps car-hire cover, and they behave independently in every calculation. *The question:* what exactly did we promise, and up to how much?

A note on the name. Some modellers prefer **Agreement** to Policy, on the grounds that an insurer's contracts include treaties, binders, scheme arrangements and service agreements, and that one supertype with Policy as a subtype is cleaner. They are right about the structure and I still do not do it, for one reason: nobody in an insurance company says "agreement" when they mean a motor policy. Call the box Policy, note on the model that reinsurance arrangements and intermediary agreements are contracts too, and leave the generalisation to the logical layer where it costs nothing to add and confuses nobody.

Risk Object. The thing exposed to loss: a vehicle, a building, a shipment, a fleet. Oracle's published health insurance components use the same abstraction under the name insurable entity, paired with an insurance type [6], which is helpful evidence that this is industry practice rather than your preference. In life, there is no object — the risk is a party in the role of life assured — and in liability there is no object either, only an activity. *The question:* what is exposed, and have we ever covered it before?

The cardinality that carries meaning

Most of the cardinality on a conceptual model is decoration. Three of these are not.

Party to Party Role is one-to-many, and the many is the point. One human being can be a policyholder, a life assured on a policy their employer owns, a beneficiary on a spouse's policy and a third-party claimant against someone else's liability cover, all on the same Tuesday.

Risk Object to Cover is many-to-many. One cover answers for several insured objects; one object is covered by several covers, often under different policies and sometimes from different insurers. This is routinely drawn one-to-many and that error propagates all the way to the catastrophe accumulation report.

Product to Policy is one-to-many, and the one is versioned. The conceptual model does not say "versioned" — that is Chapter 10's problem — but it must not say that a policy *is* a product, because then nothing downstream can separate the definition from the instance.

The money and the loss

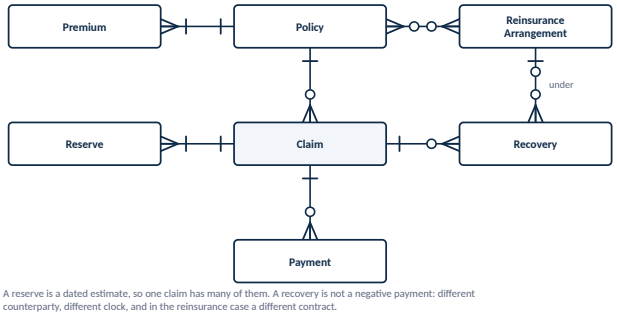


Figure 9.3 The money and the loss. Note that premium hangs off cover rather than policy, and that recovery is its own box.

Premium. The consideration for cover. Conceptually it is one entity, and resisting the urge to decompose it here is a discipline worth keeping: the risk premium, the loadings, the policy fee, the commission, the binder fee and the VAT are all components, and components are a logical-model concern. What the conceptual model must get right is that premium hangs off cover rather than off policy, because that is the grain at which it is priced and at which a conduct regulator asks about it.

Claim. An assertion against a policy that a covered event has happened. *The question:* what are we being asked to pay, and under which promise?

Loss Event. The thing that happened in the world. A hailstorm, a collision, a death. One event yields many claims: a hailstorm hits four thousand roofs, a single collision yields an own-damage claim, a third-party property claim and three bodily-injury claims.

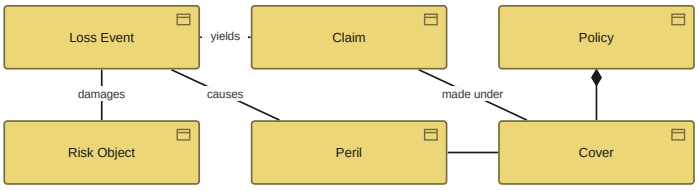


Figure 9.4 Loss event, claim, peril and cover as business objects. One event gives rise to many claims, and one peril is answered by covers across several sections of one policy.

That separation is the single most consequential entity decision in this chapter after Party Role, and it is the one most often missed, because in personal lines the two look identical nine times in ten. Catastrophe aggregation, reinsurance event limits and hours clauses all operate on the *event*. An insurer with no Loss Event entity cannot calculate a catastrophe recovery, and will not find out until the catastrophe.

Peril earns its place on the same figure. A peril is a *cause* — fire, flood, hail, theft. A cover is a *promise*. They are related many-to-many: one all-risks household cover answers fire, theft, storm and accidental damage, while one hail event triggers covers across the buildings, contents and motor sections of a single policy. Draw that relationship as one-to-many and peril-level loss analysis has to be reconstructed from free-text claim descriptions for ever after.

Reserve. An estimate, as at a date, of what a claim will cost. Not cash, not a payment, not a prediction the business can be held to. One claim has many reserves over its life, and the series of them — not the latest value — is what the reserving actuary works from. Putting Reserve on a conceptual model as a box with a many relationship to Claim is the cheapest way to win an argument you will otherwise have in year two of a data programme.

Payment. An outflow against a claim, to a payee who is a party in a role.

Recovery. An inflow that reduces the net cost of a claim: salvage, subrogation, contribution from another insurer, reinsurance. It is a separate box and not a negative payment, because it has a different counterparty, a different timeline and different accounting, and because an insurer that models it as a negative payment cannot age a reinsurance recoverable or tell you how much money it is leaving on the table.

Reinsurance Arrangement. The insurer's own cover, under which both the cession of a risk and the recovery on a loss happen. One box at this level, treaty and facultative both; the structure underneath is Chapter 10's.

The paper, the channel and the complaint

Three more entities that look administrative and are not.

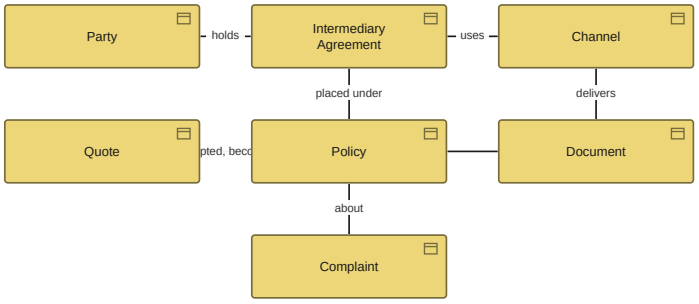


Figure 9.5 The distribution and paper entities. Channel and intermediary agreement are objects rather than attributes because both are dated and both carry authority.

Intermediary Agreement. The mandate between the insurer and an intermediary: what that party may do, on what classes, up to what limits, for what remuneration. It is on the conceptual model because in a binder-heavy market it governs who makes the decisions the insurer is liable for, and because South Africa's conduct reforms limit who may hold a binder, which activities may be outsourced under one, and what may be paid for it [15]. It also sets the data boundary: binder agreements cover data ownership and reporting, and a UMA must keep auditable records for at least five years so the insurer can oversee its work [14].

Channel. The route to market: tied adviser, broker, bank, direct, aggregator, embedded partner. Keep it as an entity and not an attribute of Policy, because channel entitlements, remuneration and data latency all hang off it, and because an attribute cannot be given a start and an end date without becoming an entity anyway.

Document. The schedule, the wording, the renewal notice, the claim letter. Insurance is documentary: the schedule and the wording *are* the contract, and the archive of what was actually sent is the evidential backbone of any complaint or ombud matter. There is a tax reason too. Under the South African rules a policy document may serve as a tax invoice, a debit note and a credit note, with similar relief for intermediary bordereaux [11] — which means the bordereau arriving monthly from a UMA is part of the tax record and not an operational convenience.

Complaint. An expression of dissatisfaction, with a category, a root cause, an outcome and a link to a policy, claim, product and intermediary. It is on the model because conduct supervision made it a reportable fact rather than an internal note: the regulator's conduct-of-business return requires standardised conduct indicators, segmented client data and claims outcomes by reason, so firms can show fair outcomes through their data rather than through a policy statement [18]. An insurer that discovers Complaint is not an entity discovers it during a return.

Quote. A priced offer made and not yet accepted. Worth a box in any insurer that sells through aggregators or direct, because the unconverted quote is the pricing feedback loop and the conduct evidence, and because an entity that is thrown away after thirty days is one nobody has modelled.

The whole thing on one page

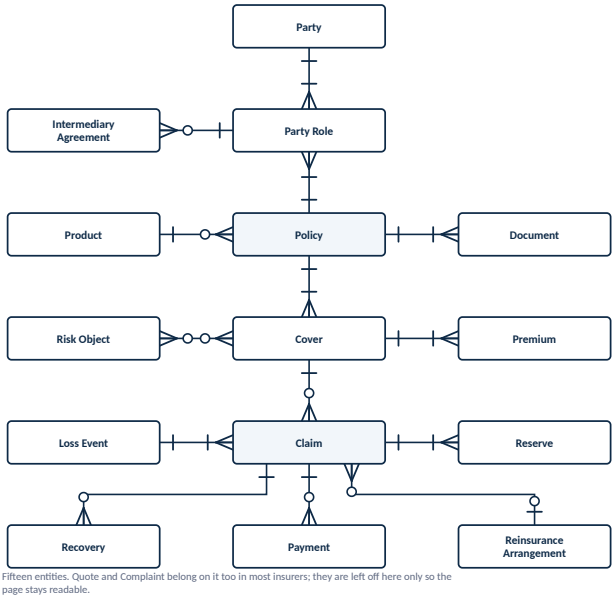


Figure 9.6 The whole conceptual model on one page. This is the figure to photograph and put on the wall; the preceding figures are the readable decompositions of it.

That is the model. Fifteen boxes, readable in a meeting, and no word on it that a claims manager would have to be taught.

Entity	What it means in business language	The question it answers
Party	A person or an organisation we deal with	Who is this, and have we dealt with them before?
Party Role	What a party is to us, between two dates	In what capacity is this party involved?
Intermediary Agreement	What an intermediary may do on our paper	Who was authorised to do what, on what date?
Product	The thing we manufacture and sell	What do we sell?
Policy	One sold instance of a product	What have we promised, to whom, for how long?
Cover	A specific promise with its own ceiling	What exactly is covered, and up to how much?
Risk Object	The thing exposed to loss	What is exposed, and have we covered it before?
Premium	The price of a cover	What are we charging, and for which promise?
Loss Event	The thing that happened in the world	What happened, and how much of our book did it touch?
Claim	An assertion that a covered event occurred	What are we being asked to pay, and under which promise?
Reserve	What we currently think a claim will cost	What did we think it would cost, and when did we think it?
Payment	Money out against a claim	Who did we pay, how much, and when?
Recovery	Money back in, reducing a claim's cost	What did we get back, from whom, and what is still owed?
Reinsurance Arrangement	Our own cover against our own book	Which of this loss is somebody else's?
Document	What we sent, and what it said	What did the customer actually receive?

Two notes on that table. First, every entry in the right-hand column is a question somebody in the business has asked out loud; that is the test I use for whether an entity has earned its box. Second, none of the questions mentions a system, and all of them are answerable in 2050.

The variants

The model above is the composite default. Three variants shift it, and an architect who cannot draw all three has only modelled a third of a South African insurer.

Life, where the risk object is a person

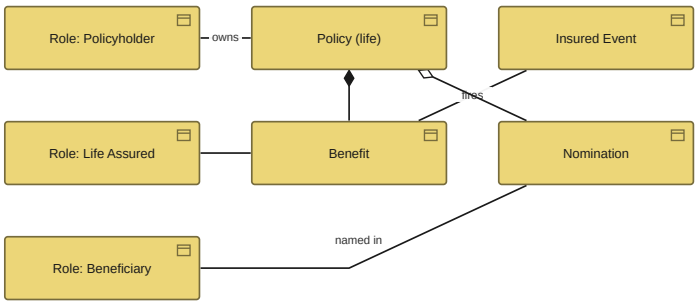


Figure 9.7 The life variant. The risk object is gone; its place is taken by a party in the role of life assured, and nomination becomes an object in its own right.

In life insurance Risk Object disappears, and its place is taken by a party in the role of life assured. Cover becomes Benefit, Peril becomes Insured Event, and two new things earn boxes.

Nomination is the beneficiary designation: a dated set of shares summing to one hundred per cent, with classes — primary and contingent — and sometimes irrevocability. It is an entity rather than a relationship because superseded nominations must remain queryable; the family will dispute the payout and the nomination in force at the date of death is the one that governs.

Multi-life cover breaks the one-policy-one-life assumption outright. South African funeral cover is routinely extended-family cover, so a single policy can carry a dozen lives assured with different ages, waiting periods and benefit amounts. This is not an edge case: of the 10.8 million new individual recurring-premium risk policies written in 2025, 6.2 million — fifty-seven per cent — were funeral [12]. A model that cannot represent the majority product is not a model of that insurer.

Group, where the member is not the contracting party

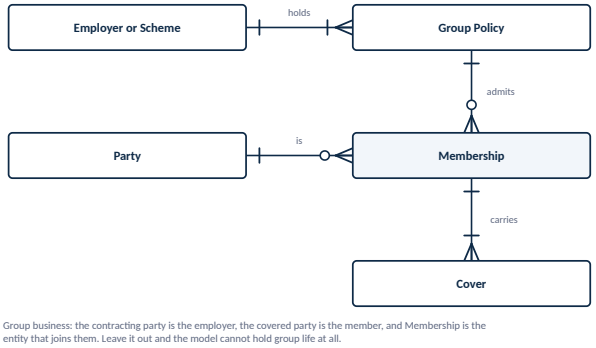


Figure 9.8 Group business. Membership is the entity that joins a covered party to a contract held by somebody else, and it is the box retail-built models lack.

This is the variant that most often defeats a model built for retail. The employer or scheme holds the contract and pays the premium. The member is covered and usually has no contract with the insurer at all. Between them sits **Membership** — a dated fact that joins a party to a group policy, carries the member's own cover levels and category, and ends when they leave the employer.

Three consequences fall out of that one box. Billing is at the scheme level and cover is at the member level, so premium and exposure sit at different grains. Claims are made by or on behalf of a party the insurer may never have underwritten individually. And data protection posture differs by role: the employer consented to things the member did not, which matters under POPIA, and particularly so for health information, where regulations in force since 6 March 2026 apply expressly to insurance companies [20].

Delegated authority, where the UMA holds the detail

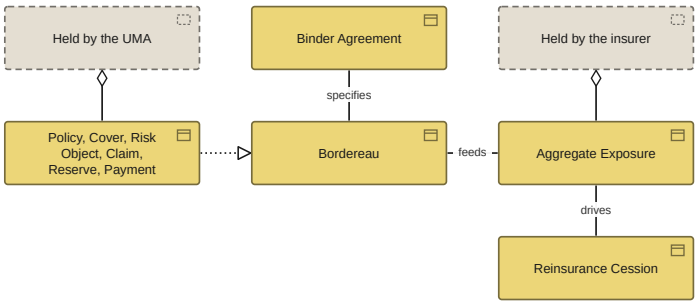


Figure 9.9 Delegated authority. Draw the standard model, then draw the boundary: for a binder book the insurer holds a bordereau, not the transactions behind it.

The hardest variant, and the most South African. Where a UMA or binder holder runs the policy and claims systems, the insurer does not hold Policy, Cover, Risk Object, Claim, Reserve or Payment as transactions at all. It holds a periodic summary.

A bordereau is a detailed report furnished periodically by the reinsured — or, in the delegated-authority direction, by the binder holder to the insurer — of premiums or losses [16]. The insurer's conceptual model for that book is therefore not the model above. It is the model above, *as reported*, at whatever grain the agreement specified, with whatever lag the agreement allowed.

The practical instruction is blunt. Draw the standard model, then draw a boundary around the part of it the insurer does not own, and write the reporting grain and lag on the boundary. That one annotation explains, in a single picture, why aggregate exposure cannot be computed daily, why the IFRS 17 grain is a negotiation rather than a query, and why single view of customer stops at the binder. Everything else about delegated authority is a consequence of it.

Two things follow that are worth saying to a steering committee. First, the grain and the lag are *contractual*, not technical: they are set in the binder agreement, which covers data ownership and reporting and obliges the holder to keep auditable records for at least five years [14]. If the architecture needs daily risk-object detail, the place to get it is the next contract renegotiation, not the integration backlog. Second, the boundary is the right place to hang the regulatory questions, because IFRS 17 groups are fixed at initial recognition and never reassessed [17] — so a cohort assignment made from a monthly bordereau is a cohort assignment you are stuck with for the life of the contract.

The entities people add that should not be there

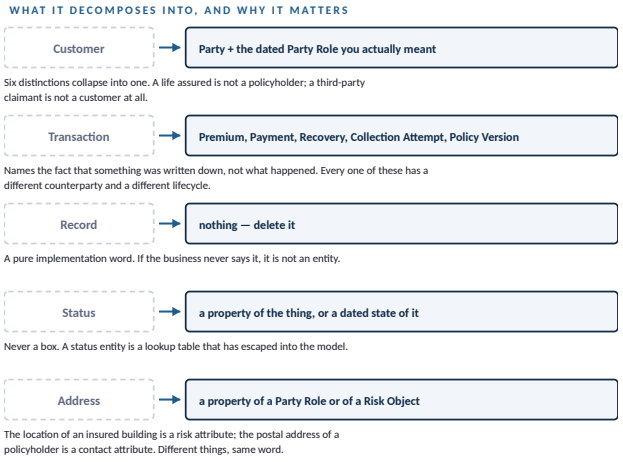


Figure 9.10 The five entities that turn up on every first draft and should not be there, and what each one is hiding.

Customer is the big one, and Chapter 10 defends the position at length. The short version: there is no customer in insurance, there are parties playing dated roles, and the word collapses six distinctions the business, the regulator and the actuary all need. When somebody insists, do not argue the abstraction. Ask them which customer they mean on a group life policy: the employer who pays, or the member who dies.

Transaction is the subtle one, because it sounds technical and respectable. It names the fact that something was written down rather than what happened. Premium, Payment, Recovery, Collection Attempt and Policy Version are all "transactions" and they have nothing in common except that an accountant once needed a word for them. A Transaction box on a conceptual model is almost always a sign that the person drawing it came from the general ledger.

Record should be deleted outright. It is a pure implementation word. If the business never says it, it is not an entity.

Two more worth watching. **Status** is a property of a thing or a dated state of it, never a box; a Status entity is a lookup table that has escaped. And **Address** is a property of a party role or of a risk object, and conflating the two is how the location of an insured building ends up in the same structure as

a policyholder's postal address, which is how an insurer discovers its flood accumulation after the flood.

There is one entity I would defend adding even though it fails test three. **Intermediary Agreement** would, strictly, not survive a change of commercial model. I keep it anyway, because in this market the delegated-authority boundary determines more about the architecture than almost anything else on the page, and a model that hides it is a model that will mislead.

Three that are borderline, and when to let them in

Collection Mandate. The customer's authorisation for the insurer to take money from their account. It looks like an operational detail and in a direct-debit market it is one. In South Africa it is not, because the mandate is now authenticated once with the debtor's own bank and held there, with a lifecycle of its own that is independent of the policy's. If the insurer's book is mass-market and monthly, put Mandate on the page; it is the entity that determines whether cover is in force.

Advice Record. What was recommended, to whom, on what analysis of their needs. It belongs on the model wherever advice is given, because conduct supervision asks whether advice was suitable and takes account of the customer's circumstances, and an advice record is the only evidence that exists. Leave it off in a direct-only insurer.

Cession. The record joining one direct policy to one reinsurance arrangement. Strictly it is the resolution of a many-to-many relationship, which this chapter has just forbidden. I let it in when reinsurance is the subject of the conversation, because the business says the word constantly and because a box named Cession ends an argument that an unnamed line does not.

The pattern in all three is the same: an entity earns its place when the business argues about it, even if a purist would say it is below the line. Be able to explain why each exception is there.

Where this sits relative to the logical model and to ACORD

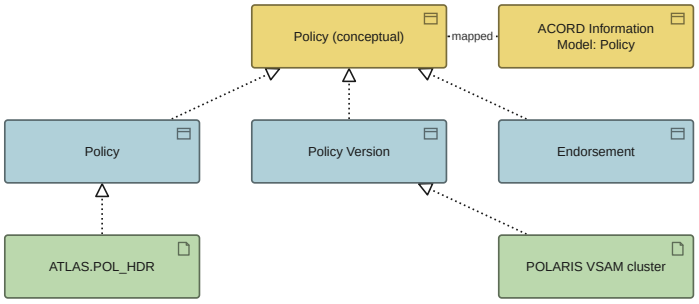


Figure 9.11 The realisation ladder. One conceptual box, three logical entities, several physical artefacts — and the conceptual box does not change when the layers below it do.

The relationship to Chapter 10's information model is a realisation ladder, and it is many-to-one upwards. Conceptual **Policy** is realised by logical Policy, Policy Version and Endorsement; those in turn are realised by physical tables in Atlas and by a VSAM cluster in Polaris. The conceptual box does not change when the logical model splits it, and that is the test of whether the split was right.

Three rules for keeping the ladder honest.

Every logical entity realises exactly one conceptual entity, or you have found something. A logical entity that cannot be traced upwards is either missing from the conceptual model or is a design artefact that should not be in the logical model either.

The conceptual model never grows to accommodate the logical one. When Chapter 10 adds bitemporal columns to everything, the conceptual model is unchanged, because valid time and record time are how you *implement* truth, not things the business talks about.

Cardinality may tighten downward but never contradict. If the conceptual model says a risk object may be covered by many covers and the logical model has a single foreign key, the logical model is wrong, not the conceptual one.

Adopt an industry model, or draw your own?

ACORD's Reference Architecture has seven facets, of which the Information Model is a conceptual representation of data elements across all lines of business covering Policy, Product, Party and Claims, with a logical entity-relationship Data Model generated from it [1]. Current releases put the Data Model and Information Model at version 2.13 and the Business Glossary at 2.16 as at December 2025 [2], with interactive online versions of the glossary and capability model released to members in January 2026 [3]. IBM's Insurance Information Warehouse offers a comparable business model plus analytical requirement templates [9], and there are open options: the first free, open, production-ready insurance data standard, openIDS Homeowners v1.0, landed on 20 November 2025 [7], and the European supervisor's reporting data point model is openly published and defines every regulatory data point with its dimensional breakdown [8].

The rule I use is narrow and I think defensible.

Draw the conceptual model yourself. Always. It takes an afternoon, it is the only artefact the business will own, and the value is in the argument, not in the diagram. Adopting somebody else's fifteen boxes skips the argument, and the argument is where you learn that this insurer calls a binder book "the agency channel" and means something different by "lapse" in two divisions.

Take the vocabulary from the industry model immediately. ACORD's glossary runs to over 6,000 terms with synonyms and line-specific usage [3]. A shared definition of "policy count" ends a decade of quarrel between finance, actuarial and operations, and nobody should be writing those definitions from scratch.

Use the industry conceptual model as a checklist, not a blueprint. For each entity in it that you lack, ask whether you genuinely do not need it. Risk object, loss event, cession, party role and policy version are the usual discoveries.

Adopt at the logical layer, not the conceptual one. That is where the industry models earn their licence fee, and it is also where Chapter 13 picks the story up.

One caution that applies to all of them. None of these models will hand you the three things that are actually hard: the two time axes that let an insurer say both what was true and what it believed [10], the IFRS 17 unit of account, and a consent model that satisfies South African data protection. Those are overlays, they are yours, and together they are most of the work. The conceptual model is not where that shows, which is exactly why it only takes an afternoon — and also why nobody should mistake finishing it for having done the data architecture.

It is worth knowing the scale of what the conceptual model has to cover before you declare it finished. The Prudential Authority counted 60,197,102 life policies in force at March 2026, up 23.2% in a year, across 159 registered insurance entities [13]. Fifteen boxes have to be true of all of them: funeral policies with eight lives assured, group schemes with fifty thousand members, single commercial property risks worth more than a small insurer's annual premium. If a box only works for the retail motor book, it is not an entity, it is an assumption.

The afternoon that produces one



Figure 9.12 An afternoon that produces a conceptual model. The deletion hour is the one that cannot be shortened.

A conceptual model is a half-day of work and a decade of use, which is the best ratio in architecture. A few things make the difference between a model and a wall of cards.

Do not invite IT. Not out of spite: because the moment a database person is in the room the vocabulary shifts, the nouns acquire suffixes and the business people defer. Hold a separate session afterwards to map what you drew onto what exists; that is a different meeting with a different purpose.

Make them write silently first. Ten minutes, one noun per card, no discussion. It gets you the words people actually use rather than the words the loudest person uses, and the synonyms it surfaces — risk item, insured item, subject matter — are themselves a finding.

Run the three tests out loud, on every card. Most of the afternoon is deletion. When somebody defends a card, write the sentence they used to defend it; that sentence is the entity definition and it is better than anything you would have written.

Test the model against three real cases before you leave. A funeral policy with eight lives assured. A group scheme where the employer pays and a member claims. A binder book where the broker issues and settles. Walk each one through the boxes out loud. Whatever breaks is next week's work, and in my experience at least one of the three always breaks, which is why I run all three.

Photograph it and stop. Do not tidy it that evening. A tidy diagram looks finished and invites approval instead of argument. Circulate the photograph, let people mark it up, and draw the clean version only once somebody has disagreed with it in writing.

The week after

The afternoon produces the model. The week after produces its authority, and that is a different job.

Write a one-sentence definition for every box, in the words the person who defended it used, and circulate those rather than the diagram — people argue with sentences and nod at pictures. Then hold the session you did not invite IT to: walk the fifteen boxes against the actual systems and record, per box, which system is the system of record and which ones hold a copy. You will find boxes with four claimants and boxes with none, and the ones with none are the finding.

Finally, put the model somewhere it can be cited and keep it there. The value of a conceptual model compounds: in month three it settles an argument about what "policy count" means, in month nine it scopes a procurement, in year two it is the thing a new architect reads on their fourth day instead of 412 pages of process diagrams. None of that happens if it lives in somebody's slide deck.

Cape Mutual is invented and nothing in this chapter describes a real insurer. The fifteen boxes are not invented. They are, as near as I can make them, the smallest set that lets an architect hold a conversation with anybody in an insurance company without having to ask what a word means — which is the only thing a conceptual model is for.

THE CHAPTER IN TEN LINES

- A conceptual model is twelve to twenty entities in business language, with no attributes, no keys, no tables and no system names — and the prohibitions are what keep the business able to read it.
- Three tests decide whether something is an entity: can you define it in a business sentence, does the business count or argue about it, and would it survive replacing every system tomorrow?
- The spine is Party, Party Role, Product, Policy, Cover and Risk Object; get those six right and the rest of the model follows.
- Three cardinalities carry real meaning: party to role is many, risk object to cover is many-to-many, and product to policy is definition to instance.
- A loss event is not a claim, and an insurer with no Loss Event entity cannot compute a catastrophe recovery — which it finds out during the catastrophe.
- A reserve is an estimate as at a date, so one claim has many; a recovery is not a negative payment, so it gets its own box.
- Three variants shift the model: life replaces the risk object with a party in a role, group inserts Membership between a contract and the person covered, and delegated authority replaces the transactions with a bordereau.
- Customer, Transaction and Record do not belong on the page; Status and Address are properties, not entities.
- Draw the conceptual model yourself because the argument is the value, but take the industry glossary immediately and adopt an industry model at the logical layer, not this one.
- The whole thing is an afternoon with six business people and no IT, and the output is a photograph that somebody has argued with.

SOURCES FOR THIS CHAPTER

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CHAPTER 10

The information model

Fifteen entities, four of them unexpected, arranged in patterns that are the same in every insurer on earth and almost never drawn correctly by an outsider. This is the chapter that stops you putting Customer on the page.

The model you will be asked to draw

Six weeks into a core replacement programme, somebody will ask you for the conceptual data model. They will want one page. They will want *Customer* on it.

Do not give them *Customer*. There is no such thing in insurance, and the remainder of this chapter is largely a defence of that sentence.

What there is instead is a small set of entities — perhaps fifteen that matter — arranged in patterns that are stable across every insurer in the world and almost never drawn correctly by an outsider. Parties playing dated roles. A product that is a definition and a policy that is an instance. A risk object that outlives every policy covering it. A loss event that is not a claim. A reserve that is an estimate with a date attached. A cession that joins a direct policy to a reinsurance treaty. And underneath all of it, two independent time axes, because an insurer must be able to say both what was true and what it believed, years after the fact, about the same policy.

This chapter builds that model. Then it tells you where to get one instead of drawing it yourself, because insurance has had published cross-line data models for thirty years and nobody should be inventing `Policy` from first principles.

> Insurance data modelling is not hard because the entities are exotic. It is hard because four things are true at once: the same party plays many roles, the same policy has many states, the same event becomes many claims, and every number has to be reproducible as at a date in the past that is no longer the current state of anything.

Party, and why there is no customer

A **party** is any legal person the insurer deals with — natural or juristic. A **role** is something a party does, *to something*, between two dates. That is the whole pattern, and it is the classic Party pattern from the data-model literature, with `Party` supertyped to `Person` and `Organisation` and all relationships carried by role-type associative entities [12]. ACORD's Information Model names Party as one of four core concepts beside Policy, Product and Claims [1].

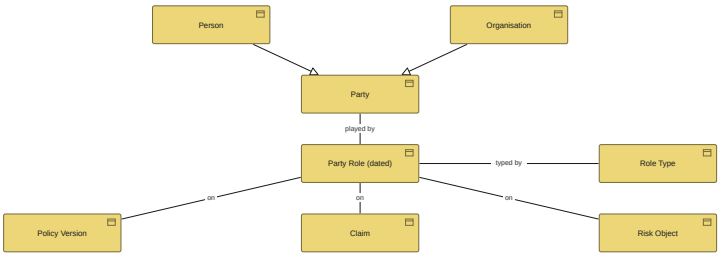


Figure 10.1 The party-role pattern. Party holds identity only; every relationship is a dated role played against a policy version, a claim or a risk object. The same human appears once as a party and many times as roles.

One human being, at Cape Mutual, on one Tuesday, can be all of the following simultaneously:

Role	What the party does	Where it is held
Policyholder / owner	Contracts with the insurer, owes the premium, may cancel, amend, assign or surrender	Atlas, on their own motor policy
Insured	The party whose interest is protected; policies also carry additional insureds and interested parties	Atlas, with the vehicle financier noted
Life assured	The party on whose death, disability or illness the benefit rests	Polaris, on a policy their employer owns
Beneficiary	Receives the proceeds; has a share, a class (primary or contingent), sometimes irrevocable	Polaris, on a spouse's policy
Claimant	Asserts a claim; may be a complete stranger to the insurer	Atlas, against a third party's liability cover
Intermediary	Licensed to sell or service; has a mandate, a hierarchy and remuneration	Beacon, as a part-time adviser
Cedant	An insurer when it buys reinsurance; the reinsurer's "customer"	Not a person at all — Cape Mutual itself

Flatten that into a `Customer` table and you have destroyed six distinctions that the business, the regulator and the actuary all need.

The five errors

In order of how much they cost.

1. **Policyholder assumed identical to insured.** False in group schemes, corporate cover, keyman policies, bancassurance and cell-captive business. Every one of those is material in South Africa.
2. **One life assured per policy.** Joint-life policies have two. Funeral cover in South Africa is routinely multi-life extended-family cover, so a single funeral policy can carry a dozen lives assured with different ages, waiting periods and benefit amounts. A one-to-one `policy.life_assured_id` makes the dominant retail product unrepresentable.
3. **Beneficiary as a foreign key.** A beneficiary nomination is a *dated set summing to 100%*, with classes and sometimes irrevocability. Superseded nominations must remain queryable, because the family will dispute the payout and the nomination in force at date of death is the one that governs.
4. **Claimant assumed to be the policyholder.** In liability business the claimant has no relationship with the insurer at all.
5. **Intermediary as a channel attribute.** An intermediary is a party with a licence, a mandate scope, a hierarchy and a remuneration arrangement. Reduce it to `channel_code = 'BROKER'` and you cannot answer what a particular intermediary was authorised to do on a particular date, which is the first question in any conduct review.

Three rules

Party holds identity, not relationships. Name, identifiers, date of birth or registration, nothing else. The moment `Party` has a `policy_number` column the pattern has collapsed.

A role is always to something. Party-in-role-on-Policy, Party-in-role-on-Claim, Party-in-role-on-RiskObject, Party-in-role-on-Intermediary-Agreement. Four separate role contexts, not one. The health standards make the same move explicitly — HL7's role vocabulary carries a `RoleClassNamedInsured` class rather than a named-insured person type [20], which is a useful citation when someone insists the pattern is over-engineering.

Roles are time-bound, and each carries its own data-protection posture. This is the part outsiders miss. A third-party claimant never consented to anything. Under POPIA their lawful basis, purpose limitation and retention are different from a policyholder's, and because life and health underwriting *is* health data, the regulations on processing health information — in force since 6 March 2026 and applying expressly to insurance companies — attach differently to different roles on the same file [27]. Consent and purpose therefore sit on the party-role, dated, not as a flag on a policy.

Policy, product, version, endorsement, schedule

> *A product is a definition. A policy is an instance.*

Everything in this section follows from that one line. The product is the manufactured template: covers, limits, options, rating factors, eligibility rules, wording and pricing algorithm, versioned independently of any sale. Vendors make the same split — a product model is "a blueprint" from which a quote and then a configured product instance called a policy is produced [13].

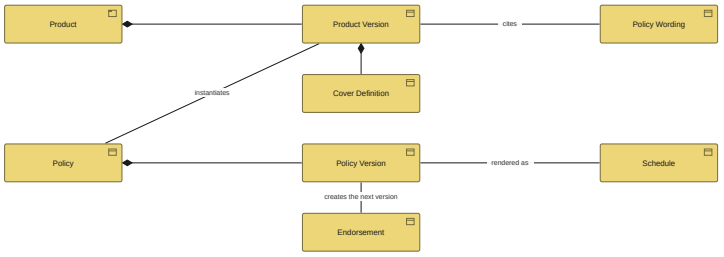


Figure 10.2 Product is a definition on its own version clock; policy is an instance with its own. An endorsement creates a new policy version, and the schedule is that version rendered — which is why storing a template reference instead of the rendered artefact silently rewrites history.

Artefact	What it is	Common failure
Product	The definition. Versioned on its own clock	Held as code, so a variant is a release
Policy	The instance: number, inception, expiry, status, parties in roles, risk objects, covers, premium	Treated as the only entity, with product rules copied into it
Policy version	A dated state of the policy; prior versions retained	Overwritten, so the policy as at date of loss cannot be reconstructed
Endorsement	The instrument effecting a mid-term change — add a vehicle, raise a sum insured, change an excess — producing a new version and usually a premium adjustment	Implemented as cancel-and-rewrite, destroying continuous cover history
Schedule	The personalised generated pages: who and what is insured, for how much, at what premium. The schedule *is* the data, rendered	Stored as a template reference plus data, so a later template change silently rewrites history
Policy wording	Standardised legal terms. A versioned document reference, not a structure	No version history, so a 2019 claim is adjudicated on 2026 terms
Cover note	Interim evidence of cover pending policy issue	Not modelled, so the gap between bind and issue is invisible

A policy number does not identify a policy

This is the sentence to write on the whiteboard. A motor policy with twelve endorsements has one number and thirteen versions. Claims adjudication, pricing, reinsurance cession and IFRS 17 measurement all need *the policy as it stood on a particular date* — which is a version, not a policy.

The architectural consequence is that the policy lifecycle is an effective-dated transaction log, not a mutable row. "What did this policy look like on 14 March 2023, and who changed it?" must be answerable, and the only designs that answer it are version-per-change or event-sourced.

Why the separation pays for itself

When an insurer says a product variant takes nine months to launch, the cause is almost always that product rules live in policy tables and application code. The definition cannot be changed without touching every instance, so every change is a release, a regression test cycle and a migration.

There is a secondary benefit that programmes consistently undervalue. South African tax law draws the same artefact-versus-event line the model does: a policy, renewal notice or endorsement that does not request payment is **not** an invoice for VAT purposes [14]. A model that distinguishes the contractual artefact from the billing event gets that right by construction. A model that treats the renewal notice as the premium transaction gets both the tax treatment and the premium register wrong.

Product, plan, cover, benefit, peril, limit, excess, sum assured

Below the product sits the cover structure, and this is where the life and short-term dialects diverge most while describing the same shape.

Concept	Short-term	Life and health
The manufactured definition	Product	Plan
A division of a multi-peril policy	Section (Motor, Buildings, Liability)	—
A protection with its own limit and conditions	Cover	—
The payout	Benefit (add-ons such as car hire)	Benefit (death, disability, critical illness, income)
The *cause* of loss	Peril — fire, flood, hail, theft, storm	Rare; the equivalent is the insured event
The ceiling	Limit, sum insured	—
The agreed amount	—	Sum assured
The insured's first-loss share	Excess, "first amount payable"	Deductible in health; life generally has none

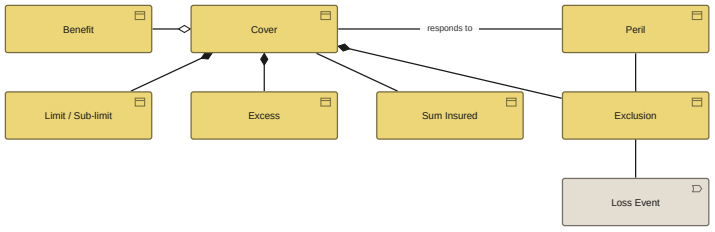


Figure 10.3 Cover carries limit, excess, sum insured, exclusions and benefits. Peril is the cause of loss and sits outside the cover: the relationship is many-to-many, and modelling it one-to-many is why peril-level loss analysis has to be reconstructed from free text.

Four distinctions carry real modelling weight.

Peril is not cover. A peril is a cause; a cover is a promise. One cover answers many perils — an "all risks" household cover responds to fire, theft, storm and accidental damage. One peril triggers covers across sections — a hail event hits the Buildings section, the Motor section and the Contents section of one policy. The relationship is many-to-many and is routinely modelled one-to-many, which is why peril-level loss analysis so often has to be reconstructed from free-text claim descriptions.

Limit is not sum insured. For a single item they coincide. But *limit* also expresses per-occurrence limits, annual aggregates, sub-limits inside a larger limit, and whether a limit reinstates after a loss or does not. That is a small structure, not an amount.

Sum insured is not sum assured. A sum insured is an *indemnity* ceiling: the insurer pays the lesser of the loss and the sum insured, and if the sum insured falls short of value, the condition of average reduces the payment proportionately. A sum assured is an *agreed value*, paid in full on the insured event regardless of economic loss. They behave differently in every calculation and they are usually the same column in a bad warehouse.

A rating factor is not a cover attribute. Claims history, credit score, telematics score and vehicle data are pricing inputs. Vendor models separate them explicitly as rating facts that "don't carry through to quote line items or the policy" [13]. Mix them into the cover structure and you have made pricing data contractual, which is both a conduct problem and an impossible migration.

The risk object — the insured thing

The **risk object** — also insured object, insurable entity, subject matter of insurance, risk item — is the thing exposed to loss. It is a first-class entity and it is absent from most bad models.

Line of business	The risk object
Motor	A vehicle
Household	The building, the contents, and specified items individually
Commercial property	A location, with plant and stock inside it
Marine cargo	A shipment
Motor trade / fleet	A *fleet* — a collection whose membership changes daily
Life	The life assured: a **party in a role**, not a thing
Liability	No object at all — an activity or legal exposure, measured by turnover, payroll, discipline and jurisdiction

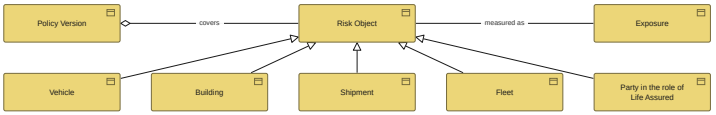


Figure 10.4 The risk object, with its subtypes by line of business. A fleet is a risk object that aggregates risk objects; in life the subtype is a party in the role of life assured; in liability there is no object at all. Exposure hangs off the risk object, not the policy.

Oracle's published Health Insurance Components use exactly this concept, naming an **insurable entity** paired with an insurance type [7] — a usable open example when you need to show that the abstraction is industry practice rather than your preference.

Four consequences

The risk object's lifecycle is independent of any policy. A vehicle exists before anyone insures it, is insured by successive insurers, and is eventually written off. Make `Vehicle` a child of `Policy` and you cannot answer "have we ever insured this VIN before?" — which is the first question in fraud detection and the first question at renewal.

Exposure lives on the risk object. Premium earns on risk-object-days, not policy-days. Add a vehicle mid-term and exposure changes without any policy date changing. An earned-premium calculation driven off policy inception and expiry is therefore wrong for every policy that was ever endorsed, which in commercial lines is most of them.

Catastrophe aggregation is a geocoded risk-object query. "What is our total sum insured within two kilometres of this river?" is answerable at the risk object and nowhere else. If addresses are held as free text on the policy, the question cannot be asked at all, which is how insurers discover their flood accumulation after the flood.

Fleets and schedules are the hard case. A fleet is a risk object that aggregates risk objects, with daily membership changes and a premium adjusted in arrears on declared exposure. In my experience this is the commonest reason an insurer cannot reconcile exposure to premium, though I have not found a study that ranks it, so treat that as judgement rather than fact.

Premium and its components

Premium is not an amount. It is a set of dated components in one of five states.

State	Meaning
Quoted	Offered, not sold
Written	Contracted in the period, before earning. Gross before reinsurance, net after
Billed	An instalment raised against the premium
Collected	Cash received — and in South Africa, the VAT event [14]
Earned	The portion attributable to expired exposure, calculated from exposure dates. Written minus earned is the unearned premium: a liability

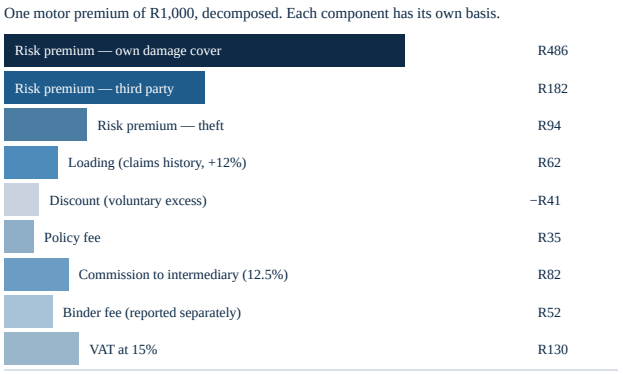


Figure 10.5 Premium modelled as dated components rather than a total. Only this shape can answer what the commission was on the theft cover of the third vehicle in March — which is the shape of question a conduct supervisor asks.

Model premium as components, not a total:

Component	Note
Risk premium per cover	The actuarial price of the promise, at cover level, not policy level
Loadings	Added for assessed extra risk — a medical loading, a claims-history loading
Discounts	Multi-policy, no-claim, voluntary-excess
Policy fee	A flat administration charge, often with a different tax treatment
Commission	Usually a percentage of premium, often capped. Paid to a party in the intermediary role, who may not be the seller
Profit / contingency commission	Contingent on a book's loss experience, calculated retrospectively across a book. Breaks policy-centric models outright
Binder fee	South Africa: a fee for performing insurer functions under a binder. Must be separable from commission, because the regulator asks about them separately [25]
VAT	In South Africa, ordinary VAT. There is no separate insurance premium tax
Instalment	A scheduled collection *against* premium — not premium. Model an instalment as premium and endorsements become unhandleable

A single `premium_amount` column cannot answer "what was the commission on the theft cover of the third vehicle in March". That is not a hypothetical reporting nicety; it is the shape of question a conduct supervisor asks, and the omni-CBR's conduct indicators push granularity in exactly that direction [26].

THE VAT FINDING: A CLAIM PAYMENT'S TAX TREATMENT DEPENDS ON DATA THE CLAIMS MODEL DOES NOT HOLD

South Africa brings insurance into the ordinary VAT system. Short-term insurance is a **taxable supply** — premium and intermediary services standard-rated unless zero-rated [14][15]. Most architects expect that. What they do not expect is that the **indemnity side** is taxed too.

Under section 8(8), a **monetary indemnity payment** to a vendor insured for a loss incurred in the course of its enterprise is a **deemed taxable supply by the insured**: the insured declares output tax, net of the excess, and the insurer takes a section 16(3)(c) deduction [14][16]. Under section 8(8A), in force from 1 January 2024, where the insurer instead pays a third party to reinstate — a **trade payment** — the insured has no deemed supply; the supplier invoices insurer and insured separately and the insurer claims ordinary input tax [14]. Recoveries from subrogation carry no output tax. And the excess is not consideration for a taxable supply at all, so booking it as income is wrong twice [14].

So the VAT treatment of a single claim payment turns on **five attributes**:

1. Is the policy a taxable supply, or zero-rated under section 11(2)?
2. Is the insured a registered vendor?
3. Was the loss incurred in the course of an enterprise?
4. Is this a cash settlement or a reinstatement paid to a supplier?
5. Was input tax denied on the original acquisition of the asset?

Now look at a claims data model. Attributes 2, 3 and 5 are almost never present. Attribute 4 is usually inferable from the payee type but not asserted. Attribute 1 requires the risk object's location, the insured's residence and vendor status, and evidence held — because zero-rating under section 11(2) covers international travel and transport insurance, stock-throughput cover while goods are in international transit, hull cover for foreign-going aircraft and ships supplied to non-resident non-vendors, and cover on property abroad where the insurer holds proof the property is outside South Africa [14].

Worse, **mixed-rate policies must be apportioned**. Where one stock-throughput premium spans standard-rated and zero-rated risk, the insurer must allocate the premium between them and notify the insured [14]. That is impossible with one premium amount and one tax code, which is what most policy administration systems hold.

The practical consequence: five attributes, three of them about the *insured's* tax position rather than the policy's, must be captured at underwriting and carried to claim settlement. Retrofitting them after a SARS query means re-underwriting a book. Add them to the party and risk-object models on day one — the cost is three columns and a validation rule.

One more detail that surprises people: **bordereaux are tax documents.** SARS allows a policy document to serve as tax invoice, debit note and credit note, with similar relief for intermediary bordereaux [14]. The spreadsheet arriving monthly from a UMA is not an operational convenience; it is part of the tax record, and it needs the retention, immutability and audit treatment that implies.

Claim, loss event, reserve, payment, recovery

A claim is not a loss event

A **loss event** — occurrence, incident — happened in the world: a hailstorm, a collision, a death. A **claim** is an assertion against a policy arising from it. One event yields many claims. A hailstorm hits 4,000 roofs. One collision yields an own-damage claim, a third-party property claim and three bodily-injury claims.

Catastrophe aggregation, reinsurance event limits and hours clauses all operate on the **event**. No `LossEvent` entity, no catastrophe recovery calculation — and since excess-of-loss recovery is computable only per loss event, after aggregating claims across policies within an hours clause and a geography, this single omission is where most reinsurance recovery leakage lives.

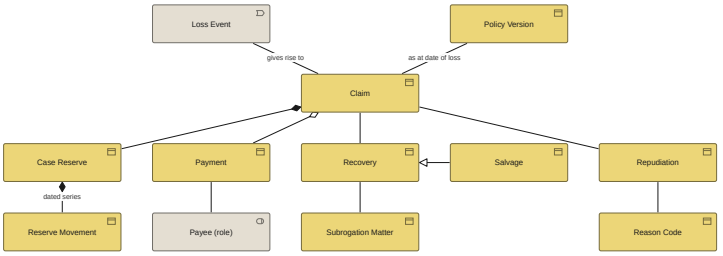


Figure 10.6 Loss event, claim, reserve, payment and recovery. The reserve is a dated series of movements, not a value; salvage is a kind of recovery and not a negative payment; and repudiation is an outcome with a reason code.

Four dates, all different

Date	What it drives
Date of loss	Which policy version responds; accident-year reporting; reinsurance attachment
Notification date	IBNR estimation; conduct turnaround metrics
Registration date	Operational workflow
Settlement date	Cash flow and development

Long-tail business adds manifestation and first-awareness dates; liability adds the date of the act or omission. I would not claim that list is exhaustive — it is the set that recurs.

Reserves are estimates with a date

A **reserve** is an estimate of amounts payable: an accounting liability, not cash set aside. A **case reserve** is a handler's estimate on one known claim; the **outstanding claims reserve** is their sum across open claims. **IBNR** is the liability for unpaid claims not reflected in the case reserve estimates for individual losses, which splits into *pure IBNR* — losses that occurred but are unreported at the evaluation date — and *broad* or *bulk IBNR*, meaning additional development on known claims [18]. That second part also travels under **IBNER**: incurred but not *enough* reserved. **ALAE** and **ULAE** are loss adjustment expenses allocated and unallocated to a specific claim.

Here is the requirement that follows, and it is the single most expensive thing in this chapter to get wrong:

> The claims mart must be a **time series of reserve movements**, not a current-state table. A reserve without an evaluation date is not a number. If reserve revisions overwrite their predecessors, the insurer has lost the ability to reserve, because every reserving method in use — chain ladder, Bornhuetter- Ferguson, Benktander, Cape Cod — works by comparing what was known at successive evaluation dates about one accident period [24].

An incurred-claims development triangle, R million.
Each cell is one accident year as it was understood at one evaluation date.

Accident year	dev 0	dev 1	dev 2	dev 3	dev 4
2021	143	171	179	181	181
2022	158	190	201	204	
2023	180	214	208		
2024	196	239			
2025	212				

The dashed step is the diagonal of "what we knew at each year-end". Overwrite it and reserving stops.

Figure 10.7 The artefact every reserving method consumes. The dashed step is the diagonal of what was known at each successive year-end — which exists only if reserve history was never overwritten.

In my experience this is the commonest insurance warehouse failure, and the commonest reason the actuarial function builds a shadow data estate on its own infrastructure and quietly becomes the de facto system of record. I have not found a survey that ranks warehouse failures, so take the ranking as judgement.

Money out, money in, and the things that are not payments

Entity	Definition and data consequence
Payment	An outflow against a claim. Many per claim, each with a payee (a party in a role), date, currency, VAT treatment and allocation to a head of damage
Recovery	An inflow reducing net claim cost. **Not a negative payment**: different counterparty, timeline and accounting. Model separately or you cannot age a reinsurance recoverable
Salvage	Proceeds from selling property taken over after a total loss. Implies a transfer-of-title event, which is itself a modelled fact
Subrogation	The insurer's right, after paying, to pursue the responsible party. A new legal matter with its own parties and timeline. No South African output tax [14]
Contribution	The proportionate share where two policies cover one loss. Needs other insurers' policies represented against the risk object
Ex gratia	A payment made without admitting liability. Must be flagged: conduct regulators and reinsurers both ask
Repudiation	Declining a claim. An outcome with a reason code, and now conduct-reporting data [26]

Reinsurance: treaty, cession, bordereau

Insurance bought by insurers — a parallel contract universe sitting over the direct book, and in most architectures an Excel-shaped hole.

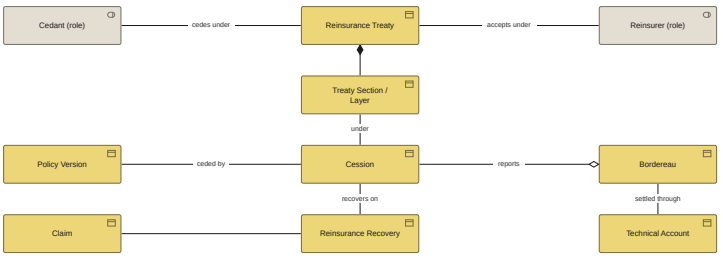


Figure 10.8 Treaty, section, cession, bordereau and recovery. The cession is the join between a direct policy version and a reinsurance contract; the bordereau reports cessions periodically and is also a tax document.

Term	Meaning
Cedant / reinsurer / retrocession	The insurer transferring risk / the party accepting it / reinsurance of reinsurance
Treaty	A standing agreement: the cedant must cede and the reinsurer must accept all business within defined terms, including policies not yet written [19]
Facultative	Risk by risk; the reinsurer may accept or decline each one [19]
Cession	The act and the record of ceding a risk or premium share — the join between direct policy and reinsurance contract
Proportional	The same share of premium and losses. **Quota share** is a fixed percentage of every risk; **surplus** means the cedant keeps a "line" and cedes the surplus above it, so the ceded percentage varies with risk size [19]
Non-proportional	**Excess of loss**: pays above an attachment point up to a limit, for a separate premium — per risk, per occurrence, or aggregate [19]
Retention / ceding commission	What the cedant keeps / the reinsurer's contribution to the cedant's acquisition costs on proportional treaties [19]

A **bordereau** is a detailed report furnished periodically by the reinsured, of reinsurance premiums or reinsurance losses [17]. The **premium bordereau** lists policies reinsured with policy-level gross premium, brokerage and ceded premium. The **loss bordereau** lists claims and expenses outstanding and paid, with insured and claimant names, policy and claim numbers, dates of loss, reserves and paid amounts — and on a quota share it may be the reinsurer's only source of information on ceded losses [17]. Reporting of this kind is primarily applicable to pro rata arrangements [17].

Why this is genuinely hard

Cessions need retrospective granularity. The treaty year depends on date of loss under a losses-occurring treaty, or on the date the policy attached under a risks-attaching one. Both require the policy and claim as they validly stood, not as they currently stand — which is the bitemporal requirement in the next section, arriving here first.

Recoveries are a second reserving exercise. Reinsurance IBNR, bad-debt provisioning and reinsurer credit exposure are all separate estimates on the asset side, aged and credit-assessed.

The bordereau is bilaterally agreed and never identical between two reinsurers. ACORD's Delegated Authority standards exist precisely to standardise this, covering data from initial risk capture through binding, premium reporting and claims reporting [1].

Delegated authority inverts the data flow. Where a UMA, MGA or coverholder holds the policy and claims system, the insurer receives summarised, lagged, inconsistently coded data about its own book [25]. In a South African non-life insurer writing substantial binder business this is not an edge case; it is the central design problem. Risk aggregation, IFRS 17 granularity, claims analytics and single view of customer all break at the bordereau boundary.

Bitemporality, and why this is the hardest temporal problem you will meet

Insurance needs **both** time axes.

- **Valid time** — Fowler calls it *actual time*; also effective or business time — is when a fact was true in the world.
- **Transaction time** — Fowler's *record time*; also knowledge or as-at time — is when the insurer knew it.

Fowler's worked example: a salary is paid at 6,000 on 25 February; on 15 March HR says a raise to 6,500 took effect on 15 February; on 5 April HR corrects it to 6,400. One timeline cannot answer both "what was the salary on 25 February?" and "what did we believe, and pay, on 25 February?" — and as he notes, the tax authority may ask the second question, so both answers matter [11]. Valid history is mutable, because 6,500 was never really true; record history is append-only [11].

Substitute sum insured for salary, broker for HR, and regulator, auditor and reinsurer for the tax authority. That is insurance.

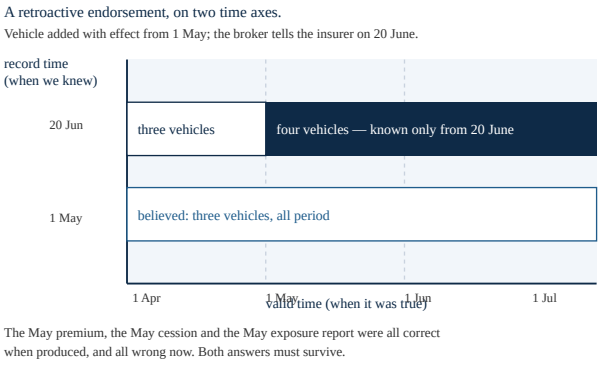


Figure 10.9 The retroactive endorsement on two axes. Valid time says four vehicles from 1 May; record time says the insurer only knew it on 20 June. Every number produced in between was correct when produced and is wrong now.

Situation	Why one axis fails
Retroactive endorsement	A broker advises on 20 June that a vehicle was added on 1 May. In between, the records said there was no cover, and the insurer priced, reported and ceded on that basis. Both facts must survive
Date-of-loss adjudication	A September claim for a May loss is assessed against the policy *as it validly stood* in May, not as it was recorded then
As-at reporting	"The position as at 31 December as reported" and "as now understood" are different numbers. Both must be reproducible years later
Development triangles	The method *is* comparing what was known at successive evaluation dates about one accident period. Overwriting reserves destroys it
Cession recalculation	A retroactive sum-insured increase changes the cession percentage, the ceded premium and the recoveries for periods already reported
IFRS 17 groups	Groups are set at initial recognition and are not reassessed during the coverage period [21], so facts as at inception must survive indefinitely
Conduct and POPIA	"What terms was the customer shown at point of sale, and what did we know then?" Consent state is itself bitemporal

Implementation, with the traps

Four columns on every versioned entity: `valid_from`, `valid_to`, `recorded_from`, `recorded_to`. Record history is append-only; a correction closes a record-time interval and inserts a new one. Fowler's mitigation — default the record date to "now" — keeps the ninety per cent of queries that want current state cheap [11].

Three warnings.

A bitemporal join is a four-predicate join and is performance-hostile. Two such entities joined across both axes will not perform at reporting volumes. Expect to materialise as-at snapshots, and design them deliberately rather than discovering the need in month nine.

Do not effective-date everything. Claim payments and premium collections are *events*: immutable, append-only, with one date each. Mis-identifying events as versioned states doubles the model and buys nothing. The test is whether the fact can be *corrected* as opposed to *reversed*. A sum insured is corrected. A payment is reversed by another payment.

The cruder option is often the one that works. Many insurers take immutable monthly snapshots of the whole policy and claims estate instead of implementing bitemporal tables. It is a blunt instrument — month-end granularity only, large storage, no intra-month reconstruction — and in practice it is frequently more reliable than a half-built bitemporal model, because it has one failure mode rather than twenty. I offer that as judgement, not as a recommendation I can source.

Master data and the single view of customer

Insurers are structurally bad at this, for five reasons that are all architectural rather than cultural.

Line-of-business silos. Motor, household, life, funeral, investment and health sit on different platforms, usually acquired rather than built, each with its own party store and key. At Cape Mutual that is Polaris, Atlas and whatever came with the last acquisition.

Weak identifiers at the point of sale. Mass-market funeral business is sold fast, through channels with minimal data capture, so matching is probabilistic rather than deterministic.

Intermediated distribution. The broker owns the relationship and often the data. Under a binder the insurer may not hold the customer record at all.

Roles, not customers. One human appearing as policyholder, life assured and third-party claimant on three systems is *correctly* three roles. Master data management must still resolve them to one identity — and collapsing identity into role is how insurers end up marketing to third-party claimants.

The commercially meaningful unit is often not the individual. It is the household or the group scheme, both of which are dated party-to-party relationships rather than attributes.

A workable target has five parts [22][23]:

1. A **party master** holding identity and identifiers only — South African ID, passport, company registration, insurer-assigned identifier — with match keys and survivorship rules.
2. A **role store** of dated roles against policies, claims, risk objects and intermediary agreements.
3. Golden records with **attribute-level lineage**, because POPIA correction rights require knowing which source contributed which value.
4. **Consent and preference as dated party-level data**, not a flag on a policy.
5. An explicit, written decision on probabilistic matching — because a wrong merge exposes one customer's data to another and is worse than no merge. That last point is a governance position I hold rather than a sourced standard, but it is the one I would put in front of a steering committee.

Do not draw this yourself: the industry data models

Insurance has had mature, published, cross-line data models for thirty years. Nobody should be deriving `Party`, `Policy`, `Claim` and `Coverage` from first principles. The question is not whether to use one, but in what role: vocabulary, conceptual reference, logical starting point, or physical schema.

ACORD

Two families, routinely conflated. The **Data Standards** are messaging formats for trading-partner exchange: AL3, the old one-way batch format for policy and commission data that still carries large agency-channel volume; ACORD's P&C XML for real-time request and response (ACORD keeps the North American "P&C" naming where South Africa says short-term); a separate Life and Annuity family; NGDS for microservices and fine-grained

transactions; and Delegated Authority for coverholder reporting. There is a Southern Africa release too, built around short-term insurance standards with a business process guide covering the policy life cycle [29].

The **Reference Architecture** is what an architect actually wants. ACORD describes seven interrelated facets, usable separately or together, across lines of business and geographies [1]:

Facet	What it is
Capability Model	A functional breakdown of value-chain activities, deliberately unordered, for process engineering, merger evaluation and operations analysis [1]
Information Model	A conceptual representation of data elements across all lines of business, covering Policy, Product, Party and Claims and their relationships, linked to all ACORD Standards [1]
Data Model	A logical entity-relationship structure generated from the Information Model, usable in any database — for physical design, warehouse design, or validating an in-house model [1][28]
Component Model	The information needed for specific business activities, linking capability to information in technology-neutral terms, for build-or-buy decisions [1]
Process Model	Workflow-oriented reference processes aligned to the Capability Model, with messages mapped to the Data Standards. The current site says Process Model; the marketplace still says Process Framework, so expect both names [1]
Business Glossary	Standardised non-technical definitions — over 6,000 insurance terms [3]
Product Model	A graphical notation plus an XML-based Product Schema [1]

Currency, as at this writing: Data Model and Information Model at v2.13 and Business Glossary at v2.16, released 12 December 2025, adding Marketing Activity as a subtype of Activity, a Lead Management sub-package and a Sales Appetite component [2]; and interactive online versions of the Business Glossary and Capability Model for members from 29 January 2026 [3].

It is not open. Members and non-members may subscribe to the full Reference Architecture or to individual components; training materials are free to members and previous versions are available by request [1]. Pricing is not published, so treat it as procurement.

IBM Insurance Information Warehouse

IBM's insurance industry model, sibling of its banking models, positioned as a fully realised development blueprint for insurance data warehouses [4]. Four layers: a **Business Model** (conceptual, implementation-independent, to agree common definitions across systems); an **Enterprise Model** (an enterprise-wide entity-relationship model with predefined warehouse structures); **Business Solution Templates**, which map every measure and dimension to the warehouse; and a generated physical environment [4].

The Business Solution Templates are the distinguishing feature, because they ship the *questions* rather than only the schema — analytical requirement definitions covering analytical CRM, profitability, risk and compliance, and financial reporting [4][5].

Note the vintage. The shipped regulatory content covers Solvency I, Solvency II at QIS5 and CP58 template level, Sarbanes-Oxley and IAS 39 [5] — which is the Solvency II *preparatory* phase, not the final supervisory taxonomies, and predates IFRS 17 and IFRS 9 entirely. Whether current releases added IFRS 17 content is something I could not establish; check before relying on it.

Oracle Insurance Data Foundation

The OIDF Application Pack sits within the OFSAA stack as a data foundation beneath Oracle's analytical applications. It is not a standalone published entity-relationship model, and publicly verifiable content is thin: versioned application packs with download specifications, changelogs, run charts, data model difference reports, slowly-changing-dimension metadata and table-to-table technical metadata [6]. The substantive documentation sits behind support entitlement.

The architectural shape is inferable from those document titles — a staging area fed by table-to-table mappings, a results area consumed by analytical applications, conformed dimensions as slowly changing dimensions — but that is inference from titles, not Oracle's own description, so do not quote it back to Oracle. The commercially important point is plain: unlike ACORD, **you cannot evaluate the model content before buying it.**

openIDS and openIDL — the open alternative

openIDL (Open Insurance Data Link) is a Linux Foundation project with AAIS, a United States non-profit statistical advisory organisation, as principal industry sponsor, bringing together carriers, regulators and technology partners [8][9]. **openIDS** (Open Insurance Data Standards) is a community initiative hosted by openIDL, aiming at an open-source data model architecture for the property and casualty insurance industry [8].

On **20 November 2025** it released **openIDS Homeowners Standard v1.0**, described as the first free, open, production-ready insurance data standard, introducing a consistent framework for capturing and exchanging property risk data [8]. It came from the openIDS Homeowners Workstream and incorporates contributed intellectual property from Cloverleaf Analytics, and is first in a planned series by line of business [8].

It is described as free, open and adoptable at no cost under Linux Foundation governance. I could not establish the licence identifier or terms — neither the Linux Foundation release nor the AAIS page names one — so if the licence matters to your use, ask before you plan around it. And be clear about scope: one line of business, one jurisdiction's regulatory context, brand new. It is not a cross-line enterprise model today. It is the first credible open competitor to ACORD, which is a different and still significant claim.

Three more open assets worth knowing:

Asset	Nature and use
EIOPA Solvency II DPM and XBRL taxonomies	Openly published by the European supervisor [10]. Not an entity-relationship model but a *reporting* model defining every quantitative reporting template data point with its dimensional breakdown — the definitive open artefact for what a prudential regulator means by each number, and mineable even outside the EU
FIBO	An open OWL financial ontology: deep on instruments, parties and legal entities, shallow on insurance coverage
HL7 and FHIR terminologies	Open health standards including insurance-adjacent role vocabularies [20]; relevant to health insurance and life medical evidence, and a citable example of the role-not-person pattern

How to use one rather than reinvent

The failure mode is binary. Either "we will adopt ACORD", followed by two years mapping 6,000 glossary terms nobody asked for; or "ours is different", followed by rebuilding the party-role pattern badly.

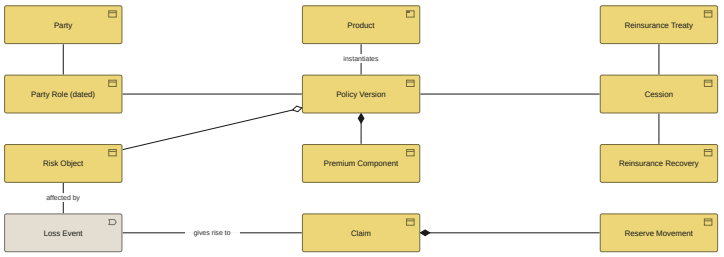


Figure 10.10 The whole core on one page: party and role, product and policy version, risk object and premium component, loss event and claim, cession and recovery. Fifteen entities, no Customer.

Eight rules, in the order you will need them.

1. **Always take the vocabulary.** A shared business glossary ends a decade of definitional argument about whether a cancelled-and-rewritten policy is a renewal. ACORD's is the obvious candidate [3]; the open EIOPA data point model is a free fallback [10].
2. **Use the conceptual model as a checklist, not a blueprint.** For each industry entity you lack, ask whether you genuinely do not need it. Risk object, loss event, cession, party role and policy version are the usual discoveries.
3. **Never adopt a logical industry model as a physical schema unmodified.** These models are exhaustive by design. Implemented literally they yield hundreds of near-empty tables and unusable performance. Subset aggressively and document every exclusion, because the exclusions are the design.
4. **Use it for the canonical integration layer, not for application databases.** The value is a shared semantic contract between heterogeneous policy administration systems — which is exactly the binder-heavy South African problem.
5. **Take the analytical requirements wherever they are offered.** IBM's Business Solution Templates [4] and EIOPA's data point model [10] encode the *questions*. Reverse-engineering those from regulation costs more than licensing them.
6. **Map, do not migrate, for messaging standards.** AL3, XML and NGDS are interchange formats. Translate at the boundary; never restructure internal data to match a message schema.
7. **Decide the licensing posture early.** ACORD requires subscription; IBM and Oracle require product licences, with Oracle unevaluable before purchase; openIDS is free but narrow; EIOPA and FIBO are open. The terms determine whether you may expose the model to partners or ship it in a product.
8. **Budget for the overlay regardless.** No industry model hands you a bitemporal implementation, an IFRS 17 unit-of-account structure or a POPIA consent model. Those three are yours, and together they are most of the work this chapter describes.

That last rule is the honest summary of the chapter. The entities are a solved problem and you can buy them. The time dimension, the regulatory grain and the consent model are not solved, cannot be bought, and are where an insurer's data architecture is actually won or lost.

THE CHAPTER IN TEN LINES

- There is no customer: there are parties, and dated roles those parties play against a policy version, a claim, a risk object or an intermediary agreement.
- A product is a definition with its own version clock and a policy is an instance with its own; a policy number identifies a policy, not a policy state.
- Peril is a cause and cover is a promise, related many-to-many; sum insured is an indemnity ceiling and sum assured an agreed value.
- The risk object is a first-class entity with a lifecycle independent of any policy, and exposure, catastrophe aggregation and fraud detection all hang off it.
- Model premium as dated components — risk premium per cover, loadings, discounts, fees, commission, binder fee, VAT — because no single total can answer a conduct question.
- South African VAT on a claim payment turns on five attributes, three of which concern the insured's own tax position and are almost never in the claims model.
- A loss event is not a claim; without a loss event entity there is no catastrophe or excess-of-loss recovery calculation.
- A reserve is an estimate as at a date, so the claims mart must be a time series of movements or the insurer has lost the ability to reserve.
- Insurance data is bitemporal — valid time and record time — and every retroactive endorsement, as-at report, development triangle and cession recalculation needs both.
- Industry models (ACORD, IBM IIW, Oracle OIDF, openIDS) give you the entities and the vocabulary; the bitemporal implementation, the IFRS 17 grain and the POPIA consent model are yours and are most of the work.

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CHAPTER 11

The conceptual application model

Chapter 12 names the systems an insurer runs. This chapter is the layer above: the applications an insurer needs, named by what they do, with nobody's product name anywhere in the model. It is the one you draw when you are designing a target rather than describing an estate.

The model you draw before anyone has chosen a vendor

Fatima Patel has eight weeks to produce a target application architecture for the Cape Mutual group. Two artefacts arrive to help her: the application inventory, 1,430 rows exported from the service management tool and sorted by owner, and a slide from a vendor with the vendor's module names in tidy coloured boxes.

Neither is a target architecture, and the reasons differ. The inventory describes what is there, at the granularity of whatever somebody once raised a ticket about, so Polaris appears four times and a departmental Access database appears once. The vendor slide describes what that vendor sells, which is a commercial boundary rather than an architectural one.

Between them sits the artefact nobody handed her: a list of the applications the group *needs*, named by what they do, with no product names in it anywhere. That is the conceptual application model, and it earns its place three ways.

It survives vendor change. Cape Mutual Short-Term will replace Atlas at some point in the next decade. If the architecture says "Atlas", every diagram in the estate is wrong the day the contract is signed. If it says "policy administration", the boxes stay and one label changes. The same holds for organisational change, which happens oftener: a model named after departments dies at the next reorganisation.

It is the thing you compare a package against. A vendor demonstration answers the questions the vendor chose. A conceptual model reverses that. For each box, three questions in order: does your product perform this function, does it *master* this data or read it from somewhere else, and which of my named roles would be working in it. The third finds the gaps, because a product can tick a function and still have no screen for the person who does the job.

It is how you discover that two systems do the same job. Duplicate function is nearly invisible in an inventory: the two systems have different names, owners, cost centres and ages, so nobody sees them side by side. Map both onto a conceptual model and they land in the same box. In my experience that is the only reliable duplicate detector an architect has, and it routinely finds three or four pairs in a composite insurer.

Three artefacts that get confused

Keep these separate, in your head and in the repository. Conflating them is the commonest reason a target architecture cannot be used.

Artefact	The question it answers	What the boxes are named after
Business capability model	What must this business be able to do?	The business's own words: *underwrite risk*, *settle claims*
Conceptual application model	Which applications must exist for it to do that?	Function: *rating and pricing*, *policy administration*
Application landscape and portfolio	Which systems exist, what do they cost, where do they run?	Products and local names: Atlas, Polaris, the vendor's product

The test for the first two is easier than the literature makes it sound. A business capability is something the organisation does, and it stays true if you switch off every computer. An application capability is something software does on the organisation's behalf. "Handle a claim" is work people do; "claims management" is a thing you can buy.

ACORD has published something close to this layer for years. Its Reference Architecture carries a **Capability Model**, deliberately a functional decomposition with no predefined workflow, and a **Component Model** describing the information and properties needed by the business activities that compose insurance processes, bridging capability to information in technology-neutral terms [1]. The Component Model is the nearest published equivalent to the model in this chapter. It is also subscription-gated, like almost everything in the Reference Architecture [1], which is why most insurers build their own. Chapter 13 covers how to use ACORD and where it stops.

> **The naming rule, and it is absolute.** No vendor name, no product name, no project name, no department name, no technology. "Rating and pricing", not "Radar Live". "Intermediary engagement", not "the broker portal programme". If a box cannot be named without one of those, it is not an application capability — it is an implementation, and it belongs in Chapter 12's layer.

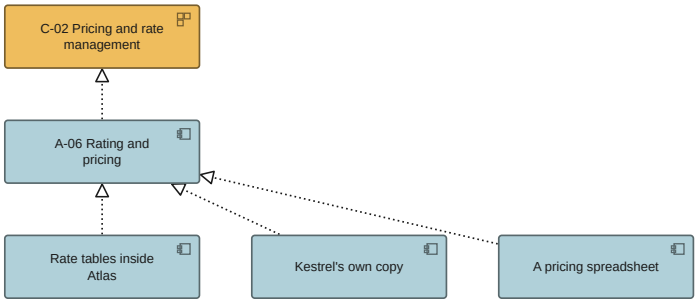


Figure 11.1 The three layers, with one capability threaded through them. Realization runs upward: three real systems realise one conceptual application, which realises one business capability. Each layer changes on its own clock, which is the whole reason for keeping them apart — and the three boxes on the bottom row are the finding.

What goes in a box, and what does not

Four tests. A candidate failing any of them is not an application capability.

1. **It masters data, or it makes a decision.** Either it is authoritative for some set of facts, or it applies rules to produce an outcome nothing else produces. A box that only moves data between two other boxes is an integration pattern.
2. **Somebody would notice if it vanished.** Name the role. If no role from Chapter 5 works in it or consumes its output directly, the box is a feature of something else.
3. **It could be bought, built or outsourced as one thing.** The procurement test, and the thing that sets granularity. "Fraud detection" passes; "red-flag rule evaluation" does not, because nobody sells it alone.
4. **It can be named without borrowing a word from a vendor, a department or a technology.**

Granularity follows from the third test. For an insurer, **eighteen to twenty-five application capabilities is the useful range.** Below fifteen, policy administration and claims collapse into "core" and the interesting argument disappears. Above forty you are modelling product features and the model will be abandoned inside a year. This chapter uses twenty-five.

> **THREE BOXES THAT ARE NOT APPLICATION CAPABILITIES**
>
> **"Digital."** Not a function but a channel preference, and on an application model it is a budget line wearing a box. Split it into customer engagement, intermediary engagement and the service layer underneath. **"Integration."** A pattern. The gateway, the streaming platform and the residual middleware belong in the technology architecture. **"AI."** A technique that will end up inside eight of these boxes. Modelling it as a twenty-sixth produces a programme with nothing to deliver.

The conceptual application model

Twenty-five application capabilities in fourteen domains. The grouping is for reading, not for ownership — several domains cut across two or three business units, and that is a feature. The table at the end of this section carries the detail: purpose, the roles who use each one, the data it masters, and the Chapter 12 category that usually implements it. What follows is the part a table cannot hold — which business capability each one serves, and what actually breaks.

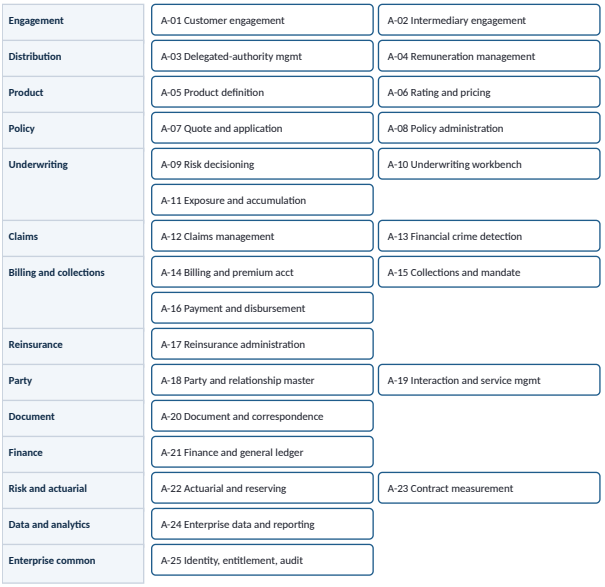


Figure 11.2 Twenty-five application capabilities in fourteen domains. The grouping is for reading; several of these domains cut across two or three business units.

Engagement and distribution

A-01 Customer engagement is what the policyholder touches; **A-02 Intermediary engagement** is the equivalent for brokers and advisers. Both serve sales and customer service, and both master almost nothing of consequence — a session, a saved basket, a submission draft. That poverty is the test of whether an engagement layer has stayed honest. The characteristic failure is accumulation: logic belonging in the core settles here, creating a second, undeclared system of record nobody governs.

A-03 Intermediary and delegated-authority management accredits an intermediary and controls what authority it holds. It serves distribution management and, directly, conduct compliance: the insurer stays responsible for what a UMA decides, so oversight is continuous rather than annual, with records retained at least five years and reporting reliable enough for insurers and regulators [2]. It is the capability most often missing altogether, and its absence is why a licence is not checked at bind.

A-04 Remuneration management pays what the distribution network earned. It is a **multi-party, time-shifted, reversible** calculation — paid up a hierarchy, often advanced before earned, clawed back when a policy lapses early. The named failure mode is **shadow accounting**, where advisers and internal teams keep parallel spreadsheets to check official payouts [3]. That source is vendor-sponsored, which does not make it wrong, and finding the shadow accounts is the fastest route to the truth about commission data.

Product, policy and underwriting

A-05 Product definition holds what the company sells, versioned and effective-dated, and serves product management. In most insurers it is not mastered anywhere — spread across the policy system, rating, the portal, the templates and the commission scheme. That fragmentation, not the age of the core, is usually why a launch takes nine months. Count the systems that must change for a launch before accepting any other explanation.

A-06 Rating and pricing serves pricing and rate management, and masters the **rating audit trail**: which version, with which inputs, produced which premium. That trail is a regulatory artefact — an ombud complaint about a quote given fourteen months ago must be answered with that quote, exactly, and many insurers cannot. The argument for pulling rating out of the core is testable: one engine behind several entry points rather than four that drift, the actuary's logic deployed as that logic, and isolation of the most volatile code from the most stable [4].

A-07 Quote and application management carries a case from interest to a decidable application, serving sales and quotation. In short-term it takes minutes; in life it is a **case management problem** with third parties the insurer does not control. Build it as a long transaction and you get a system that cannot represent a case waiting eleven days for a doctor's letter.

A-08 Policy administration holds the contract and serves contract administration. It is not a database of policies but an **event processor over time**, and it decides whether the rest of the estate can work: if it cannot reconstruct exact policy state at an arbitrary past date, no claim decision and no dispute can be defended. A life platform here is partly a ledger — it accrues contractual values, charges surrender penalties, converts to paid-up, allocates units and annuitises [5] — and a short-term platform does none of that. One capability, at least two products, and treating them as one is the commonest mistake an architect new to insurance makes.

A-09 Risk decisioning and **A-10 Underwriting workbench** both serve risk selection and underwriting, and conflating them is the second commonest mistake. The engine decides the straightforward risks; the workbench is the cockpit for the risks the engine could not or should not decide. The honest design note sits in the first: the outcome is **three-way, not two-way** — a final decision, a referral for review, or no decision at all because the data was wrong or the case too complex [6]. That third exit is where straight-through processing leaks back into manual work, and where a reported rate of eighty per cent hides a queue nobody manages. In the workbench, rules that can be scoped, prioritised and **set to expire** are the only mechanism anyone has invented that stops a rule population growing without limit [6]; the expiry feature is almost never switched on.

A-11 Exposure and accumulation management answers a question no other application can: how much are we on the hook for, where, if one event happens. It serves exposure and capital management, and it is a live requirement rather than a sophistication — weather catastrophe was the largest single force on South African short-term claims costs in the first half of 2026 [8].

Claims and financial crime

A-12 Claims management serves claims management and recovery management. The published capability set worth checking an insurer against covers multi-channel first notification with attributes analysed at notification, coverage verification against the policy **as at date of loss**, reserving, adjudication under authority limits, payment, litigation management, and subrogation, recovery and salvage with early intervention rather than after-the-fact discovery [9]. Read it, then ask which of it is done in a system.

> Reserve movement history is the reserving actuary's primary raw material, because development triangles are built from it. Treat it as immutable, correctly dated, append-only data, and say so in the model. An insurer that overwrites reserve movements is reserving from fiction, and the error will not surface for two years.

A-13 Financial crime detection serves financial crime management, and what insurers forget it masters is the **outcome labels**: confirmed fraud, suspicion unproven, claim declined. Those labels are the training data for every future model, so an insurer that does not capture them has capped its detection without knowing. The highest-value control point in the estate is the pre-payment hold; a score arriving after the money has left is a report, not a control.

Money: billing, collections and payment

Three boxes rather than one, because in South Africa they fail separately and are frequently owned by different parties. All three serve premium billing and collection, and financial management.

A-14 Billing and premium accounting says what is owed and records what arrived. Suspense is its diagnostic: how large, and how old the oldest item.

A-15 Collections and mandate management gets the money. The mandate is now a first-class, bank-held, long-lived object with its own lifecycle: under DebiCheck the creditor registers it through its own bank, it is stored at the **debtor's** bank, the debtor authenticates once, and every later collection is validated against its approved terms [10]. Something in the estate has to master it. AEDO and NAEDO were switched off on 31 October 2021 [11], so a design document naming them is stale.

A-16 Payment and disbursement moves money out, and is a controls problem more than an integration one. A change of payee bank details is one of the highest-risk events in an insurer — the mechanism behind most settlement-diversion fraud — and it belongs in the architecture, not buried in a maintenance screen.

Reinsurance, party and document

A-17 Reinsurance administration serves reinsurance management. Scoped properly it covers treaty and facultative terms, cession calculation, ceding and profit commission, reinstatement premium, commutations, and bill and bordereau production [12]. Scoped as it usually is, it is two spreadsheets maintained by two people — routinely the largest concentration of knowledge risk in the insurer.

A-18 Party and relationship master serves customer and party management, and its decisive choice is whether it is **operational** — called at transaction time, authoritative — or **analytical**, reconciling identities afterwards for reporting. Build the second, call it a single view of customer, and the contact centre agent is still looking at three screens with parts of the customer on them. **A-19 Interaction and service management** serves customer service and complaints, and holds the complaint with its root cause: a regulated record since the conduct reforms made complaints monitoring, analysis and reporting an obligation [7], not an internal note. A licence for A-19 does not buy you A-18, and the number of insurers who believed it did is remarkable.

A-20 Document and correspondence management serves contract administration and conduct compliance. It is treated as a printing utility when it is a compliance system: the schedule and the wording *are* the contract. Test whether the archive stores the **rendered artefact** or the data plus a template reference. If it is the second, a template change silently rewrites history, and the document you produce in 2027 for a 2019 complaint is not the document the customer received.

Finance, risk, data and the common layer

A-21 Finance and general ledger serves financial management and reporting, and architects skip it. The posting-granularity decision inside it is one of the most consequential in the estate, because a chart of accounts that cannot serve financial *and* management reporting forces a permanent allocation layer with its own team [13].

A-22 Actuarial modelling and reserving serves actuarial valuation and capital management. It almost always needs a policy-level actuarial data store beside it, because the policy system cannot serve projection workloads — and that store, built as an extract, quietly becomes the de facto system of record for anything the actuaries trust. **A-23 Insurance contract measurement** serves financial reporting, and the pattern to learn is the **sub-ledger**: it sits between the actuarial and source systems and the general ledger, absorbing the new requirement so administrative, actuarial and finance systems stay largely intact. Inside it are data management, calculation, and accounting-and-posting, and the published guidance is explicit about configuration over customisation [13].

A-24 Enterprise data and reporting serves information management and regulatory compliance. It masters nothing operational — it masters **history**, including the only surviving record of books whose source systems have been switched off, which quietly makes it tier-one. Note what vendor arguments for the modern platform tend to omit entirely: regulatory reporting, the workload the thing exists to serve [15]. In South Africa that means the Quantitative Reporting Template, issued and amended by prudential notice [14].

A-25 Identity, entitlement and audit serves every capability and is owned by none. It belongs on this model rather than quietly in the technology architecture because in insurance the authority limits are *business* rules — who may bind R5 million of commercial property, who may authorise a R400,000 claim, which broker may settle under a binder — and when they are enforced in a user interface instead of a service, a direct integration walks past them.

The table

Role names are the jobs from Chapter 5. "Typical real-world system" points at the category from Chapter 12, not at a product.

ID	Application capability	Purpose	Primary user roles	Data mastered	Typical real-world system
A-01	Customer engagement	Lets a policyholder quote, buy, view, change and claim without an employee	Policyholder, contact centre agent	Session and saved-quote state, device registration	Digital quote-and-buy stack, app, self-service portal
A-02	Intermediary engagement	Lets a broker or adviser quote, bind within mandate, service and claim	Independent broker, tied agent, field agent	Intermediary working state, submission drafts	Broker and adviser portal
A-03	Intermediary and delegated-authority management	Accredits intermediaries and controls what authority each one holds	Distribution manager, compliance officer, key individual	Intermediary agreement, licence and appointment status, binder scope, bordereau receipts	Producer and UMA management, bordereau ingestion
A-04	Remuneration management	Calculates and pays commission, binder fees, overrides, profit share and clawback	Financial accountant, brokerage principal, tied agent	Commission scheme, earned and paid position, clawback exposure	Commission management system
A-05	Product definition	Holds what the company sells, versioned and effective-dated	Product manager, product actuary, legal counsel	Product and plan definitions, versions, cover and benefit structure, question sets	Product configuration, product factory
A-06	Rating and pricing	Returns a premium from risk attributes, reproducibly	Pricing actuary, and every channel as consumer	Rate tables, algorithm versions, rating audit trail	Rating and pricing engine
A-07	Quote and application management	Carries a case from interest to a decidable application	Independent broker, tied agent, underwriting assistant	Quote, application answer set, requirements and evidence, case state	New business and onboarding platform
A-08	Policy administration	Holds the contract and processes every event against it over time	Policy servicing clerk, underwriter, administrator or TPA	Policy, policy version, risk detail, limits, premium components, beneficiaries, event history	Policy administration system, life and short-term
A-09	Risk decisioning	Decides the straightforward risks with no human involved	Chief underwriter as rule owner, referral desk on the exits	Underwriting rules and versions, automated decision and reasons	Automated underwriting rules engine
A-10	Underwriting workbench	Gives an underwriter the cockpit for cases the rules would not take	Underwriter (commercial), referral desk, medical underwriter	Underwriting decision, loadings, exclusions, authority used, referral history	Underwriting workbench
A-11	Exposure and accumulation management	Says how much is at risk, where, from a single event	Underwriter (commercial), treaty negotiator, capital actuary, Chief Risk Officer	Modelled exposure, aggregates by peril and geography, remaining treaty capacity	Catastrophe and exposure management platform
A-12	Claims management	Registers a loss, decides cover, quantifies, pays and recovers	Claims handler, claims manager, assessor, loss adjuster, recoveries clerk	Claim, loss event, the four dates, cause coding, reserve movements, payments, recoveries, claim file	Claims management system
A-13	Financial crime detection	Scores and investigates suspected fraud before the money leaves	Forensic investigator, claims handler	Scores and contributing factors, cases, entity links, watchlists, outcome labels	Fraud scoring and case management, link analysis
A-14	Billing and premium accounting	Says what is owed and when, and records what arrived	Financial accountant, collections clerk	Premium receivable, instalment schedule, cash allocation, suspense, arrears	Billing and premium accounting module
A-15	Collections and mandate management	Gets the money, and holds the mandate over the contract's life	Collections clerk	Mandate and status, collection attempts and reason codes, derived cover-in-force	Collections module plus a mandate management layer
A-16	Payment and disbursement	Moves money out, under control	Claims manager, financial accountant, compliance officer	Payee bank detail and verification, instruction and approval trail, payment status	Payment hub and bank integration
A-17	Reinsurance administration	Records cessions, settles with reinsurers, recovers their share	Reinsurance manager, treaty negotiator, reserving actuary	Treaty and facultative terms, cessions, ceded premium, recoverable, programme state	Reinsurance management system
A-18	Party and relationship master	Says who this is, and whether it is the same person as that one	Head of data, and every role through other applications	Party identity and identifiers, contact detail, relationships, dated roles, consent	Party master, master data management
A-19	Interaction and service management	Manages the conversation, the service case and the complaint	Contact centre agent, compliance officer, ombud liaison	Interaction history, service case, complaint and root cause	CRM and case management
A-20	Document and correspondence management	Produces the documents that are the contract, and keeps them	Policy servicing clerk, legal counsel, compliance officer	Templates and versions, correspondence record, archive of the rendered artefact	Document composition and output management
A-21	Finance and general ledger	Holds the journals, the chart of accounts and the statutory accounts	Financial accountant, Chief Financial Officer	Financial posting, account structure	General ledger and finance suite
A-22	Actuarial modelling and reserving	Values the liabilities and models the capital	Reserving actuary, capital actuary, Head of Actuarial Function	Assumptions and versions, model audit trail, technical provisions	Actuarial modelling and reserving platform
A-23	Insurance contract measurement	Turns actuarial output into accounting entries	IFRS 17 team, financial accountant	Contractual service margin and roll-forward, risk adjustment, coverage units, groups and cohorts	IFRS 17 engine and sub-ledger
A-24	Enterprise data and reporting	Holds conformed history and produces management and regulatory output	Head of data, reserving actuary, pricing actuary, compliance officer	Conformed history, reconciled reporting facts, lineage	Warehouse or lakehouse plus regulatory reporting
A-25	Identity, entitlement and audit	Says who may act, within what limit, on whose behalf, and records it	Compliance officer, operations, information officer	User and service identity, entitlement and authority limit, delegation, audit trail	Identity and access management plus an

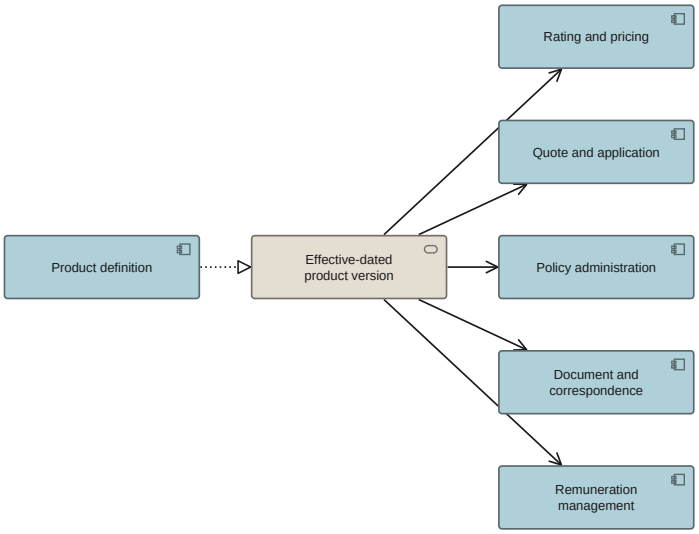


Figure 11.3 Why product definition is the capability with the largest payoff: five others consume it. When it is not mastered, each of the five holds its own copy and one regulatory change is applied five times.

The patterns

Twenty-five boxes imply nothing about how many systems implement them. Four patterns tell you what to expect, and they are what makes the model usable in a procurement conversation.

Usually exactly one. Finance and general ledger, because finance will not tolerate two. Party and relationship master, by aspiration — in practice one system half the estate ignores. Identity, entitlement and audit in theory; three in practice, which is the finding.

Usually several, and correctly so. Policy administration, one per line of business and often one per acquired book, because a life platform and a short-term platform share no data model. Rating, claims, product definition and both engagement capabilities multiply the same way. The number of policy administration instances is, in my experience, the best single proxy for architectural debt; that is a heuristic, not a measurement. The discipline is not to collapse them but to **declare** them. Three instances of one capability is an architecture; three undeclared instances is a surprise.

Usually bought. Policy administration, claims, rating, document composition, the general ledger, actuarial modelling, the contract measurement engine, fraud scoring, reinsurance administration. Mature categories with real products, where building needs defending rather than assuming.

Usually built. Customer engagement and the service layer under it, bordereau ingestion and normalisation, the orchestration holding a multi-channel quote path together, and the reporting on the data platform. Notice the pattern: insurers buy the applications that master data and build the ones that mediate experience. That is the right way round, and it reverses when a programme is under pressure.

Usually missing entirely. A short list, and the most valuable part of this chapter.

Missing capability	What you find instead	What it costs
A-03 Intermediary and delegated authority	Intermediary identity mastered in the commission system; licence status in a spreadsheet; mandates in email	Licence not checked at bind, which is live conduct exposure; one brokerage existing five times with five spellings
A-05 Product definition	Product spread across the policy system, rating, portal, templates and commission scheme	A nine-month launch cycle blamed on the core
A-11 Exposure and accumulation	A quarterly extract into a spreadsheet, or nothing	Accumulation blindness at exactly the moment the weather changed [8]
A-17 Reinsurance administration	Two spreadsheets and two people	Usually the largest concentration of knowledge risk in the insurer; recoveries that cannot be evidenced to a treaty clause
A-25 Identity and entitlement as a service	Authority limits enforced in user interfaces	A direct integration that walks past every limit, and a standing audit finding

Why these five and not five others? Not accident. Each is owned by somebody without a seat at the architecture table — distribution operations, the product committee, the chief risk officer, one reinsurance manager, internal audit. Applications get built for the people who attend the prioritisation meeting.

How the model changes with the line of business

The twenty-five boxes are stable. Their weight is not, and an architect carrying one weighting into a different insurer will design the wrong thing.

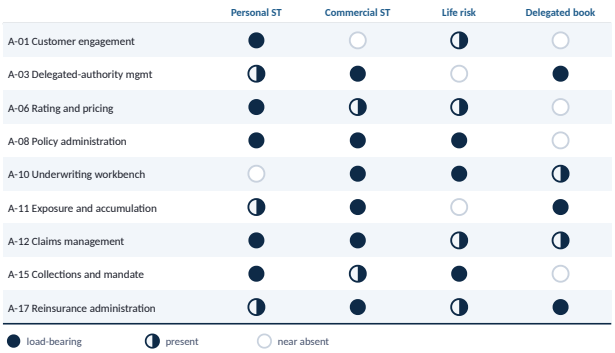


Figure 11.4 The same twenty-five boxes, weighted four ways. An architect who carries one column's weighting into another column's insurer will design the wrong thing.

Line of business	Dominant capabilities	Nearly absent	The shape of the problem
Personal short-term	A-01, A-06, A-08, A-12, A-15	A-10, A-11 at risk level	Volume and cost to serve. Rating sophistication is the competitive position; collection success is the retention lever
Commercial short-term	A-02, A-07, A-10, A-11, A-17	A-01	Schedule-bearing contracts, negotiated terms, accumulation. The broker holds the relationship and often the administration
Individual life risk	A-07, A-08, A-10, A-20, A-22	A-11	Case management with external evidence providers; a contract outliving several technology generations [16]
Group and employee benefits	A-08, A-18, A-24	A-01, A-06	Membership arrives as a monthly schedule from a third party; the member is a life assured, not a policyholder
Funeral and microinsurance	A-01, A-12, A-15, A-20	A-10, A-11, A-17	Cost to serve near zero, multi-life contracts, collections intelligence as core, claims paid in hours
Delegated authority and UMA books	A-03, A-11, A-17, A-24	A-05, A-06, A-07, A-08	The model inverts — see below
Reinsurance assumed	A-17, A-21, A-22	A-01, A-06, A-15	Contract, section, layer, cession. No quote, no bind, no policyholder servicing

The delegated-authority row deserves a paragraph, because it most often wrecks a target architecture built from a global reference model. **A UMA runs its own policy administration.** For that book the insurer does not have a policy administration capability; it has an ingestion capability. Policy data arrives as a bordereau, periodically, in somebody else's format, after the fact. Everything needing policy-level, near-real-time data breaks at that boundary: accumulation, contract measurement granularity, claims analytics, single view of customer, conduct monitoring. On the model, A-03 and A-24 move from the edge to the centre and A-05 through A-08 are somebody else's. I would argue this is the most under-appreciated integration problem in the local market; that is an argued position, and no credible public figure exists for how much premium sits behind delegated authority, so describe the model and do not quote a share. Two coming changes press on it: the COFI Bill consolidating the conduct regime, and the Omni-Risk Return expanding data and reporting requirements across the value chain — which means aggregating UMA data to a standard the insurer does not control, from systems it does not own [2].

How the model changes with size

Insurer	Boxes you will find	How they are implemented	The real constraint
Microinsurer or monoline start-up	Eight to ten	One administration suite, a payment provider, a spreadsheet	Nobody masters product, so the suite's limits are the product's limits
Mid-size single-licence insurer	Eighteen to twenty-five	A bought core plus six to ten surround systems	The surround was bought one system at a time, so A-05 and A-18 are contested
Large composite group	All twenty-five, several times over	150 to 400 real applications against 25 capabilities	Nothing missing, nothing single. The work is deciding which duplicate survives
Group with a large delegated book	All twenty-five, plus partners' estates you cannot see	Your 25 plus somebody else's 25, joined by a file	Oversight and reconciliation, not transaction processing

The small end is where architects are most often wrong, and the error is always the same: assuming the microinsurer's model is a simplification of the large one. It is not. A microinsurer with ten boxes has made real choices — not to master product separately, not to own its payment rails, not to have an underwriting workbench at all — each defensible at that scale and indefensible at ten times the volume. So the useful question at a small insurer is not "which boxes are missing" but "which of them becomes load-bearing at three times this size". That is the growth roadmap, and a conversation the executive will actually have.

How to use it

As a target architecture skeleton

Start from the twenty-five, not from the estate, then make four passes.

1. **Mark the system of record for each data entity, once.** One application masters a party, one a policy, one a mandate. Where two candidates exist you have found the programme's first real decision.
2. **Mark the capability each box serves, and look for orphans.** A box serving no business capability is a system looking for a purpose; a capability served by no box is a gap, and the gaps are the deliverables.
3. **Mark the roles.** Walk the role list from Chapter 5 and ask where that person spends the day. A role with no application is working in email and Excel, which is a finding. An application with no role is a platform somebody bought.
4. **Only then map the real systems on**, many-to-many in both directions. Boxes with three systems are rationalisation candidates; boxes with none are build-or-buy decisions; a system landing in four boxes is a monolith whose replacement is a programme rather than a project.

The output is a target that can be argued about on substance, because every box has an owner, a data assignment and a user.

As a package-fit assessment

This is where the model pays for itself fastest. A package evaluation run from a requirements list produces a spreadsheet nobody believes, because the vendor answers every line "yes, configurable". Run it from the conceptual model, scoring each box on four axes.

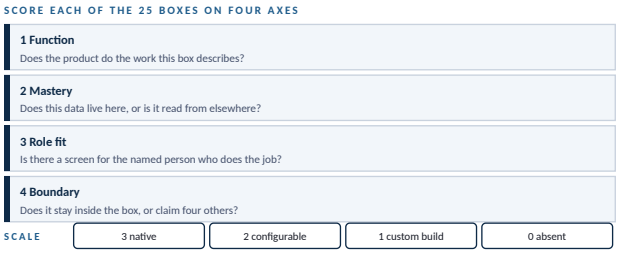


Figure 11.5 The package-fit scorecard. Axis two is the one vendors answer badly and axis three is the one they are never asked.

Three traps. **Demo coverage**, where a product shows a screen for a function it does not master — ask where the data lives when the product is switched off. **Scope creep by suite**, where a bought core quietly claims eight boxes and you accept it, then find its product definition cannot express your funeral range. And the expensive one: **assessing fit against the model you have rather than the model you need**, because a package scored against the current estate's boundaries will always favour the vendor whose boundaries match the incumbent's.

As the basis for the application portfolio

The portfolio is the inventory with decisions attached, and the conceptual model makes the decisions comparable. Each real application gets a primary conceptual capability, a secondary list where it genuinely spans, a system-of-record flag per data entity, and a disposition — invest, maintain, consolidate, retire. Two reports then become possible. **Capability coverage**: for each of the twenty-five, how many systems, how old, how much a year. And **duplication by capability**, the report that funds rationalisation, because it is the only one showing a finance director two systems doing the same job. Keep it to one primary capability per application; a model where everything maps to everything produces a picture with no decisions in it.

Cape Mutual against the model

Nothing about Cape Mutual is real. It is the worked example carried through this book, and Polaris, Atlas, Beacon, Sentinel, Compass, Ledger and Kestrel are not products you can buy.

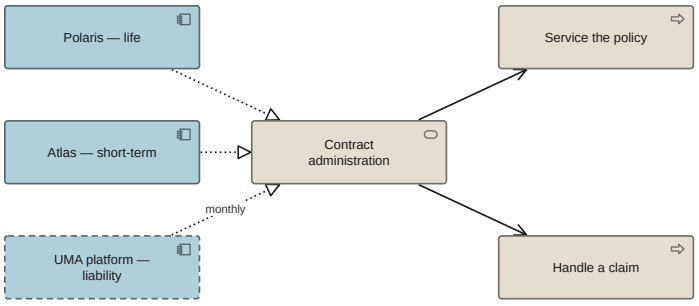


Figure 11.6 One conceptual capability, three implementations, one of them somebody else's and a month behind. Drawing the capability as a service and the systems as components is what makes the asymmetry visible. Cape Mutual is invented.

Conceptual capability	What implements it at Cape Mutual	The finding
A-05 Product definition	Polaris tables, Atlas tables, a rates spreadsheet, templates in legal's file share	Not mastered anywhere. Four sources, no authoritative list of what the group sells
A-06 Rating and pricing	Inside Polaris, inside Atlas, again inside Kestrel	Three implementations. Kestrel and Atlas can price one risk differently
A-08 Policy administration	Polaris, Atlas, and the liability UMA's own platform	Three instances of one capability, one of which the group does not own
A-11 Exposure and accumulation	Nothing	The gap the next flood will price
A-17 Reinsurance administration	One spreadsheet, one person	The largest single-point knowledge risk in the group
A-18 Party and relationship master	Compass, analytically	A reporting artefact. The contact centre agent is still looking at Polaris
A-25 Identity and entitlement	Enforced in Beacon's user interface	Boland Brokers' R5 million bind limit lives in an email

Seven rows, and between them they are the group's technology strategy: two duplicates to collapse, two capabilities to build from nothing, one party master to promote from analytical to operational, one authority model to move out of a screen and into a service. None of that was visible in the 1,430-row inventory. All of it is visible on one page with twenty-five boxes.

Where this goes wrong

The model is drawn at the vendor's granularity. Somebody starts from the incumbent suite's module list — a commercial decomposition — and the result cannot be used to evaluate anyone else. The tell is boxes whose names you have heard in a sales meeting.

The model has no owner, so it rots. It needs one named architect, a quarterly review, and a rule that no system enters the portfolio without a capability assignment.

Capability and application get fused. You see this when a box called "underwriting" serves a capability called "underwriting" and is realised by a system called "the underwriting system". Three layers collapsed into one, nothing can be reasoned about, and the only remaining question is which vendor.

The model becomes the argument instead of informing it. Twenty-five boxes exist to make a conversation about duplication, system of record and sequencing possible in one meeting. An architect defending the boxes rather than the decisions has mistaken the map for the country. If the business says exposure management belongs inside underwriting rather than beside it, move it. The box is cheap. The decision underneath it is not.

THE CHAPTER IN TEN LINES

- A conceptual application model names the applications an insurer needs by function, never by product, department or technology.
- It earns its place three ways: it survives vendor and organisational change, it is what you score a package against, and it is the only reliable way to find two systems doing one job.
- Keep business capability, conceptual application and real system in three separate layers; collapsing them leaves only the question of which vendor.
- Eighteen to twenty-five boxes is the useful range for an insurer. Below fifteen you cannot assess a package; above forty the model will be abandoned within a year.
- Five capabilities are missing in most insurers — delegated-authority management, product definition, exposure and accumulation, reinsurance administration, and entitlement as a service — and each is owned by somebody with no seat at the architecture table.
- Insurers buy the applications that master data and build the ones that mediate experience; when that reverses, a programme is under pressure.
- Policy administration, rating, claims and product definition are usually several systems, and that is correct. Declare the instances rather than pretending to collapse them.
- Where authority is delegated, the model inverts: ingestion, oversight and reconciliation replace the capabilities a UMA performs on your behalf.
- A microinsurer's ten-box model is a set of deliberate choices, not a simplification of a large insurer's; the useful question is which missing box becomes load-bearing at three times the volume.
- Mark the system of record once per data entity, mark the roles, and only then map the real systems on. Boxes with three systems are rationalisation; boxes with none are build-or-buy.

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CHAPTER 12

The application landscape

This is the chapter you will come back to. It names the systems an insurer actually runs, says what each one is authoritative for, how each one goes bad, and what to ask about it in your first week.

The shape of the estate

Ask a CIO for the application inventory and you will get a spreadsheet of two hundred to two thousand rows. The part that matters is small, and falls into four groups. The boundaries between the groups are where the architecture lives.

The core. One system holds the contract. It knows that policy 40188221 is in force, covers a 2019 Polo for R280,000, costs R612 a month and carries a R6,500 excess, changed on 1 March because the customer wanted a lower premium. Next to it sit the system that pays claims against that contract and the one that collects the money. Policy administration, claims, billing: the oldest systems in the building, the most expensive to change, the ones nobody volunteers to replace.

The surround. Around the core sits everything that decides. What do we sell — product configuration. What does it cost — rating. Will we accept this risk — underwriting. What do we send the customer — documents. Who is this person — party master and CRM. Is this claim honest — fraud detection. Every one began life *inside* the core and was pulled out at a different time, for a different reason. Which ones have been pulled out tells you most of the insurer's technology history.

The channel layer. Brokers, independent advisers, tied agents, call centres, websites, apps, aggregators, bancassurance partners, worksite teams. Each is a front door with its own sales process and regulatory treatment. This layer changes fastest and matters most to the people who set the budget, which is why its architecture is usually the worst.

The finance and risk stack. Actuarial models, reserving, capital modelling, the IFRS 17 engine and sub-ledger, the general ledger, the warehouse, regulatory reporting. Its deadlines are external and immovable, which changes how you treat it. A broker portal can slip a quarter. A Prudential Authority return cannot slip at all.

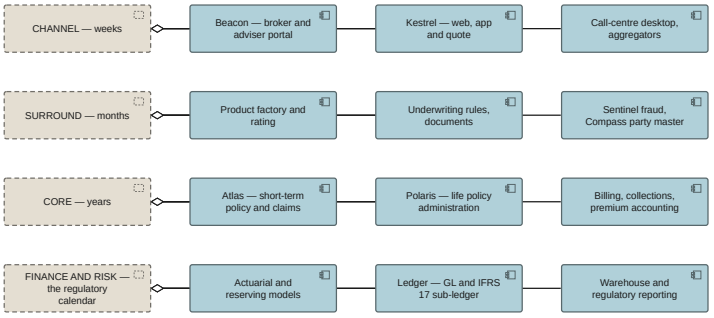


Figure 12.1 The four groups of an insurer's estate as groupings over application components, with the change cadence each group is allowed on the grouping itself. The core is not where the interesting work is; it is where the expensive work is.

Why does it look like this? Because of a decision taken before anyone reading this was working. **The unit of record is the policy, not the customer.** For fifty years the policy was what the business counted, priced, reserved against, reinsured and reported on, and the customer was an attribute of it — a name, an identity number, an address. Almost everything that now feels awkward follows from that choice, and every single-view-of-customer programme is a retrofit of it. It was not a mistake: it is the correct model for the business as run in 1975, still carrying the load in 2026.

The second structural fact is that **insurance is two industries** and the systems do not interchange. Life and long-term on one side; short-term — motor, property, liability, what North Americans call P&C — on the other. South Africa licenses them separately under the Insurance Act 18 of 2017, so a composite group is two regulated entities and at least two cores. Any statement about "the policy system" is probably true of one side of the house and false of the other.

Here is the map: sixteen categories, and what each is supposed to be authoritative about.

#	Category	Authoritative for
1	Policy administration (PAS)	Policy, coverage, premium, policy events
2	Product configuration / product factory	Product definitions and versions
3	Rating and pricing engine	Rate tables, algorithms, rate versions
4	Underwriting workbench and rules engine	Decisions, loadings, exclusions, authority
5	New business and onboarding	Application, requirements, evidence
6	Claims management	Claim, reserve, payment, recovery
7	Billing, collections, premium accounting	Premium receivable, mandate, arrears
8	Reinsurance management	Cessions, treaties, recoverables
9	Actuarial, reserving, IFRS 17	Provisions, CSM, risk adjustment
10	Distribution: portals, commission, UMA	Intermediary, licence, commission
11	CRM and party master	Party, relationship, consent
12	Document generation, output management	Correspondence record, archive
13	Fraud detection and claims analytics	Scores, cases, networks, outcomes
14	Warehouse, lake, regulatory reporting	Conformed history
15	Payments and disbursements	Payment instruction, bank details
16	General ledger, HR, procurement	Journals, employees, suppliers

Expect multiple instances of one, three, four, six and nine — one per acquired book, line or country — and exactly one general ledger, because finance will not tolerate two. In my experience the number of policy administration instances is the best single proxy for architectural debt, and a better opening question than anything about cloud. That is a heuristic, not a measurement.

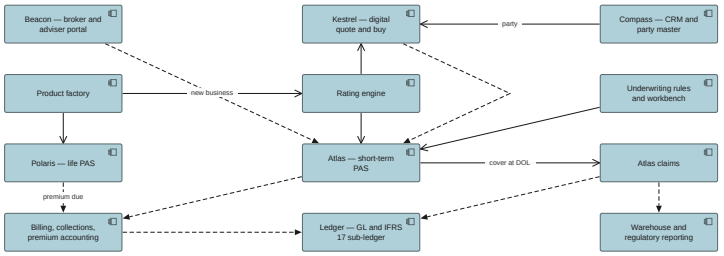


Figure 12.2 Cape Mutual's core estate: channels on top, the deciding systems in the middle, two policy administration systems side by side, finance and data at the bottom. Cape Mutual is a fictional insurer used throughout this book.

> Cape Mutual Insurance Group, its systems and its people are invented for this book and carried over from *Modelling the Enterprise* so the examples stay consistent. Polaris, Atlas, Beacon, Sentinel, Compass, Ledger and Kestrel are not products you can buy, and Cape Mutual is not a disguised case study.

1. Policy administration — the core

The policy administration system holds the contract. If it says the policy is in force at R2m sum assured and R450 a month, that is the truth and everything else is a copy. It has been called the brains of the operation for twenty years [1], and is usually the oldest and most expensive system present.

The mechanism matters more than the label. A PAS is not a database of policies but an **event processor over time**: issue, endorse, alter, renew, lapse, reinstate, cancel, mature, claim, surrender, terminate. Each event recalculates premium, re-dates cover, emits documents, triggers commission and posts transactions. Each event also creates a new version of the contract with an effective date, so the system can answer "what did this policy cover on 14 June 2023". It is the question every claim decision and every dispute turns on. A PAS that cannot reconstruct past state exactly is not fit for purpose, and some cannot. It also produces the premium register, so it starts the financial close.

Inside it: the coverage data model — policy, then section, then insured item or life assured, then benefit — event processing, effective dating, billing schedules, document and commission triggering, financial posting and bulk runs. It **masters** policy number and version, status history, risk detail, limits, deductibles, premium components, dates, beneficiary nominations, intermediary of record and the event history. What it should *not* master is customer identity, product definition or cash received: those belong to the party master, the product factory and billing. Legacy systems absorb all three anyway, which is why those three categories cannot be fixed without touching the core. It **integrates with** everything, and in most insurers it is the hub literally, because downstream systems read its tables.

Life PAS versus short-term PAS

These are different animals, and treating them as one category is the commonest mistake an architect new to insurance makes. It is that a life PAS is partly a bank ledger and a short-term PAS is not.

A life policy may run for sixty years. Over that time it can accrue a cash value, pay bonuses, charge a surrender penalty, escalate annually and turn into an annuity. Every one of those values is **contractual**: if the system says the surrender value is R184,322.17, the insurer owes R184,322.17. The system runs an account. A motor policy runs for twelve months — in South African personal lines, often monthly. It accrues nothing. It is a price attached to a risk, re-priced at renewal and pro-rated when the customer adds a car in month seven. The system runs a price.

Dimension	Life / long-term	Short-term
Term	Decades	12 months; often monthly-renewable in South African personal lines
Character	A savings vehicle as well as cover — runs an **account**	Pure risk transfer — runs a **price**
Distinctive functions	Cash and surrender value accrual with interest, policy loans, surrender charges, paid-up conversion, riders, beneficiary logic, annuitisation, bonuses, unit-linked allocation and unit pricing [2]	Multi-risk schedules for fleet and property, per-item rating, no-claim bonus, pro-rata endorsement, excess structures
Typical technology	Mainframe; VSAM, IMS and DB2 still common [3]	More often packaged Java or.NET
Replacement	Brutal — the book outlives several technology generations	Feasible — a twelve-month book runs off

The consequence decides programme strategy. A short-term insurer can put new business on a new platform, stop writing on the old one, and run the old book off in eighteen to twenty-four months. That is a real strangler-fig and it works. A life insurer cannot, because the book will still be paying claims in 2070. It has three options and no fourth: convert the data, which is hard because contractual values must reproduce to the cent; keep the legacy behind a façade indefinitely; or outsource the closed book to a third-party administrator. Each is a different architecture. Do not let a programme start before someone has chosen.

How it becomes legacy. Not through age, but through four things. Product logic gets hard-coded, so every product is a code release [2]. Batch lands on the critical path, so the business day is shaped by the overnight run. And there is no API surface, so every downstream system couples to the PAS schema, which means the schema can never change, which means the system can never improve. Add mergers that multiply instances with no decommissioning, and the loss of the three people who understood the code [3]. Oliver Wyman puts the average UK life and pensions conversion at **36 to 48 months** [3]. Plan against that number, not the vendor's.

Day one.

1. How many PAS instances, per line and legal entity, and which books sit on each?
2. Is product definition configuration or code — walk me through the last launch, with dates.
3. Can it reconstruct exact policy state at an arbitrary past date, and does claims use that?

2. Product configuration and the product factory

A product is a set of coverages, benefits, options, eligibility rules, limits, defaults, question sets and document templates, with rules saying which combinations are legal. Where they do it is the product factory, and "factory" is the aspiration: launching a product is configuration, not a software project.

Inside it: a product hierarchy with inheritance, coverage and benefit definitions, riders with eligibility rules, question sets, limit structures, versioning and effective dating, and deployment to the consuming systems. It **masters** product definitions and versions, and the mapping from a version to its rating algorithm, templates and rules.

Here is the finding to expect. **In most insurers product definition is not mastered anywhere.** It is spread across the PAS, the rating engine, the portal, the templates and the commission scheme, with no authoritative list of what the company sells. That fragmentation, not the PAS, is usually the root cause of a nine-month launch cycle. When someone says launches are slow because the core is old, count the systems that must change.

How it becomes legacy. Configuration grows expressive enough to be a programming language, without a programming language's tooling — no version control, no tests, no diff. Old versions cannot be retired because in-force policies reference them, so the catalogue only grows. And two products that should share a benefit definition do not, so one regulatory change is applied twenty times and the twentieth is applied wrongly.

Day one.

1. Where is the authoritative list of what we sell, and if there isn't one, how many places hold it?
2. How many systems change for a launch, and how long did the last new product take?
3. How many product versions are live, and how many have in-force policies?

3. Rating and pricing engines

A rating engine takes risk attributes in and returns a premium: rate tables, the algorithm combining them, factor interactions, discounts and loadings, minimum premiums and taxes. Architecturally it is one of the most consequential boundary decisions in the estate. Three patterns, in the order they arrived.

Pattern	Where the rates live	What it costs you
Embedded	Rate tables are PAS tables; the algorithm is PAS code	Simplest to run. Every rate change queues behind a PAS release, capping cadence and sophistication
Externalised, called by the PAS	A separate engine over REST; the PAS calls it and supplies mapping data	The dominant modern pattern. Needs a disciplined contract and a mapping layer someone owns
Rating as a service	One engine serving several PAS platforms, portals and B2B connections	The target state. Demands strict version control and quote reproducibility, and makes the engine a tier-one dependency [4]

WTW makes the externalised-rating argument in four testable terms [4]. **Multiple entry points:** one engine behind the portal, the aggregator feed, the call centre and the PAS, not four that drift. **Consistency:** the logic the actuaries built in analytics is deployed as that logic, not re-implemented by a developer who is afterwards told which policies were mis-rated. And **isolation:** pulling the most volatile logic out of the core keeps it stable while the core is replaced. Architects undersell that last point; it is the strongest. WTW sells Radar and Radar Live — WTW products, not Verisk's, a confusion in circulation — so treat the benefit claims as vendor-sourced.

It **masters** rate tables, algorithm versions, effective-dated deployments and the **rating audit trail**: which version, with which inputs, produced which premium. That trail is a regulatory artefact — if a customer complains to the ombud about a quote given fourteen months ago you must reproduce it exactly, and many insurers cannot. It **integrates with** the PAS, quote and aggregator front ends, the underwriting workbench, broker portals, the pricing environment and the warehouse.

How it becomes legacy. Logic drifts between engine and PAS until the two price differently and nobody can say which is right. Rate versions are not retained.

Day one.

1. Where is premium calculated per channel, and is it the same code path for quote, issue, endorsement and renewal?
2. Can you reproduce a specific quote from fourteen months ago, exactly, today?
3. Is there a second rating implementation anywhere — portal, aggregator adapter, spreadsheet?

4. Underwriting: rules engine, workbench, straight-through processing

Two things get conflated and you must separate them. The **automated underwriting rules engine** decides the straightforward risks. The **underwriting workbench** is the underwriter's cockpit for the risks the engine could not or should not decide.

The size of that second population makes underwriting interesting. Underwriting a R45-a-month funeral policy is almost nothing: an age band, a waiting period, a cover limit, done in milliseconds. Underwriting a R400 million property risk with business-interruption cover takes six weeks, a surveyor's visit, a facultative referral to a reinsurer and three rounds on deductibles. Both are called underwriting; only one can be automated, and the systems serving them share almost nothing.

Inside a workbench. Newgen describes submission intake, risk prioritisation, triage, clearance and policy generation on one platform, with a rules engine whose rules can be scoped, prioritised and **set to expire** [5]. Notice the expiry feature: the only mechanism anyone has invented that stops rule populations growing without limit, and in my experience almost never switched on.

The design point most architects miss is in the same source: the outcome is **three-way, not two-way** — a final decision without review, a referral for human review, or **no decision at all**, because the data was wrong or the case too complex [6]. Ask what happens in that third case: it is where straight-through processing leaks back into manual process, and where a reported STP rate of eighty per cent hides a referral queue nobody manages. A workbench also holds evidence ordering, authority limits, counter-offer and loading, and accumulation views.

It **masters** the decision and its reasons, loadings and exclusions, the authority used, the evidence received and referral history — commonly mastered badly, as free text in a PAS note field, which destroys later analysis of underwriting quality. I would call that the normal case, but I cannot source it as a prevalence claim.

How it becomes legacy. Rules proliferate unowned until nobody can state the underwriting appetite. The same rule exists in the engine, the PAS and the portal, in different versions. And model decisions become unexplainable at exactly the moment a regulator asks.

Day one.

1. What share of new business is decided without human touch, measured how, trending which way?
2. Where do rules live, how many are there, and who retires them?
3. When the engine cannot decide, who picks the case up, and is that queue measured?

5. New business and onboarding

From "someone wants cover" to "a policy exists". In short-term it takes minutes. In life it takes weeks, involves third parties the insurer does not control, and is a **case management problem** rather than a transaction. Build it as a long transaction and you get a system that cannot represent a case waiting eleven days for a doctor's letter.

Inside it: quote capture and illustration — for life it must match the PAS's own calculations to the cent — application capture with **reflexive questioning**, where answering yes about blood pressure opens six more questions [6], requirements chasing, sanctions screening, mandate capture, and acceptance, collection and issue.

The life medical evidence set explains where the weeks go. The SOA survey lists each source and how automated its arrival is [6]: the application; a paramedical report from a nurse examination; laboratory results and physical measurements — bloods, urine, body mass index, blood pressure, cotinine

for smoking; the **Attending Physician's Statement**, a report from the applicant's own doctor and the largest source of case delay; a full medical examination; additional requirements such as an ECG; **MIB**, the Medical Information Bureau disclosure exchange; **MVR**, the Motor Vehicle Record, used as a mortality proxy; and the prescription profile from pharmacy dispensing history.

> MIB, MVR and the US prescription databases are North American infrastructure. Do not design a South African life onboarding flow around them. South African insurers work from the application, tele-underwriting, pathology and paramedical providers, and industry data-sharing arrangements. I could not source a definitive description of the South African evidence landscape; go to ASISA or a local reinsurer's underwriting manual rather than a global vendor's reference architecture.

It **masters** the application answer set — the evidential basis of any later non-disclosure dispute — plus requirements and their status, evidence, consent and case state. For a life policy that record must survive the term plus the prescription period — decades. Ask the question in those terms and watch the retention policy fall over.

How it becomes legacy. The front end is rebuilt every three years while the application data model is never touched. Three channels run three onboarding stacks writing to one PAS with three sets of validation, so one risk is accepted on one channel and declined on another. Not-taken-up cases are untracked, so conversion leakage is invisible. And consent is captured per channel and never reaches the party master.

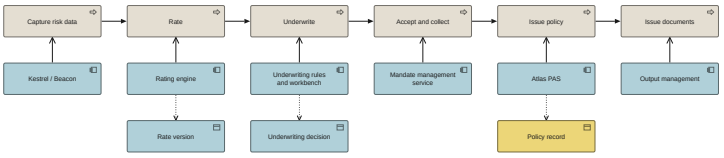


Figure 12.3 Quote to bind across systems. Each step in the chain is served by a different component, and the three objects at the bottom are the ones a regulator or an ombud will later ask you to reproduce.

Day one.

1. How many onboarding paths write to the PAS, and do they enforce the same rules?
2. Can you retrieve the application answer set for a policy sold in 2004, today?
3. What is the not-taken-up rate by channel and reason?

6. Claims management

Everything upstream is a promise; claims is the keeping of it. A claims system records a loss, decides whether it is covered, estimates the cost, pays, and recovers what it can from whoever else should bear it.

Inside it. Guidewire's published capability set is a good checklist [7]: multi-channel first notification of loss — FNOL — with claim attributes analysed *at notification*, so a severe claim is flagged in the first minutes, not the third month; coverage verification against the policy **as at date of loss**, where PAS point-in-time reconstruction stops being theoretical; **reserving**, where accuracy is the headline operational KPI; adjudication with authority limits; payment of indemnity, expenses and benefits; **litigation management**, including flagging non-litigated claims whose characteristics suggest an attorney is about to appear; **subrogation, recovery and salvage**, with early intervention to create and preserve the opportunity rather than discover it after settlement; and reinsurance recovery.

It **masters** claim number and status, date and circumstances of loss, cause and peril coding, participants, case reserve by head of damage, every payment and recovery, **reserve movement history**, and the claim file. It **integrates with** the PAS in both directions — coverage at date of loss out, claim history back for renewal rating and no-claim bonus — plus fraud scoring, payments, reinsurance, documents, the warehouse and external data.

> Reserve movement history is the actuary's primary raw material. Claims development triangles are built from it. If the dating is wrong, or movements are overwritten rather than versioned, reserving is working from fiction and the error will not surface for two years. Treat it as immutable, correctly dated, append-only data, and say so in the model.

How it becomes legacy. Cause-of-loss coding is free text, or a two-hundred-value list used inconsistently, which quietly destroys claims analytics downstream. The claim links to the *current* policy version instead of the one in force at date of loss, producing wrong coverage decisions that are hard to detect. Recoveries are tracked in spreadsheets outside the system, so leakage cannot be measured, cannot be argued about, and persists.

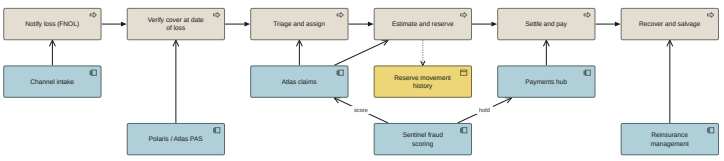


Figure 12.4 The claims chain from first notification to recovery. Note that the policy administration system is consulted for cover as at date of loss, not as at today, and that fraud scoring touches the chain twice.

Day one.

1. How is coverage at date of loss determined, and has it ever been wrong?
2. Is reserve movement history immutable and correctly dated — and do the actuaries trust it?
3. Where are subrogation, salvage and reinsurance recoveries tracked, and can you quantify leakage?

7. Billing, collections and premium accounting

Billing produces what is owed and when; collections attempts to get it; premium accounting records the result. In long-term insurance this is not a back-office function but operationally dominant: the whole model depends on a recurring monthly collection succeeding at scale, and the lapse rate is a direct function of collection success.

Inside it: schedule generation, mandate management, collection file submission, unpaid handling and retry strategy, arrears workflow, lapse and reinstatement triggers to the PAS, cash allocation and suspense, and the entries that reach the ledger. It **masters** premium due and received, mandate detail and status, collection attempt history, unallocated cash — suspense — arrears position and the receivable ledger. Suspense is a good diagnostic: ask how large it is and how old the oldest item.

The South African collections picture

This is where South Africa genuinely differs, and getting it wrong is expensive.

Plain **EFT debit order** still exists: a standard instruction, not authenticated, easy to dispute. **AEDO and NAEDO** — authenticated and non-authenticated early debit orders — were **switched off on 31 October 2021** under the Reserve Bank's second Amended Directive 1 of 2017. From 1 November 2021 those instructions could not be submitted and migration had to be complete [8]. If AEDO or NAEDO appear in a current design document, the document is stale.

DebiCheck is the replacement, and it changes the data model rather than just the file format. It is an authenticated collection: the debtor confirms the arrangement with **their own bank**, once, at the start of the contract, after which the bank permits only collections falling **within the approved terms** [9]. It exists because debit orders were increasingly processed without a mandate, and because debtors also disputed orders that did have valid mandates; the project launched in June 2017 [9]. The lifecycle has four steps: the creditor **registers** the mandate through its own bank, and it is stored at the **debtor's** bank; the debtor **approves** it electronically; the debtor's bank **validates** each subsequent collection against the agreed terms; and somebody has to **manage** it over the life of the contract [9]. Three authentication types exist.

Type	How the customer authenticates	What to expect
TT1	Real-time request pushed to the banking app, USSD menu or an ATM during the sale. Expires if unanswered	Altron FinTech reports 35–40% of attempts failing monthly — customers miss the notification or cannot reach the app in time — with a merchant fee per failure. A vendor figure from a company selling the alternative [10]
TT2	Same mechanism, longer window: roughly two days. Expires if unauthenticated	Workable where you will contact the customer again. The policy cannot start until the mandate exists, which creates a pending-business state the PAS may not have
TT3	Face to face: card into a TT3-enabled device, PIN entered, bank confirms in real time	Claimed success above 99.5%, again vendor-sourced. Named target case: field insurance and funeral cover on portable devices [10]

> **The mandate is now a first-class, bank-held, long-lived object with its own lifecycle, and something in your estate must master it.** Expect a named mandate management component between billing and the banks — Hyphen, RealPay, Netcash, Altron FinTech and Stitch occupy this space. These services consolidate differing bank formats into one and typically accept collections in the existing FACS format, so the creditor system needs few changes [9]. A mandate that cannot be amended without re-authentication is a product constraint, not an IT issue.

How it becomes legacy. Mandate data diverges from policy data after an alteration, so the escalated premium is collected at the old amount or fails silently. Unpaid reason codes are not mapped to actions, so one retry strategy is applied to everything and is therefore wrong for everything. And commission is paid on premium *due* rather than *received*, building a clawback exposure nobody has quantified.

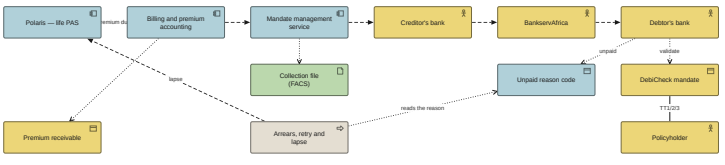


Figure 12.5 The South African collections chain under DebiCheck. The mandate is held at the debtor's bank, authenticated once by the policyholder through TT1, TT2 or TT3, and validated against every subsequent collection.

Day one.

1. What share of collections are DebiCheck rather than plain EFT, split by authentication type?
2. Who masters the mandate, and what happens to it when the premium escalates?
3. Can you reconcile premium due to received at policy level, and how old is the oldest suspense item?

8. Reinsurance management

Reinsurance is insurance bought by the insurer. The system records what was ceded to whom under which contract, settles with reinsurers and recovers their share of claims. Duck Creek scopes it as managing reinsurance partners, contracts, bills, recoveries and payables [11]. Learn the vocabulary before the first workshop; nobody will explain it.

Term	Meaning
Treaty / facultative	A standing contract covering a class automatically, versus risk-by-risk placement
Proportional	The reinsurer shares premium and loss: **quota share** is a fixed percentage, **surplus** is the cession above a retention line
Non-proportional	The reinsurer pays above an attachment point: **excess of loss** per risk or occurrence, **catastrophe**, **aggregate**
Cession / retention	Passing a share of a specific risk on, versus what the insurer keeps
Bordereau	A schedule of risks, premiums or claims reported to a reinsurer — or by a delegated-authority partner to an insurer

Inside it: treaty and facultative terms, cession calculation, auto-inuring, ceding and profit commission, reinstatement premium, commutations, and bill and bordereau production [11]. It **masters** contract terms, cession records per risk and per claim, ceded premium, the **reinsurance recoverable** — both a balance-sheet asset and a credit exposure — and the programme structure. It **integrates with** the PAS, claims, the IFRS 17 stack and the ledger.

How it becomes legacy. It never gets built. The programme lives in two spreadsheets maintained by two people, frequently the largest single concentration of knowledge risk in the insurer. Cessions are calculated in aggregate rather than at policy level, so a recovery cannot be evidenced against a treaty clause when the reinsurer queries it. Aggregate erosion is tracked by hand, and recoverable is unaged, so a slow-paying reinsurer is invisible until the auditors find it.

Day one.

1. Is cession calculated at risk level, and can you evidence one recovery against one treaty clause?
2. Where does the programme structure actually live — expect a spreadsheet, then confirm it.
3. How is reinsurance recoverable aged and credit-assessed, and who owns that?

9. Actuarial, reserving and the IFRS 17 engine

The actuarial estate values liabilities. Short-term reserving builds claims development triangles and projects ultimate losses — IBNR and IBNER. Life valuation projects per-policy cash flows on mortality, morbidity, lapse and economic assumptions. IFRS 17 converts that output into accounting entries, and the conversion is where architecture is needed.

The pattern to learn is the sub-ledger, and the Society of Actuaries gives the clearest public account of it [12]. **The IFRS 17 sub-ledger sits between the actuarial and source systems and the general ledger**, which is the whole point: administrative, actuarial and finance systems stay largely intact while a new component absorbs the new requirement. Upstream of it: policy, claims, expense and reinsurance systems, actuarial models, ETL. Downstream: the general ledger, receiving **interface files**, and IFRS 17 reporting. Three processes sit inside it [12]. **Data management** — a staging area converting source data to the granularity calculation and posting need, with typical inputs being actual and expected cash flows, **coverage units**, discount and FX rates, and the **links between direct contracts and reinsurance held**. **Calculation** — contractual service margin roll-forward, meaning interest accretion, adjustment for experience variance and assumption changes, and amortisation. **Accounting, posting and reporting** — accounting events, recovery of acquisition cash flows, loss-component allocation, posting to the GL, and the controls that reconcile the flows. A **policy-level actuarial data store** is a common extra component, because the PAS cannot serve projection workloads.

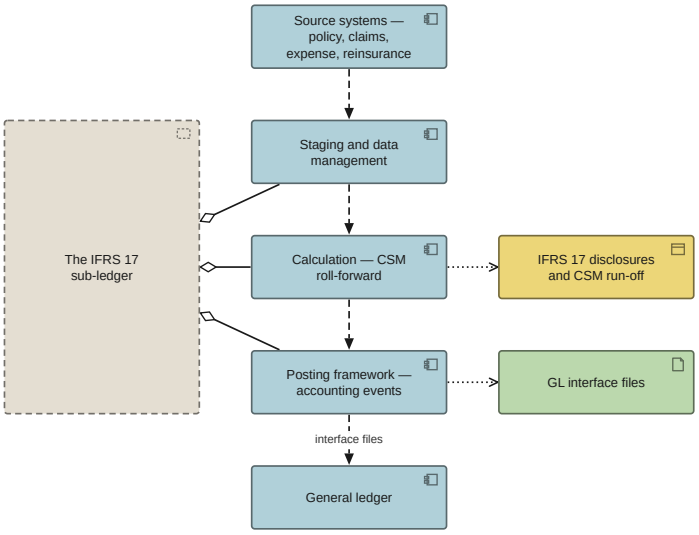


Figure 12.6 The IFRS 17 sub-ledger sits between the source and actuarial systems and the general ledger, which receives interface files — an Artifact, not a direct posting [12]. The grouping is the sub-ledger boundary: three components inside it, and everything upstream and downstream left intact.

Two design points carry forward. **Posting granularity must serve financial *and* management reporting**, and the chart of accounts must carry enough for both [12]; get this wrong and you build a permanent allocation layer. The source is also explicit about **configuration over customisation** [12] — advice routinely ignored.

It **masters** technical provisions and their components, best-estimate liability, risk adjustment, CSM, coverage units, groups of contracts and cohorts, assumption sets and versions, and the model audit trail.

How it becomes legacy. Valuation runs on extracts nobody can reproduce. The actuarial model turns out to be the only place certain product logic is right, so it diverges from the PAS and the divergence is found by accident. And sub-ledger-to-GL reconciliation stays manual and eats the close window.

Day one.

1. Where does the sub-ledger sit, and does the GL receive interface files or direct postings?
2. Is discounting applied upstream or inside the sub-ledger, and does posting granularity serve management reporting?
3. Is product logic duplicated between PAS and actuarial model, and who would notice if the two drifted?

10. Distribution: portals, commission, UMAs and binders

Insurance is overwhelmingly intermediated, so this is where insurers compete and where the architecture is thinnest. It covers intermediary onboarding and licence management, broker portals, commission calculation and payment, and — where authority is delegated — the systems a UMA runs for the insurer.

Commission is harder than it looks. One payment depends on producer hierarchy, overrides, product, premium, policy and appointment status, splits, bonus eligibility and retroactive adjustments [13]. It is a **multi-party, time-shifted, reversible** calculation: paid up a hierarchy, often advanced before earned, clawed back when the policy lapses early. That advance-and-clawback framing is mine and is standard practitioner understanding; the sources cover hierarchy, overrides and retroactive adjustment only. The named failure mode is worth memorising: **shadow accounting**, where advisers and internal teams keep a second set of records in spreadsheets to check or challenge official payouts — "a patchwork system that's not a system at all" [13]. That source is vendor-sponsored, which does not make it wrong: shadow accounting exists at almost every insurer, and finding it is the fastest route to the truth about commission data.

UMAs in South Africa. A UMA represents an insurer under a **binder agreement** and performs specialist technical functions — underwriting, claims management, policy administration — for a class of business the insurer will not staff internally. It needs a **Category I FSP licence under FAIS** and approval as a **binder holder under the Insurance Act**. The binder agreement sets authority and compliance obligations and covers remuneration, indemnity, **data ownership** and reporting; because the insurer stays responsible for the UMA's decisions, oversight is continuous, and the named pitfalls are exceeding binder authority, delayed reporting and conflicts of interest where the UMA is linked to a brokerage [14]. Its system obligations are specific: data accurate and auditable, **records retained for at least five years**, reliable reports for insurers and regulators, and POPIA compliance [14]. It **masters** intermediary identity, licence and appointment status, hierarchy and splits, commission schemes, commission earned and paid, and clawback positions.

> **The pattern to recognise: a UMA runs its own policy administration system.** The insurer's "policy data" for that book therefore arrives as a **bordereau** — a periodic schedule — not as live records. Everything needing policy-level, near-real-time data breaks at that boundary: catastrophe aggregation, IFRS 17 granularity, claims analytics, single view of customer, conduct monitoring. I would argue this is the most under-appreciated integration problem in the South African short-term market. That is an argued position, not a sourced finding — deliberately so, because no credible public figure exists for how much premium sits behind delegated authority. Describe the model; do not quote a share.

Two regulatory changes are coming and both bite architecturally [14]. The **COFI Bill** will replace and consolidate existing legislation including the FAIS Act and the Long-Term and Short-Term Insurance Acts, in phases, moving to a single principles-based market-conduct regime; the date is unconfirmed. The **Omni-Risk Return** will expand data and reporting requirements across the value chain, which means aggregating UMA data to a standard the insurer does not control, from systems it does not own.

How it becomes legacy. Intermediary identity is mastered in the commission system rather than the party master, so one brokerage exists five times, with five spellings. Licence status is not checked at bind, which is live conduct exposure. Bordereau ingestion is manual with no reconciliation to premium received. And a portal is built per channel, each with its own copy of the quote logic.

Day one.

1. Who is system of record for intermediary identity and licence status, and is licence checked at bind?
2. How much of the book sits behind delegated authority, and at what granularity and latency?
3. Where is the commission shadow accounting — it exists, so find it.

11. CRM, party master and the single view of customer

The **party master** answers "who is this, and is this the same person as that one?". The **CRM** manages interactions with them. A CRM licence does not buy you a party master.

Inside a party master: a party model for individuals and organisations; identity resolution with matching, merge, unmerge and survivorship rules; relationship and consent modelling; and **role modelling**, the piece most often built too thin, because one person is policyholder on one contract, life assured on another and claimant on a third. It **masters** party identity, identifiers, contact details, relationships, roles and consent, and **integrates with** every system that names a person.

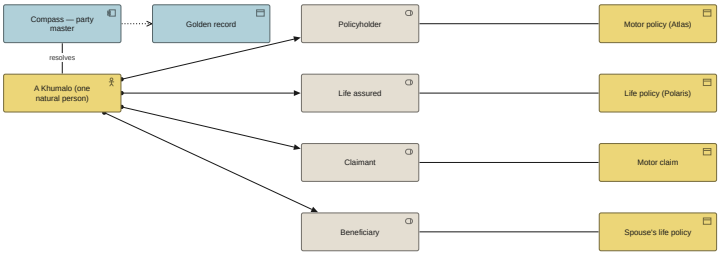


Figure 12.7 One natural person, four roles, four contracts. A party master that cannot model this without duplicating the person will never deliver a single view of customer.

The decisive architectural choice is whether it is **operational** — systems call it at transaction time and it is authoritative — or **analytical**, reconciling identities after the fact for reporting. The published case-study material describes the operational pattern: a central operational customer master across all lines [15]. My observation is that many build the analytical one, call it a single view of customer, then find it does not solve the servicing problem because the call-centre agent is still looking at the PAS. I cannot source that as a prevalence claim, but it is the first thing to check.

How it becomes legacy. The master is populated but no operational system is changed to call it, so it becomes a reporting artefact with a governance board. Matching over-merges, turning two people into one with a privacy incident attached, or under-merges, so the single view never arrives. The identity number is used as a primary key and turns out to be neither unique nor always present. And consent captured in the CRM is not honoured on the outbound path.

Day one.

1. Operational or analytical, and which systems actually call it at transaction time?
2. Can one party hold multiple roles across contracts without duplication?
3. Can you state total exposure to a single party across all lines today, without a project?

12. Document generation and output management

Insurance is documentary. The schedule and the wording *are* the contract, and much regulation concerns what must be disclosed, when and in what form. The category that produces and keeps them — customer communications management — is treated as a printing utility. It is a compliance system.

Inside it: template authoring a business user can use; data merge from PAS, claims and billing; assembly, on-demand and batch generation, and routing to print, email, SMS, portal and WhatsApp; and archive and retrieval of the **exact rendered artefact**. It **masters** templates and versions, the **correspondence record** — what was sent, to whom, when, by what channel, in response to which event — and the archive. That record is the evidential backbone of every conduct complaint and ombud matter, which is why it **integrates with** the party master for language and channel preference as much as with the PAS.

> The trap to test for: does the archive store the rendered document, or the data plus a *reference* to the template? If it is the second, a template change silently rewrites history, and the document you produce in 2027 for a 2019 complaint is not the document the customer received. This is common, almost never on a risk register, and a bad week when it surfaces.

How it becomes legacy. Templates proliferate — one per product per variant per channel per language — with no inventory, so one wording change is applied hundreds of times. The batch print run sits on the overnight critical path. And digital preference is captured but print is still produced, so the insurer pays twice.

Day one.

1. How many templates exist, and can a business user change one without a release?
2. Can you produce the exact document a customer received in 2014?
3. Who honours channel and language preference, and is it one place or several?

13. Fraud detection and claims analytics

Scores claims — and increasingly applications — for fraud likelihood, routes the suspicious to investigators, and runs the investigation. The techniques are red-flag rules, predictive and machine-learning scoring, **link and network analysis** to find organised rings across apparently unrelated claims, and industry exchanges such as ISO ClaimSearch in the US [16][17].

Inside it: rule-based red flags; predictive scoring at FNOL, pre-payment and pre-renewal; entity resolution across claims, parties, addresses, bank accounts, vehicles and service providers; and network analysis. It **masters** scores and their contributing factors, cases, networks and entity links, watchlists — and, crucially, **outcome labels**: confirmed fraud, suspicion unproven, claim declined. Those labels are the training data for every future model; an insurer that does not capture them has capped its fraud detection without knowing. It **integrates with** claims in both directions, the party master, on which entity resolution depends, and payments, where the pre-payment hold is the highest-value control point.

How it becomes legacy. Scoring happens and nothing operational changes, so hits are ignored. The score arrives after the payment has left. Outcomes are never fed back, so models decay. Link analysis depends on party data quality the party master never delivered. Rules accrete and false positives rise until investigators stop trusting the queue, at which point the system is worse than nothing because it provides false assurance. And explainability is insufficient to defend a repudiation to a regulator or an ombud.

Day one.

1. At what points is scoring applied, and does a score ever actually stop a payment?
2. Are investigation outcomes captured as structured labels and fed back into the model?
3. What is the false-positive rate, and do the investigators trust the queue?

14. Warehouse, lakehouse and regulatory reporting

Conformed, historised data for management reporting, actuarial analysis, pricing, claims analytics — and the one with a hard deadline, regulatory returns. That deadline is why the warehouse is structured around regulatory line-of-business taxonomies rather than how the business thinks of itself.

The lakehouse argument, as Databricks frames it: traditional architecture requires copies into separate stores and handles streaming and unstructured input poorly, while a lakehouse gives one source of truth, faster integration of sources such as weather, traffic and wearables, and machine learning directly on lake data [18]. Treat the cost and loss-ratio claims as vendor-sourced, and notice what the argument omits entirely: **regulatory reporting** [18]. That omission is the finding, because regulatory reporting is the workload the warehouse exists to serve.

South African reporting. Insurers report to the **Prudential Authority**, inside the Reserve Bank, and to the **FSCA** for market conduct. The prudential instrument is the **Quantitative Reporting Template**, issued and amended by prudential notice, including a solo template [19]; the **Omni-Risk Return** will expand requirements materially [14]. The consequence: QRT reporting needs granularity the operational systems were never designed to produce, especially for business written under delegated authority, and reconciliation between the QRT, the IFRS 17 output and the statutory accounts becomes a standing control activity rather than a project. That characterisation is my reading; the sourcing covers the existence and amendment of the templates, not their granularity.

It **masters** nothing operational — it masters **history**: policy and claim state at each reporting date, and the reconciled version of facts whose sources have been switched off. For an insurer that has retired a PAS, the warehouse is often the only remaining record of that book, which quietly makes it tier-one.

How it becomes legacy. Two warehouses, old and new, both load-bearing, forever. The regulatory return comes from a parallel manual process because the warehouse is not trusted. And finance, actuarial and operations hold three definitions of "policy count" with nothing to arbitrate — the problem ACORD's Business Glossary exists to solve [22].

Day one.

1. Which returns come from the warehouse, and which from a parallel manual process?
2. Is lineage documented from return line back to source field?
3. What history exists only here, because its source system is gone?

15. Payments and disbursements

Money in — premium, covered under collections — and money out: claim settlements, refunds, commission, reinsurance premium, suppliers. Outbound payment is a controls problem more than an integration one: it is where operational error and internal fraud turn into cash loss. **Inside it:** payee and bank-detail management with account-holder verification, instruction approval with segregation of duties, per-bank file generation, real-time initiation, reconciliation, and sanctions screening at payment.

South African rails. EFT credit is batch — same-day or dated — and the historic default for claim payments. Debit order and DebiCheck handle collections. **PayShap** is the real-time option, launched in **March 2023** as part of the Reserve Bank's Rapid Payments Programme under Vision 2025, built through BankservAfrica, PASA and the banks, and processed by BankservAfrica. It is a **push-payment** model; the payee is identified by bank details or

by a **ShapID** proxy such as a mobile number; and it is **ISO 20022** compliant. At launch, real-time clearance applied to low-value transactions **up to R3,000** [20].

> That R3,000 figure is the launch-era limit and I could not verify whether it has since been raised — check current BankservAfrica and PayShap scheme rules before designing against it. The limit matters architecturally, not just commercially: funeral and small personal-lines claims fall inside it and could settle in seconds, while motor and property claims still need a batch rail. That splits your payment architecture into two paths whether you planned it or not.

It **masters** payee bank details and their verification status, instructions and their approval trail, payment status and returns. **A change of bank details is one of the highest-risk events in an insurer** — the mechanism behind most settlement-diversion fraud — and it belongs in the architecture, not buried in a maintenance screen.

How it becomes legacy. Each source system holds its own bank integration and format, multiplying the controls surface. Bank details are mastered per policy rather than per party, so one change must be made eleven times. Approval is enforced in the user interface rather than the service, so a direct integration bypasses it. And payments-made to payments-instructed reconciliation is manual.

Day one.

1. How many distinct paths run from a system to a bank, and is approval enforced in the service or the UI?
2. Where are bank details mastered, and what is the controlled change process?
3. Can every payment be traced to an authorised instruction and a business event?

16. The general ledger

The general ledger does not know what a policy is, and the vendor market is mature. Architects skip it, and that is a mistake, because three things make it matter more than its place in the inventory suggests.

First, **the GL interface is the insurer's most consequential integration**. Under IFRS 17 the sub-ledger posts to the GL through interface files, and the chart of accounts must carry enough to serve financial *and* management reporting [12]. A chart designed before IFRS 17 forces an allocation layer, which becomes a permanent reconciliation cost with its own team. The GL is boring; the posting-granularity decision is not, and it is usually taken by someone who was not told it was architectural.

Second, **expense data is actuarial input**. Expense cash flows are an explicit input to the IFRS 17 sub-ledger [12], so a cost-centre structure designed for finance's convenience can make per-product expense attribution impossible — and that attribution is what tells the business which products make money.

Third, **tied-agent identity overlaps HR**. Tied agents are employees, so the boundary between HR, the intermediary register and the commission engine is real, and it is botched in both directions: agents in HR but not in the register, and agents still earning commission after they have left.

Day one.

1. Does the chart of accounts support IFRS 17 posting granularity and management reporting?
2. How does expense allocation reach the actuarial and IFRS 17 processes?
3. How many systems post to the GL, and how is that interface controlled?

Integration patterns that actually show up

Everything above is a box. What follows is the wiring, and the wiring is where insurance differs most from what you are used to. Except where cited, this closing section is practitioner consensus rather than sourced fact, and I have marked it so because you will be asked.

The overnight batch, and why it still exists

The batch cycle is the defining operational rhythm of an insurer. Overnight the PAS processes what it cannot do interactively: renewal invitations, escalation and indexation, anniversary processing, lapse identification, value accrual for life, collection file generation, document batch production, and the interfaces to the ledger and the warehouse, from which downstream systems then load.

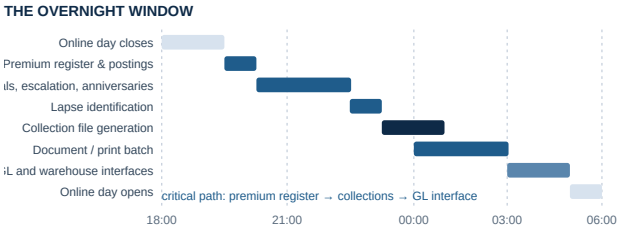


Figure 12.8 A representative overnight batch window. The shape is typical rather than measured; what matters is that the business day is bounded by it and the window only ever shrinks.

Four consequences to design around. **Data is a day old everywhere except the PAS**, so any "real-time" experience built on warehouse or replica data is lying by up to twenty-four hours, and the customer will find the discrepancy before you do. **The batch window only ever shrinks**, because the business wants longer online hours while the book grows. **Dependency chains are brittle and usually undocumented**, so a failure at step 40 of 120 forces a restart decision at three in the morning, taken by whoever is on call and may not know what step 41 does. And **month-end and year-end runs** exercise code

paths used twelve times or once a year: the paths with the fewest tests and the least living knowledge. Ask what is in the batch, what the critical path and window are, and what the restart procedure is.

System of record versus system of engagement

The legacy PAS remains the **system of record**. New digital capability is built in front of it as a **system of engagement** — portal, app, aggregator integration, service layer — owning experience, orchestration and often a read-optimised projection of policy data. It works, and the reason is honest: it defers a core replacement nobody can afford. It fails in three characteristic ways. The engagement layer accumulates business logic belonging in the core, creating a second, undeclared system of record nobody governs. The projection drifts, because no one owns reconciliation against the PAS. **Write-back is the hard part** — reads project easily, but a self-service change must eventually land in the PAS, often through overnight batch or a manual queue, so the customer sees a change the record does not yet reflect. Two sources of truth then become visible in one interaction: the app says R612, the letter says R648. Ask what logic has accumulated in that layer, how writes reach the record and with what latency, and whether the layer is on a path to becoming the record or permanently subordinate.

What replaced the service bus

From roughly 2005 to 2015, insurance integration meant service-oriented architecture with an **enterprise service bus**: a central product mediating between systems, with canonical message formats and orchestration in the middle. Insurers invested more heavily than most industries, because the bus appeared to solve the N-to-N problem across a sprawling estate and the canonical model mapped neatly onto ACORD XML, which already existed [22]. Its failure mode was centralisation, not technology: one team, one release schedule, one point of contention, and orchestration logic accumulating in middleware with no business owner.

What replaced it is a layering, not a product swap.

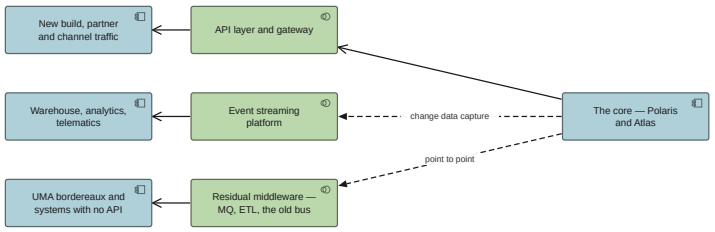


Figure 12.9 What replaced the enterprise service bus: three pieces of system software and one relationship. Point integration is drawn as a Flow rather than a box, because that is what it is — a direct dependency with no component in the middle. Every real insurer has all four, and the bottom two carry more traffic than anyone admits.

An **API layer or gateway** handles synchronous exposure, security, throttling and versioning. **Event streaming** carries asynchronous state-change propagation, decoupling and replay. **Point integrations and change data capture** serve systems that will never have an API. And **residual middleware** — message queues, ETL, the old bus — persists, because switching it off is a programme nobody will fund. Three of those four are things you can point at and name an owner for. The third is not: a point integration is a dependency between two systems with nothing in the middle, which is why the figure draws it as a line rather than a box, and why nobody is funded to look after it. Kai Waehner is explicit that the practical starting point is **brownfield**, that existing MQ, ETL, ESB and API tooling remains in place, that streaming is complementary to it rather than a replacement, and that proprietary interfaces — COBOL, EDI, SAP BAPI and iDoc — are the real obstacles [21].

The streaming use cases published for insurance are event-driven claims processing, telematics feeding real-time price adjustment, and **legacy integration and offloading** [21] — and that last one is the pattern you will actually use. Change data capture pushes changes out of the core into a stream, and consumers are served from the stream rather than the core. Note the gap — **mainframe CDC against IMS, VSAM or DB2 for z/OS is materially harder and more expensive**. That is the practitioner view and not something I can source, but do not let a plan assume the two are equivalent.

The mainframe reality in life insurance

What is sourced: Oliver Wyman describes UK life and pensions policy administration as aging, on legacy infrastructure that blocks better propositions and cost savings, with a shortage of legacy experts and migrations surfacing critical problems late. It names **VSAM, IMS and DB2** as extraction sources, all mainframe technologies, and puts average conversion at **36 to 48 months** [3].

Four further claims are **practitioner consensus rather than sourced fact**, and should be presented that way in your own documents: that a large life insurer typically runs between five and twenty policy administration instances; that systems written in the 1970s and 1980s remain in production; that COBOL remains the dominant language in life administration; and that closed blocks are commonly outsourced to third-party administrators on their own platforms. None is in the sources above — Oliver Wyman never names COBOL or the mainframe at all [3]. Say "this is the practitioner view" and you are defensible; say "research shows" and someone will ask which research.

Why it persists is not inertia. A life policy can be in force for sixty years, so the system must outlive any technology generation — the replacement problem recurs, it does not resolve. Policy values are contractual and must reproduce to the cent, which makes conversion a *correctness* problem rather than a data-movement one, and correctness problems do not respond to more contractors. And modernising a closed book is uneconomic: there is no future revenue to pay for it.

Ask which books are open and which closed; for each closed book, whether the strategy is run-off, conversion or outsourcing; and whether the legacy policy value calculations can be independently reproduced. The last question decides whether a conversion is a project or a fantasy.

THE CHAPTER IN TEN LINES

- An insurer's estate has four parts — the core, the surround, the channel layer and the finance and risk stack — and the boundaries between them are the architecture.
- The unit of record is the policy, not the customer; every single view of customer is a retrofit of a decision taken fifty years ago.
- A life policy administration system is partly a bank ledger and a short-term one is not, which is why a short-term insurer can run a book off in two years and a life insurer cannot.
- Point-in-time reconstruction of policy state is not a reporting feature; claims decisions and disputes depend on it.
- In most insurers product definition is mastered nowhere, and that fragmentation — not the age of the core — is why launches take nine months.
- Under DebiCheck the mandate is a bank-held, long-lived object with its own lifecycle, and something in the estate must master it; AEDO and NAEDO were switched off on 31 October 2021.
- The IFRS 17 sub-ledger sits between the actuarial and source systems and the general ledger, and posting granularity has to serve both financial and management reporting.
- A UMA runs its own policy administration system, so that book arrives as a bordereau, and everything needing policy-level data breaks at that boundary.
- The overnight batch means data is a day old everywhere except the core, and the batch window only ever shrinks.
- Nothing replaced the service bus as a product; an API layer, event streaming, point integrations with change data capture, and residual middleware coexist in every real insurer.

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- **19.** SARB Prudential Authority, "Prudential Communication 6 of 2025: Replacement of Quantitative Reporting Template" https://www.resbank.co.za/content/dam/sarb/publications/prudential-authority/pa-public-awareness/covid-19-response/2025/Prudential%20Communication%206%20of%202025__%20Replacement%20of%20Quantitative%20Reporting%20Template.pdf
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CHAPTER 13

The reference architecture

Insurance has a published answer to "what are the words" and "what is the list of things we do". It is called ACORD, it is mostly not free, and almost everyone misuses it as a target architecture.

Three teams, three numbers

In her second week as lead enterprise architect at Cape Mutual, Fatima Patel asked a simple question: how many policies does the group have? Finance said 2.9 million. Actuarial said 3.4 million. Distribution said 4.1 million.

Nobody was lying. Finance counted contracts that had produced a premium in the last quarter. Actuarial counted contracts carrying a reserve, which includes paid-up life policies producing no premium at all. Distribution counted *lives covered*, because a funeral policy with a spouse, four children and two parents is eight lives and eight sets of commission-bearing cover on one contract. Four weeks of meetings produced a reconciliation spreadsheet that nobody trusted and that broke the following quarter.

Cape Mutual is a fiction carried through this book, and nothing about it is a real case study. The argument is not.

This is the problem a reference architecture solves. Not the big problem — not *what should our architecture be*. The small, grinding, expensive one: what are the words, and what is the list of things we do. An insurer that cannot answer those two questions consistently cannot be modelled, measured or modernised, and will spend the first eight months of every programme re-deriving them.

Insurance has a published answer to both. It is called ACORD. It is good, it is large, it is mostly not free, and it is routinely misused. This chapter is about what it actually contains, what it cannot do, and how to use it without starting a two-year mapping exercise that produces nothing.

What a reference architecture is for, and what it is not

A reference architecture is a published, industry-level description of a sector's concepts, capabilities, processes and data, produced so that individual firms do not have to invent them. In insurance it means: here is what a policy is, here is the list of things an insurer does, here is how those things relate, here are the messages you exchange with counterparties.

There are three honest uses.

A shared vocabulary. The most valuable and the least glamorous. When finance, actuarial and operations each hold a private definition of "policy count", "lapse" or "earned premium", every number in the organisation is contestable, and the argument recurs at every month-end. An external, published definition settles it — not because it is better than yours, but because it is *nobody's*. Definitional disputes are political; an industry glossary de-politicises them.

A coverage check. Given a complete published list of what insurers do, you can ask of your own estate: which of these can we actually do, where, by whom, and how well. The value is in the *absences*. Nobody writes down the capability they have never had. A published list makes the hole visible.

Boundary language. When you write a request for proposal, integrate with a reinsurer, or negotiate a binder with a UMA, a shared term for the thing being exchanged saves weeks. In some markets it is not optional: Lloyd's uses ACORD standards for counterparty messaging and the Lloyd's Core Data Record is based on them [5].

Three things a reference architecture is not.

It is not a target architecture. It has no opinion about your strategy, your cost base, your distribution mix or your sequencing. It cannot have one — a model that told every insurer what to build would be wrong for all of them. The reference model describes the *problem space*; your target architecture is a set of choices within it. Confusing the two is the single most common failure, and it is expensive because it looks like progress.

It is not an implementation. An industry logical data model is exhaustive by design. Implemented literally it yields hundreds of near-empty tables and performance nobody will defend.

It is not a substitute for knowing your own estate. The model will not tell you that Polaris cannot represent a second life assured without an operational workaround. Only reading Polaris will tell you that.

If you work in TOGAF terms, the fit is straightforward and worth saying out loud, because it stops the "where does this go" argument. An industry reference architecture is an input to the Architecture Repository — specifically to the reference library — and it is consumed in Phase B when you need a business capability baseline and a glossary, and in Phase C when you need a conceptual information model to check your own against. It is not an Architecture Vision, it is not a Target Architecture, and it produces no Architecture Definition Document. The deliverables remain yours. What the reference model removes is the blank page, and the three months of committee work that a blank page generates.

> A reference architecture gives you the vocabulary and the list. Your strategy, your sequence and your choices are yours. If a programme's deliverable is "ACORD adoption", the programme has no deliverable.

ACORD as an organisation

ACORD was founded in 1970 as a non-profit by carriers and agents — the name was originally Agent Company Operations Research and Development. It began by standardising the paper: the proprietary new-business and claims forms each carrier had invented for itself. In the late 1970s it added electronic data standards alongside the forms, later extending both to life and annuity, surety and reinsurance. It has published over 850 form variants

and maintains a library of more than 1,200 electronic transaction types. It is headquartered in Little Falls, New Jersey, with a London office, and its subsidiary ACORD Solutions Group runs the ADEPT exchange platform [5].

That history explains the shape of the organisation and the two things an architect will find confusing about it. ACORD's centre of gravity is *messaging between trading partners*, because that is what it was built to do. The modelling assets — the part an enterprise architect actually wants — are a later, smaller limb grafted onto a messaging body. They are published together, described in similar language, and sold through the same marketplace, which is why people conflate them constantly.

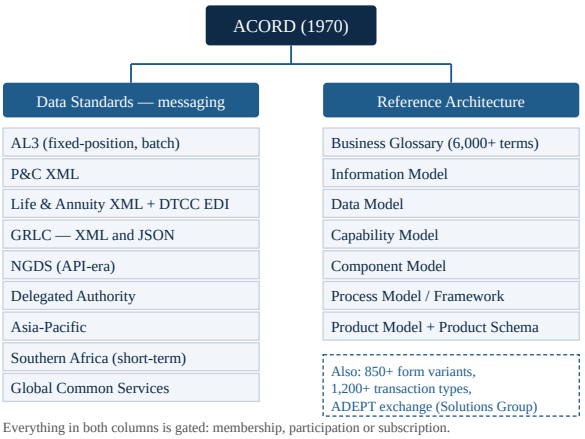


Figure 13.1 What ACORD publishes. Two families, both gated. The messaging standards on the left are the organisation's original and larger business; the Reference Architecture on the right is what an enterprise architect wants.

Two families, routinely confused

Family	What it is	Who needs it
Data Standards (messaging)	Wire formats and message sets for exchanging policy, claims, premium and placing data with trading partners. AL3, ACORD XML, GRLC, NGDS, Delegated Authority, regional families	Integration architects, anyone with a counterparty that mandates it
Reference Architecture (modelling)	Seven interrelated models describing insurance concepts, data, capabilities, components, processes and products	Enterprise and data architects, programme and procurement teams

Both are gated. Neither is a substitute for the other. The messaging standards tell you what to put on the wire; the Reference Architecture tells you what the things on the wire *mean*, and the two are explicitly linked — which is the most interesting property ACORD has, and the one most often wasted.

AL3, and why insurance integration looks the way it does

Start with the oldest surviving messaging standard, because it teaches more than the modern ones.

AL3 is still the workhorse of US agency-to-carrier "download" — the nightly transfer of policy and commission data from a carrier into an agency management system. It is positional and fixed-length. Elements belong to groups identified by codes such as `6POL` for policy, `6VEH` for vehicle, `6BSI` for basic policy information. Every transmission opens with a Transmittal Header Group and a Message Header Group carrying routing and synchronisation data. Groups nest sub-groups for repeating items — schedule items, additional interests. Each element has a starting byte offset and a fixed length within its group. There are no tags and no delimiters: **byte position is the only identifier**. Short values are padded, with trailing spaces for alphanumeric fields and leading zeros for numeric ones. A field written one character too long shifts and corrupts every element after it [6].

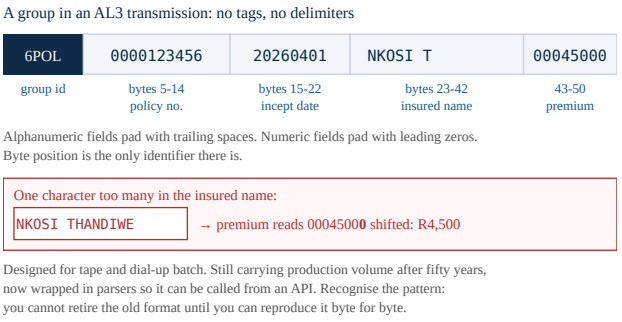


Figure 13.2 AL3 is positional and fixed-length. The field values here are invented. The failure mode is not: a value one byte too long shifts every element after it, and the receiving system has no way to detect that it happened.

An architect may never touch AL3. It is worth ten minutes anyway, for three reasons.

It is the clearest available evidence of *why* insurance integration is as it is. This is a format designed for magnetic tape and dial-up batch transfer, still carrying production volume more than fifty years later, now wrapped in parsers so it can pretend to be an API. That pattern — a 1970s artefact with a 2020s façade — recurs throughout this industry, and recognising it will save you from proposing the obvious thing and being told, correctly, that it cannot be done.

It explains why the industry's canonical models are shaped as they are. A model whose first job was to describe fixed-position records will carry that influence in its granularity.

And it sets the realistic test for any "modernisation" of an interface: can you reproduce the old format, byte for byte, from the new one? Until you can, you cannot switch off the old one, and the old one is where the counterparties are.

The rest of the messaging families

Family	Formats	Notes
Property & Casualty	XML and AL3	Also relevant to Delegated Authority and Workers' Compensation
Life & Annuity	XML plus DTCC EDI specification files	The US life distribution spine
Global Reinsurance & Large Commercial (GRLC)	XML **and JSON**	Claims, Placing, Accounting & Settlement. The most modern line-of-business family
NGDS	Object resource definitions, schemas, templates	Fine-grained transactions for APIs, microservices and IoT; aligned to the Reference Architecture
Multi-Functional	XML	Employee Benefits, Producer Licensing, Delegated Authority
Asia-Pacific	XML	Asia-Pacific General Insurance; Australia/NZ Life
Southern Africa	XML	Short-Term Insurance and Exposure Reporting
Global Common Services	Rules and guides	Naming & Design, DRI, AMS SOAP — cross-standard conventions

Source for the family list and formats: ACORD's own data standards page [3].

Two naming corrections, because both appear in circulation and both will make you look careless. The reinsurance family is **GRLC**, not "RLC". And JSON appears in GRLC, not in the general-insurance families — if someone tells you ACORD is "XML only", they are a release or two behind.

NGDS deserves a specific warning. It is ACORD's API-era answer: a lightweight, technology-agnostic standard for fine-grained transactions, with an Object Model linking data exchange to business processes, and online help files available to members with NGDS access [7]. A July 2026 ACORD note describes it as supporting RESTful APIs, IoT, microservices and mobile, and announces a "Getting Started with NGDS" fact sheet readable by non-members [8]. What neither page states is whether NGDS is specified as JSON Schema or as OpenAPI, which lines of business it covers, or what the release timeline is — the material references current and prior releases without naming them [7]. If you are evaluating NGDS, you cannot answer basic technical questions from public material. That is itself a finding, and worth writing into your assessment rather than glossing over.

ACORD Southern Africa

There is a Southern Africa family. The **Southern Africa Short-Term Insurance Standards** are the main standard and the only line of business named. A release comprises release notes, a help file, schema files for XML validation, a business process guide covering the policy life cycle, and XML sample files. A separate **Exposure Reporting Standards** is named without further detail. Standards training materials are free to download for ACORD members [4].

The history is thin but real: a South African ACORD standards pilot and the subsequent availability of a South African ACORD XML standard were reported locally [12], and ACORD published a Southern Africa release in April 2015 [13].

> What I could not establish: the current release version, the specific message set, and — the question an architect actually cares about — **how widely South African insurers, brokers and UMAs have adopted it**. No primary source gives an adoption level. Treat any figure you are quoted as hearsay until someone shows you a schema in production. If this matters to your programme, ask ACORD or SAIA directly, and ask your largest three broker counterparties what they can send you today.

This matters because the Southern African integration problem is the bordereau — dozens of UMA and binder holders submitting policy and claims data in dozens of formats. A regional XML standard with a policy-lifecycle process guide is precisely the right shape of answer. Whether it is the answer in practice is an empirical question about your counterparties, not a question about the standard.

The Reference Architecture: seven facets

ACORD's own Reference Architecture page names seven components and gives no version numbers [1].

Facet	What it contains
Business Glossary	Standardised, non-technical definitions of insurance concepts, with synonyms, line-specific usage and cross-references. Over 6,000 terms as at January 2026 [2]
Information Model	A formal conceptual representation of data elements across all lines of business, covering Policy, Product, Party and Claims, linked to all ACORD Standards
Data Model	A logical entity-relationship structure **generated from** the Information Model, usable for any database implementation, with an implementation guide and a fictional carrier use case
Capability Model	A baseline of activities across the insurance value chain, as a functional decomposition **without predefined workflow**; supports process engineering and merger or divestiture evaluation
Component Model	The information and properties for the business activities that compose insurance processes; **bridges** the Capability and Information Models, technology-neutral
Process Model	Workflow-oriented reference processes aligned to the Capability Model, with **messages mapped to the ACORD Data Standards**
Product Model	A graphical notation for representing insurance products, plus the ACORD Product Schema, an XML representation for interchange

> NAMING
>
> ACORD's current overview page calls the sixth facet the **Process Model** [1]. The ACORD marketplace lists a product page titled "Process Framework (ACORD Reference Architecture)" and a "Process Model V2.3" [9]. Both names are in circulation and refer to the same facet. Write it as "the Process Model, also published as the Process Framework" and check the marketplace product page for the live version before you cite one.

Versions are at product level, not overview level. The overview page omits them; the marketplace carries them. On 12 December 2025 ACORD released Data Model v2.13, Information Model v2.13 and Business Glossary v2.16, adding refinements to the Activity domain, a Marketing Activity subtype, a new Lead Management sub-package and a Sales Appetite component [11]. The marketplace separately lists Component Model V2.4 [10] and Process Model V2.3 [9]. If you need to pin a version in a contract or an architecture decision record, go to the marketplace page for that facet; do not quote the overview.

Now the property that makes this more than seven unrelated documents. The Data Model is *generated from* the Information Model. The Component Model *bridges* the Capability Model and the Information Model. The Process Model is *aligned to* the Capability Model and maps its messages to the Data Standards [1]. Put end to end, that is a traceability chain:

capability → component → process → message → schema

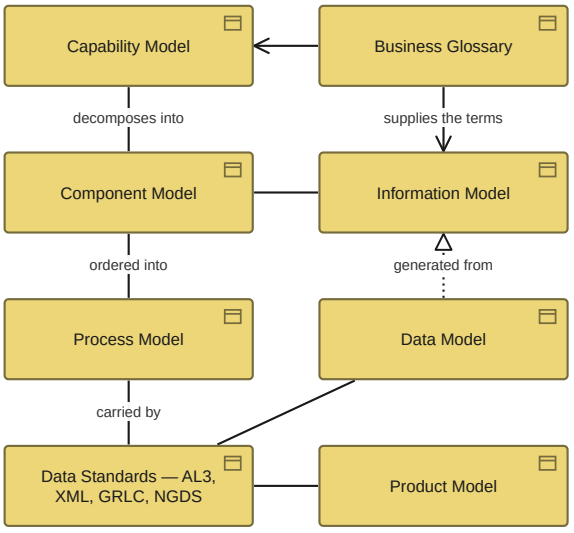


Figure 13.3 The seven facets and the chain that connects them, drawn as business objects because that is what they are: published information products, not systems. Read the left column top to bottom — capability, component, process, message — and that is the traceability from a capability to the schema that validates the message realising it. The glossary serves every other facet; the Data Model realises the Information Model, which is why it can be generated from it.

That chain is the real value proposition, and it is the thing almost every team misses by licensing one facet in isolation. A capability model alone is a poster. A data model alone is an argument with your DBA. The chain lets you start from "we need to be able to handle a third-party recovery", arrive at the information that capability needs, the process that performs it, the message that carries it and the schema that validates it — and, crucially, trace back the other way when someone asks why a field exists.

In practice you will not use all seven, and you should decide which you need before you talk to procurement.

If the question is…	The facet is…	And the test of success is…
What do these words mean, and who disagrees?	Business Glossary	Month-end stops producing two versions of the same number
What do we do, and what are we missing?	Capability Model	You can name an absence nobody had written down
What entities must our model contain?	Information Model	Your own conceptual model gains risk object, loss event, cession and policy version
What should the canonical integration layer look like?	Data Model, subset hard	A UMA bordereau and a core extract map to the same structure
What information does this activity need?	Component Model	A build-or-buy decision stops being a feature-list comparison
What does the end-to-end flow look like, and what message carries each step?	Process Model	An interface contract is derived, not invented
How do we express a product without writing code?	Product Model	A product definition survives a change of policy administration system

The sequencing advice follows from that table. Start with the Glossary and the Capability Model, because they are cheap to use, politically useful and immediately legible to the business. Take the Information Model next, as a checklist against your own conceptual model rather than as a replacement for it. Take the Data Model only when you have a concrete canonical layer or warehouse to build, and subset it aggressively, documenting every exclusion — an undocumented exclusion becomes an argument in two years. Take the Component and Process Models when you are about to make interface or sourcing decisions and want traceability you can defend. Take the Product Model when you are specifying a product factory.

January 2026: the models go interactive

On 29 January 2026 ACORD released enhanced online, interactive versions of two facets: the Business Glossary, with over 6,000 terms, and the Capability Model, covering core value-chain activities such as underwriting and policy administration — **with tools to measure current capabilities**. Access is members only, on the ACORD Standards Online Applications site. No public access is mentioned and no versions are given [2].

The measurement tooling is the signal worth reading. ACORD is repositioning the Capability Model from a taxonomy into a self-assessment instrument. That is useful and also a hazard: a scored capability assessment becomes a political artefact the moment it has numbers on it, and the numbers will be used in budget arguments by people who did not read the definitions. More on that below.

What is free, and what is not

Be blunt about this, because programme plans die here.

Asset	Access
Reference Architecture, whole or individual facets	**Subscription.** Members and non-members may subscribe; some organisations get access through membership terms. No component is described as free [1]
Interactive Business Glossary and Capability Model (Jan 2026)	**Members only** [2]
Data Standards, all families	**Members and participants only** — join the relevant programme, then sign in [3]
NGDS online help files	Members **with NGDS access** [7]
Standards training materials	**Free to ACORD members** [4][7] — free *within* membership, not publicly free
"Getting Started with NGDS" fact sheet	Readable by non-members [8]

The honest summary: almost nothing substantive is free. Marketing descriptions, news items and one fact sheet are public. Everything you would actually build from requires membership or a subscription, and pricing is not published, so it is a procurement exercise with a lead time.

> A programme planned on the assumption that the ACORD models can be downloaded will fail, and it will fail late — at the point where someone needs the Information Model to start data design, three months into a plan that assumed it was already in hand. Put the subscription on the budget and in the critical path, and settle the licensing posture early: can you show the model to a UMA? Embed it in a product you sell? Give it to a systems integrator? Those answers determine whether the model can do the boundary-language job at all.

How to use a capability model properly

A capability is a statement of what an organisation is *able to do*, independent of who does it, how, in what order, or with which system. "Claim Adjudication" is a capability. "The claims team adjudicates claims in Atlas within 48 hours" is not — that is an organisation, a system and a service level wearing a capability's clothes.

ACORD's Capability Model is deliberately a functional decomposition **without predefined workflow** [1]. Architects from process-heavy backgrounds read that as incompleteness. It is the opposite: it is what makes the model reusable. The moment you fix the order, you have chosen a process, and a process is a design decision that differs between a direct motor insurer and a specialty liability UMA even though their capability sets largely overlap.

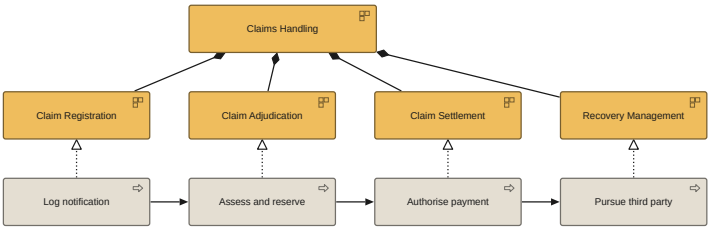


Figure 13.4 A capability decomposition carries no order; a process does. Both describe claims handling. Only one of them is a design decision, and it is not the top two rows.

So: three legitimate uses, and three ways it goes wrong.

Use one — a shared vocabulary for scope. When a programme says "we are replacing claims", the capability list converts that into a boundary: these eleven capabilities are in, these four are out, recovery management is deliberately out and here is who still owns it. Scope arguments at month nine are almost always capability arguments that nobody had at month one.

Use two — a coverage check. Walk the published list against your estate and record, for each capability, whether you can do it, where, and how well. The findings that pay for the exercise are the absences and the duplications, not the confirmations.

Use three — an investment conversation with people who do not model. A capability list is the only architecture artefact a chief operating officer will read voluntarily, because it is written in business language and contains no boxes labelled "ESB".

And the failure modes.

Colouring in. The map becomes red, amber and green, the colours become a scorecard, the scorecard becomes a budget weapon, and within two cycles the colours are negotiated rather than assessed. If you score, write the scoring criteria down first, publish the evidence per score, and expect to defend both.

Mistaking a system for a capability. "We have a fraud system, so we have fraud detection" is false whenever the scores never stop a payment. Capability assessment must ask what *changes* as a result of the capability, not what is installed.

Treating the model as the target. A reference capability model lists what insurers in general do. It does not say which of them *you* should do, or do yourself. A deliberate decision not to hold a capability — to delegate exposure accumulation to a reinsurance broker, say — is a legitimate architecture, and a map that marks it red is lying.

Mapping ACORD capabilities onto an estate

Here is the procedure that works, in the order that works.

1. **Take the published decomposition as given and do not edit it.** Resist renaming. The value is that it is external. You may *add* a capability the model lacks, and you will find one or two in South Africa — binder administration and Sasria coupon handling are candidates — but mark additions clearly as local extensions.
2. **Agree the depth once.** Two levels for an executive conversation, three for a programme scope, four only where you are about to make a build-or-buy decision. Mixed depth makes the map unreadable and the gaps unprovable.
3. **Map applications to capabilities, not capabilities to applications.** Start from each system you actually have and ask which capabilities it realises. This direction finds duplication; the other direction hides it.
4. **Record a verdict per capability with evidence.** Not a colour. A sentence and a source.
5. **Then, and only then, look at the gaps.**

Verdict	Means	What it implies
Owned	One system realises it, one accountable owner	Nothing. Leave it alone
Duplicated	Two or more systems realise it independently	Divergent business rules, two sets of numbers, double change cost. The most common finding in a composite group
Partial	Realised for some lines, channels or products only	A hidden manual process somewhere. Find it before you plan
Absent	No system, no owner	Either a genuine gap or a capability performed on spreadsheets by one person
Delegated	Performed by a UMA, TPA, broker or reinsurer	The insurer still carries the regulatory accountability. Needs oversight-by-design, not absence

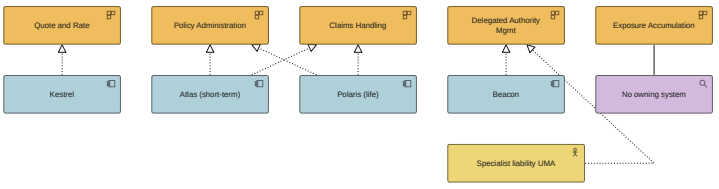


Figure 13.5 Cape Mutual's first honest coverage map (the group is fictional). Two cores realise the same two capabilities — the duplication that produces three policy counts. Delegated authority is performed by the UMA, so it is delegated, not absent. Exposure accumulation has no realising system at all.

The figure shows the shape of Cape Mutual's first honest map. Two cores — Polaris for life, Atlas for short-term — both realise policy administration and claims handling, which is the duplication you would expect in a composite group and which is the reason the group has three policy counts. Delegated authority management is realised only by Beacon, the broker and adviser portal, with the specialist liability UMA performing the underwriting itself; the capability is therefore *delegated*, not absent, and the architectural work is oversight rather than build. Exposure accumulation has no realising system at all. Nobody had ever written that down, because nobody had ever been handed a list that contained the words.

That last point is the whole argument for the exercise. The gap was not discovered by analysis. It was discovered by comparison against an external list.

What happened next at Cape Mutual is also instructive, because the map changed the conversation rather than the architecture. Thandi Nkosi, as chief operating officer, did not ask for the duplication to be removed — removing it means retiring a core, and nobody retires a forty-year-old life platform because a diagram is untidy. She asked a narrower and better question: which of the two cores is authoritative for each number we publish, and who signs that off. That is a data ownership decision, it costs nothing, and it fixed the three-policy-counts problem within a quarter. The exposure accumulation gap went to Riaan Botha as chief risk officer, because an absent capability with a regulatory consequence is a risk item before it is a project. And the delegated authority finding went into the next binder renewal, because the oversight obligation sits with the insurer whatever the UMA's systems do.

Three different owners, three different instruments, one map. An architect who presents a coverage map as a shopping list of projects will get none of them funded. An architect who presents it as a set of questions addressed to the people who can answer them will get most of them resolved without a programme at all.

> The map is a snapshot of a moving estate. Date it, name the evidence behind each verdict, and re-run it when something material changes — a new core, an acquisition, a binder taken back in-house. An undated capability map is quoted for years after it stopped being true, and it will be quoted at you.

What ACORD does not give you

A short, honest inventory, because expectations set here save a year.

- **Bitemporality.** Insurance data needs both the date a fact was true and the date you knew it. No industry model hands you a working bitemporal implementation.
- **An IFRS 17 unit of account.** The portfolio, profitability bucket and annual cohort structure is yours to build, and it is not a reporting view — it is a key on every cash flow.
- **A POPIA lawful-basis and consent model.** Purpose, basis, source and date, per party per purpose. Absent from every industry model.
- **South African structures at the depth you need.** Binder and UMA delegated authority, the Sasria coupon attached to an underlying policy, DebiCheck mandates, cell captive ring-fencing. Some are representable; none are given to you.
- **Non-functional requirements.** Batch windows, recovery objectives, the fact that your overnight run takes six hours and the business wants eighteen hours of online time.
- **Sequencing, cost and feasibility.** The model is a map, not a route.
- **Your products.** The Product Model gives you a notation and a schema for expressing products. It does not contain yours, and the only complete specification of a product sold in 1987 is probably the code in Polaris.

The characteristic failure is treating ACORD as a download rather than a programme. An insurer that licenses the Information Model, maps 40% of it and stops has bought a liability: a half-canonical model now competing with the policy administration system as a source of truth, with no owner and no reconciliation.

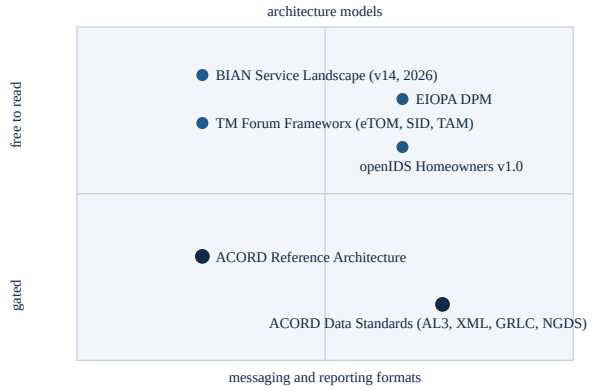
Calibrate: what other industries have

If you have worked in banking or telecoms you will arrive expecting more than insurance gives you. Knowing why protects you from proposing the wrong thing.

Banking has BIAN. The Banking Industry Architecture Network was established in 2008 by financial institutions, software vendors and integrators, publishes a UML-based metamodel and service definitions, maps its work to TOGAF, and runs its own foundation certification [16]. Its central artefact is the **Service Landscape**: version 14.0 was released in February 2026, and a practitioner account puts it at around 340 modelled service domains, of which 242 carry Semantic APIs, some with ISO 20022-aligned variants and AsyncAPI definitions for event-driven use [14][15]. Critically, the same account states that the Service Landscape, Business Object Model and Semantic APIs are **free online**, with only the BIAN Guide paid for [15]. That source is a BIAN consultancy promoting its own services, so treat the adoption claims cautiously — but the access model is checkable, and it is the decisive difference.

Telecoms has TM Forum Frameworx, a longer-standing and more granular set: the Business Process Framework (eTOM) for process and activity descriptions, the Information Framework (SID) as a unified reference data model, the Application Framework (TAM) describing OSS and BSS application functions, and integration frameworks [17]. Telecoms architects routinely specify procurements in TAM terms. Even there, the record is uneven — the Information Framework has seen poor uptake in some domains, such as inventory management [17].

Insurance's open options are early. On 20 November 2025 openIDS, a community initiative hosted by the Linux Foundation's openIDL project, released the **openIDS Homeowners Standard v1.0**, described as the first free, open, production-ready insurance data standard, with further lines of business planned [18]. It is one line of business in one jurisdiction's regulatory context. It is also the first credible open competitor ACORD has had. Separately, EIOPA publishes the Solvency II Data Point Model and XBRL taxonomies openly [19] — not an entity-relationship model, but the definitive public statement of what a prudential regulator means by each number, and mineable even outside Europe.



Insurance's architecture models sit in the bottom-left quadrant; banking's and telecoms' sit top-left. That single difference explains most of the gap in how often each gets used.

Figure 13.6 Calibration. Placement reflects the access terms each body publishes and whether the artefact is an architecture model or a wire format; it is not a quality judgement. Insurance's architecture assets are the gated ones.

So why is insurance's reference architecture less dominant in insurance than BIAN is in banking? Five reasons that hold up, and one that is my reading rather than a sourced fact.

1. **Access.** BIAN's landscape and APIs are free at the point of use [15]; ACORD's facets are subscription-gated with unpublished pricing [1]. Diffusion follows friction. A standard a graduate can read on a Tuesday afternoon ends up in more designs than one requiring a procurement cycle.
2. **Centre of gravity.** ACORD started with forms and grew into messaging; the Reference Architecture is the newer, smaller limb. BIAN was an architecture body from the first day. When the organisation's main business is wire formats, the architecture assets compete internally for attention.
3. **Product heterogeneity.** A bank's retail products are variations on deposit, loan, payment and custody. Insurance spans a sixty-year with-profits whole-life contract, a R400 million property risk placed in London, and a R35-a-month device policy sold at a till. A model general enough to cover all three must abstract until it stops being directly useful — so insurance reference models are either enormous or line-specific.
4. **No forcing function.** Open banking mandates and PSD2 obliged banks to expose standard interfaces, and ISO 20022 gave payments a common spine. Insurance has no equivalent horizontal mandate. The exception proves the mechanism: in the London market, where Lloyd's requires ACORD messaging and bases the Core Data Record on it [5], ACORD *is* dominant.
5. **Vendor incentives.** Core banking vendors advertise BIAN alignment as a procurement advantage. Insurance core vendors sell their own data model as differentiation, and an insurer's extension model is usually vendor-specific. Alignment to a neutral model reduces lock-in, which is not what a vendor is selling.

And the reading, offered as judgement rather than fact: much of ACORD's modelling content arrived by industry contribution rather than ground-up committee design — PRIMA contributed a model in 2008, IBM contributed assets in 2009, and Information Model 2.0 followed in 2011 [20][21]. That plausibly accounts for the model's scale and for the occasional stylistic inconsistency across facets. I cannot source the causal link; the contributions are documented, the inference is mine.

The practical consequence for you is calibration. A banking architect can lean on BIAN and expect the room to accept it. In insurance you will have to argue for the reference architecture, pay for it, and demonstrate value from one facet before you get the others. Plan the argument, not just the adoption.

What to do on Monday

- Write down your own three definitions of "policy count" and who holds each. That is your business case for a glossary, and it takes an afternoon.
- Price the subscription and settle the licensing posture before you design anything that depends on the model.
- Do the capability coverage check at two levels, with evidence sentences rather than colours, and present the absences first.
- If you have UMA or binder counterparties, ask the three largest what formats they can send today, and whether any of them has ever heard of the Southern Africa standards. The answer shapes your integration strategy more than the standard does.
- Keep the traceability chain in view. Licensing one facet is defensible; licensing one facet and then claiming the benefits of the chain is not.

THE CHAPTER IN TEN LINES

- A reference architecture gives you a vocabulary and a coverage check; it is not a target architecture, and a programme whose deliverable is "ACORD adoption" has no deliverable.
- ACORD publishes two families that get conflated constantly: the messaging Data Standards, which are its original and larger business, and the seven-facet Reference Architecture, which is what an enterprise architect wants.
- The seven facets are Business Glossary, Information Model, Data Model, Capability Model, Component Model, Process Model (also published as the Process Framework) and Product Model — and their value is the traceability chain from capability through component and process to message and schema.
- Versions live on ACORD's marketplace product pages, not the overview: Data Model v2.13, Information Model v2.13 and Business Glossary v2.16 were released on 12 December 2025, and interactive Glossary and Capability Model versions followed on 29 January 2026, members only.
- Almost nothing substantive is free. Put the subscription in the budget and on the critical path, and settle the licensing posture — whether you may show the model to a UMA or ship it in a product — before you design anything that depends on it.
- A capability model carries no workflow on purpose; map applications to capabilities rather than the reverse, record a verdict with evidence rather than a colour, and present the absences and duplications first.
- The ACORD Southern Africa short-term standards exist, with schemas, a policy-lifecycle process guide and samples — but local adoption levels are not something I could source, so ask your three largest counterparties what they can actually send you.
- ACORD gives you no bitemporality, no IFRS 17 unit of account, no POPIA lawful-basis model and no South African binder, Sasria or DebiCheck constructs. Budget for the overlay regardless.
- Banking's BIAN is free to read and telecoms' Frameworx is deeply embedded in procurement language; insurance's architecture assets are gated, line-heterogeneous and lack a regulatory forcing function — except in the London market, where ACORD does dominate.
- Calibrate your expectations: in insurance you will have to argue for the reference architecture, pay for it, and prove value from one facet before you get the rest.

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CHAPTER 14

Technology providers

Somebody will ask you which system to buy before you know enough to answer. This chapter will not tell you — it will teach you what kind of thing each vendor sells, what question that kind of thing exists to answer, and why in South Africa the shortlist is now a regulatory filing as well as an architecture decision.

Read this before you read anything else

In your first month at an insurer somebody will ask which policy administration system to buy. You will not know; nobody three weeks into the industry does. What you *can* learn quickly is what **kind** of thing each vendor sells, and what question it exists to answer.

> FRAMING, AND IT IS NOT NEGOTIABLE
>
> **Every company named here is an example of who plays in a category. Nothing in this chapter is a recommendation. Nothing is an endorsement. It is not a shortlist and it must not be used as one.**
>
> Inclusion means a vendor was recognisable enough to illustrate a category and had a source behind it. Omission means nothing: some of the strongest products here are missing because no primary source could be found. There are no rankings or quadrants, because analyst positions in this market are purchased research and age badly.

What rots first

The research behind this chapter was done in **October 2026**, and knowing which parts to distrust first is more useful than the chapter itself.



Figure 14.1 What goes stale first. Distrust this chapter from the top down.

Ownership rots fastest. In the eighteen months before this was written, **Sapiens** went from NASDAQ-listed to privately held, Advent International completing the purchase on 17 December 2025 [2][3]; **Duck Creek** was taken private by Vista Equity Partners [4]; and **Arius**, the short-term reserving product, stopped being Milliman's when Akur8 bought it in September 2024 [19]. Three categories, wrong the moment they printed.

Ownership is an architectural attribute: a platform under private-equity margin pressure behaves differently in a roadmap conversation than a founder-led or listed one. Product names rot almost as fast. What does not rot is the shape of the problem — these eight categories have been recognisably the same eight for twenty years.

Part one: the global market, by capability

Vendors sell suites, built so you cannot see where one capability stops and the next begins. Invisible boundaries are harder to argue with and harder to leave.

Capability	The question it exists to answer
Core platform	What cover, to whom, on what terms, for what premium, with what claims?
Rating and pricing	What is the price, and who can change it without a release?
Underwriting and risk data	What must we know that we cannot know ourselves?
Claims and fraud analytics	Which claims deserve more effort, and which less?
Reinsurance	What have we ceded, what can we recover, what do we retain?
Distribution and broker technology	Whose system holds the client, and how does premium reach us?
Actuarial, reserving and IFRS 17	What is this book worth, and can we reproduce the number?
Data platform	Where does all of the above reconcile?

Now the same market the other way round, as a readable subset.

Vendor	Core	Rating	Risk data	Fraud	Reins.	Distrib.	Actuarial	Data
Guidewire	•	•						
Sapiens	•	•			•			•
Socotra	•	•						
FINEOS	•							
WTW		•					•	
Akur8		•					•	
Verisk		•	•	•	•	•		
LexisNexis Risk			•					
FRISS			•	•				
Acturis						•		
Moody's			•				•	

Read the rows, not the dots. A vendor in one column is a product decision. **A vendor in three or more columns is a concentration decision** — different risk, and in South Africa a regulatory conversation. Verisk's row is the clearest case.

Core platforms: policy administration, billing and claims

The system of record for the contract and for the money: what cover, to whom, on what terms, at what premium, with what claims. It decides what the insurer can sell, how fast and at what cost.

Three axes separate the category: **short-term versus life, annuity and group**, different data models over different time horizons; **integrated suite versus assembled components**; and **licensed software you run versus software the vendor operates**, which decides who carries the regulatory outsourcing obligation.

Core platforms — examples, not recommendations

- **Guidewire** — the short-term reference point, strongest in North America, now mostly cloud subscription; its quarter to 30 April 2026 shows subscription revenue up 35% against **licence revenue down 2%**, a vendor mid-transition [1].
- **Sapiens** — the broadest multi-line portfolio among the large independents and the most geographically spread; Advent-owned since December 2025 [2].
- **Socotra** — API-first and developer-oriented, positioned for greenfield rather than estate replacement [6].
- **FINEOS** — a specialist in life, health, group and disability claims, where a claim is a recurring income stream with repeated eligibility assessment, not a one-off indemnity. It is ASX-listed, not private-equity held [8][9].
- **Oracle** — OIPA for life and annuity, the option that arrives inside an existing Oracle relationship [10].

Also here: **Duck Creek** [4], **Insurity** [5], **INSTANDA** [7], **EIS** [11], **Majesco** [12], **msg life** [13], **DXC**.

The questions that matter.

- *Product-configuration* or *process* platform? Demand a live new-product build in the room, by a business configurer, no vendor engineer at the keyboard.
- **Unit of deployment** — can one line go live alone?
- Where does **rating** live? That one answer decides whether pricing can act without a core release.
- Is **billing and collections** in scope, and does it know how premium is collected where you operate?
- **Data egress**: incremental feeds you run yourself, or your analytics strategy is a hostage.
- **References in your jurisdiction and size band.** North American logos say nothing about your statutory returns.

Rating and pricing

This category exists to separate the *price* from the *policy system*. An insurer that waits for a core release to change a rate has lost the ability to compete, and usually does not notice until a competitor reprices a segment in a week and takes the profitable half of the book. Three jobs hide under one word — **model building**, **rate deployment**, **portfolio monitoring** — and products are strong at different ones.

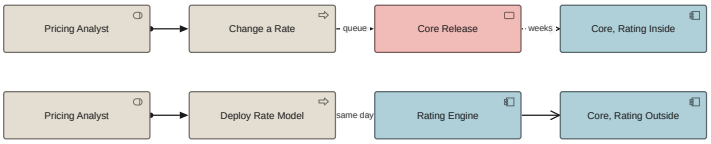


Figure 14.2 Where rating lives decides who can change a price. Top: rating inside the core, so a rate change waits for a release. Bottom: rating outside, so pricing can act.

Rating and pricing — examples, not recommendations

- **WTW Radar and Radar Live** — the long-established build-and-deploy pair; Radar 5, October 2025, added generative-AI querying of portfolio performance, machine-led model fitting and native Databricks and Snowflake connectors [14][15]. **These are WTW products, not Verisk's**; the two have collaborated on commercial pricing, probably the source of the confusion [16].
- **Earnix** — positions as AI decisioning rather than rating alone, with pricing, rating and underwriting sharing data in real time, and explainability built in [17].
- **Akur8** — the distinctive claim here: machine-learning-assisted fitting that produces **transparent** generalised linear and additive models, so the output stays filable and explainable [18][19].
- **hyperexponential** — built for **specialty and commercial**, where the unit of work is a bespoke submission priced against a model, not a retail rate table [20].

The questions that matter.

- **Personal lines or specialty?** hyperexponential sits on one side of that line, Earnix and Radar on the other.
- **Who owns rate deployment?** If a pricing analyst cannot push a change to production themselves, with approval and audit and no core release, the tool has not solved the problem it was bought for.
- **Model governance.** Can you reproduce the model version that priced a quote eighteen months ago? Where rates are filed, that is why transparent-model approaches persist.
- **Does it overlap your core?** Buying rating separately means switching off something you have paid for: fine as a written decision, bad as an accident of two procurements.

Underwriting and risk data

Everything here is information the insurer cannot generate itself: industry loss history, property and vehicle characteristics, identity signals, medical evidence, catastrophe exposure. Also here are the **automated underwriting rules engines** for life. Architecturally these are one shape — **an external dependency in the quote path** — bringing latency, availability, cost per call and data-protection obligations. It is where "we just call the API" hides most of the work.

Underwriting and risk data — examples, not recommendations

- **Verisk** — the broadest United States short-term data estate: industry loss data, policy forms and prospective loss costs, catastrophe modelling, a claims data exchange [30].
- **LexisNexis Risk Solutions** — identity, prior-carrier, vehicle and property attributes as scores straight into the quote; recent additions include vehicle mileage and a Current Carrier Commercial product, June 2026 [23][24].
- **Swiss Re Magnum** — automated life and health underwriting from a reinsurer, shipping a rule base derived from Swiss Re's medical manual: the vendor states 2,322 risk factors and 30,000-plus rules [21].
- **Munich Re Automation Solutions** (ALLFINANZ) — the equivalent from the other major life reinsurer [22].

> **THE STRUCTURAL POINT ABOUT LIFE UNDERWRITING ENGINES**
>
> The two leading automated life underwriting engines belong to the two largest life reinsurers, and that is not incidental. **The rule base *is* the reinsurer's underwriting manual.** Adopting it aligns your risk selection with that reinsurer's appetite — frequently what the insurer wants, because it buys automatic reinsurance acceptance of the automated decision. A sound trade, and a dependency most architecture diagrams do not show. Ask who owns the rule base, and what happens when you change reinsurer.

The questions that matter.

- **Is the data available in your country?** Almost all the richest short-term data assets here are United States-specific, so a South African architect should assume most of this category does not apply.
- **Cost per call against conversion**, not the rate card — a per-quote variable cost, paid on the quotes that never convert.
- **Consent and retention.** Under POPIA, who is the responsible party for data a vendor pulls on your behalf?

Claims and fraud analytics

This category directs claims effort: fast-track this one, investigate that one, assign a specialist, pursue recovery. The marketing moved from rules-and-scores to agents in three years; the architecture has not moved. It is still **a decisioning layer that reads claims data and writes a recommendation back**

into the workflow, so its value is capped by the claims data model it reads — a brilliant model over a poor data model produces confident nonsense.

Note where this is going. FRISS putting *underwriting* analytics beside claims analytics, and Shift's references to suspicious *policy* activity, both say detection is migrating from the claim back to the application — a fraudulent claim on a fraudulent policy is cheapest to stop at inception. That needs a read path into new business which many insurers never built.

Claims and fraud analytics — examples, not recommendations

- **Shift Technology** — AI agents across short-term and health claims, covering coverage and liability, fraud, subrogation, injury and process, plus a cross-carrier claim-history network [25][26].
- **FRISS** — short-term "trust automation" spanning underwriting analytics, claims analytics, investigations, sanctions screening, and media checking that detects manipulated or AI-generated images in claims, a genuinely new control [27][28].
- **Verisk** — here too, because an industry-wide claims-history exchange is itself an anti-fraud control: the strongest signal is usually that the same loss has been claimed elsewhere [30].

The questions that matter.

- **False-positive economics.** The cost of a fraud model is handler time wasted and honest claimants delayed, so demand false-positive and fast-track rates measured on a back-test of *your* book.
- **Where does the score land?** A score in a vendor dashboard changes nothing; it must arrive in the handler's queue, inside the claims system, with the reason attached.
- **Explainability for disputes.** Can you justify a partly model-driven declinature to an ombudsman?
- **Does it need a consortium?** Cross-carrier networks are worth only what local participation gives them, and in a smaller market one may not exist.
- **Who owns model drift** — who retrains, how often, on whose data?

Reinsurance

These systems calculate cessions across treaties and facultative placements, track recoveries and retrocession, produce bordereaux and statutory schedules, and show what the insurer **actually retains**. This is where many cores are thinnest, and the gap appears at year-end close or after the first large event.

Reinsurance — examples, not recommendations

- **Sapiens ReinsuranceMaster** — a single platform for large and multinational programmes; version 9, April 2024, added inter-company retrocession, NAIC Schedule F reporting and real-time allocation [29].
- **Verisk Sequel** — specialty, excess-and-surplus and London-market underwriting, broking and claims, with Rulebook for pricing, Sequel Impact for exposure management and Whitespace for placement; Hg sold Sequel to Verisk, which separately bought Rulebook for about $87m [30][31].

An older product, **SICS**, is associated with the Sapiens portfolio; its current name and position could not be confirmed as at October 2026, so nothing more is claimed.

The questions that matter.

- **Treaty complexity against volume.** A proportional treaty over a homogeneous personal-lines book is arithmetic; a multi-layer non-proportional programme with reinstatements and inuring covers is not.
- **Where is cession calculated** — in the core at policy level, or from a periodic feed? That decides whether net exposure is knowable in real time.
- **Event response** — how long after a catastrophe until you have a reliable net-loss estimate. That is the real test.
- **Statutory schedule coverage** for every jurisdiction you report in: for a group with African subsidiaries, more than anyone first counts.

Distribution and broker technology

Insurance is mostly sold by intermediaries, so an entire software market has the **broker, agency or managing general agent** as its customer. You need it for two reasons: it is the surface through which most premium arrives, and its data standards and extranets constrain what you can offer digitally.

Distribution and broker technology — examples, not recommendations

- **Applied Systems** — North American and United Kingdom agency management on Applied Epic, with a brand family covering insurer connectivity, submissions and rating [32].
- **Vertafore** — the other large North American incumbent, a Roper Technologies subsidiary since 2020, spanning agency management, licensing, rating and analytics [33].
- **Acturis** — the dominant United Kingdom broker platform, also serving insurers and agencies; its own site claims it built eight of the top ten United Kingdom insurer extranets, which is the architecturally interesting statistic: one vendor on **both sides** of the insurer–broker boundary shapes what is possible for everyone [34].
- **Novidea** — broker, agency and coverholder management **built on Salesforce**, the defining fact about it: it inherits Salesforce's data model, extensibility and licensing economics, for better and worse [35].

The questions that matter.

- **Whose system of record holds the policy?** Broker platform holding the client record and insurer holding the policy record means permanent reconciliation. Decide where truth lives.
- **Commission and premium reconciliation** is the hardest part here and the commonest source of leakage. Test it with real intermediary statements.
- **Delegated authority controls.** If agencies or binder holders bind on your paper, the system must *enforce* limits, not merely record breaches.

Actuarial, reserving and IFRS 17

Three overlapping jobs: **actuarial modelling** (cash-flow projection for pricing, valuation and capital), **reserving** (what we owe on claims that have happened, including the ones we have not heard about), and **financial reporting** under IFRS 17. IFRS 17 is why architects get dragged in: it demands granular, auditable, reproducible cash-flow data at a "unit of account" no legacy core was designed to produce, which makes it **a data-lineage problem wearing an accounting costume**. Hence the programmes that bought an engine and spent two years on the feeds.

Actuarial, reserving and IFRS 17 — examples, not recommendations

- **FIS Insurance Risk Suite — Prophet** — the life actuarial incumbent, now modular across model development, enterprise execution, assumptions management, data integration, results aggregated to an IFRS 17 unit of account, and a separate regression-testing module whose existence shows how much work governance is [36].
- **Moody's** — AXIS for life actuarial modelling, RMS for catastrophe risk after an acquisition of about $2bn [37].
- **WTW ResQ** — short-term loss reserving; in October 2024 WTW launched machine-led reserving, applying its own algorithms to traditional methods to generate automated independent loss estimates that remain interpretable [38].
- **Akur8 Reserving** — Arius and Arius Enterprise, acquired from Milliman in September 2024; Milliman remains a major actuarial presence, but **do not attribute Arius to Milliman in the present tense** [18][19]. **SAS** packages an IFRS 17 solution [39].

The questions that matter.

- **Is your IFRS 17 answer an engine or a data problem?** Most failures are upstream: the core cannot produce cash flows at the needed granularity with stable keys, and an engine does not fix that.
- **Reproducibility.** Can you re-run a prior period with that period's model, assumptions and data and get the same number? Auditors will ask; many estates cannot.
- **Model risk governance** — version control, change approval, regression testing.
- **Who runs it?** Actuarial prefers tools it controls, IT prefers managed platforms. Decide early, or acquire a shadow estate.

Data platforms and the hyperscalers

Every category above produces or consumes data, and none is the insurer's analytical system of record. The data platform is where policy, claim, payment and external data reconcile. The hyperscalers address insurance less through distinct insurance products than through **industry framing of general-purpose services plus a partner ecosystem**, and understanding that is the point.

Data platforms and the hyperscalers — examples, not recommendations

- **Microsoft** — the most explicitly documented insurance framing of the three, organised by business scenario and referencing European operational-resilience regulation; its named partner list is itself a map of this chapter [40].
- **Amazon Web Services** — visible in insurance mainly through large carrier migrations; the best-documented African case is **Old Mutual**, over 2,000 servers and 800 applications in roughly three years, South African estate complete March 2022, mostly lift-and-shift, with Tamr replacing legacy master data management [41].
- A current named catalogue of AWS and Google Cloud *insurance-specific* accelerators **could not be confirmed** here, so no product names are given.

The questions that matter.

- **Is the accelerator a product or a slide?** Ask for the artefact: a repository, a reference architecture, a production customer you can speak to.
- **Where does your core's data land, and in what form?** A software-as-a-service core with a weak export story quietly becomes the constraint on your data strategy, and you will not notice for two years.
- **Concentration.** Core on one hyperscaler, data on another, vendors on a third is now a regulatory conversation as well as an engineering one.
- **Egress and gravity.** Model the cost before designing multi-cloud analytics.
- **Who holds the model?** Inference in a vendor's tenant against policyholder data is an outsourcing arrangement with notification obligations, not a feature.

Part two: South Africa and Africa

This is the half you cannot get from a global analyst report. Three structural features make South Africa different, and they need establishing before any vendor is named.

A large, mature intermediated market with a formal delegated-authority layer. Binder holders and underwriting management agencies perform insurer functions under agreement, and cell captives — pioneered in South Africa in 1993 — sit alongside them [68]. This creates a software category that barely exists elsewhere: multi-tenant administration platforms whose *user* is the agency or administrator, with the licensed insurer as a reporting consumer rather than an operator.

Collections is a first-class architectural problem. Premium is collected largely by debit order against thin-margin accounts, through a bank-mandated authentication regime that has changed materially. Lapse risk here is a payments-infrastructure problem before a customer-behaviour one.

Outsourcing is now a named prudential standard with notification deadlines — which makes the vendor landscape a regulatory matter, not only a procurement one.

Local platform vendors and administrators

South Africa has a real domestic insurance-software industry, mostly serving brokers, agencies, administrators and mid-sized insurers rather than the largest carriers. Several predate most of the companies the word "insurtech" brings to mind.

Vendor	Primary user	What distinguishes it
Silverbridge	Life, group schemes, pensions, benefits	Exergy policy administration; go-lives at Absa Life, Standard General, BetterLife [42][43][44][45]
Genasys Technologies	Insurers, agencies, brokers, administrators holding delegated authority	South African-founded, now London-headquartered; explicit bordereaux and delegated-authority features [47]
Cardinal (Cardinal Insurance Management Systems)	Short-term insurers and intermediaries	C360 and C360i with a pre-built application layer; states 300-plus clients [49][50]
OWLS Software	Insurers, agencies and brokers together	Cloud-native, short- and long-term, with automatic broker remittance calculation [51]
TIAL Technologies	Short-term insurers, administrators, underwriting managers, brokers, SADC-wide	Established 1999, South Africa and Mauritius; policy, financials, claims, CRM, reinsurance [52][53]
PROFIDA	Small insurers, agencies, cell captives, administrators	Marketing through to collections, commission and accounting in one product [54][53]
Brolink	Insurers needing binder services; brokers needing administration	Not a software vendor first — an administrator whose system comes bundled with the authority [55]
Fulcrum Group	Intermediaries	The same bundle on the money side: collection, premium financing, treasury [56]
Turnkey Africa (Kenya)	Insurers, bancassurers, brokers, pension administrators	The clearest pan-African core vendor; TurnQuest spans life, short-term and bancassurance [61]

Two smaller entries reveal more than the big ones. **Teessen Consulting's Insure-IV** targets short-term brokerages with a no-programming product factory [53]. And **Legacy Underwriting Managers' Legacy On-Line**, built for local insurers since 2003, automates **debit-order follow-up and resubmission** and fully automates windscreen, geyser and fast-track claims [57].

Stop at that last one. It is a *product feature* in a local vendor's marketing, and no global core brochure says anything like it. That tells you what hurts here.

> **WHAT COULD NOT BE CONFIRMED**
>
> No primary source as at October 2026 could settle: **Silverbridge's** ownership after its deletion from the FTSE/JSE Africa indices in September 2022 [46]; **Genasys Technologies'** cap table after the R260m Frog Capital investment of December 2021 [48], or whether its SKi branding is still used [88]; **Brolink's** ownership; **Flexi's** install base, where verified figures date from 2014; **Innovation Group's** ownership and the 1insurer brand after the Carlyle acquisition [58][59]; or **SSP's** current South African presence, where the verified local material — a platform supplied to Natsure for its agency network — is years old [60].
>
> None of this criticises a vendor. Verify ownership yourself before it reaches a board paper.

Which global platforms have actually landed at South African insurers

The honest answer: **fewer than vendor marketing implies, and mostly in claims or finance rather than policy administration.**

Insurer	Vendor and scope	Status as sourced
Santam	Guidewire — underwriting selected December 2012, claims management February 2019	Guidewire announced "Santam goes live on Guidewire" on 9 June 2026: a cloud deployment covering policy management, documents, claims submission and tracking, underwriting and customer service [62][63][64]
Discovery Health	SAP FS-CD for collections and disbursements, via Serole Technologies; **policy administration deliberately kept in-house**	Programme from 2014, request for proposal 2016, kick-off February 2019, build under way in November 2019 [65]
Old Mutual	Amazon Web Services — infrastructure, not core application	2,000-plus servers, 800 applications, roughly three years; South African estate complete March 2022 [41]
King Price, Discovery, Guardrisk, GENRIC, SA Home Loans	Named as clients by Genasys Technologies	Which systems or lines these cover is **not stated** by the source [47]
Hollard International, Absa Life, Standard General, BetterLife	Silverbridge Exergy	One a client since 2016; the others reported as Exergy go-lives [42][43][44][45]

Any count of South African insurers running a given global core platform is not available from a reliable source, and none is estimated here.

The thinness has consequences: the local implementation-skills pool is smaller than the reference list implies, statutory reporting is likely custom work, and support differs when the nearest delivery centre is in another timezone.

Santam and Guidewire: fourteen years of elapsed market reality



Figure 14.3 The three public dates in the Santam–Guidewire arc. Read the warning before you quote this at anybody.

Read those three public dates slowly. Guidewire selected for **underwriting** in December 2012 [63]. Guidewire selected for **claims management** in February 2019 [64]. "Santam goes live on Guidewire" announced on **9 June 2026** [62].

Santam is no minnow: the release describes it as South Africa's leading general insurer with market share above 22%, Africa's largest general insurer, distributing through more than 3,600 intermediaries as well as directly [62].

> READ THESE DATES CORRECTLY
>
> Those dates **do not** establish how long any single implementation took. Between 2012 and 2026 sit re-scoping, phasing, a second product-line selection, and things no press release mentions. Which Guidewire products Santam went live on, the exact date, the lines of business and the implementation partner are **not named in the release** and are not asserted here.
>
> Use the arc for **elapsed market reality**, not as a benchmark. It is not "a Guidewire implementation takes thirteen years". It is: *from the day a large South African insurer first chose this platform to the day the vendor announced it live, thirteen and a half years of calendar passed.*

That matters because somebody will put a three-year core replacement plan in front of you. Discovery Health's timeline says the same thing more quietly: 2014 to a February 2019 kick-off, build still under way that November — for a *component*, not a suite [65]. Neither arc is evidence of incompetence. Both show what core change is: a decade-scale organisational undertaking with a software project inside it.

Discovery Health: buying a component on purpose

Discovery Health replaced its collections and disbursements with SAP FS-CD — broker collections, claims payment and premium collection, invoicing, automated overdue reminders, payable and receivable accounting and dunning, integrated to policy management and finance. And it **declined a wall-to-wall implementation**, explicitly taking what the reporting calls a *"build and buy"* approach: it kept its own policy onboarding, its own policy administration and its own bespoke systems [65].

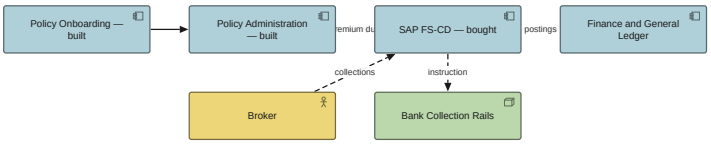


Figure 14.4 Discovery Health's build-and-buy boundary as reported: a bought collections and disbursements component, with policy onboarding and administration kept in-house.

A large, technically confident insurer looked at a very capable suite and bought one component. Not timidity but a capability boundary drawn deliberately: money movement is commodity and heavily regulated, so buy it; the product and member experience are how Discovery competes, so build

and keep them.

The UMA, binder and administrator software market

Underwriting management agencies are specialist underwriting businesses writing and administering niche classes on an insurer's licence. Local trade coverage places them in agriculture, marine, motor gap, cyber liability, renewable energy, and bonds and construction guarantees; brokers use them for smaller clients a mainstream insurer would not look at. There is a visible trend of insurers buying out or "divisionalising" agencies, which commentators argue erodes the independence that made them valuable [66].

> **A NUMBER YOU WILL BE ASKED FOR AND MUST NOT INVENT**
>
> There is **no sourceable count of underwriting management agencies in South Africa**, and **no sourceable figure for their share of short-term premium**. Both get quoted confidently in presentations; neither could be verified for this chapter. Describe the model, do not publish a size. If somebody hands you a percentage, ask for the primary source and expect not to get one.

An agency needs policy administration, claims, billing, commission, bordereaux and reporting — a small insurer's core at a fraction of the scale, with an obligation to report into the licensed carrier that owns the risk.

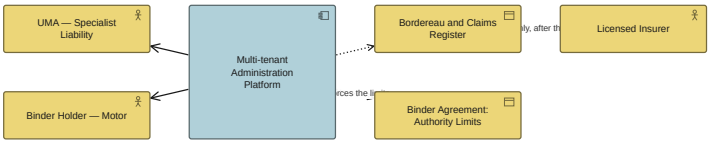


Figure 14.5 The South African administration platform: the agency or binder holder is the user, the licensed insurer is a reporting consumer, and the binder agreement is a control the platform must enforce.

The local trade press has a better checklist than most consulting frameworks. A FAnews digital magazine on broker, agency and insurer administration systems asked: which modules suit the target market; how much control the client has; what self-service and data-extract tools exist; what integration is supported; and how the vendor keeps pace with compliance change [67]. Cell captives are the adjacent structure, pioneered locally in 1993 and large enough that the two biggest rank fourth and eighth among local non-life insurers [68].

Joint Standard 1 of 2024: the vendor landscape becomes a filing

Joint Standard 1 of 2024 was issued by the Prudential Authority and the Financial Sector Conduct Authority under the Financial Sector Regulation Act of 2017. It replaced Prudential Standard GOI 5 and took effect on **1 December 2024**, applying to insurers licensed under the Insurance Act of 2017, Lloyd's and foreign reinsurer branches excluded [69][70][71]. It requires **due diligence before outsourcing any activity**, proceeding only where benefits demonstrably outweigh costs and risks; **explicit consideration of concentration risk** from a provider serving multiple insurers; **notification to both regulators at least 30 days before a material outsourcing** and within **seven days of termination**; and existing arrangements compliant within **24 months, by 30 November 2026** [69].

> **WHAT IS GENUINELY UNSETTLED HERE**
>
> The legal commentary used here **does not define "material"**, and addresses neither binder and agency arrangements nor cloud and IT services explicitly; one note also carries an unreconciled six-month reference alongside the 24-month deadline [69]. So **whether a software-as-a-service core or a binder arrangement is "material outsourcing" is not something this chapter can tell you.** Go to the standard, then to a lawyer.
>
> Raise it as a question, because the answer changes the programme plan: *if* a cloud core is material outsourcing, vendor selection carries 30-day regulatory pre-notification and a documented concentration-risk assessment. The shortlist stops being a decision and becomes a filing.

Note the date: 30 November 2026 falls weeks after this research, so if you are reading this later there is a first wave of compliance practice to learn from.

Digital natives, and what they built on

The interesting question about South African insurtech is not who they are but **did they buy a platform or build one?** The answer, overwhelmingly, is build — and that is the finding.

Naked, founded by actuaries, runs car, home and valuables cover with automation across buying and claims. In May 2026 it released an application inside OpenAI's ChatGPT giving a **final, binding** car insurance quote — a first for South Africa — connected to "the same live underwriting and rating technology that runs Naked's website and app" [72]. That clause is the architectural content: the novelty is the channel, the asset is the rating stack they own [73].

Root is the most architecturally interesting local entry, because its customer is other insurers and distributors: an insurance platform sold as a product — policy administration, a product workbench, claims tooling, data access and integrations behind documented APIs, plus embedded insurance. Founded 2016, API-first, naming Admiral, Octopus EV, FinChoice and Momentum Health [74]. A South African team built a core for itself, then turned the build into something others buy.

Simply, in mass-market life, disability and funeral cover, is described as both a registered financial services provider **and an insurance software developer**; Hollard invested in 2023 [75].

LUMKANI: CONTEXT-SPECIFIC PRODUCT DESIGN

Lumkani bundles **low-cost networked smart fire detectors** with inclusive insurance for informal settlements: when one detector senses fire it alerts homes within a 60-metre radius by alarm and SMS. Cover is fire and funeral up to R40,000 first-loss at R88 a month, underwritten by Hollard, distributed through female agents living in the communities they serve [76][77]. The product, the device, the distribution model and the policy were designed as one thing, and no platform in the first half of this chapter has a module for that.

Pineapple reportedly raised a Series B described locally as a record for South African insurtech, but **the amount and date rest on secondary coverage only, and what it built on could not be established** [78]. **Click2Sure** and **Yalu** belong to this market's history, but **all sources for both are years old and their status, ownership and platforms could not be confirmed**. Private funding into local insurtech was reported at around R680m in November 2023 [87].

The lesson for an incumbent is not "build like Naked", which built a narrow stack for a narrow book with no legacy. It is narrower: **a greenfield digital core is achievable at startup scale in this market, and the local talent to do it exists.**

Payments and collections: where global platforms get South Africa wrong

If you take one thing from this half of the chapter, take this section.

A large share of local personal-lines and mass-market life premium is collected by **debit order**, against accounts where timing relative to payday decides success. A failed collection triggers a grace period, reinstatement and eventually a **lapse**, costing the premium, the lapse administration and the re-acquisition. Collections is a **retention system**, not a finance back-office function.

Rail	Mandate authenticated by	Position in the day	Consequence
EFT debit order	Nobody; the insurer holds the mandate	Standard debit window	A consumer can have a collection reversed **without proving it was unauthorised** [79]
DebiCheck	The consumer, **through their bank, before collection**	Earlier than EFT debits	The bank holds a digital mandate record, so fraudulent disputes are harder and your evidence stronger [79]
RM (Registered Mandates)	**Not** required before registration	**Evening**, before EFT debits	Replaced the temporary RMS allowance; same-day collection possible, juristic accounts usable, credit tracking weaker. **All RMS mandates converted on 10 March 2025** [80]

DATES MATTER IN THIS TABLE

AEDO and NAEDO were switched off as live collection products on 31 October 2021, DebiCheck replacing them. And the source for the RMS-to-RM shift is dated 2 August 2024 — **before** the conversion it describes — reporting roughly 16.7m DebiCheck collections a month at a 77% success rate [80]. **That conversion's outcome and current success rates could not be verified.**

The reported causes of failure: insufficient funds, worst when the debit lands before salary; **disputes, growing and mainly affecting EFT**; changed banking details; and mandate fraud. Reversal abuse has reached the courts [84].

The mitigations are architecture, not operations. **Align collection dates with salary dates.** Offer **weekly or fortnightly cycles** for hourly and informal workers — a billing engine no monthly-premium data model anticipated. Drive **retry logic from the failure reason**, not blind resubmission. Offer **alternative payment methods at the point of failure**.

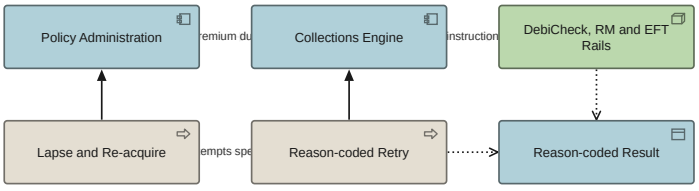


Figure 14.6 Collections as a retention system. If the reason code is not in the data model, the retry policy and the lapse decision cannot exist.

That diagram requires a reason code from the rails, a retry policy that reads it, a lapse process that knows how many attempts have happened and why, and a route back into the policy record. If the reason code is not in your data model, none of the rest can exist — and many estates store the result as success or failure, nothing more.

Payments and collections — examples, not recommendations

- **Stitch** — recurring collections, DebiCheck, variable recurring payments and pay-by-bank, with card and wallets for catch-up; it cites an undated pilot collecting on 74% of previously uncollectable mandates, a vendor figure [79][81].
- **Amplifin** — the AMP platform; its commentary on the RMS-to-RM shift noted RMS portfolios at 50% to 80% of some debt collectors' volumes [80].
- **Netcash** and **DirectDebit** — established local collection providers with DebiCheck offerings [82][83].
- **Fulcrum Collections** — the same capability from the intermediary side [56].

THE SINGLE MOST RELIABLE GAP IN A GLOBAL CORE

Almost no global core natively models DebiCheck mandate authentication, RM presentment windows, reason-coded retries or salary-aligned collection cycles. **This is the most dependable place a global core selection will require local build or a local collections partner — and it is almost always discovered late, after signature.** It is also the strongest single argument in this book for component architecture over suite monoliths here.

Systems integrators

A short section, because honesty requires it. What can be sourced: **Serole Technologies**, an integrator with a niche in SAP for insurance, implemented SAP FS-CD at Discovery Health [65]. **Deloitte South Africa** has partnered with international insurance IT specialists [85]. **Ovations** was appointed local partner for an international insurance firm [86]. **Solveco** took BetterLife live on Exergy [45]. **AWS Professional Services** worked on the Old Mutual migration with Tamr [41]. And **Innovation Group** operates here as a claims-operations outsourcer, not a core vendor [58].

WHAT THIS CHAPTER WILL NOT TELL YOU

The roles of BSG, Entelect, Synthesis, DVT, iOCO, EOH, Accenture, PwC, EY and KPMG in South African insurance **core implementations** could not be confirmed against any source. **This chapter does not assert which integrator implemented what**, and nor should you: "I heard they did Santam" is how procurement gets misled.

The honest framing is structural. **The South African insurance integration market is thinner and more specialised than the global tier-one list suggests, and for global cores the implementation capability is frequently imported or vendor-supplied.** That is a programme risk worth naming, and it shows up as schedule rather than cost.

How to read a vendor landscape as an architect

Everything above is raw material. This part transfers.

Build versus buy, per capability — and the forgotten third option

The question is never build or buy, but **where on the spectrum each capability sits**, and the answer differs between capabilities inside one insurer.



Figure 14.7 Four positions on the build-and-buy spectrum, with the question each answers.

Buy where the capability is commodity and the regulator effectively specifies the answer — general ledger, statutory reporting, sanctions screening. **Build** where it is how you compete and no product is shaped like your intent — Naked's rating stack, Lumkani's detector-and-policy model. **Buy, then extend** where the capability is standard but your distribution is not: most policy administration in an intermediated market.

The failure mode is **building the commodity and buying the differentiation**, which is what happens when procurement rather than architecture sets the boundary, because procurement optimises per contract and the boundary only makes sense across contracts.

The forgotten option: **buy the platform, build the product layer on its API**, owning product definition and experience while renting the record-keeping machinery — what Root sells and what Socotra and INSTANDA assume [74][6][7].

What a core replacement actually costs

The licence line is the smallest number in the business case. Enumerate the rest, in descending order of how badly each is under-estimated:

1. **Duration and opportunity cost.** The Santam arc is the number to show anyone who thinks this is a two-year programme [62][63][64].
2. **Data migration** — usually the largest package, because legacy policy data encodes decades of undocumented product variation nobody can explain.
3. **Parallel run** — both estates live, staffed and reconciled, for longer than planned.
4. **Integration rebuild** — every downstream report, broker feed, payment interface and actuarial extract.
5. **Capability lost during the change freeze**, since your competitors do not freeze.
6. **The local gap** — collections, statutory reporting, the implementation-skills pool.
7. **Exit cost**, written into the contract, with data egress rights treated as an architectural control rather than a legal nicety.

The one-platform trap

The trap is believing a suite removes integration. It does not. It **hides** integration inside a vendor boundary you cannot see, test independently or replace a piece at a time. The symptoms:

- **A rate change requires a core release**, so pricing runs at the core's cadence. The costliest consequence.
- **The weakest module sets the ceiling** — in an intermediated debit-order market, frequently collections or commission, which decide retention and intermediary relationships.
- **Analytics is hostage** wherever data egress is poor, and **concentration is now a regulatory position**, because Joint Standard 1 of 2024 requires an insurer to consider whether a provider's work for other insurers increases risk to it [69].
- **Ownership change resets the roadmap.** Sapiens, Duck Creek and Majesco are all privately held — not a criticism, a reason to hold roadmap commitments contractually [2][4][12].

So define your capability boundaries **before** you look at vendors, and insist any suite be describable as components with contracts between them. A vendor who cannot describe their own product that way has told you something for free. And remember Discovery Health, which bought one component and kept policy administration [65].

Why a shortlist is an architecture decision

Because the shortlist **is** the decision. By the time three vendors are in a procurement, the architecture has been chosen: data model, extension model, release cadence, integration style, cloud posture, the skills market you hire from, and the exit cost you will one day pay. Negotiation moves price and dates, not those. So:

- Write the **capability model first** and shortlist against it, not against vendor feature lists written to make boundaries invisible.
- Fix the **non-negotiable constraints before the market scan**: who changes a rate and how fast; where data of record lives and how it leaves; the recovery objective; the outsourcing posture; what must stay replaceable.
- Shortlist for **the shape of the problem, not the size of the vendor**. hyperexponential against Earnix is specialty versus retail; Socotra against Sapiens is greenfield versus estate; OWLS or TIAL against Guidewire is a question about *who the user is*.
- **Reference in your own jurisdiction and size band**, asking what the client built that they expected to configure.
- **Record it as an architecture decision record** with the alternatives and why they were excluded. In three years somebody will ask, and the regulator may be among them.
- **Name the reversibility.** A decision that cannot be undone belongs at board level.

And now it is also a regulatory filing

If the arrangement is **material outsourcing** under Joint Standard 1 of 2024, you owe both regulators 30 days' notice, board-policy confirmation, documented due diligence and a concentration-risk assessment before entering it [69][70][71]. Whether a cloud core or a binder arrangement is "material" is **not settled in the material used here**. But the consequence holds either way: the day you finalise a shortlist, you may be starting a regulatory clock. Plan for it in the schedule, and put the concentration-risk assessment in the architecture decision record — it is an architecture argument before it is a compliance artefact.

And once more, because it matters

Every company named in this chapter is an example of who plays in a category. Nothing here is a recommendation, a ranking, an endorsement or a shortlist.

The research was done in **October 2026**. Ownership will be wrong first, product names second, client references third. The category boundaries will hold a few years; the questions each category exists to answer will outlast this book.

One sentence to remember: **decompose the suite, name the capability boundaries before you meet the vendors, and ask who the user of each system actually is.**

THE CHAPTER IN TEN LINES

- Every vendor named in this chapter is an example of who plays in a category, not a recommendation, an endorsement or a shortlist.
- The research cut-off is October 2026: ownership goes stale first, then product names, then client references; the questions each category exists to answer outlast all of them.
- Organise the market by capability, not by vendor, because suites are built to hide the boundaries an architect needs to see.
- A vendor appearing in three or more capability columns is a concentration decision, not a product decision.
- Where rating lives decides whether pricing can act without a core release, and that is usually the most consequential single boundary in the estate.
- Santam's Guidewire arc — underwriting selected December 2012, claims February 2019, go-live announced June 2026 — is elapsed market reality, not an implementation benchmark, and it should end any three-year core replacement plan.
- Discovery Health deliberately bought a collections and disbursements component and kept its own policy administration: that is what refusing the one-platform trap looks like.
- South Africa's delegated-authority layer creates a software category whose user is the agency or administrator, with the insurer as a reporting consumer.
- Collections is a retention system in this market, and almost no global core handles DebiCheck, RM presentment windows or reason-coded retries natively.
- Under Joint Standard 1 of 2024 a material outsourcing carries 30-day pre-notification to both regulators and a documented concentration-risk assessment, so the shortlist may also be a filing.

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CHAPTER 15

Regulation, risk and what the regulator inspects

A regulator never asks how your systems are built. It asks questions that only a particular kind of estate can answer — twice, on time, with evidence. Every rule in this chapter lands somewhere in your model.

What inspection actually looks like

Riaan Botha, Cape Mutual's chief risk officer, has sat through four supervisory engagements. (Cape Mutual is invented for this book; the pattern is not.) Not one contained a question about technology.

The Prudential Authority asked this. Show us the solvency capital requirement for the quarter, the data it ran on, the assumption set and its version, who signed it off and when. Now show us the same calculation for the previous quarter, and explain the movement.

The FSCA, in a separate engagement, asked something with nothing in common. Show us every claim repudiated for non-disclosure in the last twelve months, by product, channel and reason code; the complaints that followed, how long each took and what redress was paid; and the advice records for policies sold by the three advisers with the highest early lapse rates.

Neither regulator asked how the systems were built. Both asked questions only a particular kind of estate can answer. That is this chapter: regulation does not demand architecture, it demands **answers that reproduce** — and reproducible answers have three properties most estates lack.

- **Reproducibility.** The same question, asked twice over the same period, returns the same number — after a restatement, a model change or a staff departure.
- **Lineage.** You can walk from the published number back to the transaction through every transformation, and the walk is recorded rather than remembered.
- **Attribution.** For any judgement inside the number — a reserve estimate, an underwriting decline, a repudiation — you can name who or what made it, on what inputs, under which version of which rule.

Everything below lands on one of those.

Twin Peaks, and why South Africa has two regulators

South Africa supervises financial institutions through two authorities established by the Financial Sector Regulation Act of 2017. The **Prudential Authority**, inside the South African Reserve Bank, supervises safety and soundness: can the insurer pay what it has promised. The **Financial Sector Conduct Authority** supervises market conduct: does the insurer treat its customers fairly. Where a matter touches both, they issue **Joint Standards** together [1][9].

The split is not administrative tidiness. Prudential and conduct failure have opposite incentives: declining valid claims improves solvency, and an insurer can be generous to the point of insolvency. A single supervisor holding both mandates trades them off invisibly. Two force the trade-off into the open.

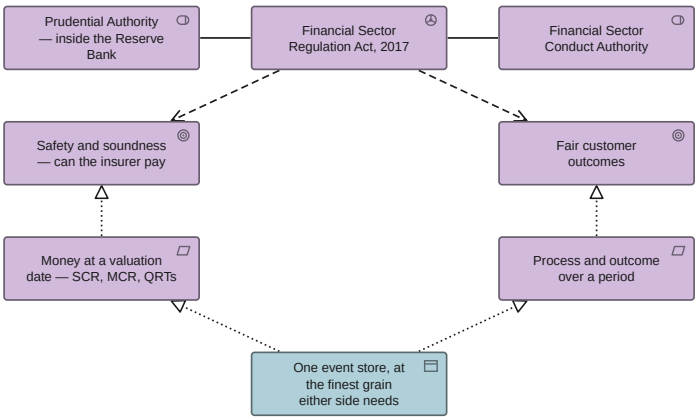


Figure 15.1 Twin Peaks as a motivation view: one Act, two supervisors, two goals, two returns at two different grains. The design rule is the Data Object at the bottom — build the event record once, at the finest grain either side needs, and realise both returns from it.

For an architect this has one dominant consequence, and getting it wrong is expensive. The two regulators want data at fundamentally different grains. The Prudential Authority wants **money at a valuation date**: balances, provisions, capital, exposures, as at a point in time, consolidated. The FSCA wants **process and outcome over a period**: who was advised, what was sold, how long it took, what the customer asked for, what they got, who complained.

The temptation is to build two reporting estates, because the demands look unrelated. They are not: they are two derivations from the same events. A claim payment is a cash flow for the Prudential Authority and a turnaround time with an outcome code for the FSCA. Build the event record once, at the finest grain any consumer needs, and generate both returns. Reconciling two independently sourced regulatory datasets is a permanent cost with no upside.

The Prudential Authority: licence classes are a data attribute

The **Insurance Act 18 of 2017** commenced on 1 July 2018 with 42 Prudential Standards covering governance, risk management, internal controls and financial soundness, with dedicated standards for insurers, microinsurers, branches of foreign reinsurers and Lloyd's, and insurance groups. Requirements apply **proportionally** to the nature, scale and complexity of each insurer's risks. The Act also created a microinsurance framework aimed at financial inclusion; a microinsurance licence may include both life and non-life products [1].

The part architects skip, and should not, is the licensing structure. The Act restructured licensing into **classes and sub-classes** set out in Schedule 2. Life business comprises Risk, Fund Risk, Credit Life, Funeral, Life Annuities, Individual Investment, Fund Investment, Income Drawdown and Reinsurance — Risk splitting into individual and group death, health, lump-sum disability and recurring-payment disability, and the investment classes into guaranteed, market-related, linked and with-discretionary-participation [2]. Non-life business comprises Motor, Property, Agriculture, Engineering, Marine, Aviation, Transport, Rail, Legal Expense, Liability, Consumer Credit, Trade Credit, Guarantee, Accident and Health, Travel, Miscellaneous and Reinsurance, mostly split into personal and commercial lines [3].

> Licence class and sub-class is an attribute of every product, derivable for every contract in force. Not a reference-data footnote. A product that drifts outside its sub-class is unlicensed business, which is a licence matter rather than a reporting error; returns are reported by class, so the mapping drives the numbers; and licence conditions can restrict particular sub-classes for a particular insurer.

Where this breaks is predictable. Product taxonomies get invented by marketing and by whoever configured the product factory, with no mapping to the statutory classes, and a "miscellaneous" bucket appears and grows because mapping an awkward product is harder than parking it. Then a configurable product platform — the thing everyone wants, for good reasons — lets a business user launch a variant in a fortnight with no class decision attached.

The demarcation boundary makes the stakes concrete. Since 1 April 2017 health insurance products are prohibited unless the demarcation regulations exempt them: gap cover and hospital cash plans are permitted, primary healthcare insurance is not. A product that drifts across that line is not mis-classified, it is prohibited.

SAM, the solvency regime, and the QRT burden

South Africa's prudential regime is **Solvency Assessment and Management**, Solvency II-aligned, developed from 2009 through three Quantitative Impact Studies and a parallel reporting period before go-live, and implemented through the Insurance Act and its Prudential Standards [1].

The mechanics, in the order they matter to a data architect. Valuation is **economic, not accounting**: assets and liabilities at market-consistent value. **Technical provisions** are the best estimate liability — the probability-weighted present value of future cash flows — plus a **risk margin**. The **solvency capital requirement** is calculated by standard formula or an approved internal model, decomposed by risk module: market, credit, life underwriting, non-life underwriting, operational. Below it sits a lower **minimum capital requirement**. Against both stand **own funds**, tiered by loss-absorbency. Alongside the numbers runs the **own risk and solvency assessment** and a set of statutory control functions including a Head of Actuarial Function.

At March 2026 the Prudential Authority reported median solvency cover of 1.8 times for life primary insurers and 1.7 times for non-life primary insurers — and 1.2 times for non-life cell captives, a thinner cushion on a fast-growing segment [4].

Insurers file **quantitative reporting templates**, at solo and group level. The architectural fact about QRTs is not their content but their mutability: the set is amended by Prudential Notices and Communications continuously. A 2022 Prudential Notice amended the solo template [7]; Prudential Communication 6 of 2025 replaced one outright [6].

> Never build a regulatory report to the shape of the current template. Hold the data at a grain finer than any template needs, with the dimensional breakdowns the regulator reasons in, and generate the template as a presentation step. Insurers that hard-coded QRT structures into extract logic have paid for the same programme three times. For a free, rigorous statement of what a prudential regulator means by each data point, the European supervisor publishes its Data Point Model and XBRL taxonomies openly — not a schema to adopt, but a specification to mine [24].

What SAM demands of data: asset holdings at market value with look-through into collective vehicles; liability **cash flows projected forward**, not merely balances carried; exposure and risk-characteristic data at the grain each risk module needs; counterparty data for credit risk; and all of it as at a valuation date with a recorded, re-runnable assumption set. The failures are familiar enough to have been catalogued [23] — actuarial models fed by hand-built spreadsheets, a quarterly extract reconciled to the general ledger with a balancing plug nobody owns, and the one that ends careers, an assumption set overwritten in place so that last quarter's number cannot be reproduced at all.

A worked example of regulatory lag: one cash flow, three bases

This is the most useful single thing in the chapter, because it is the shape of the problem rather than an instance.

IFRS 17 took effect for annual periods beginning on or after 1 January 2023. The Prudential Authority's QRTs were designed under IFRS 4. They are therefore not aligned with the accounting basis the insurer now reports on, and the Authority issued interim guidance saying so explicitly. Under Prudential Communication 7 of 2023 [5]:

- Premium terms — written, earned, unearned — **keep their pre-IFRS 17 meaning** in the QRT, though IFRS 17 abolished premium as a revenue measure.
- Deferred acquisition costs are not shown separately; they sit inside the insurance contract liability.
- Gross technical provisions are reported as one total, without the breakdown previously given.
- Where the liability for remaining coverage is an **asset**, it goes under "Other Assets" with a disclosing comment.

- The first post-implementation submission reports the current period on IFRS 17 with the prior period left on IFRS 4, comparatives not restated; subsequent submissions are all IFRS 17.
- Enabling assumptions must be explained in the QRT comment sections.

Read that as an architect. A single premium instalment, collected once from one policyholder, must simultaneously support an IFRS 17 measurement at group grain, a pre-IFRS 17 "written premium" definition for the prudential return, and a SAM economic valuation that is neither. Three bases over one event.

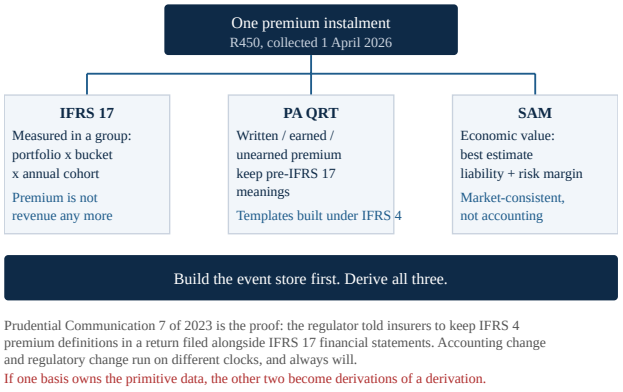


Figure 15.2 Regulatory lag, made concrete. One cash flow, three simultaneous measurement bases, none of which can be derived reliably from either of the others.

This is not a transitional nuisance that resolves when the templates catch up. Regulatory and accounting change arrive on different clocks, from different bodies, with different transition rules, forever. The rule that follows is non-negotiable: **build the event store first, and derive every basis from it.** The moment the IFRS 17 engine becomes the only place the cash flow exists at full grain, the prudential return is a derivation of a derivation, and nobody can explain why two numbers differ.

One honesty note: I could not pin down the operative 2026 South African QRT set from public material. Later Communications replace templates, so the Authority's site is the only reliable source for what is live today [6]. Use this section for the mechanism, not the template list.

The FSCA: conduct, and why it is the hardest thing to retrofit

Final insurance conduct regulations and the **Policyholder Protection Rules** took effect on 1 January 2018. They introduced principles for premiums and premium reviews; **minimum data governance requirements**; rules on negative-option marketing; minimum claims-management requirements; complaints monitoring, analysis and reporting to the regulator and the Ombuds; limits on who may hold a **binder**, which activities may be outsourced under one and what remuneration may be paid; the principle that an insurer outsourcing critical functions remains ultimately responsible; and a progressive reduction of maximum **causal event charges** on pre-2009 legacy life policies on lapse, surrender, transfer or premium reduction [8].

Two of those deserve an architect's attention.

Minimum data governance requirements, written into insurance conduct regulation in 2018. A conduct regulator legislating data governance is the clearest signal available that conduct supervision is a data problem, not a policy-document problem. If your data governance programme is still justified on internal grounds, it has a statutory basis you are not using.

Causal event charges, which look like an actuarial curiosity and are a textbook architecture problem. The rule applies only to contracts issued before a date, bites on a specific set of events, and its maximum **changes over time**. That is version-dated product logic on a closed book, resolved as at the event date, in a system whose product logic was never designed to be versioned — and the reason you cannot freeze legacy product rules and walk away.

Treating Customers Fairly: six outcomes, six design consequences

TCF is the outcomes-based philosophy under which the FSCA supervises conduct, adopted from the UK and carried into South African practice, and the policy spine of the COFI reform intended to give it statutory teeth [10][11]. Architects are usually shown the six outcomes as a poster. Here is what each demands of a system.

TCF outcome	What it demands
1. Fair treatment is central to the firm's culture	Evidence, not assertion: decision logs, exception approvals, who overrode what and why. A culture claim you cannot evidence from data is discounted
2. Products are designed to meet the needs of identified customer groups	**Design assumptions must be data, not a PDF** — target market, expected claims experience and value proposition, machine-readable so experience can be compared against them
3. Customers get clear information before, during and after sale	Every communication reproducible as issued: template version, merged data, date, channel. Reconstructing it from today's template is not evidence
4. Advice is suitable and takes account of circumstances	The record of advice as a first-class entity, linked to the policy sold and the adviser who sold it
5. Products perform as customers were led to expect	Experience data comparable to outcome 2's assumptions, at product and cohort grain. Almost nobody can evidence this, because the assumptions were never stored
6. No unreasonable post-sale barriers to change, switch, claim or complain	Measured friction: cancellation reasons, replacement linkage, switching metrics, and event-level timestamps across the claim and complaint lifecycle rather than status flags

Outcomes 2 and 5 form a loop — design assumptions in, experience out, compared — broken in most insurers because the assumptions live in a pricing document. Fixing it is a modelling decision, not a reporting one.

Omni-CBR, the FSCA's cross-sectoral Conduct of Business Return, is how the regulator collects this. It defines standardised conduct indicators so conduct can be compared across the sector and supervision targeted, and is reported by providers, banks, insurers and microinsurers, investment providers, co-operative financial institutions, retirement funds and benefit administrators — **quarterly in the end state**. Its structure is general sheets every institution completes plus sector-specific business-composition sheets. The granularity is a step change: rather than a yes-or-no answer on conflicts of interest, providers must disclose affiliated parties, relationship details, financial arrangements and marketing associations, and institutions must **segment and report detailed client data including demographics and product holdings** [12][13].

That last requirement collides with POPIA minimisation and the rules on special personal information. Reconciling them is architectural work: demographic attributes consistently coded across every line, held under a lawful basis covering regulatory reporting, and tagged so the different legal gate on special categories is enforced automatically rather than by convention.

> The omni-CBR phase-in has slipped at least once. Full quarterly statutory conduct reporting was targeted for June 2026, but the operative status, template version and submission dates as at late 2026 are not something I could establish from public material. Check the FSCA's own publications before committing a date to a steering committee.

Modelling a regulatory requirement properly

This is where ArchiMate's Motivation elements earn their place, and where most models misuse them. A regulatory requirement is not a box labelled "compliance" bolted to an application. It is a chain from somebody who cares, through what they care about, to a specific demand on a specific capability — and the chain is what makes it arguable, traceable and testable.

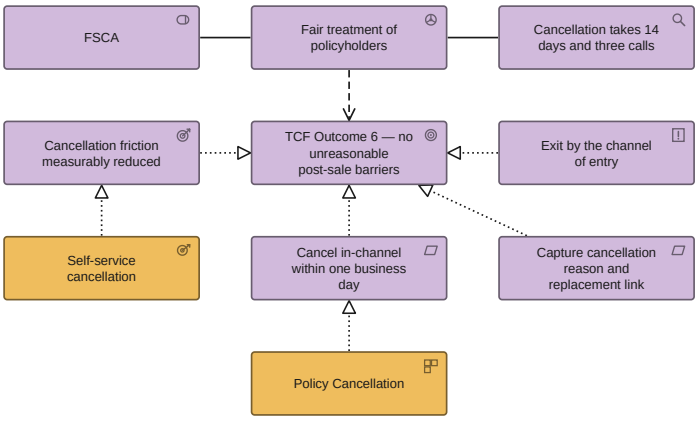


Figure 15.3 A regulatory requirement modelled properly. Realization points upward from the specific to the general; Influence carries contribution without guarantee; Association carries the stakeholder and the assessment. ArchiMate 4 has no Constraint element — a restriction is a Requirement.

Read the figure as a sentence. The **Stakeholder** is the party with an interest — the FSCA. The **Driver** is what that stakeholder cares about, stated as a concern rather than a target. The **Assessment** is an evidential finding about the driver: cancelling takes fourteen days and three telephone calls. The **Goal** is the intended end state. The **Outcome** is an achieved, measurable result, distinct from the goal and realised by a **Course of Action**. The **Principle** is a general rule you have adopted that holds beyond this case. The **Requirements** are the specific, testable demands that realise the goal.

Four conventions separate a motivation model from a mind map.

- **Realization points upward**, from specific to general: requirements realise goals, outcomes realise goals, and core elements — a capability, a process, an application component — realise requirements. If an arrow does not read "contributes to the achievement of", it is probably the wrong

relationship.

- **Influence is for effect without guarantee.** A driver influences a goal; a principle influences a requirement. Use the optional strength attribute where you need to say whether the influence is positive or negative.
- **Association carries the stakeholder and the assessment.** A stakeholder is associated with a driver; an assessment is associated with the driver it assesses. Neither realises anything.
- **ArchiMate 4 has no separate Constraint element.** A constraint is a Requirement — one that restricts rather than demands. Modellers carrying habits from earlier versions reach for Constraint and invent something; do not.

The payoff is practical. When the business asks why self-service cancellation is funded ahead of a new quote screen, the model answers in one picture: a named regulator, a documented finding, a testable requirement. Challenge the requirement later and you can see which goal it realises, and therefore what else must change if it is dropped.

FAIS and the adviser

The **Financial Advisory and Intermediary Services Act 37 of 2002** licenses financial services providers and their representatives: fit and proper requirements, competence, record-keeping on advice, disclosure. Brokers and UMAs are authorised providers under it. The FSCA plans no insurance- or FAIS-specific interventions for 2026 to 2029 — both go through the transition of sectoral laws into COFI instead [9].

Three design consequences, commonly missed:

- **A representative is a party with a licensed scope, not a name on a commission record.** Licence category, authorised product sub-categories, supervision status, and effective date ranges for all of it.
- **Point-of-sale authorisation is a check, not a report.** Is this representative authorised, today, for this product category? If the answer arrives in a monthly exception report, the policy is already sold.
- **The record of advice is an entity**, linked to the needs analysis, the options considered, the product recommended, the product actually sold and the adviser. A PDF with no link to the policy satisfies a filing requirement and fails an inspection.

What breaks: the representative register kept in a spreadsheet the sales system cannot read; commission paid to a representative whose authorisation lapsed two months ago; and a record of advice that exists for every sale except the ones where the customer declined the recommendation — exactly the ones a regulator asks about.

FICA, and its argument with POPIA

Under the **Financial Intelligence Centre Act**, accountable institutions — long-term insurers among them — must perform risk-based customer due diligence, maintain a Risk Management and Compliance Programme, keep records, and report suspicious and cash-threshold transactions. Verify specifics against the Act and the Centre's guidance: my sources for the detail redirected.

The architectural points hold regardless. Screening is not an onboarding event: the events that matter are the ones that change who benefits — a change of beneficiary, payee or bank account, a cession, a single-premium top-up. An estate that screens at onboarding and nowhere else is screening the least interesting moment in the contract's life.

Then the tension. POPIA says minimise and delete; FICA says retain and report. Both are law, and the Financial Intelligence Centre has had to confirm that privacy law does not prevent regulatory reporting. The resolution is never one corporate retention period. It is **retention per purpose**, with the lawful basis recorded alongside, so a deletion request can be answered with "these elements are retained under statutory obligation, these are deleted" — and evidenced.

POPIA, and the two sections that catch insurers

POPIA term	Means	Elsewhere
Data subject	The person the data is about: policyholder, life assured, beneficiary, third-party claimant, witness, dependant	Data subject
Responsible party	Determines purpose and means — the insurer	Controller
Operator	Processes on behalf of and under the authority of a responsible party — the UMA, the TPA, the cloud host, the call centre	Processor
Special personal information	Health, sex life, religious belief, race, trade union membership, political persuasion, criminal behaviour, biometrics. Prohibited under section 26 unless section 27 or an authorisation applies	Special category data

Insurance is a hard case because the business *requires* special personal information. Life and medical underwriting is health data processing; fraud investigation is criminal-behaviour and biometric data processing. No version of this industry avoids the prohibited categories, only versions that handle the gates properly.

The health regulations are now in force. The *Regulations Relating to the Processing of Data Subjects' Health Information by Certain Responsible Parties, 2026* took effect on **6 March 2026**, made under section 112(2)(c) to assist interpretation of section 32(6). They apply to a named list that expressly **includes insurance companies**, alongside medical schemes, administrators, managed healthcare organisations, pension funds and employers. The obligations: process health information only where section 27 permits; technical and organisational measures against loss, damage, unauthorised destruction and unlawful access; security and confidentiality of health records both **physical and electronic**; disposal that prevents later unauthorised access; a legal, employment, professional or written duty of confidentiality; alignment to generally accepted sectoral information-security practice; and cross-border transfer only where section 72(1) is satisfied [14].

Note what that touches: the paper file in the underwriting office, the scanned claim attachment, the backup, the offshore group reporting extract, and the sectoral security baseline you are now measured against. Health data becomes a separately regulated class with its own disposal and transfer regime.

Section 71(1): automated decisions. A data subject is protected from a decision based **solely** on automated processing that has legal consequences or significantly affects them, including profiling, unless section 71(2) applies. The published worked example concerns lending — a bank cannot grant or reject a loan application based solely on an AI-generated creditworthiness profile [15]. The insurance analogues are obvious: automated underwriting declines, algorithmic pricing, repudiation driven by a fraud score. I found no authoritative insurance-specific reading, so treat the application as inference rather than settled law. The design implication is unambiguous either way: **record whether a decision was wholly automated**, and preserve the inputs and the rule or model version that produced it. That is the attribution property from the start of this chapter — cheap at design time, close to impossible to retrofit.

Section 57(1)(a): unique identifiers. Prior authorisation from the Information Regulator is required to process a unique identifier for a purpose other than that for which it was collected, where it will be **linked with information held by other responsible parties**. Read that against what a cross-insurer fraud consortium does. Shared claims databases, industry data exchanges and bureau enrichment all work by taking an identity key and matching it against data held by other insurers — the described activity, almost word for word.

> Joining your data to another insurer's on a shared identifier is an authorisation-gated flow, not an integration. Keep a register of which identifiers leave for which purpose under which authorisation, make each flow independently switchable, and never let a consortium feed become a silent dependency of claims decisioning. If the authorisation lapses or is refused, claims must still work.

Two more shape the model. **Third-party claimants** are data subjects whose personal and often health data you hold although you have no contract with them and took no consent — so their lawful basis, purpose limitation and retention all differ from a policyholder's, which is the practical argument for modelling parties by role rather than as customers. And **purpose compatibility** governs secondary use: training or testing a model on production personal data must be compatible with the original purpose or carry its own basis, making de-identification an architecture capability rather than a data-science convenience.

What POPIA makes architectural: lawful basis, purpose, consent and source recorded per party per purpose with dates; special-personal-information elements tagged in the catalogue, because a different legal gate applies; retention enforceable at field level, with deletion that works through warehouse, lake, backups and reinsurer bordereaux; an operator inventory with contracts, because every UMA, TPA, assessor, panel supplier and cloud service is an operator; a cross-border transfer register with a section 72 analysis per flow; and data-subject access, correction and objection as **supported operations** rather than manual projects.

IFRS 17

IFRS 17 took effect for annual periods beginning on or after 1 January 2023, replacing IFRS 4. For an accountant it is a measurement standard. For an architect it is a **mandatory change to the unit of account**, and therefore to the grain of every finance number the insurer produces.

The General Measurement Model and its four building blocks

The default model — also called the building-block approach — measures a group of contracts as four components.

1. **Fulfilment cash flows.** Probability-weighted estimates of future cash flows within the contract boundary: premiums in, claims and benefits out, directly attributable expenses, acquisition cash flows. Explicit, current and unbiased — no prudential margins hidden inside.
2. **Discounting.** The time value of money, on a curve reflecting the liability's characteristics. The inception, or **locked-in**, rate must be preserved per group indefinitely, because the contractual service margin accretes interest at it.
3. **Risk adjustment.** Explicit compensation for bearing non-financial risk, separately determined and disclosed. Under IFRS 4 this was an implicit margin inside the reserve; now it is a number with its own methodology and audit trail.
4. **Contractual service margin.** Unearned profit, held at inception and released to profit as service is delivered, measured by **coverage units**. It cannot be negative: where the group is onerous, a loss is recognised immediately through a loss component.

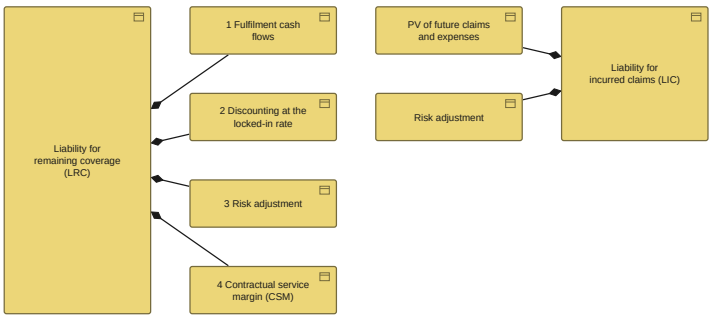


Figure 15.4 The four building blocks, and the two liabilities they compose — Composition because a block exists only inside the measurement it belongs to, and because the risk adjustment on the left is not the same object as the one on the right. Blocks two and four are the ones that create data obligations rather than accounting entries: a locked-in rate and a coverage-unit basis must survive every migration.

The liability splits in two. The **liability for remaining coverage** covers service not yet provided and, under the general model, equals the present value of future cash flows plus risk adjustment plus CSM. The **liability for incurred claims** covers claims already incurred and has two components — the present value of future expected claims and expenses, and a risk adjustment — spanning IBNR, IBNER and case reserves for expired risk. Cash flows expected more than twelve months after the incurred date are **required to be discounted**, a change from much existing short-term reserving practice [17].

The Premium Allocation Approach, and its eligibility test

PAA is the simplification for the liability for remaining coverage. It is permitted where the result would not differ materially from the general model, **or** where the coverage period is one year or less. Under it the liability is premiums received less premiums allocated to the period, and acquisition costs paid less the amortised portion, with no need to separate the components. Where coverage is one year or less, discounting may be omitted and acquisition costs expensed as incurred [18][19].

> Here is the trap, and it has caught a good many short-term programmes. PAA is not a free pass on data. You still need general-model data to run the **eligibility test** — projecting the liability for remaining coverage under both approaches, on base and sensitivity assumptions, to show the results do not differ materially — and for **onerous-contract testing** and loss-component tracking. Two sets of data are generally needed. An insurer that elected PAA "to avoid the complexity" is running a shadow general-model calculation whether it planned one or not.

The Variable Fee Approach

VFA is **mandatory**, not optional, where three conditions hold at inception: the policyholder participates in a clearly identified pool of underlying items, receives a substantial share of their returns, and the benefits vary with those items. Reinsurance held cannot use it. It requires the fair value of the underlying items and the entity's share, plus documentation for any risk-mitigation exception. Its defining mechanic: changes in financial risk and time value relating to future service **adjust the CSM** instead of hitting profit or loss. It targets unit-linked and with-profits business [18].

The architectural consequence is underestimated: under VFA the **asset side becomes an input to liability measurement**. Fund valuations, unit prices and asset allocation must be available at the measurement grain and date, coupling the investment platform into the finance close in a way IFRS 4 never did. If unit pricing runs on a different calendar from the reporting close, that is now a reporting defect.

Level of aggregation: the thing that changes your data model

IFRS 17 measures **groups**, not contracts. The grouping is built in three steps [16].

A **portfolio** is contracts exposed to similar risks and managed together — typically one product line: whole life, annuities, motor. Each portfolio divides at initial recognition into **profitability buckets**: contracts onerous at inception, which recognise a loss immediately and carry no CSM; contracts with no significant possibility of becoming onerous; and the remainder. The division exists so that losses cannot be hidden by offsetting them against profitable contracts in a larger pool. Each bucket then divides by issue period into **annual cohorts**, generally one year or less, keeping profitability trends visible as pricing and conditions shift.

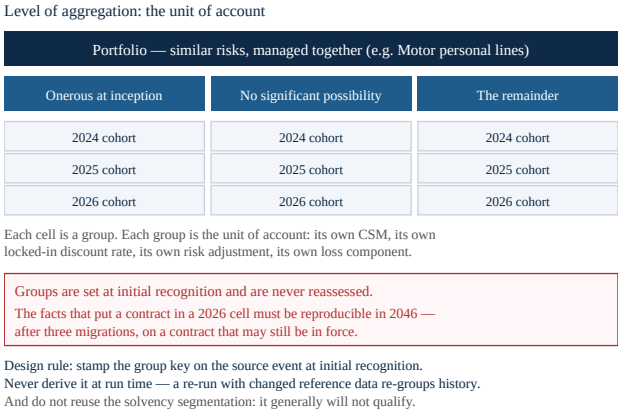


Figure 15.5 Portfolio, profitability bucket, annual cohort. Multiply those three dimensions and you have the grain at which every cash flow must be keyed, permanently.

Two constraints follow that are easy to read past and expensive to miss.

Groups are set at initial recognition and are not reassessed during the coverage period [16] — a permanent retrospective obligation. The facts that determined a 2023 assignment must be reproducible in 2043, on a contract that may still be in force, after three migrations. Most insurers therefore compute at the minimum permissible level, because finer granularity multiplies volume and cost with no accounting benefit.

Do not reuse solvency groupings. They generally will not work for IFRS 17, so a new grouping policy and a data remediation exercise are usually needed [16]. If someone proposes deriving IFRS 17 groups from the existing SAM segmentation to save time, push back.

What all of that means for data

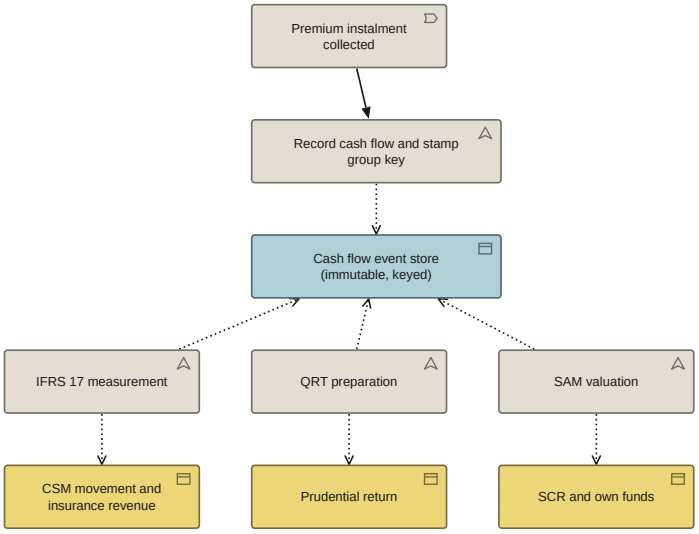


Figure 15.6 One event, keyed once, read three ways. The group key is stamped where the cash flow is recorded, not computed inside the measurement engine — which is the difference between lineage you can walk and a number you have to defend from memory.

As design rules:

- **Every cash flow carries an immutable group key** — portfolio, profitability bucket, annual cohort — stamped at initial recognition on the source event, not computed later by the measurement engine. Derive the key at run time and a re-run with changed reference data silently re-groups history.
- **Inception assumptions are preserved indefinitely**, including the locked-in discount rate per group, and **coverage units** — the basis on which CSM releases to profit — must be defined, stored and auditable per group.
- Claim payment patterns and outstanding reserves must be **resolvable by group**, usually a new requirement on claims data; direct and reinsurance cash flows must be **linked**, with reinsurance held measured separately group by group.
- Premium receipt timing and **instalment schedules** are measurement inputs, not billing details; acquisition cash flows and renewal expectations must support allocation between current and future renewal groups, with impairment testing.
- The insurance result must be separable from the finance result, and current-service changes from past-service changes.

Where it breaks: a sub-ledger reconciled to the general ledger by hand each month; group assignment computed in the engine rather than stamped on the source event; coverage units defined in one actuary's spreadsheet; and onerous-contract testing that needs data the policy administration system never captured. None of that is theoretical — Momentum Group disclosed an onerous contract value of R349 million for the 2026 financial year, down 39% [22]. Published, audited, moving, and something in the estate must produce it every six months.

Joint Standard 1 of 2024: vendor selection becomes a regulatory filing

Joint Standard 1 of 2024, issued by the Prudential Authority and the FSCA under the Financial Sector Regulation Act of 2017, replaced Prudential Standard GOI 5 and took effect on **1 December 2024**. It applies to all insurers licensed under the Insurance Act 2017, including micro-insurers and reinsurers; Lloyd's and branches of foreign reinsurers are excluded. Its obligations [20][21]:

- **Due diligence before outsourcing any activity**, proceeding only where benefits demonstrably outweigh costs and risks.
- Explicit consideration of **concentration risk** from a provider serving multiple insurers.
- A bar where the provider's key persons fail the governance standard's competence and integrity requirements.
- **Notification to both regulators at least 30 days before entering a material outsourcing arrangement**, confirming compliance with the board-approved outsourcing policy and risk appetite; and notification within **seven days of termination**.
- **Existing arrangements must comply within 24 months — by 30 November 2026** — or earlier on renewal.

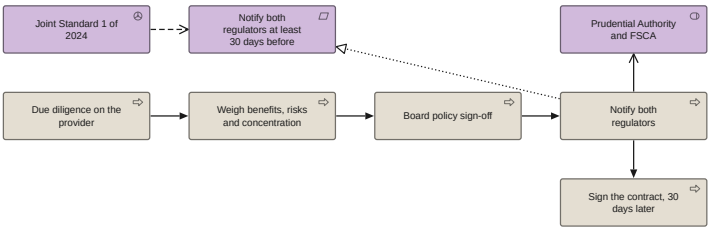


Figure 15.7 Joint Standard 1 of 2024 inserts a regulatory gate into procurement: a Driver that influences a Requirement, and a process that realises it before the contract can be signed. The 30 days is the visible cost; the board-policy sign-off that must precede the notification is usually the longer one.

If a software-as-a-service core is a material outsourcing arrangement, the shortlist is a regulatory filing. That changes procurement four ways. The timeline gains 30 days plus the board-policy sign-off preceding the notification — in most insurers a committee cycle, not a signature. The **concentration assessment becomes evidence** rather than a paragraph, and since one hyperscaler and a handful of core vendors serve most of the market, concentration is the default you must argue against. The **exit plan becomes a selection deliverable**, because you cannot show benefits outweigh risks without one. And the key-persons test makes the provider's own governance a gating item no feature-led evaluation will surface.

> Be careful here, and so was I. The practitioner analyses I relied on do not define "material", do not address binder and UMA arrangements or cloud and IT services explicitly, and one contains an unreconciled six-month reference alongside the 24-month deadline [20][21]. Whether a SaaS core, a cloud platform or a binder arrangement falls inside "material outsourcing" is a question for the standard's own text and for legal advice, not for an architecture book. Ask it early, in writing, and record the answer in the decision record — because if the answer is yes and you did not file, the remedy is not technical.

Draft Joint Standards on **operational risk and resilience** and on **cloud computing and the offshoring of data** sit in the pipeline alongside the COFI transition [9]. Cloud region, data residency, exit capability and provider concentration are becoming regulated matters rather than procurement preferences — and retrofitting data residency is a migration, not a configuration change.

A calendar worth carrying

Date	What	Why it matters
1 Jan 2018	Policyholder Protection Rules [8]	Data governance, complaints, binder limits, causal event charges
1 Jul 2018	Insurance Act 2017, 42 Prudential Standards [1]	Licence classes become product attributes
1 Jan 2023	IFRS 17 effective	New unit of account on every cash flow
1 Dec 2024	Joint Standard 1 of 2024 [20]	30-day pre-notification on material outsourcing
6 Mar 2026	POPIA health information regulations [14]	Health data separately regulated
30 Nov 2026	Existing outsourcing arrangements deadline [20]	The whole vendor estate, not the next deal
31 Dec 2026	JIBAR cessation planned; ZARONIA [9]	Discount curves, index-linked products, contract references
Pending	COFI Bill — gazetted 17 Apr 2026, awaiting tabling as at 3 Jul 2026 [9]	Conduct moves from rules to outcomes

What the regulator actually inspects

Six questions. If the estate answers them all without a project, the architecture is sound enough.

1. Can you reproduce last quarter's published number, from the data as it stood, with the assumption set as it stood?
2. Can you walk any published number back to the transactions behind it, through recorded transformations?
3. For any decision affecting a customer, can you say whether it was wholly automated, on what inputs, under which rule version, approved by whom?
4. Can you report by licence class, IFRS 17 group, distribution channel and product without a reconciliation step?
5. Can you name every operator holding policyholder data, under what contract, in which country, on which transfer basis?
6. Can you switch off a single external data flow without stopping business?

None of those is a technology question. All are answered by architecture, and all are cheap at design time. That asymmetry — cheap before, ruinous after — is why this chapter sits in a book for architects rather than one for compliance officers.

THE CHAPTER IN TEN LINES

- Regulation does not demand architecture; it demands answers that reproduce. Reproducibility, lineage and attribution are the three properties every rule in this chapter reduces to.
- Twin Peaks gives you two supervisors wanting data at two grains — money at a valuation date for the Prudential Authority, process and outcome over a period for the FSCA. Derive both from one event store; two warehouses means permanent reconciliation.
- Licence class and sub-class under Schedule 2 of the Insurance Act is an attribute of every product and must be derivable for every contract in force, because a product that drifts out of its sub-class is unlicensed business.
- Never build a regulatory report to the shape of the current QRT. Templates are replaced by Prudential Communications; hold the data finer than the template and generate it.
- The worked lesson in regulatory lag: the QRTs were designed under IFRS 4 and keep pre-IFRS 17 premium meanings, so one premium instalment must support an IFRS 17 measurement, a pre-IFRS 17 premium definition and a SAM economic valuation at once.
- Each of the six TCF outcomes is a data requirement. Outcomes 2 and 5 form a loop that is broken in most insurers, because product design assumptions live in a pricing document rather than as data.
- POPIA's health information regulations took effect on 6 March 2026 and name insurers expressly; section 57(1)(a) makes a cross-insurer fraud consortium an authorisation-gated flow rather than an integration; section 71 means you must record whether a decision was wholly automated, with its inputs and rule version.
- IFRS 17 measures groups, not contracts — portfolio, profitability bucket, annual cohort, set at initial recognition and never reassessed. Stamp the group key on the source event; electing the Premium Allocation Approach does not remove the need for general-model data.
- Joint Standard 1 of 2024 turns vendor selection into a regulatory filing: 30 days' notice to both regulators before a material outsourcing, and every existing arrangement compliant by 30 November 2026.
- Model a regulatory requirement with ArchiMate's Motivation elements properly — Realization upward, Influence for contribution without guarantee, Association for stakeholder and assessment — and note that ArchiMate 4 has no Constraint element: a restriction is a Requirement.

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CHAPTER 16

Modelling insurance in ArchiMate

Forty-two elements, a few hundred insurance concepts, and no dictionary. This chapter is the mapping — and the reasoning behind every choice, because the reasoning is what you will need for the concept this chapter missed.

Forty-two elements, a few hundred concepts

An architect arriving in an insurer wants a dictionary. Policy equals this element, claim equals that one. There is no such dictionary, and the reason is arithmetic. ArchiMate 4 gives you 42 elements, nine of which carry no layer at all: Role, Collaboration, Process, Function, Service, Event, Path, Grouping and Location. Insurance brings a few hundred terms. The glossary two chapters back ran to eighty-eight, and it was deliberately short; ACORD's business glossary passed six thousand terms in its January 2026 release [2].

So the mapping cannot be one concept to one element, and three things happen instead.

Many concepts collapse onto one element. Policyholder, life assured, beneficiary, claimant, cedant and intermediary of record are all Role. That is the notation telling you something true: these are positions a party occupies, and the differences live in the data, not the structure.

Some concepts are not elements at all. Sum insured, excess, limit, waiting period, attachment point and retention are attributes. Drawn as elements they produce a diagram that is unarguably complete and that nobody will read.

A few need a relationship rather than an element. Premium moving from a broker's trust account to the insurer is a Flow. A peril triggering covers across three sections of one policy is an Association whose cardinality ArchiMate cannot express.

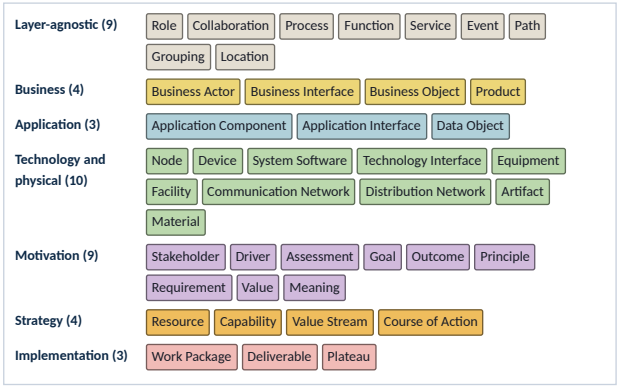


Figure 16.1 The whole vocabulary, by domain. Nine elements carry no layer, so in this notation a Process is the same colour whether it is business or application behaviour. Everything an insurance model needs is in here, and nothing in here is insurance-specific.

> **THE ONE RULE THAT PREVENTS MOST OF THE DAMAGE**
>
> ArchiMate models **types**, not instances. There is no element for policy 4471920, no element for Mrs Dlamini, no cardinality, no effective dating. If you need to say "a policy has one or more risk objects, each with a dated sum insured", you need a data model, and you should build one. The ArchiMate model answers a different question: what kinds of thing exist, what behaviour acts on them, and which system is responsible. Architects who try to make ArchiMate do entity-relationship work produce something that fails as both.

One more thing before the decisions. The layer-agnostic elements are drawn in a single neutral colour, so a Process looks the same whether it is business or application behaviour. Colour no longer tells you the layer; context does — what the element is assigned from, what it realises, what it accesses. That is uncomfortable for a few weeks and then it is useful, because insurance is full of behaviour that sits across the seam. Automated underwriting is one decision implemented partly as an application function and partly as a human referral, and a notation that forces you to pick one is lying for you.

Five decisions that settle most of the model

Almost every argument in an insurance model reduces to one of five questions. Learn each test and you will answer the rest by analogy.

Is it a business object or a product?

A **Product** is what the insurer sells, priced and bundled, with a contract attached. A **Business Object** is a thing the insurer holds information about and acts on. The test: can you buy it? Motor Comprehensive is a product. The policy that results is a business object. The schedule, the wording, the claim, the cession and the premium are all business objects.

The trap is the other direction. Insurers habitually say "product" when they mean the policy ("the product lapsed"), and architects obligingly draw a Product element with a lifecycle. A Product has no lifecycle. It has versions, and versions are data (see below). If you find yourself wanting to say "this product was cancelled on 3 March", you are talking about a policy.

Is it a service or a process?

A **Service** is behaviour visible from outside, with a consumer who would notice its absence. A **Process** is the sequence inside that produces it. The test has two parts: would someone outside the boundary ask for this by name, and could you reorganise it completely without them caring? If both, it is a service and the sequence behind it is a process.

Insurance gets this wrong in a specific way. Departments are named after functions — Underwriting, Claims, Collections — and those names arrive in the model as processes. They are not processes. "Underwriting" has no sequence; it is behaviour with no fixed order, which is exactly what a **Function** is for. "Underwrite a commercial property risk" does have a sequence, and it is a process. Draw the function for the operating model, the process for the flow of work, and link them with Realization.

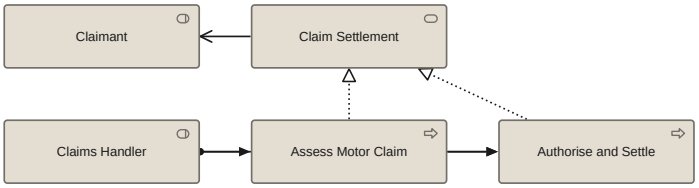


Figure 16.2 Service or process. Claim Settlement is what the claimant consumes and would notice the absence of; the two processes are how the insurer produces it and can be reorganised without telling anyone. The Claims Handler role is assigned to the processes, never to the service.

Is a channel an interface or an actor?

"Channel" conflates three different things and the model must separate them: the **party** (a broker, a bank, a retailer), the **access point** (a portal, a branch counter, USSD, a WhatsApp number), and the **agreement** (a binder, an affinity contract, a mandate).

The party is a Business Actor in a Role. The access point is a Business Interface, with whatever Application Component sits behind it. The agreement is a Business Object, and sometimes a Collaboration (next decision). The test for interface versus actor is whether the thing can decide. A broker decides which insurer to approach; a portal cannot decide anything.

Why it matters more here than elsewhere: the real architectural question about a channel is an entitlement question. May it quote? Bind? Issue? Endorse? Settle a claim up to what value? That can only be asked of a party in a role under an agreement. Draw "Digital Channel" as a Business Actor and the question becomes unaskable, which is roughly how insurers come to have four quote paths with three sets of rules.

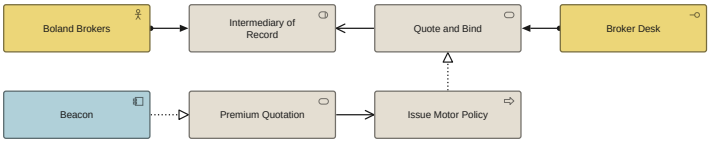


Figure 16.3 A channel is three things. Boland Brokers is the actor; Intermediary of Record is what it is on this contract; Broker Desk is the access point exposing the service. The entitlement question — may this party bind? — can only be asked of the role. Note the two Service elements: Premium Quotation is an application service, Quote and Bind a business service, and only the context says so.

Is a UMA an actor or a collaboration?

Both, and the distinction is where the oversight problem becomes visible.

An underwriting management agency is a company. That is a Business Actor. Within its mandate it holds a position — binder holder — and that is a Role, assigned to the processes it performs: underwrite, issue, service, and often settle claims. So far, ordinary.

But the insurer remains liable for every policy the UMA issues and must pay valid claims, and an insurer outsourcing critical functions remains ultimately responsible [17][12]. That liability is not performed by either party alone. It is performed by the arrangement: the UMA exercising authority, the insurer monitoring, both bound by the binder agreement. That is a **Collaboration** — two or more actors co-operating to perform behaviour neither performs alone.

The arrangement carries obligations too: auditable records kept five years, reporting to a standard the insurer does not control [16]. The practical test: if the behaviour is wholly inside the mandate, model the Role. If it crosses the mandate — a referral above authority, a claim above the settlement limit, the quarterly performance review, the aggregate exposure the insurer cannot derive from its own transactions — model the Collaboration, and give the insurer a visible oversight process. A model with a UMA and no oversight process is describing an arrangement that the regulator would not recognise.

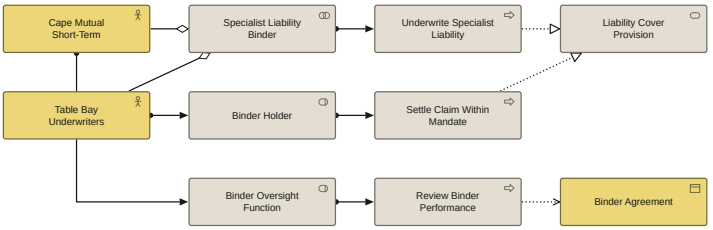


Figure 16.4 A UMA is an actor, a role and a collaboration. Inside the mandate, Table Bay acts alone in the binder-holder role. The underwriting authority itself belongs to the arrangement, which is why the Collaboration is assigned to it — and why the oversight process and the binder agreement have to appear. A UMA drawn without oversight describes an arrangement no regulator would recognise.

Is it an element at all?

The last and most useful decision. An element earns its place if behaviour acts on it, something depends on it, or someone owns it. If a concept is a property of another concept, write it down in the data model and keep it out of the picture.

Excess, sum insured, limit, sub-limit, waiting period, retention, attachment point, no-claim bonus step, commission rate: attributes. Peril: an element, but reference data rather than an event. Policy version: an element, but only because cover validation reads the version rather than the policy, and that Access relationship is the whole point.

> **THE COMPLETENESS TRAP**
>
> There is a kind of insurance model, produced around month four of a programme, that contains every noun in the policy wording. Six hundred elements, it validates, and it answers no question anyone asked. The test for an element is not "is it real" but "does a decision change if it is here".

The mapping

Here is the mapping itself, in six parts. The third column is the part that matters: it is the reasoning, and the reasoning is what you will use on the concept this table does not contain.

Parties and distribution

Insurance concept	ArchiMate element	Reasoning
Party — a natural or juristic person	Business Actor	Identity, not behaviour. One actor, many roles, modelled once. There is no Customer element because there is no customer.
Policyholder / owner	Role	A position on one contract for a period: owes premium, may cancel, amend or assign.
Insured, additional insured, life assured	Role	The interest protected, or the life the benefit rests on. Routinely not the policyholder — group, keyman and bancassurance all break that assumption.
Beneficiary	Role	Carries a share, a class and sometimes irrevocability. Those are data; the role is the structure.
Claimant and third party	Role	May be a stranger to the insurer with no contract and no consent given. Naming the role is what makes that visible.
Broker, bank, retailer, employer — the firm	Business Actor	A legal entity with its own licence, mandate and remuneration.
Intermediary of record	Role	What that firm is on this contract. Commission attaches to the role, not to the actor, which is why the seller and the payee can differ.
UMA, MGA, binder holder	Business Actor plus Role	The firm is the actor; binder holder is the bounded position it occupies.
The binder or outsourcing arrangement	Collaboration	Behaviour neither party performs alone, with the insurer still accountable. Add an oversight process or the model is wrong.
Underwriter, claims handler, assessor, actuary	Role	Never a Business Actor. People move; the process needs the position.

The contract and what it promises

Insurance concept	ArchiMate element	Reasoning
Product, or plan in life	Product	The one composite in the business domain: it aggregates the services it promises and composes the contract that governs them.
Product variant	Product, with Specialization	Family funeral plan specialises the base plan. Inherit; do not clone.
Product version	Not an element — data in the product factory	ArchiMate has no effective dating. One Product per rate revision gives you four hundred elements and no insight.
Policy	Business Object	The sold, in-force instance type. Not an instance: policy 4471920 is a row, not an element.
Policy version	Business Object	Earns its place only because behaviour distinguishes it: cover validation reads the version in force at date of loss.
Endorsement	Event, plus a Business Object	The event changes the contract; the document evidences it. Two different things, routinely drawn as one.
Schedule and policy wording	Business Object, realised by an Artifact	The schedule is the data; the rendered document is the artifact, and the archive must hold the artifact, not a re-render instruction.
Cover (short-term) / benefit (life)	Service	A promise of behaviour to a named consumer. Claims realise it. That edge is why cover is not an object.
Peril, and the insured event	Business Object for the peril, Event for the occurrence	A peril is a cause and a classification; the hailstorm is the occurrence. Confusing them destroys catastrophe analysis.
Limit, sum insured, excess, exclusion, waiting period	Not elements — attributes of a cover	Terms, not structure. Model them here and the diagram grows without becoming truer.
Risk object — vehicle, building, consignment, fleet	Business Object	First-class, with a life independent of any policy, as Oracle's published insurable-entity concept has it [18]. Make it a child of Policy and you cannot answer "have we insured this vehicle before".

Money

Insurance concept	ArchiMate element	Reasoning
Premium	Business Object, as dated components	Risk premium per cover, loadings, discounts, policy fee, commission, binder fee, tax. One premium amount answers no regulator's question.
Instalment	Business Object plus Event	A scheduled collection against premium, not premium. Model it as premium and endorsements become unhandleable.
Collection mandate (DebiCheck)	Business Object held by an external Business Actor	The authenticated mandate lives at the debtor's bank. Drawing it inside the insurer misstates who controls it.
Commission, binder fee, profit commission	Business Object; the calculation is a Function	Multi-party, time-shifted and reversible, and calculated over a book rather than a policy. A Function, because there is no fixed sequence.
VAT	Business Object as a premium component, plus a Requirement	South Africa runs insurance inside ordinary VAT, with zero-rating that turns on risk-object location and the insured's residence and vendor status [3].
Earned premium, loss ratio, combined ratio	Not elements — derived values	They belong in the Function that calculates them, with the definition pinned in a glossary. Two insurers' combined ratios are not the same number.
IFRS 17 group — portfolio, profitability bucket, annual cohort	Business Object	The unit of account, fixed at initial recognition and never reassessed [15]. Every cash flow carries its key into the sub-ledger [20], so the model must name it.

Claims

Insurance concept	ArchiMate element	Reasoning
Loss event, occurrence	Event	One event, thousands of claims. Catastrophe aggregation, event limits and hours clauses all key off the event, not the claim.
Claim	Business Object	An assertion against a policy arising from an event. Not the event, and not one per event.
Cover validation	Process, with Access to Policy Version	The Access edge is the only thing in the model that forces the as-at-date-of-loss question, which is where wrong claim decisions come from.
Case reserve, and its movements	Business Object; movements as Events	An estimate as at a date. The movement history is the actuary's raw material, so date it exactly and never overwrite it.
Claim payment	Business Object; paying is a Process	An immutable event, append-only. Do not effective-date it; that doubles the model for nothing.
Recovery, salvage, subrogation	Process plus Business Object	Not negative payments. Different counterparty, timeline and tax treatment, and in subrogation a new legal matter with its own parties.
Repudiation	Event, with a coded Business Object for the reason	A technical outcome, and now conduct-reporting data. Free-text reasons are a reporting failure waiting to happen [9].

Reinsurance

Insurance concept	ArchiMate element	Reasoning
Treaty	Business Object	A standing contract whose terms must be modelled data, not a PDF in a folder, if cession is to be computed rather than typed [5].
Cession	Business Object, written by a Process	The join between a direct policy version and a reinsurance contract. Without it, recoveries cannot be evidenced claim by claim.
Reinsurance cover	Service, realised by the reinsurer	This one edge reframes the picture: the insurer is a consumer of someone else's service, and that is what reinsurance is.
Bordereau	Business Object, realised by an Artifact, moved by a Flow	Periodic, itemised, bilaterally agreed and never identical between two counterparties [4]. The Artifact is the file; the Flow is the relationship that matters.
Reinsurance recoverable	Business Object	A balance-sheet asset and a credit exposure to a named actor. Model the actor or you cannot see the concentration.
Retention, attachment point, limit, reinstatement	Not elements — attributes of the treaty	Structure lives in the data. What belongs in the model is the behaviour that reads them.

Systems, strategy and the rules

Insurance concept	ArchiMate element	Reasoning
Policy administration system	Several Application Components	Never one. See the traps at the end of this chapter; this single choice decides whether a modernisation business case is arguable.
Rating engine	Application Component, plus a Service for premium calculation	The service is the thing four quote paths share. Draw the component only and the duplication stays invisible.
Product factory	Application Component; its content is a Data Object	The Data Object realises the contract the Product composes, and holds the versions and effective dates ArchiMate cannot.
Delegated-authority data feed	Artifact crossing a boundary, consumed by a Process	The bordereau boundary is the most consequential line in a South African non-life model. Draw the boundary, not just the file.
Capability	Capability	An ability the enterprise has, with an owner and a maturity. Not an activity, and not a department.
Operating-model function — Underwriting, Claims, Collections	Function	Behaviour with no fixed sequence, which realises a capability and is realised by processes.
Value stream — quote to bind, loss to settlement	Value Stream	Stages that deliver value to a named consumer, served by capabilities. The COO's view of the same facts.
Regulation — POPIA, the PPRs, IFRS 17	Driver, with the specific duty as a Requirement	A Driver cannot be realised by a system; a Requirement can. The Realization edge is the trace an auditor asks for.
Regulator — FSCA, Prudential Authority	Stakeholder, and Business Actor where it consumes a service	It is both: a stakeholder with concerns, and a counterparty that receives returns on a schedule.

Product, done properly

Product is the only composite element in the business domain, and insurance is the industry that needs it most. It exists to say: this bundle of promises, governed by this contract, is the thing we sell. Nothing else in the notation does that.

The structure is two relationships. The Product **aggregates** the services it promises and **composes** the contract that governs them. Aggregation for the services, because the same cover sells under several products — third-party liability appears in the cheap one and the expensive one, and composition would force a copy. Composition for the contract, because it exists only as part of that product.

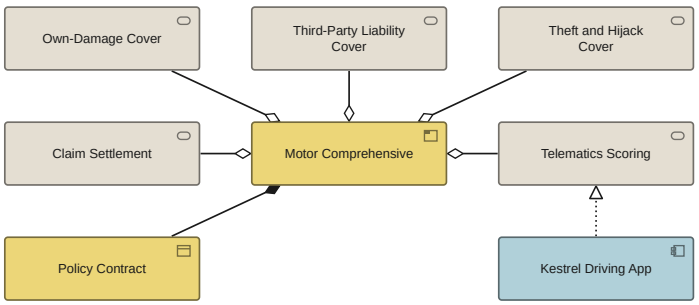


Figure 16.5 Product is the one composite in the business domain. It aggregates the promises — three covers, settlement, and the telematics scoring the customer also bought — and composes the contract that governs them. Aggregation for the services because the same cover sells under several products; composition for the contract because it exists only as part of this one.

Three choices in that figure are worth arguing, because each will be challenged.

Cover is a Service, not a Business Object. This is the one people fight about. The argument for an object is that cover looks like data: it has a limit, an excess, a start date. But those are the *terms* of the cover, which belong on the schedule, and the schedule is already an object. The cover itself is a promise of behaviour toward a named consumer, and the claims process is what makes good on it. Model cover as a Service and you can draw the Realization from the claims process to the promise, which is the single most useful edge in an insurance model: it is the answer to "what actually happens when we have to pay". Model cover as an object and that edge is not expressible, and the claims process ends up connected to nothing a customer bought.

The contract is one object, not three. In practice it is the wording plus the schedule plus any endorsements, and in a documents-heavy insurer it is worth composing all three, because output management, archiving and ombud evidence act on each separately. In a first pass, one Policy Contract object with the parts noted is enough. Say which you did and why.

Application and technology services belong in the product. A behaviour-based motor product priced off a telematics score is not a cover plus an app; the scoring is part of what the customer bought and part of what the insurer must deliver. The Product aggregates it and the Application Component realises it. That is the cleanest justification for the Product element in the notation, and the one that makes a product manager look at an architecture diagram with interest.

Product versus plan, and what to do about versions

Life says plan, short-term says product, and they mean the same construct. Use the Product element for both and stop worrying about it. The distinction that does matter, and that insurers lose money on, is the three-way split between **family**, **variant** and **version**.

A family is a set of products sharing structure: funeral cover in all its forms. A variant is a product in its own right, sold under its own terms — Family Funeral Plan, Extended-Family Funeral Plan, the retail-partner version carrying the partner's brand. A version is the same variant with different rates, wording or rules, effective from a date.

Model the variants as Products with a Specialization to the base, so you can say what they share. Model the family as a Grouping if you need to talk about it. Do **not** model versions. ArchiMate cannot effective-date anything, and a model with one Product per rate revision becomes unreadable inside a year while still answering nothing, because the question people actually ask — "what were this policy's terms on 14 March 2023?" — is answered by the product factory's version store, not by a diagram.

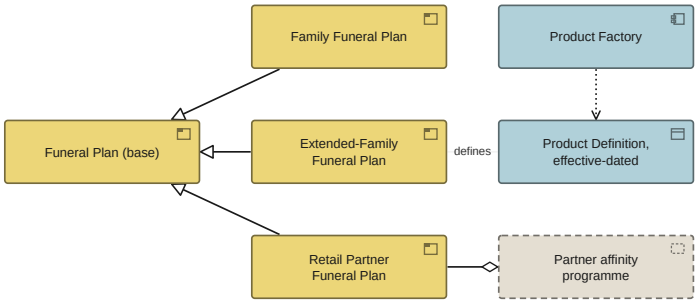


Figure 16.6 Variants inherit; versions are data. Three funeral variants specialise one base plan, and the retail-partner variant is aggregated into the partner's programme. The effective-dated definition lives as a Data Object in the product factory, because ArchiMate cannot date anything and a model with one Product per rate revision answers nothing.

> **WHAT THIS BUYS THE PROGRAMME**
>
> The separation above is the architectural form of the industry's oldest complaint: nine months to launch a product variant. If variants are Products with inherited structure and versions are data in one place, a variant is a configuration exercise. If variants are copies and versions are code, a variant is a release. The model does not fix that, but it makes the difference visible to the people who fund it.

Party and role, and why getting it wrong poisons a model

The costliest mistake an outsider makes in insurance is building Customer as an entity. The architectural version of the same mistake is drawing Policyholder as a Business Actor.

Here is what is actually true. One human being can be, at the same moment: policyholder on their own motor policy; life assured on a policy their employer owns; beneficiary on a spouse's policy; claimant against a stranger's liability cover; third party in another insurer's claim file; and, if licensed, an intermediary. Six positions, one party. The Party pattern — a supertype with roles carried on dated associations — is thirty years old in the data-model literature, and ACORD's Information Model names Party as one of its four core concepts beside Policy, Product and Claims [1].

Health informatics reached the same conclusion: HL7 names the insured as a role class, not a person [19]. In ArchiMate the party is a **Business Actor** and each position a **Role**, with Assignment from actor to role. The role is always a role *on something*: on a policy, a claim, a risk object, or toward the insurer itself.

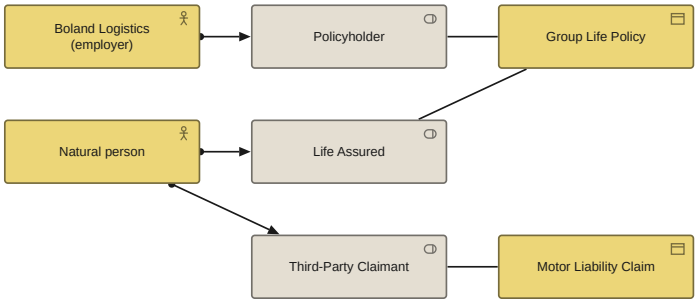


Figure 16.7 Parties play roles. The employer holds the group policy, the employee is the life assured on it, and the same human being is a third-party claimant on a motor liability claim where there is no contract and no consent. One actor, three positions, each a role on something.

Note what that figure says that the wrong version cannot. The employer is the policyholder and the employee is the life assured — the normal shape of group business, and the shape that breaks every model assuming policyholder equals insured. The same person is also a third-party claimant on a motor liability claim, with no contract and no consent given to anyone.

Now the version to refuse.

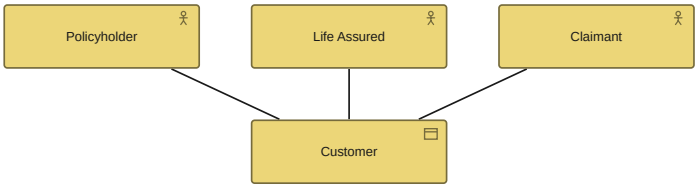


Figure 16.8 The version to refuse. Three actors and a Customer object. It validates, it is wrong, and it propagates: the Business Actor becomes an application entity, which becomes a table, which becomes the reason a third-party claimant gets a marketing letter.

Three things go wrong when roles are drawn as actors, and they compound.

First, you cannot say that one party holds several positions, so the model implies that policyholders, claimants and life assureds are different populations. They are not, and every cross-sell, fraud and single-view programme depends on knowing it.

Second, the error propagates downward. A Business Actor called Policyholder becomes an application entity called Customer, which becomes a table, which becomes the reason the insurer markets retirement annuities to a third-party claimant whose data it holds only because an insured vehicle hit them. That is the normal consequence of collapsing identity and role, not a hypothetical.

Third, you lose the data-protection distinction. A policyholder consented. A third-party claimant did not, has no contract, and often has health data in the file. Their lawful basis, purpose limitation and retention are all different, and after the 2026 health-information regulations brought insurers expressly into a named list of responsible parties with specific duties over health records, that difference is enforceable [10]. A role-based model is the only one in which that distinction is even sayable.

> **TWO THINGS ARCHIMATE WILL NOT DO FOR YOU HERE**
>
> Roles are dated in reality and undated in ArchiMate: a superseded beneficiary nomination must stay queryable, and the model cannot express that. Put it in the data model and say so in the view's notes. Second, internal positions (claims handler) and party positions (policyholder) are both Role, which is correct and confusing. The convention that works: keep party-on-contract roles in the information views and internal roles in the process views, and never put both on one diagram without a legend.

The claim, as process, event and object

Claims is where an insurance model either earns its keep or reveals that nobody checked it. It is also the best place to show the three-way split between event, process and object, because all three are present and insurers habitually merge the first two.

Vendors build for it: claim attributes are analysed at notification so severe claims surface early rather than late [13]. A **loss event** happened in the world: a hailstorm over Centurion, a collision on the N1, a death. A **claim** is an assertion against a policy arising from it. One event yields many claims — a hailstorm hits four thousand roofs; one collision yields an own-damage claim, a third-party property claim and three bodily-injury claims. If the model has no Loss Event element, the insurer cannot aggregate a catastrophe, cannot apply an hours clause, and cannot compute a catastrophe reinsurance recovery. That is where the money goes missing.

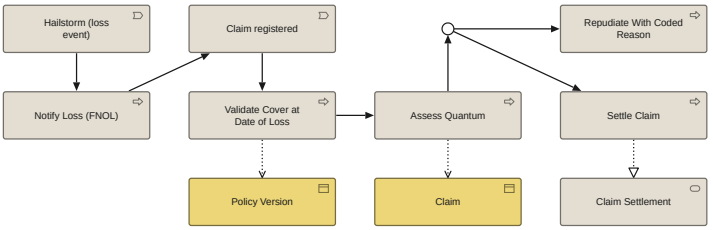


Figure 16.9 Event, process, object, service in one picture. The hailstorm triggers notification; cover validation accesses the policy *version*, not the policy; the or-junction treats repudiation as a business outcome rather than an error path; and settlement realises the service the customer bought.

Read the figure left to right and notice four choices.

The loss event **triggers** notification; it does not realise it. Triggering is right because the event is outside the insurer's control and the process starts in response to it.

Cover validation **accesses the policy version**, not the policy. This is the edge that pays for the whole diagram. Ask an insurer how coverage at date of loss is determined and you will get one of three answers: from the policy as it stands today (wrong, and the source of repudiations that get overturned); from a point-in-time reconstruction in the administration system (right, and expensive); or "the handler looks at the file" (common, and unmeasurable). Drawing the Access to a version forces that conversation in the first hour of the engagement.

The authorise step branches through an **or junction** to settlement or repudiation. Repudiation is not an error path. It is a business outcome with a reason code, reported to a conduct regulator and counted in the complaints statistics, and a model that treats it as an exception has already mis-specified the requirement.

Settlement **realises** the service the customer bought. Trace that Realization back through the Product's aggregation and you can say which cover this process makes good on. That is the chain that lets you answer a product manager's question — "if we change the excess structure, what changes operationally?" — without a workshop.

Now the information side.

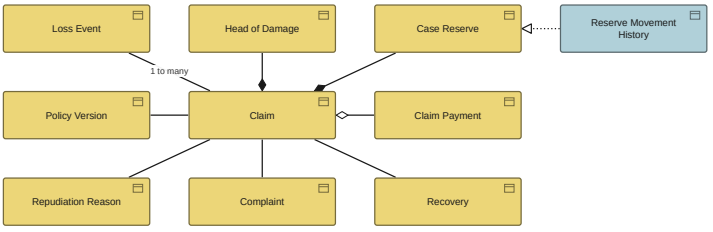


Figure 16.10 The claim, as information. Recovery hangs off the claim separately from payment because it is not a negative payment. The case reserve is one object with a movement history, realised by the Data Object the actuaries build triangles from. The loss event is associated with the claim, not composed of it, because one event has many claims and the notation cannot say "many".

Three deliberate decisions in that structure. Recovery hangs off the claim separately from payment, because a recovery is not a negative payment: different counterparty, different timeline, different accounting, and under South African VAT rules a subrogation recovery carries no output tax while a cash indemnity payment to a vendor insured triggers a deemed supply by the insured [3]. Case reserve is one object with a movement history rather than a current value, because the actuarial function builds development triangles from movements, and an insurer that overwrites reserves has quietly lost the ability to reserve. And the loss event is associated with the claim rather than composing it, because one event has many claims and ArchiMate cannot say "many" — so the honest thing is an Association with the cardinality recorded where cardinality belongs.

Reinsurance, which most models leave out

Open ten insurance architecture models and nine have no reinsurance in them: policy administration, claims, billing, CRM, a warehouse, and nothing about contracts carrying a large share of the risk and margin. Usually because reinsurance lives in two people's spreadsheets, which is why it should be modelled.

The shape is a second contract universe sitting over the first.

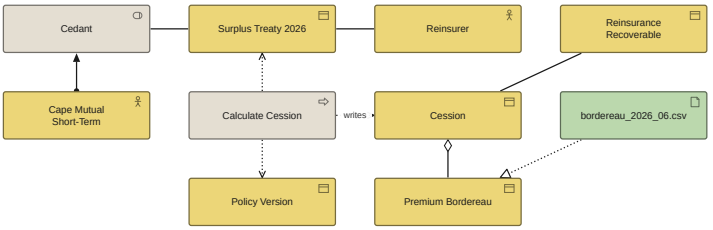


Figure 16.11 The second contract universe. The insurer appears as cedant — the one role it holds nowhere else in the model. Cession is the join between a policy version and a treaty, written by a process rather than typed into a spreadsheet, and the bordereau is an object realised by a file that two counterparties will never format the same way.

The insurer appears here in a role it holds nowhere else in the model: **cedant**. That is worth a sentence in any review, because it is the moment a business audience realises the insurer is also a buyer of insurance, with the same problems its own customers have — a contract it did not write, a counterparty whose solvency it depends on, and a claim it has to prove.

Cession is the join. A Process calculates it, reading the policy version and the treaty, and writes a Cession object. For proportional treaties that calculation can run at bind time, per policy, which is why a well-modelled insurer can tell you its net exposure today. For a surplus treaty the ceded share varies with the size of the risk, so the calculation needs the sum insured as it validly stood, which means a retroactive endorsement changes cessions for periods already reported. Martin Fowler's bitemporal pattern is the usual answer, and reinsurance is the clearest insurance case for it [6].

Excess of loss is a different data problem and needs a different picture.

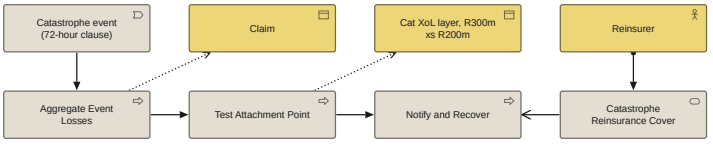


Figure 16.12 Why the loss event earned its place. A proportional cession is computable per policy at bind time. An excess-of-loss recovery is computable only per event, after claims are aggregated inside an hours clause and tested against an attachment point. No Loss Event element, no catastrophe recovery — and that is where recovery leakage lives.

A proportional cession is computable per policy. An excess-of-loss recovery is computable only per **loss event**, after aggregating claims across policies within an hours clause and a geography, and testing the total against an attachment point. That is why the Loss Event element earned its place three sections ago, and it is where most reinsurance recovery leakage happens: not in fraud, not in over-settlement, but in recoveries nobody computed because no system held the event.

Two more notes an architect should carry into the room. Reinsurance modules exist to do exactly this — contract terms, cession calculation, bills and bordereaux, recoveries and payables [14] — and an insurer without one is doing it in a spreadsheet. The **bordereau** is a Business Object realised by an Artifact and moved by a Flow — periodic, itemised, bilaterally agreed, and never identical between two reinsurers [4]. And under delegated authority the flow reverses: where a UMA holds the policy and claims system, the insurer *receives* summarised, lagged, inconsistently coded data about its own book. Draw that reversal. It is the most consequential line in a South African non-life model, and it is invisible unless somebody draws it.

Motivation: from a rule to the system that satisfies it

Chapter 15 built the chain from a regulator's concern down to a testable requirement. This section does the half that usually gets skipped: the Realization edges that run from the requirement down into the systems. Its job is one question — which system satisfies which rule, and what breaks if we switch that system off.

Work it with Treating Customers Fairly, the outcomes-based framework the conduct regulator supervises against — six outcomes, from products designed for identified customer groups through to no unreasonable post-sale barriers to switching, claiming or complaining [7]. Outcome six is the interesting one for a model, because it is measured by friction: cancellation reasons, replacement linkage, switching metrics. And the conduct-of-business return now asks for segmented client data and detailed conduct indicators, so firms demonstrate fair outcomes through their data rather than through a policy document [8][9].

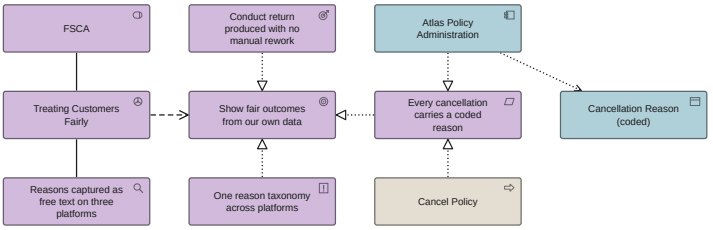


Figure 16.13 From a conduct outcome to the thing that satisfies it. The driver cannot be realised by anything; the requirement can, and is — by a component, a process and the coded data object between them. This is the view that answers the only question that justifies a motivation layer: if Atlas goes, which rule loses its realisation?

The chain is Stakeholder, Driver, Assessment, Goal, Outcome, Requirement, and then — the part people skip — Realization from a core element to the Requirement.

The elements were defined in the previous chapter; what matters here is the discipline each one enforces in an insurer. Writing "TCF" as a requirement rather than a driver is the commonest error in this layer, and it makes the model untestable. An outcome you cannot state as a number with a date is a goal in the wrong box. And a requirement is a sentence a system can satisfy or fail: "every cancellation carries a coded reason and, where applicable, a link to the replacing policy".

Then the realisation. An Application Component realises the requirement, a Process realises it, a Data Object realises it. Now you can answer the question that justifies the entire layer: if Atlas is decommissioned in 2029, which regulatory requirements lose their realisation, and what has to take them over? No other artefact in an insurer answers that. Not the control register, not the application inventory, not the programme plan.

The same chain works for privacy, and the privacy case is sharper because the duty is specific.

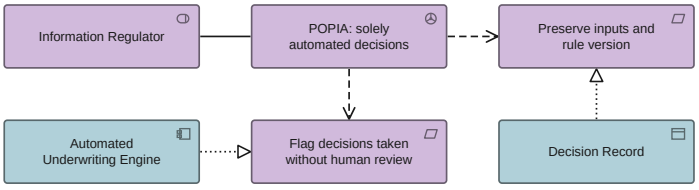


Figure 16.14 A privacy duty, made testable. Two requirements a system can satisfy or fail, realised by the engine that takes the decision and by the record it writes. Whether the section applies to underwriting as read here is commonly argued rather than settled; the architectural obligation is the same either way.

POPIA protects a data subject from a decision based solely on automated processing where it has legal consequences or significantly affects them, profiling included, unless an exception applies. The published illustration concerns credit, and I could not source an authoritative insurance-specific reading — treat the application to underwriting as commonly argued rather than settled [11]. But the architectural implication does not depend on the finer reading: an insurer must be able to say whether a decline was wholly automated, and reproduce the inputs and the rule version that produced it. As a requirement that is unambiguous, testable, and almost never implemented, because automated underwriting engines were built to decide, not to explain.

> **THE MOTIVATION TEST**
>
> A motivation view that contains no Realization edges from core elements is decoration. A motivation view where every Requirement is realised by at least one named component or process is the most valuable thing an enterprise architect can hand a Chief Risk Officer, and the only version of it that survives an audit.

Strategy: capabilities, and what ACORD actually gives you

Every insurer has a capability model. Most of them are wallpaper. The difference between the two is whether anything hangs off each box.

In ArchiMate a **Capability** is an ability the enterprise possesses, with an owner. It is realised by core behaviour — functions and processes — and enabled by **Resources**. It is not an activity and not a department. "Claims Management" is a capability; "Assess Motor Claim" is a process; "Claims" is a department and belongs nowhere in the model.

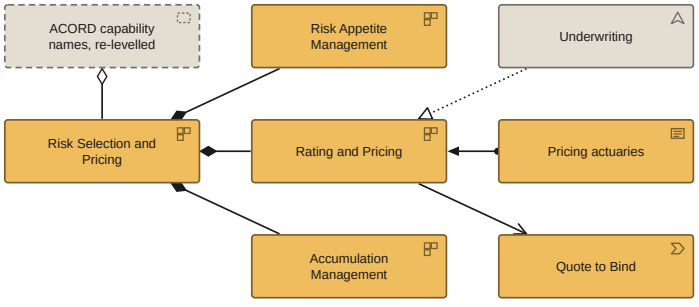


Figure 16.15 A capability that carries something. Three sub-capabilities, a resource that is the real constraint, the function that realises the ability in the operating model, and the value stream it serves. The grouping holds ACORD's names, re-levelled to how this insurer is managed — a property of the capability, not a relationship to it.

ACORD's Reference Architecture includes a Capability Model: a baseline of activities across the insurance value chain, published deliberately as a functional decomposition **without predefined workflow**, and supporting process engineering and merger or divestiture evaluation [1]. Since January 2026 there is an interactive version with tooling to measure current capabilities, which tells you where ACORD thinks the value is [2].

That property — no predefined workflow — is exactly what makes it a good source of capability names and a bad source of processes. So map it, but do not pretend the two models are the same thing.

An ACORD capability is an activity in the industry value chain. An ArchiMate Capability is an ability of *this* enterprise, with an owner, a maturity and an investment case. They are different kinds of statement, and the mapping is name-level, not semantic. Practically:

1. **Take the names, not the levelling.** ACORD's decomposition is built for cross-industry comparison. Yours has to match how your insurer is managed, which is usually by line of business and distribution channel, neither of which ACORD knows about.
2. **Record the mapping as a property, not a relationship.** Tempting to draw Specialization or Realization from your capability to ACORD's. Resist it: you will be asserting a semantic equivalence you cannot defend, and the first person to compare the two models will find twenty counter-examples.
3. **Import a subset and justify every exclusion.** The failure mode with every industry reference model is the same: adopt it whole, map forty per cent, stop, and own a half-canonical model competing with the systems for authority.
4. **Make each capability carry something** — an owner, a maturity score, the applications supporting it, the investment proposed. A capability with none of those is a label.

The other three strategy elements earn their place in an insurer more than they do in most industries. **Resource** is where you model the thing that actually constrains you: two remaining mainframe engineers, one pricing actuary who understands the motor model, a panel assessor network. **Value Stream** is the COO's view — loss to settlement, quote to bind — served by capabilities and carrying a Value at the end. And **Course of Action** is where the honest options live: replace the core, wrap the core, or run the book off. Those are strategy, they are mutually exclusive, and putting them in the model means the rejected ones survive in writing, which is how an architecture decision stays decided.

Viewpoints that earn their keep

A viewpoint is a view for a named audience with a named question. These are the ones asked for repeatedly in an insurer.

Viewpoint	What it shows	Who asks for it	What it settles
Requirements realisation	Requirement traced to the component, process or data object that satisfies it	Chief Risk Officer, compliance, internal audit, the regulator's examiners	Which system satisfies which rule, and what loses cover when a system is retired
Capability map, with maturity	Capabilities, owners, supporting applications	Executive committee, CFO	Where the next twenty million rand goes, and which capability is weakest
Value stream	Stages from a customer trigger to the value delivered, served by capabilities	COO, Head of Claims	Cost to serve and turnaround time, argued on one page
Business process co-operation	Processes, roles, handoffs, authority limits	Operations, binder oversight, the UMA relationship owner	Who may bind, settle and pay what, and where the handoff drops work
Product	A product with the services it promises and the contract that governs them	Product manager, pricing actuary, conduct compliance	What was actually promised, and what changes if the bundle changes
Information structure and data realisation	Business object to data object to artifact, with the systems that hold each	Chief Data Officer, actuarial, the IFRS 17 lead	Granularity and lineage, and whether a required number can be produced at all
Application co-operation	Components, services and the dependencies between them	CIO, integration lead	Coupling, duplication, and where the delegated-authority boundary sits
Technology and deployment	Nodes, networks, locations, what runs where	Operations, resilience and outsourcing compliance	Concentration, data residency and recovery, now regulated matters
Implementation and migration	Plateaus, work packages, deliverables	Programme board, finance	Sequence, and what is true at the end of each year
Layered	One picture from motivation to technology for one line of business	Board, a new CIO, due diligence	Whether the enterprise hangs together, and where it visibly does not

Two viewpoints are missing from the standard set and should not be. A **delegated-authority view**: for each binder and UMA relationship, who holds which authority, to what limit, over which product, reporting what to whom in what format. In an intermediated market that is the first question a new Chief Risk Officer asks. And a **reinsurance view**: programme structure, treaties, cedant and reinsurer roles, and the processes computing cession and recovery. Without them, an insurer's two largest undocumented dependencies both stay undocumented.

Four traps

Each of these is common, each produces a model that validates, and each wastes a year.

Modelling the org chart. Business Actors named after departments, composed into a hierarchy, with processes named after the departments that run them. It is quick, it is recognisable, everyone nods — and it is worthless, because the org chart changes next March and the model does not describe anything that survives the change. It also teaches the organisation that architecture is documentation. Model Roles and the behaviour assigned to them; put the hierarchy in the HR system where it belongs.

Modelling the system menu. Application functions named after screens: Policy Enquiry, Claim Capture, Bulk Print Run. That is what a model built from a user manual looks like. The test for an application service is whether something outside the component depends on it. Nobody outside depends on Policy Enquiry; four systems depend on "give me the policy as it stood on a date". The first is a menu item, the second is the architecture.

Drawing the policy as a process. A chain of boxes labelled Quote, Issue, Endorse, Renew, Lapse, Claim, Terminate, with arrows. It looks like a process and it is not one. Those are **events** in the life of a Business Object, each triggering different processes, in no fixed order, often concurrently, sometimes retroactively. Drawing them as a sequence encodes an assumption that is false — that renewal follows endorsement — and that assumption will show up later as a system that cannot process a retroactive endorsement on a lapsed policy, which is a real thing that happens every day.

Treating the policy administration system as one component. This is the expensive one.

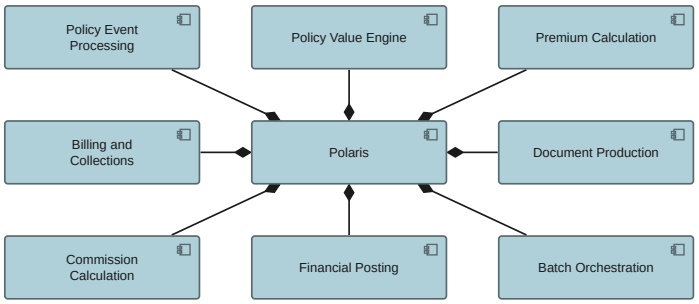


Figure 16.16 One box on the board slide, eight of twelve components here. They have different owners, different change rates and wildly different replaceability. This decomposition is what turns "replace the policy administration system" into a sequence of affordable decisions, and it is the highest-value hour an architect can spend inside an insurer.

The administration system is the oldest and most expensive thing in the estate, and on the board slide it is one box. Inside it there are at least a dozen distinguishable components: policy event processing, premium calculation, the policy value engine that accrues cash and surrender values, billing and collections, document production, commission calculation, financial posting, batch orchestration, beneficiary and nomination handling, correspondence history, product tables and the reporting extracts. They have different owners, different change rates, different failure modes and wildly different replaceability.

Why the single box costs money: every modernisation argument depends on separability. If the administration system is one thing, the only option is replacement — for a life book, a conversion of three to four years on published averages, against a book still paying claims in 2070 [21][22]. If it is twelve things there are options: externalise rating, lift collections out, put a service in front of the policy read, and leave the value engine alone for a decade because contractual values must reproduce to the cent and nothing else needs touching. The decomposition is not documentation. It converts an unaffordable programme into a sequence of affordable ones, and it is the highest-value hour an architect spends in an insurer.

THE CHAPTER IN TEN LINES

- ArchiMate 4 gives you 42 elements and insurance gives you a few hundred concepts, so the mapping collapses many concepts onto one element, demotes many to attributes, and turns a few into relationships.
- The model holds types, not instances: no cardinality, no effective dating, no policy 4471920 — so build a data model alongside it and stop asking ArchiMate to be one.
- Five decisions settle most arguments: object or product, service or process, interface or actor, actor or collaboration, and element or attribute.
- Product is the one composite in the business domain: it aggregates the services it promises and composes the contract that governs them, which is why cover is a Service and not an object.
- Variants are Products with a Specialization; versions are data in the product factory, and modelling versions as elements is how a product catalogue becomes unreadable.
- Policyholder, life assured, beneficiary and claimant are Roles on a Business Actor; drawing them as actors creates the Customer entity and everything that follows from it.
- A loss event is not a claim: without the Event element an insurer cannot aggregate a catastrophe or compute an excess-of-loss recovery, which is where the money goes missing.
- Reinsurance belongs in the model — cedant as a role, treaty and cession as objects, reinsurance cover as a service the insurer consumes, and the bordereau boundary drawn explicitly.
- A motivation view with no Realization edges from components and processes to requirements is decoration; with them, it answers which rule loses its realisation when a system is retired.
- The policy administration system is never one component, and the hour spent decomposing it is what converts an unaffordable replacement into a sequence of affordable moves.

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CHAPTER 17

Cape Mutual, end to end

One invented South African insurer, modelled from strategy to technology, a view at a time — with the argument for every modelling choice shown, and the rejected options kept in, because those are the ones that teach.

A warning before we start

Cape Mutual Insurance Group does not exist. There is no such insurer, no Polaris, no Atlas, no Thandi Nkosi. Everything in this chapter is invented, and it is invented deliberately, because a real case study would force a choice between being accurate and being useful. Where a number or a pressure is drawn from the real South African market it carries a source; everything else is a composite built to be typical rather than true.

What is typical about it: a composite group with a life business and a short-term business that grew up separately; a forty-year-old life administration system nobody can replace and nobody can leave; a delegated-authority relationship the insurer cannot see into; a conduct regulator asking for data the systems were not built to produce; and a weather pattern that has made the catastrophe line the most volatile number in the accounts [4][5].

Cape Mutual Life writes risk and funeral business on **Polaris**, in its fortieth year. Cape Mutual Short-Term writes motor, household and small commercial on **Atlas**. **Beacon** is the broker and adviser portal, **Sentinel** scores claims for fraud, **Compass** is the group CRM, **Ledger** is the general ledger and IFRS 17 sub-ledger, and **Kestrel** is the two-year-old digital quote-and-buy stack. Specialist liability is written by **Table Bay Underwriters** under a binder. Everything runs between the **Observatory** and **Bellville** data centres. **Boland Brokers** is the largest independent on the book.

The invented numbers, so the choices later have something to push against. The short-term business writes about R4.2bn of gross written premium a year, two-thirds of it motor, three quarters of it through intermediaries. The life business writes about R6.1bn, dominated by funeral cover sold in volume through a tied force and two retail partners — which is the normal South African shape, since funeral policies were 57 per cent of all new individual recurring-premium risk policies written in the country in 2025 [2]. Polaris carries 2.3 million in-force life policies and the group lapses a large fraction of new funeral business inside a year, mostly through failed collections rather than deliberate cancellation. Specialist liability, written by Table Bay, is small premium and large single risks.

Fatima Patel is the lead enterprise architect, and what follows is the model she builds, view by view, over about six weeks. At each step: what she chose, and what she rejected. The rejections matter more, because an accepted option tells you what somebody wanted and a rejected one tells you what the enterprise is actually like.

The motivation view: why the group is changing

Start here, always. Not because motivation modelling is fashionable but because an insurance estate will absorb any amount of work, and the only defence is a view that says what the work is for and who asked.

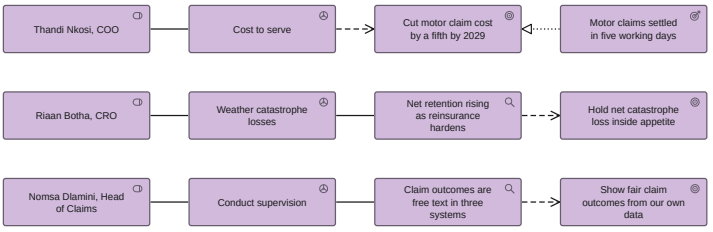


Figure 17.1 Why the group is changing. Three named executives, three drivers, and goals stated so that an outcome can be measured against them. Nothing here names a system: motivation that names solutions has stopped being analysis.

Three drivers, three executives, three different kinds of pressure. Riaan Botha's is the hardest number in the group: weather catastrophe was the largest force on short-term claims costs in the first half of 2026 across the market [4], reinsurance has hardened, and a hardening market means the insurer retains more of each event [3][4]. Thandi Nkosi's is slower and larger — non-life return on equity is forecast to fall sharply by 2028 [1], which makes cost to serve the operating metric that matters. Nomsa Dlamini's is new: the conduct-of-business return will ask for segmented client data and detailed conduct indicators, and a firm demonstrates fair outcomes through its data rather than through a policy document [6][7].

Four choices in that view are worth defending.

Stakeholders are named people, not functions. "Finance" has no concerns; Thandi Nkosi has concerns. Named stakeholders make the view checkable — she can be shown it and can say it is wrong — and they expire usefully, because when she leaves, somebody has to re-confirm that the driver still has an owner.

Assessments are dated findings, not opinions. "Repudiation reasons are free text in three systems" is a thing Fatima can evidence and somebody can refute. Without assessments a motivation view is a wish list, and reviewers treat it as one.

Rejected: modelling each TCF outcome as a goal. The first draft had six goals, one per TCF outcome. It looked thorough and it was useless: nobody at Cape Mutual is accountable for "clear information", so none of the six could be assigned, measured or funded. The second draft has one driver — conduct supervision — with goals that match how the group is actually managed. The TCF outcomes stay, as a property on the driver.

Rejected: putting the systems in. The temptation is to draw Polaris next to the cost driver, because everyone knows Polaris is the problem. Resist it for one more view. Motivation that names solutions stops being analysis and becomes advocacy, and the first executive to notice will stop trusting the rest of the model.

One element is missing from the figure and belongs in the repository behind it: a **Principle**, "one reason taxonomy across every platform". Principles are the part of this layer most often written as slogans — "customer first", "cloud by default" — and a principle that cannot be violated by a specific design decision is not a principle. This one can: the Kestrel team proposed its own cancellation reason list in week three, and the principle is why that proposal was refused in a ten-minute conversation rather than a three-week argument.

What the view settled. Two things, immediately. The catastrophe driver had no owned goal, which turned out not to be an oversight — nobody in the group was accountable for net retained catastrophe loss as distinct from gross, and the view made the gap visible to the audit committee before the model went any further. And the cost-to-serve goal was quantified on the spot, because a goal with no number attracts an Outcome element and an Outcome element has to be measurable or it is obviously a goal in the wrong box.

The capability map

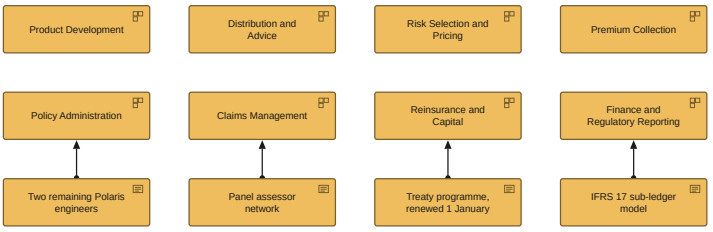


Figure 17.2 One map for both licences. Eight capabilities at level one, with the resource that actually constrains each of four of them. Two of those resources are people, one is a contract renewed every January, and none of them is software — which is usually the finding.

Eight capabilities at level one, which is about right for an insurer; fewer and they are too coarse to argue about, more and the executive committee loses the thread. Each one carries a resource where the resource is the real constraint, and three of those are worth a sentence: two remaining engineers who understand Polaris, a panel assessor network the group does not own, and a treaty programme that is renegotiated every 1 January.

Chosen: capabilities shared across both licences. Cape Mutual Life and Cape Mutual Short-Term are separately licensed and separately managed, so the obvious map is two maps. Fatima drew one. The group's investment arguments are group-level, the duplication between the two businesses is the main cost story, and a single map makes the duplication visible. The line of business becomes an attribute, not a dimension of the map.

Rejected: the ACORD capability model as published. It is a good list and a bad map for this insurer, because its decomposition is built for cross-industry comparison and Cape Mutual is managed by line of business and channel [8]. Fatima takes the names, re-levels them, and records the ACORD identifier as a property on each capability. That keeps the option of benchmarking later without pretending the two models mean the same thing.

Rejected: colour-coded maturity on the map itself. Everyone wants a heat map. ArchiMate has no maturity attribute, so a heat map is a rendering of a property held in the repository, not a model. Draw it for the executive committee by all means; do not let it become the model, because within a year the colours will be stale and nobody will know which version was scored.

Rejected: a capability for the UMA relationship. The first draft had "Delegated Authority Management" as a ninth capability. It is a real thing the group must do — oversight, bordereau ingestion, aggregate monitoring, claims authority limits — and it was moved to a sub-capability under Risk Selection and Pricing and another under Claims Management, because splitting it out implied the insurer could manage the relationship without owning the underwriting and claims abilities underneath it. It cannot. The insurer remains liable for every policy Table Bay issues and must pay valid claims on them, and an insurer outsourcing critical functions remains ultimately responsible [16]. A capability map that puts delegated authority in its own box quietly contradicts that.

The decomposition is where capability modelling earns the effort, and only one of the eight needs it now.

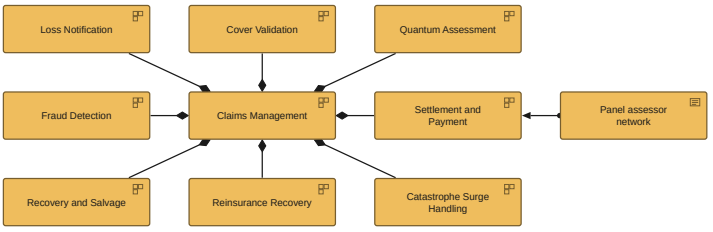


Figure 17.3 Claims Management, decomposed. Catastrophe surge handling is drawn because it does not exist: a capability with no supporting application is how an architect says "we cannot do this" without saying it about a person.

Eight sub-capabilities under claims, and the one that does not exist yet is catastrophe surge handling — ten to fifty times normal volume with no elastic capacity is the known failure mode, and after two cat events in one quarter it is Nomsa Dlamini's first ask. Modelling it as a capability rather than a

project is deliberate: capabilities survive the project that builds them, and a capability with no supporting application is the cleanest way an architect has of saying "we cannot do this" without saying it about a person.

Each box carries four properties in the repository and nothing else: an owner, a maturity score from the last assessment with its date, the applications that support it, and the investment proposed in the current cycle. That is the whole discipline. A capability with none of those is a label, and a map of labels is the artefact that gives capability modelling its reputation.

The value stream for a motor claim

The capability map is the chief financial officer's view. The value stream is the chief operating officer's view of the same facts, and it is the one that gets quoted back at you in meetings.

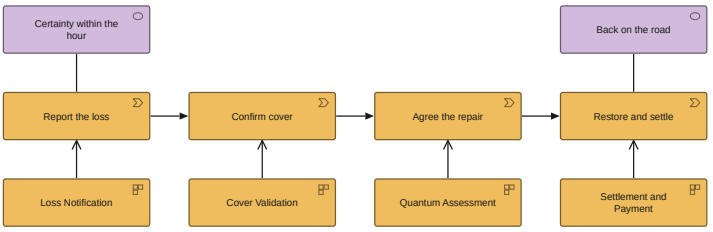


Figure 17.4 A motor claim, from the claimant's side. Four stages, not nine steps: a stage ends where the customer's state changes. Each is served by a capability, and the value is attached where the customer experiences it rather than where the insurer books it.

Four stages, each triggering the next, each served by a capability, with value attached where the customer actually experiences it. Notice that the stages are named from the customer's side — report the loss, confirm cover, agree the repair, restore and settle — not from the insurer's. That is the whole point of the construct. "FNOL", "adjudication" and "quantum" are internal words, and a value stream written in them is a process model wearing a strategy costume.

Chosen: four stages. The first draft had nine, because the process has nine steps. Stages are not steps. A stage ends where the customer's state changes, and there are four of those: they have told us, they know they are covered, they know what happens next, they have their car back.

Rejected: one value stream for all claims. A funeral claim and a commercial liability claim do not share a value stream. Funeral is paid in forty-eight hours on a death certificate; liability takes three years and a lawyer. Cape Mutual has three streams, and the motor one is modelled first because it carries the volume.

Rejected: putting the cost of each stage on the diagram. The numbers change quarterly and the diagram does not. Stage cost and cycle time are properties, reported from the repository into a dashboard the claims operation already has; the view carries the structure that makes the numbers comparable between quarters.

What the view settled. Stage two — confirm cover — takes an average of eleven hours at Cape Mutual, almost all of it waiting for a human to check the policy as it stood at the date of loss. Nobody had ever measured it, because in the process model the check is one box among nine and nobody measures a box. Drawn as a stage with its own capability and its own piece of customer value, it became a question: why does confirming cover take eleven hours when the policy system holds the answer? The answer — because Polaris and Atlas can both tell you what a policy looks like now, and neither can cheaply tell you what it looked like in March — is the finding that funded the next three views.

The business process layer

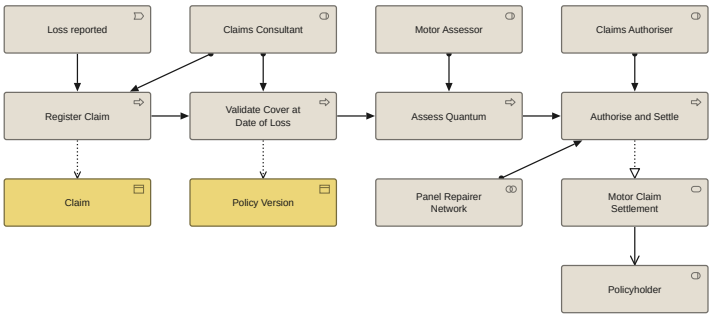


Figure 17.5 The motor claim process. Roles rather than job grades, an Event rather than a first step, the panel repairers as a Collaboration, and the Access to Policy Version that decides a quarter of the programme budget. The same view for specialist liability differs in exactly one place: who holds the authority to settle.

Now the detail, and the first view a claims manager will read without being taught the notation. Four processes in sequence, roles assigned to each, a service realised at the end and consumed by the policyholder, the objects each process touches, and the panel repairer network as a Collaboration rather than an actor, because Cape Mutual and the repairer jointly perform the repair and neither performs it alone.

The load-bearing element is the Access from cover validation to **Policy Version**. Follow it into the next views and it decides the application architecture, the data architecture and a quarter of the programme budget.

Chosen: roles, not job titles. The model says Claims Consultant, Motor Assessor and Claims Authoriser. Cape Mutual's org chart says Claims Consultant I, II and Senior, which is a grading scheme, not an architecture. Roles map to authority limits; grades map to salary bands.

Rejected: modelling the repudiation path as an exception. The first version had a happy path and a note saying "or decline". Repudiation is around one claim in eleven at Cape Mutual, it is reported to a conduct regulator with a reason code, and it generates most of the complaints. It gets a process, a coded outcome and an object, like everything else that happens routinely.

Rejected: a swimlane per department. The claims manager asked for one, and it would have been a faithful drawing of the current organisation, which is exactly what makes it the wrong artefact: the handoffs it shows are the ones being removed. Roles assigned to processes carry the same information and survive the reorganisation.

There is a second version of this view for the specialist liability book, and it looks almost identical except that the Claims Authoriser role is assigned to a Table Bay employee up to R250,000 and to a Cape Mutual employee above it. That is the only difference on the page, and it is the entire delegated-authority control. An architect who has drawn that view once will never again accept "the UMA handles claims" as a description of anything.

One more element earns its place: the **Event** at the start. Loss reported is not the first step of the process, it is the thing that triggers it, and the distinction matters operationally because the clock the conduct regulator cares about starts at notification, not at registration. Turnaround is measured between event timestamps across the whole lifecycle, not from status flags [6], and a process model with no events has no timestamps to measure between.

The application co-operation view

Here is the version Fatima drew first, which is also the version Cape Mutual actually has.

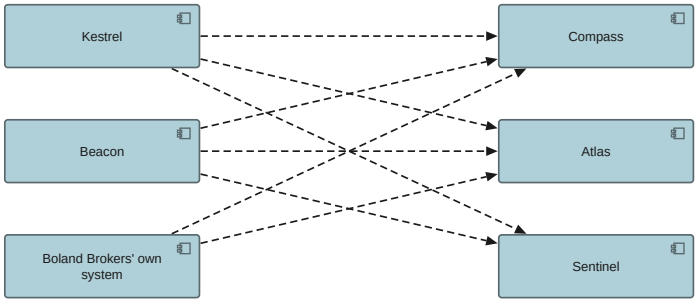


Figure 17.6 What Cape Mutual actually has, and what the target has to replace. Every front end wired to every back end, no contract between any two of them, and a change to one claim table breaking three things nobody can list. This is the normal shape of an insurer that added digital channels without changing the record system.

Every front end talks to every back end. Kestrel reads policy data straight from the Atlas database because the project had a deadline; Beacon has its own copy of the party record; Boland Brokers' own system files a claim by sending a spreadsheet. Eight integrations, no contract between any two of them, and a change to the Atlas claim table breaks three things nobody can enumerate.

This is not a strawman. It is the normal shape of an insurer that has added digital channels to a record system without changing the record system, and it is drawn here because the target only makes sense next to it.

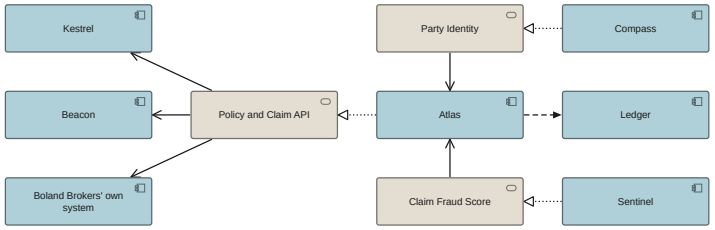


Figure 17.7 One seam. A policy and claim service that Atlas owns and every consumer uses, with party identity and fraud scoring arriving as services rather than as calls into other systems, and an asynchronous flow of postings to Ledger. The seam may transform, route and cache; it may not decide.

One seam. A policy and claim service that every consumer uses, realised by Atlas, with party identity and fraud scoring arriving as services rather than as direct calls into the respective systems. Atlas posts to Ledger as a flow, because the posting is asynchronous and nobody is waiting for it.

Chosen: a service seam over Atlas, not an integration layer in front of it. The distinction matters. A service seam is a contract Atlas owns and must honour; an integration layer is a second component that accumulates business logic belonging in the first. The characteristic failure of the system-of-engagement pattern is exactly that — the engagement layer becomes an undeclared second system of record, the read projection drifts with nobody owning reconciliation, and write-back turns out to be the hard part. Cape Mutual's rule: the seam may transform, route and cache; it may not decide.

The seam has a direction problem that is worth naming, because it is where this pattern usually fails. Reads project easily: a cached policy summary behind an API is a weekend's work. Writes do not. When a policyholder changes a debit order date in Kestrel at nine at night, that change has to land in Atlas, and until it does the customer has seen a change the record does not reflect. Cape Mutual's rule is that every write is a command the record system accepts or rejects, with a visible state in between, and that the engagement layer never shows the customer a value it has not had confirmed. It is slower to build and it is the difference between one source of truth and two.

Rejected: replacing Atlas. It was on the table, and a twelve-month short-term book can in principle be run off onto a new platform. Fatima rejected it on sequencing rather than merit: with Polaris unresolved, a claims surge capability missing and a conduct return due, a core replacement would consume the only three people who can do any of the other three things. The market's own timelines support the caution — the public record of one large South African insurer's core programme runs from a 2012 selection to a 2026 go-live announcement [10], and those dates do not measure one implementation, but they do measure elapsed reality.

Rejected: event streaming everywhere. Change data capture out of Atlas into a stream, with consumers reading from the stream, is the pattern that fits this problem and it is well evidenced in insurance [11]. It is also the second move, not the first. Streaming out of a system with no service contract just distributes the coupling more widely, and change data capture off the Polaris mainframe is a different and much harder proposition than off a relational database.

Rejected: putting Table Bay Underwriters inside the view. The UMA runs its own administration system. Cape Mutual's data for that book arrives as a bordereau, periodic and in the UMA's format, and drawing it as just another component would hide the thing that actually matters: the boundary. It gets its own view, and the boundary is drawn as a flow of an artifact across it, with the insurer's ingestion and reconciliation processes on the near side. The test of that view is whether it can answer one question — what is our aggregate exposure in the Cape Town CBD today, including the Table Bay book? At Cape Mutual the answer is "as at last month's bordereau, probably", and the view is what makes that sentence sayable out loud.

The component everyone forgets

Nothing in the figure is called collections, and in a South African insurer that is a mistake you only make once. A large share of personal-lines and mass-market life premium is collected by debit order against accounts where timing relative to payday determines whether the collection succeeds, and a failed collection becomes a grace period, then a reinstatement letter, then a lapse. The rails have changed under the industry's feet: the early-debit-order streams were switched off on 31 October 2021, and DebiCheck replaced them with a mandate the debtor authenticates once at their own bank, after which the bank permits only collections inside the agreed terms [17][18].

That makes the mandate a first-class, bank-held, long-lived object with its own lifecycle, and something in the estate has to master it. At Cape Mutual nothing does: Polaris holds a collection instruction, the mandate lives at the debtor's bank, and a third-party gateway sits between them. So the model gains a Business Object for the mandate, an external Business Actor for the bank, an Application Component for the gateway, and — the part that changes behaviour — collections moves out of the finance capability and into retention, where retry strategy driven by failure reason rather than blind resubmission is worth more to the group than anything in the digital programme [19].

The data realisation ladder

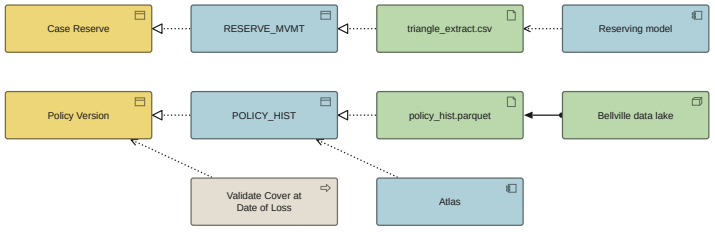


Figure 17.8 Business object, data object, artifact. Realization runs upward, Access runs from the behaviour that uses each rung. Two ladders because they carry the group's two hardest obligations: reconstructing a policy as at a past date, and holding reserves as a movement history rather than a current value.

This is the view that insurance architects skip and data people wish they had not. Three rungs: the **Business Object** is what the business means, the **Data Object** is how a system holds it, and the **Artifact** is the file that actually exists on a disk somewhere. Realization runs upward; Access runs from the behaviour that uses each.

Two ladders are drawn because they carry the two hardest obligations in the group. Policy Version is the one the claims process reads, and the question "can you reconstruct this policy as it stood on 14 March 2023" is answered on this diagram or not at all. Case Reserve is the actuarial one, and the rung that matters is that the data object is a **movement history**, not a current value: a claims mart that holds today's reserve is a claims mart from which no development triangle can be built, and at that point the actuarial function builds its own data estate and the group has two versions of the truth.

Chosen: model the artifact, not just the data object. It looks like over-modelling until the first time someone asks where a number came from. The artifact is where lineage becomes physical — a named extract, on a named node, produced by a named job — and lineage is now a regulatory expectation rather than good practice.

Rejected: a canonical data model for the whole group. The service-bus era answer, and it fails the same way every time: the canonical model becomes a committee artefact satisfying nobody, and every change queues behind the team that owns it. Cape Mutual takes the ACORD information model as a checklist instead — for each entity in it that the group lacks, ask whether it is genuinely not needed. Risk object, loss event, cession, party role and policy version are the usual discoveries, and four of the five were missing here [8].

Rejected: effective-dating everything. The group needs bitemporal history on policy, cover, party role and reserve, because the regulator can ask both what was true and what was known [9]. It does not need it on claim payments or premium collections, which are events: immutable, append-only, and doubled in size for nothing by a second time axis.

There is a third ladder Fatima drew and then deliberately left out of the published view, because it needs its own audience: the IFRS 17 one. The unit of account is a group of contracts — portfolio, profitability bucket, annual cohort — fixed at initial recognition and never reassessed [20], so every cash flow in the group has to carry an immutable key, and facts that determined a 2023 assignment must be reproducible twenty years later. The sub-ledger sits between the source and actuarial systems and the general ledger, taking cash flows, coverage units and discount rates in and producing posting files out [21]. At Cape Mutual that sub-ledger is inside Ledger, the reconciliation to the general ledger is manual, and the prudential return still speaks the pre-IFRS 17 language of written, earned and unearned premium because the templates were designed under the old standard [22]. Three measurement bases over one set of events: the only sane architecture is to build the event store once and derive all three, and the data ladder is where that argument is won or lost.

The technology view

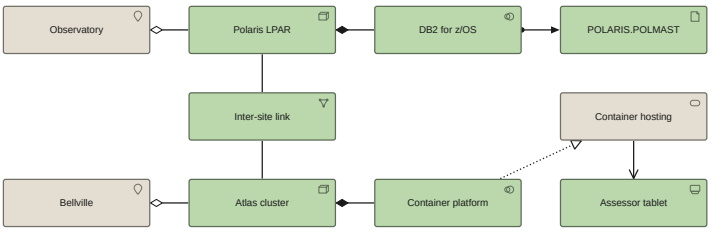


Figure 17.9 Two sites, two worlds. The sites are Locations because the view needs containers; Facility is the element for the building itself, and the group that prices flood risk for a living has not assessed the flood exposure of its own Bellville site.

Two sites, two worlds. The Polaris logical partition at Observatory with its database and its VSAM-era policy master; the Atlas cluster at Bellville on a container platform; a link between them carrying the overnight batch; and, at the edge, the assessor tablet that is the only piece of technology most claimants ever see the insurer use.

Chosen: Location for the data centres, Facility only where the building matters. Location aggregates the nodes and keeps the view readable. Facility is the element for the building itself — its power, its cooling, its flood exposure — and Cape Mutual will need it, because the group that prices flood risk for a living has not assessed the flood exposure of its own Bellville site. That is a real pattern and it is worth asking about in any insurer.

Rejected: modelling the batch schedule here. A hundred and twenty jobs with dependencies is a scheduling diagram, not an architecture view, and it belongs in the operations runbook. What the architecture view must carry is the consequence: data is a day old everywhere except Polaris, so any real-time experience built on a replica is lying by up to twenty-four hours, and the batch window is a capacity constraint that only ever shrinks.

Rejected: a single disaster-recovery statement for the group. Observatory and Bellville are drawn as peers and they are not. The Atlas cluster can fail over in hours; the Polaris partition has a recovery plan that was last exercised end to end some years ago and depends on two people. Writing one recovery objective across both sites would have been tidy and false, and the view's job here is to make the asymmetry visible to the people who sign the resilience attestation.

Rejected: cloud as a technology decision. Pieter van Zyl wanted the target state drawn in a hyperscaler's regions. Fatima kept it out of this view, for a reason that is specific to South African insurers: draft joint standards on operational resilience and on cloud computing and the offshoring of data are in the regulator's 2026–2029 plan, and material outsourcing already carries pre-notification and concentration-risk obligations under Joint Standard 1 of 2024 [12][13]. Cloud region, residency and exit are regulated matters here, which makes them motivation and implementation decisions with technology consequences, not the other way round.

The layered view

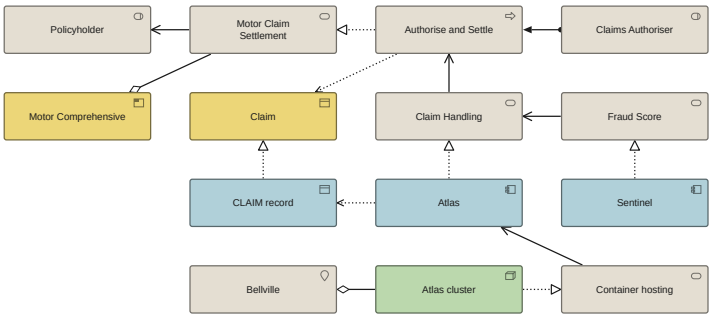


Figure 17.10 One thread, five layers. A single promise under a single product, traced to the platform it runs on. The value of this view is political rather than technical: it is what a board member or a new CIO looks at to decide whether the enterprise hangs together. Keep it to one thread or it will never be printed.

One line of business, one claim, from the policyholder at the top to the container platform at the bottom. This view has one job and it is political: it is the picture a board member, a new chief information officer or a due-diligence team looks at to decide whether the enterprise hangs together.

Keep it to one thread. The instinct is to put everything in, and the result is unreadable at A4 and therefore never printed. A layered view that traces a single valuable thing — one product, one claim, one customer promise — from motivation to metal is worth ten that try to be complete.

What this one shows that none of the earlier views do: the Product at the business layer aggregates the service the process realises, which is consumed by the policyholder, which is supported by an application service, realised by Atlas, running on a platform at Bellville. Five layers, one chain, no gaps. When Pieter van Zyl asks what breaks if the Bellville platform is unavailable for a day, the answer is on the page, and it is not "claims" — it is a named promise to a named role under a named product.

Rejected: the same view for the life business. Fatima tried it and stopped. The equivalent chain for a funeral policy runs through Polaris, and halfway down it stops being a chain and becomes a knot: the policy value engine, the collections instruction, the document production and the commission calculation are all inside one component, so the layered view degenerates into a box with eight arrows into it. That is a true picture and a useless one, which is why the life side gets the decomposition view from the previous chapter instead. Knowing which view to refuse is part of the skill.

The change view

Last, and only last, the plan. Fatima has watched programmes begin with the plan and reverse-fit the architecture to it, and the tell is always the same: a roadmap with workstreams named after vendors.

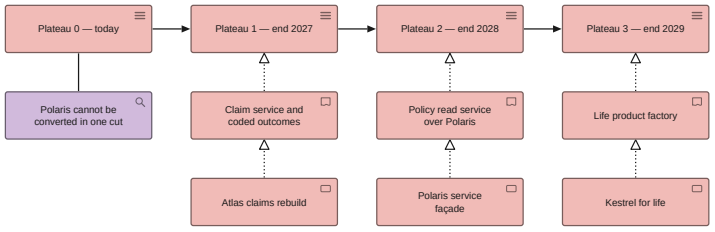


Figure 17.11 Three plateaus, not thirty work packages. A plateau is a state the group could stop at and defend if the money ran out. Work packages produce deliverables, deliverables realise plateaus — and the assessment attached to today is the reason the sequence is shaped this way.

Three plateaus and a starting position. A plateau is a state of the enterprise that is stable enough to stop at — not a milestone, not a release. The test Fatima applies: if the money ran out here, would the group be in a defensible position? If no, it is not a plateau, it is a phase of a project.

Plateau 0 — today	Plateau 1 — end 2027	Plateau 2 — end 2028	Plateau 3 — end 2029
Polaris and Atlas both load-bearing · Kestrel reads the Atlas database directly · claim outcomes in free text · the Table Bay book visible monthly	One claim service, used by every consumer · outcomes coded at source · conduct return produced from the warehouse · Polaris untouched	Policy read service in front of Polaris · engagement layer and warehouse off the mainframe · still no conversion	New life business written on Kestrel against a product factory · Polaris a closed book in run-off, not a system under change

Figure 17.12 The same four states in the form the programme board will ask for. The dates are commitments to a state of the enterprise, not to a delivery; the first one carries a regulatory obligation, which is why it is first.

Plateau 0, today. Polaris and Atlas both load-bearing, Kestrel reading the Atlas database directly, claim outcomes in free text, the Table Bay book visible only in a monthly bordereau.

Plateau 1, end of 2027. The claim service exists and every consumer uses it; claim outcomes are coded at source; the conduct return is produced from the warehouse rather than by hand. Polaris is untouched. This is the plateau that pays for the rest, because it is the one that satisfies a regulatory obligation with a date on it.

Plateau 2, end of 2028. A policy read service in front of Polaris, so the engagement layer and the warehouse stop reading the mainframe directly. Still no conversion. The life book is still administered where it has always been administered, and the façade buys the group the option to decide later.

Plateau 3, end of 2029. New life business is written on Kestrel against a product factory, and Polaris becomes a closed book in run-off rather than a system under active change. The forty-year problem is not solved. It is contained, which is the realistic outcome, because a life conversion averages three to four years on published figures and the book still pays claims decades out [14][15].

Chosen: a façade before a conversion. The alternative — convert Polaris to a modern platform — was costed and rejected on correctness risk rather than money. Contractual policy values must reproduce to the cent; the product logic encoded in that system is frequently the only complete specification of products sold in the 1990s; and the two people who understand it are the resource named on the capability map.

Chosen: a plateau that delivers a regulatory obligation first. Not because compliance work is more valuable, but because it is the only work that cannot be deferred, and a programme whose first plateau is optional will be re-planned by the first budget review. The group's regulatory load over the same window is not light — conduct reporting phasing in, a conduct bill in train, joint standards on outsourcing, operational resilience and cloud, and a reference rate cessation [12] — and a plan that treats all of that as a separate stream of work running beside the architecture is a plan that will collide with itself in year two.

Rejected: a big-bang group core. One platform for life and short-term, one party master, one claims system. It is the slide every executive wants to see and it is a decade of work with a single point of failure, in a group whose two halves have different contract durations, different mathematics and

different regulators. The sophisticated buyers in this market do the opposite: buying one component and keeping the rest is a documented choice at a large South African health insurer, which retained its own policy administration and bought only collections and disbursements [23]. That is the right instinct.

Rejected: deliverables without plateaus. The programme office's version of this view was thirty work packages on a Gantt chart. Work packages produce deliverables; deliverables realise plateaus; plateaus are states someone can be accountable for. Without the middle two, a change view is a schedule, and schedules do not survive contact with a catastrophe quarter.

One assessment is attached to Plateau 0 and it is the most important element in the view: Polaris cannot be converted in one cut. It is attached to the starting state rather than written into a risk log because risk logs are read by the people who already agree. On the change view it sits where every programme conversation starts, and it is the reason the three plateaus are shaped the way they are.

What the model is for

Six weeks, ten views, one fictional insurer. The point of the exercise is not the diagrams, which will be wrong within a year in the ordinary way. It is that at the end of it Fatima Patel can answer, in front of an executive committee and without a workshop: what are we changing and who asked; what can this group do and how well; what does a claim cost us and where does the time go; who may bind, settle and pay what; which system holds the truth and which holds a copy; where can a number be traced to; what breaks when a site goes down; what is true at the end of each year; and which of these things we decided not to do, and why.

That last clause is the one that matters. An architecture is the set of decisions you have taken and can still defend. Everything else is drawing.

THE CHAPTER IN TEN LINES

- Cape Mutual is invented; the pressures on it are not, and the order of the views is the argument of the chapter.
- Start with motivation and keep the systems out of it, because a motivation view that names solutions has stopped being analysis and become advocacy.
- One capability map for a composite group, not one per licence, because the duplication between the two businesses is the cost story the group needs to see.
- A value stream has stages, not steps: a stage ends where the customer’s state changes, and drawing the four of them is what exposed the eleven hours spent confirming cover.
- Roles, not job grades, and an Event rather than a first step — because authority limits attach to roles and conduct turnaround is measured between event timestamps.
- The target application view is one service seam the record system owns, not an integration layer in front of it, and writes are commands the record accepts rather than projections.
- Draw the delegated-authority boundary explicitly: the question it has to answer is what the group’s aggregate exposure is today, including the book it does not administer.
- Business object, data object, artifact — the ladder is where lineage becomes physical, and where the reserve movement history is either present or quietly missing.
- Plateaus are states the group could stop at and defend, and the first one carries a regulatory obligation because that is the only work that cannot be deferred.
- A façade before a conversion, a contained forty-year system rather than a solved one, and every rejected option written down — an architecture is the decisions you can still defend.

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CHAPTER 18

Your first ninety days

Ninety days is not enough to understand an insurer, but it is enough to be useful — if you read the record before the strategy, ask each specialist the one question only they can answer, and build something small enough that the people who do the work will correct it.

Ninety days is not long enough to understand an insurer. It is long enough to be useful, and long enough that people stop forgiving the obvious questions. So the problem is what to learn first, from whom, and what to build with it.

What follows is a sequence, not a method. It assumes nobody will brief you properly and that inside a month somebody will ask your opinion on something expensive. Cape Mutual Insurance Group is the worked example: a South African group with a life business, a short-term business and a UMA relationship for specialist liability. Nothing about Cape Mutual is real; the failures it exhibits are ordinary.

Week one: eight things to read, and where they live

Somebody will hand you an organisation chart, a strategy deck and a project list. Leave all three for week two. They describe intent. You need the record.

1. The last published results. Free, and unusually honest because audited. Read the segment analysis, not the chief executive's letter: which businesses make money, from what, and what the group says about its own channel mix. Santam's FY2025 report gives a combined ratio of 88.7% and direct channels at 22% against a target above 30% [1] — two numbers that say where the investment will go.

2. The sector context. The Prudential Authority publishes quarterly sector data: entity counts by licence class, assets, premium, claims ratios, solvency cover [2]. ASISA publishes the life aggregates [3]. Half an hour here sizes your employer against the market.

3. The product catalogue, asked for properly. "Send me the product catalogue" gets brochures. Ask instead for *the rate tables and the wordings, by product version, with in-force policy counts against each*. What comes back, or does not, is your first real finding.

4. Three policy documents and three claim files. One funeral, one motor, one commercial — then a whole claim file for each, file notes included. The highest-yield hour of week one: the schedule is the data model made visible, and the file notes are where the process lives.

5. The reinsurance programme. You will be given one slide. Ask for the treaty wordings and the last cession reconciliation too [4].

6. The last two regulatory returns. The prudential Quantitative Reporting Template, and whatever conduct return the group files. The QRTs were designed under IFRS 4, so the Prudential Authority issued interim guidance: premium terms keep their pre-IFRS 17 meaning [5]. Proof that one cash flow must serve three bases at once.

7. The month-end close calendar and the overnight batch schedule. Together, the true operating model.

8. The complaints register and ombud determinations. The only place where the gap between the document the customer received and the system's behaviour is written down precisely.

Published reports come from the company secretary; returns from actuarial or finance; the close calendar from financial operations; the batch schedule from the people who get paged at night, not the service catalogue; complaints from compliance. Ask for the batch schedule in week one: it arrives faster than an application inventory, and it is truer.

Read the statutory frame once in the background: the Insurance Act 18 of 2017 and its Prudential Standards, commenced 1 July 2018 [6]; the conduct regulations and Policyholder Protection Rules, effective 1 January 2018, which brought minimum data governance requirements and limits on who may hold a binder [7]; and the direction of travel through COFI [8].

The people, and the one thing each can tell you that nobody else can

Book eight conversations in three weeks. Forty-five minutes, no slides, one notebook. Ask each the specific question below, then the same closing one: *what would you fix if it were yours?*

The pricing actuary. Not a stakeholder — a source of requirements. Ask: *where is a rate change actually made, who deploys it, and how long from decision to live quote?* If the actuary builds models in one tool and somebody re-implements them in the policy system, you have found the defect behind mis-rated policies and the audit finding that follows [9]. Then: can any historical quote be reproduced exactly? A complaint fourteen months later needs the rate version that priced it.

The reserving actuary, who in a group of any size is a different person. Ask: *can the warehouse produce a loss development triangle by class, accident period and development period, gross and net of reinsurance, reproducibly?* A pause means the actuarial function has built a shadow data estate, and that estate is now the real system of record. Then: *is reserve movement history immutable and correctly dated?* Overwritten reserves destroy the triangle, and the triangle is the method [10].

The chief underwriter. Ask: *what is our underwriting appetite, and where is it written down?* Then: *what share of new business is decided with no human touch, and which way is it trending?* The gap between the stated straight-through rate and the size of the referral queue measures how honest the reporting is. Then ask what happens when the rules engine cannot decide at all: accept, refer and *no decision* is the real three-way outcome, and the third is where automated underwriting leaks back into manual work [11].

The head of claims. Ask: *how do we establish cover as at the date of loss, and has it ever been wrong?* That reveals whether the policy system can reconstruct a past state, its most consequential property. Then: *where are recoveries tracked — subrogation, salvage, third party, reinsurance — and can you quantify leakage?* If

they live in spreadsheets, leakage is unmeasurable by construction. Then surge: weather catastrophes were the largest force on South African short-term claims costs in the first half of 2026 [12], so it is a design requirement, not a contingency.

The head of distribution. Ask: *what share of the book is written or administered under delegated authority, and at what granularity does that data reach us?* In a binder-heavy market the policy administration system is often not where policies are administered; data arrives as periodic bordereaux in somebody else's format [13]. Then: is commission earned on premium due or on premium received, and where is the shadow accounting? There is always some — a second set of records used to check or challenge official payouts [14].

The CFO. Ask: *does the chart of accounts carry enough information for IFRS 17 posting and for management reporting, or is there an allocation layer?* The IFRS 17 sub-ledger sits between the actuarial and source systems and the general ledger, feeding it interface files [15]; a chart designed before 2023 forces an allocation layer that becomes a permanent reconciliation cost.

The CRO. Ask: *which outsourcing arrangements are material, and when were the regulators notified?* Joint Standard 1 of 2024 took effect on 1 December 2024: due diligence before outsourcing, concentration risk considered where a provider serves several insurers, notification to both regulators at least thirty days beforehand, and existing arrangements compliant within twenty-four months [16]. Whether a cloud core counts as material is unsettled, which is why you want the CRO's view on record before shortlisting anything.

The data people. Ask: *which regulatory returns come out of the warehouse, and which out of a parallel manual process?* And: *is lineage documented from return line to source field?* A warehouse nobody trusts for the return is a reporting artefact with a budget. Ask too who arbitrates when finance, actuarial and operations define one measure differently; ACORD's Business Glossary, over 6,000 terms as at January 2026, is subscription-gated, so that answer has a price [17][18].

The person who has been there twenty years. Usually in operations, sometimes actuarial support, occasionally a developer who will not retire. Find them by asking three people who to call when something strange happens. Ask: *what has been tried before, and why did it stop?* Then: *which system does the business rely on that is not on any diagram?* They will name a spreadsheet, and it will hold the reinsurance programme, the product rules for a closed book, or the mapping between two customer identifiers. In life insurance this person is often the only complete specification of products sold decades ago — a documented reason legacy platforms persist, not a sentimental one [19].

Twelve questions that tell you most

Ask these of everybody. Write down who disagrees; the disagreement is the finding.

1. How many policy administration systems are there, which book sits on each, and who owns each end-of-life plan?
2. Is product definition configuration or code? Walk me through the last launch, with dates.
3. Can you reconstruct a policy as it stood on an arbitrary past date — and does claims use that?
4. Where is premium calculated per channel, and is it one code path for quote, issue, endorsement and renewal?
5. What share of the book sits behind delegated authority, and at what granularity does the data arrive?
6. Can you reconcile premium due to premium received at policy level, and how old is suspense?
7. Is reserve movement history immutable and correctly dated — and do the actuaries trust it?
8. How is cover at date of loss determined, and has it ever been wrong?
9. Where does the reinsurance programme live, and can you evidence one recovery to one treaty clause?
10. What is in the overnight batch, what is the critical path, and what is the restart procedure at three in the morning?
11. Who is system of record for a person, and does any transaction actually call it?
12. Which returns come out of the warehouse, and which come out of a spreadsheet?

> Eleven of the twelve are answered by a demonstration, not an opinion. "Can you reconstruct this policy as at 14 March 2023?" is a request, and the ten minutes somebody spends attempting it teach you more than a workshop.

What you ask for, and what you will actually get

You ask for	What arrives	What to do with it
Application inventory	A spreadsheet, eighteen months old, naming decommissioned systems and missing two live ones	A list of names to verify, never a baseline
Integration catalogue	A diagram with a bus in the middle, no arrows labelled	Ask for the batch schedule and the firewall rules
Process maps	A BPMN set drawn for an audit, describing the intended path	Sit beside a claims handler
Data dictionary	The warehouse's generated table list	Ask for the lineage of one regulatory return line
Product catalogue	Marketing brochures	Ask for rate tables and in-force counts by product version
Reinsurance programme	One slide	Ask for treaty wordings and the last cession reconciliation
Target architecture	A slide with a layer labelled "API"	Ask what was decided, by whom, and what it cost
Data model	An ER diagram of the warehouse, or nothing	Build the small model yourself

None of this is anyone's fault. Insurers are old, acquisitive and heavily regulated; artefacts get produced for an audit and abandoned once it passes. Treat each one as a hypothesis, not as evidence.

The first model, and why it must be small

Resist the instinct to model the estate: a model built from documents in month one is a diagram of the documents.

Build one slice instead — **one product, one event, end to end, down to the data**. For Cape Mutual, a motor claim on a personal-lines policy: the highest-volume path, the shortest feedback loop, and the most people who can tell you when you are wrong.

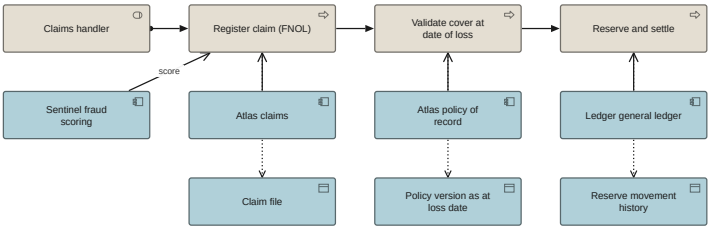


Figure 18.1 The first model: one product, one event, four layers. A personal-lines motor claim at Cape Mutual Short-Term, from handler to data object. Cape Mutual is fictional.

Model it in four layers on one page: the actor, the process steps, the applications serving each step, and the data objects each step reads or writes. Then do the thing that makes it valuable — take it to the claims handler, the pricing actuary and whoever runs the overnight batch, separately, and ask them to correct it.

Three properties earn a small model its place. It is **falsifiable**: a handler can point at a box and say that is not how it works. It is **traceable**: each step names a system and each system names a data object, so an assumption has nowhere to hide. And it is **finishable** in two weeks, which matters, because an unfinished model carries no weight.

What it will expose, in this order: cover validation reads the current policy version rather than the one in force at the date of loss; the fraud score arrives after the payment is instructed; the reserve is stored as a current value rather than a movement; and two of the four systems have their own idea of who the customer is. Each is a defect with a named owner — a better first-quarter output than a target state nobody asked for. If the group runs life and short-term, build the slice twice: a life policy administration system is partly a bank ledger, with surrender values, policy loans accruing interest and paid-up conversion, and a short-term one is not [20].

Telling within a month whether the estate is healthy

No maturity assessment required. Six tells, all answerable in four weeks.

One: does one number reconcile three ways? Take a single month's premium. Ask for it from the policy administration system's premium register, from the general ledger, and from the warehouse feed behind the regulatory return. In a healthy estate they agree, or the differences are derived and documented. In an unhealthy one, three people send three numbers and a fourth explains the bridge from memory.

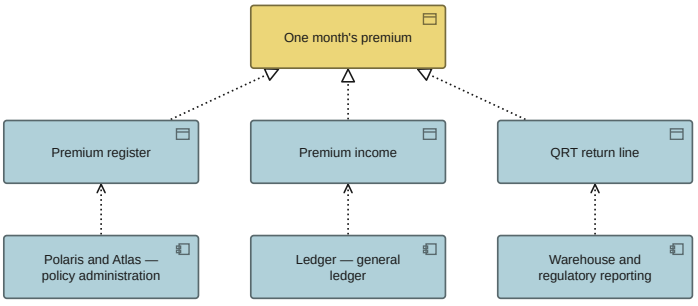


Figure 18.2 The one-month health test, drawn properly: one business fact at the top, three data objects that each claim to realise it, three systems that master them. Ask three owners for the same month's premium. In a healthy estate the differences are derived; in an unhealthy one they are explained from memory.

Two: how long from rate decision to live quote? Days is healthy. A quarter means pricing runs at the core system's release cadence — a competitive position, not an IT inconvenience, where rates may need changing more than once a year [9].

Three: can suspense be aged? Unallocated cash is normal. Unallocated cash nobody can age at policy level means premium received reconciles to premium due only in aggregate, so cover-in-force status is an opinion.

Four: what share of collections are authenticated? AEDO and NAEDO were switched off on 31 October 2021 and DebiCheck replaced them, the mandate authenticated once by the debtor at their own bank and held there [21][22]. A mandate is now a bank-held, long-lived object with its own lifecycle, and something must master it. If nobody can name that component, collection failure is being managed by retrying blindly — and in the mass market, collection failure rather than deliberate cancellation is the leading cause of lapse.

Five: is there a second implementation of anything that prices or pays? A rating spreadsheet behind an aggregator adapter, a commission calculation in the broker portal, a payment path that skips the approval service. Each is a divergence waiting for a regulator or an ombud to find it.

Six: who can still change the oldest system, and how many of them are there? Count the names. If the answer is two and both are over sixty, that outranks every item on the strategy deck. Average life and pensions conversions of thirty-six to forty-eight months are documented [19], as are selection-to-go-live arcs measured in years [23]. Stop promising a two-year core replacement.

Four mistakes newcomers make

Trusting the system inventory. It is a list of names somebody typed. Verify each entry against two independent sources: the general ledger postings and the batch schedule. Systems in neither may not exist; systems in the batch that appear in no inventory certainly do.

Believing the process documentation. Documented processes describe the path a case takes when nothing is wrong. Insurance operations are a sequence of exceptions — the referral, the retroactive endorsement, the claim on a lapsed policy, the broker who emails a spreadsheet. The exception path is the process.

Assuming the warehouse agrees with the policy system. It does not, and it is not meant to. The warehouse masters history: states at each reporting date, slowly-changing dimensions, and for retired platforms the only surviving record of a book. The policy system masters the contract now. Where they differ, ask which question each was built to answer before calling either wrong. The dangerous case is a third source — an actuarial data store, a finance extract — that quietly became authoritative because nobody could get what they needed from the first two [15].

Treating the actuary as a stakeholder. The expensive one. A stakeholder has preferences you balance. The actuarial function has *requirements* that are a condition of the insurer's licence: contract-level, movement-complete, effective-dated data, reproducible at an arbitrary past evaluation date [10][15]. Insurance data is bitemporal — what was true and what was known are two axes, and an auditor will ask the second question [24]. Design for current state, offer the actuaries an extract, and they will build their own estate; it will become the system of record for the numbers that matter.

The ninety-day plan

Every row produces an artefact somebody else can use. An architect with no output in the first quarter has spent credibility and bought nothing.

Days	Theme	What you do	What you produce	Done when
1–10	Read the record	The eight documents; three policies and three claim files; batch schedule and close calendar	One page of questions the reading could not answer	You can say what the group sells and how it makes money
11–20	Listen	The eight conversations and twelve questions; ask to be shown	Notes, plus every disagreement between two senior people	Three people have named the same spreadsheet
21–30	One number, three ways	Premium reconciliation; suspense ageing; collections authentication mix	A two-page health note: six tells, answered or explicitly unanswerable	Somebody senior disputes a figure and you can source it
31–45	Build the slice	One product, one event, four layers, down to the data	The slice model, corrected by the people who do the work	A claims handler has changed a box on it
46–60	Find the real boundaries	Delegated-authority data flows; the rate-change path; where cover-at-date-of-loss is computed; who masters a person	A capability view of the estate; ACORD's Capability Model is a defensible starting point [17]	You can name, per capability, the system of record and who disagrees
61–75	Name the defects	A register: defect, evidence, owner, consequence if ignored	Fewer than fifteen items, each with evidence	Every item has a named owner who has seen it
76–90	Say what you would do	Two or three positions, each with alternatives and what excluded them	Decision records, plus one page on sequencing	The CRO and CFO have read it and neither was surprised by a fact

The artefacts are small on purpose: a two-page health note gets read, a forty-page assessment gets filed. And resist writing a target architecture in the first ninety days. You will be asked for one, probably in week six, by somebody who needs it for a budget cycle. The capability view and the defect register come first, because a vendor shortlist is itself an architectural decision: by the time three names sit in a procurement, the data model, extension model, release cadence and exit cost have been chosen.

What you are really building in ninety days is the right to be believed. Every reconciliation you run, every box a handler corrects, every defect with a named owner buys a little of it. The big model comes after.

THE CHAPTER IN TEN LINES

- Read the record before the strategy: published results, returns, three policies, three claim files, the batch schedule and the close calendar.
- The batch schedule and the month-end close calendar are the real operating model; everything else is intent.
- Eight conversations, one specific question each, and the same closing question: what would you fix if it were yours?
- The actuarial function is a source of licence-critical requirements, not a stakeholder whose preferences you balance.
- Twelve questions tell you most, and eleven of them are answered by a demonstration rather than an opinion.
- Build one slice — one product, one event, four layers, down to the data — and let the people who do the work correct it.
- Six tells answer the health question in a month, starting with whether one month's premium reconciles three ways.
- The newcomer's mistakes are trusting the inventory, believing the process maps, expecting the warehouse to agree with the policy system, and under-serving the actuary.
- Produce something small every fortnight; an architect with no output in the first quarter has spent credibility and bought nothing.
- Decline to write a target architecture in ninety days — the capability view and the defect register have to come first.

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APPENDIX A

The cross-maps

The reference pages an architect actually uses. Five dimensions defined with IDs, then seven maps that join them, all hinged on business process — because process is the only dimension the business will argue about.

What these are, and what they are not

Everything here maps to **business process**, because process is the only dimension the business will argue about and therefore the only safe hub. A capability map alone provokes no argument. A data model alone provokes the wrong one. Put process at the centre and every other dimension has somewhere to attach.

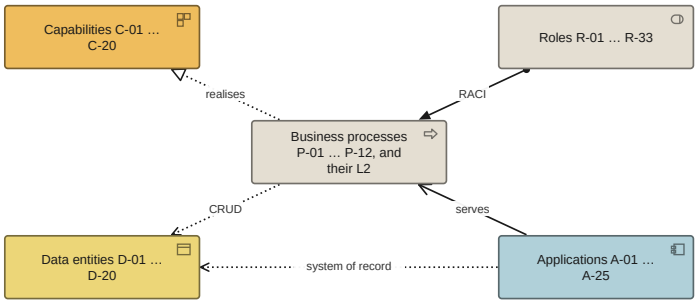


Figure 19.1 Process as the hub, and each cross-map as the ArchiMate relationship it really is: a process realises a capability, a role is assigned to it, it accesses the data, an application serves it. The one map that does not touch process — application to data — is the Access edge along the bottom, and it is the one that settles the system of record.

Five warnings before you use any of it.

A worked starting set for Cape Mutual, not a universal reference. Cape Mutual is the invented composite insurer carried through this book — Cape Mutual Life for risk and funeral, Cape Mutual Short-Term for motor, household and small commercial, and a UMA relationship for specialist liability. Nothing here describes a real company. The shapes are typical; the assignments are mine.

The IDs are local. `P-04`, `C-07`, `A-10` mean something inside this book and nothing outside it. They are not ACORD identifiers [8] and not anybody's standard. Adopt them as your own and version them.

Illustrative, not measured. A real cross-map comes out of workshops with the named role in the room, and it will differ from this one. Where my assignment is contestable the row says so, because the contestable rows are where the decisions live.

The grain is L1 and L2 only. Chapter 8 argues for stopping there, and these maps would be unmaintainable at L3. Where a map needs a step rather than a process, it names the step in the cell.

Expect gaps, and read them as findings. A capability with no application, a process with no accountable role, an entity with no system of record: not errors in the table, but the three things most worth taking to a steering committee.

Adapt it in this order. Processes first, because everything hangs off them. Then roles, the cheapest to verify — walk the list past an operations manager and they will fix a third of it in twenty minutes. Then data, then applications, and capabilities last, because a capability map is what people most enjoy arguing about and least need right early.

Half one — the definitions

Business processes

The L1 inventory is Chapter 8's: twelve end-to-end processes, one trigger, one outcome and one owner each. Three are cycles rather than triggered flows, and are marked. The L2 set uses Chapter 8's modelled L2 processes where it has them and fills the rest at the same grain.

ID	L1 process	Trigger, and the outcome that ends it	Owner
P-01	Develop and maintain a product	A market, pricing or regulatory decision → a product version live in every consuming system	Product manager. A lifecycle, not a flow
P-02	Appoint and manage an intermediary	An application to transact → an accredited intermediary with a mandate and a licence check	Distribution manager
P-03	Quote and bind	A request for a price → a priced offer, accepted or lapsed	Head of distribution
P-04	New business and underwriting	An application or accepted offer → a policy in force, or a decline with reasons	Chief underwriter
P-05	Service the policy	A request to change something → a new policy version and the document proving it	Head of operations
P-06	Collect the premium	A billing date → cash allocated and cover status derived	Head of collections
P-07	Renew or continue	A date → a contract continuing on current or revised terms	Head of distribution. No trigger in a monthly book
P-08	Handle a claim	A notification of loss → a settled or repudiated claim, closed	Claims manager
P-09	Recover	A paid claim with a recovery prospect → cash in, and a final net claim cost	Claims manager, in practice nobody
P-10	Handle a complaint	Dissatisfaction expressed → a resolved complaint with a recorded root cause	Compliance officer. Regulated [1]
P-11	Cede and recover reinsurance	A bound risk, and separately a large loss → a cession recorded; a recovery collected	Reinsurance manager. Two processes, one name
P-12	Reserve, close and report	A calendar date → a filed return and signed accounts	Chief financial officer. A schedule, not a flow

ID	L2 process	What happens
P-01.1	Define and price the cover	Events, exclusions, limits, excesses, waiting periods, algorithm, rate tables
P-01.2	Approve, configure and version	Sign-off and compliance review, then configure every consuming system and version in-force policies
P-02.1	Accredit and licence-check	Verify licence, representative register and fit-and-proper before granting authority
P-02.2	Agree mandate, binder, remuneration	Scope authority, set limits, agree fees, record data ownership and reporting [2]
P-02.3	Monitor authority, ingest bordereaux	Normalise periodic schedules, reconcile to premium received, monitor breaches
P-03.1	Capture the risk, request a price	A risk description complete enough to rate, from whichever channel it came
P-03.2	Rate, refer and set terms	Price on the current rate version; refer outside mandate or appetite
P-03.3	Present, accept and bind	Present the offer, record acceptance, create the policy and first version
P-04.1	Capture the application	The answer set, with reflexive questioning. The basis of any later dispute
P-04.2	Order, chase, receive evidence	Requirements raised, ordered, aged, chased. The main source of life cycle time
P-04.3	Underwrite, offer and issue	Automated, referred or no decision; loadings; facultative referral; then issue
P-05.1	Capture the change and its date	The request, the effective date, and whether it is inside binder authority
P-05.2	Re-rate and write the new version	Decide on re-rate, pro-rate the premium, write a dated new version
P-05.3	Compose and issue the endorsement	Send the document proving the change, and archive what was sent
P-06.1	Acquire and maintain the mandate	Register through the creditor's bank, authenticate with the debtor, amend later [3]
P-06.2	Raise and present the collection	Raise the instalment, build the file, present against approved terms
P-06.3	Retry, allocate and derive cover	Read the reason code, retry as it justifies, clear suspense, derive cover in force
P-07.1	Re-rate and invite	Re-price on current rates and experience; invite within notice requirements
P-07.2	Treat, roll or record the lapse	Intervene above a rate-increase threshold; roll the period, or record silence
P-08.1	Notify, register, validate cover	Capture on any channel; validate against the policy as at the date of loss
P-08.2	Score for fraud and route	Score at registration, route the suspicious, hold before payment
P-08.3	Assess, authorise and settle	Verify quantum, revise the reserve as a movement, authorise, pay, close
P-09.1	Identify and pursue the recovery	Salvage, subrogation, contribution or nothing; dispose, lodge, negotiate or write off
P-09.2	Allocate and finalise net cost	Allocate to the claim and restate the net cost behind the loss ratio
P-10.1	Receive, classify, investigate	Capture, classify against the taxonomy, acknowledge in time, assemble the file
P-10.2	Respond, record cause, escalate	Answer formally, record root cause for conduct reporting, answer the ombud in time [4]
P-11.1	Apply treaty terms, compute cession	Treaty, section and share, per risk, at bind and at every endorsement
P-11.2	Produce bordereaux and accounts	Premium bordereaux, statements and supporting detail per reinsurer [5]
P-11.3	Recover, age and account	Aggregate by loss event within the hours clause, test attachment, claim, age it
P-12.1	Produce the actuarial foundation	Contract-level, movement-complete, effective-dated data, gross and net
P-12.2	Reserve, value and measure	Case reserves and IBNR, or projected cash flows, then measurement [6]
P-12.3	Close, reconcile and file	Post the sub-ledger, reconcile, explain the movement, file the returns [7]

Business capabilities

Twenty capabilities in the business's words. Each stays true if you switch off every computer.

ID	Capability	What the business must be able to do
C-01	Product management	Decide what to sell, define it legally, and change it
C-02	Pricing and rate management	Set a technical price, and know whether it is adequate
C-03	Distribution channel management	Reach customers through chosen channels, and manage the mix
C-04	Intermediary and delegated authority	Appoint intermediaries, police authority, remunerate it
C-05	Party and relationship management	Know who it deals with, in what capacity, with what consent
C-06	Sales and quotation	Turn interest into a priced, bindable offer
C-07	Risk selection and underwriting	Decide which risks to accept, on what terms
C-08	Exposure and accumulation management	Know its aggregate exposure to one event, before the event
C-09	Contract administration	Hold the contract and every change to it, for its whole life
C-10	Premium billing and collection	Bill what is owed and actually collect it
C-11	Claims management	Decide, quantify and settle what it owes
C-12	Recovery management	Get back what somebody else should bear
C-13	Reinsurance management	Buy, place and operate its own cover, and collect on it
C-14	Financial crime management	Detect and act on fraud before money leaves
C-15	Customer service and complaints	Answer the customer, resolve the complaint, learn the cause
C-16	Actuarial valuation and reserving	Estimate what it owes, and explain the movement
C-17	Capital and risk management	Hold enough capital, know its risks, test them
C-18	Financial management and reporting	Close the books and report a number it can defend
C-19	Regulatory compliance and conduct	Stay licensed, and treat customers as the regime requires
C-20	Information and data management	Know what its data means, where it came from, who may see it

Roles

Chapter 5's jobs, narrowed to those appearing in the maps. Two are insurer-side counterparts Chapter 5 does not tabulate separately — distribution manager and head of operations — included because somebody must be accountable for P-02 and P-05.

ID	Role	What the person does
R-01	Product manager	Owns a product line: features, wording, performance, position
R-02	Pricing actuary	Builds and monitors rating models; owns the technical price
R-03	Independent broker	Advises clients, markets risks, services the book, often handles claims
R-04	Tied agent or adviser	Needs analysis and sale of one insurer's products
R-05	Field agent (funeral)	Signs up families at taxi ranks, workplaces and in homes
R-06	UMA or underwriting manager	Underwrites and administers a book under a mandate
R-07	Distribution manager	Appoints and oversees intermediaries; owns channel performance
R-08	Contact centre agent	Answers, identifies, resolves or escalates, measured on handling time
R-09	Underwriter (personal lines)	Works the referral queue; the engine handles the rest
R-10	Underwriter (commercial)	Reads submissions, orders surveys, negotiates, drafts terms
R-11	Chief underwriter	Sets appetite, authority limits and guidelines; signs the big ones
R-12	Referral desk	Clears exceptions against guidelines, fast
R-13	Medical underwriter (life)	Reads medical evidence, applies ratings, orders tests
R-14	Underwriting assistant	Chases documents, prepares files, issues schedules
R-15	Policy servicing clerk	Processes endorsements, address and bank changes, cancellations
R-16	Collections clerk	Chases failed premium, manages mandates and retries
R-17	Claims handler	Decides cover, quantum and case reserve within authority
R-18	Claims manager	Runs the queue, the service levels, the leakage, the panel
R-19	Assessor (motor)	Inspects vehicles; agrees the repair method and cost
R-20	Loss adjuster (commercial)	Investigates larger or disputed losses independently
R-21	Forensic investigator	Pursues suspected fraud; interviews; builds the file
R-22	Recoveries clerk	Chases third parties, salvage buyers and other insurers
R-23	Reinsurance manager	Cessions, bordereaux, technical accounts, recoveries
R-24	Treaty negotiator	Designs and places the programme; negotiates retentions
R-25	Reserving actuary	Builds triangles, estimates IBNR, explains the movement
R-26	Head of Actuarial Function	Statutory opinion on provisions, pricing, reinsurance adequacy
R-27	Capital actuary	Runs the capital model and the solvency assessment
R-28	Financial accountant	Reconciles premium, claims, commission and suspense
R-29	Compliance officer	Monitors against the conduct regime and reports breaches
R-30	Ombud liaison	Answers complaints escalated outside the company, in time
R-31	Legal counsel	Wordings, contracts, litigation, regulatory correspondence
R-32	Head of data	Warehouse, reporting, governance, the actuarial foundation
R-33	Head of operations	Owns servicing throughput, cost to serve, the authority model

Conceptual data entities

Chapter 9's fifteen, then the five it admits as borderline and this book keeps. No attributes, no keys, no system names.

ID	Entity	What it means	Attributes that matter
D-01	Party	A person or organisation we deal with	Identity, identifiers, type, contact points
D-02	Party Role	What a party is to us, between two dates	Role type, valid-from and valid-to, what it attaches to
D-03	Intermediary Agreement	What an intermediary may do on our paper	Authority scope, limits, licence status, remuneration, dates
D-04	Product	What we manufacture and sell	Version, effective dates, class and sub-class, cover structure
D-05	Policy	One sold instance of a product	Number, version, status, inception, expiry, intermediary of record
D-06	Cover	A specific promise with its own ceiling	Limit, excess, sum insured or assured, conditions, dates
D-07	Risk Object	The thing exposed to loss	Type, identity, location, value, construction
D-08	Premium	The price of a cover	Risk premium, loadings, discounts, fees, commission, tax, state
D-09	Loss Event	What happened in the world	Date, peril, geography, the hours clause it falls inside
D-10	Claim	An assertion that a covered event occurred	The four dates, cause code, status, incurred and net cost
D-11	Reserve	What we think a claim will cost	Amount, head of damage, as-at date, movement reason. Append-only
D-12	Payment	Money out against a claim	Payee role, amount, date, head of damage, tax treatment
D-13	Recovery	Money back in, reducing a claim's cost	Route, counterparty, expected, received, age
D-14	Reinsurance Arrangement	Our own cover against our own book	Type, basis, retention, limits, reinstatements, period
D-15	Document	What we sent, and what it said	Type, template version, rendered artefact, channel, date
D-16	Collection Mandate	Authorisation to take money from an account	Authentication type, approved terms, status, holding bank [3]
D-17	Advice Record	What was recommended, to whom, on what analysis	Needs analysis, recommendation, disclosures, adviser, date
D-18	Cession	The join between a policy and a reinsurance arrangement	Share, ceded premium, treaty section, effective dates
D-19	Quote	A priced offer the insurer made	Risk description, price, rate version, validity, outcome
D-20	Complaint	A recorded expression of dissatisfaction	Category, root cause, outcome, turnaround, escalation route

Conceptual applications

Chapter 11's twenty-five, by ID. That chapter carries purpose, user roles, data mastered and the real-world category; this is the index.

ID	Application capability	Domain
A-01	Customer engagement	Engagement
A-02	Intermediary engagement	Engagement
A-03	Intermediary and delegated-authority management	Distribution
A-04	Remuneration management	Distribution
A-05	Product definition	Product
A-06	Rating and pricing	Product
A-07	Quote and application management	Policy
A-08	Policy administration	Policy
A-09	Risk decisioning	Underwriting
A-10	Underwriting workbench	Underwriting
A-11	Exposure and accumulation management	Underwriting
A-12	Claims management	Claims
A-13	Financial crime detection	Claims
A-14	Billing and premium accounting	Billing
A-15	Collections and mandate management	Billing
A-16	Payment and disbursement	Billing
A-17	Reinsurance administration	Reinsurance
A-18	Party and relationship master	Party
A-19	Interaction and service management	Party
A-20	Document and correspondence management	Document
A-21	Finance and general ledger	Finance
A-22	Actuarial modelling and reserving	Risk and actuarial
A-23	Insurance contract measurement	Risk and actuarial
A-24	Enterprise data and reporting	Data and analytics
A-25	Identity, entitlement and audit	Enterprise common

Half two — the maps

Seven maps, all keyed on the IDs above. Where a matrix would be too wide for the page it is a list table of pairs, which is also the form that survives being pasted into a repository.

Capability to process

Read it both ways: a capability exercised by no process is an aspiration, and a process exercising seven capabilities is probably two processes.

Capability	Processes that exercise it
C-01	P-01.1, P-01.2
C-02	P-01.1, P-03.2, P-07.1
C-03	P-03.1, P-03.3, P-07.1
C-04	P-02.1, P-02.2, P-02.3, P-05.1
C-05	P-03.1, P-04.1, P-04.3, P-05.1, P-08.1
C-06	P-03.1, P-03.2, P-03.3, P-04.1
C-07	P-03.2, P-04.2, P-04.3, P-05.2, P-07.1
C-08	P-03.2, P-04.3, P-11.1, P-11.3
C-09	P-03.3, P-04.3, P-05.1, P-05.2, P-05.3, P-07.2
C-10	P-06.1, P-06.2, P-06.3
C-11	P-08.1, P-08.2, P-08.3
C-12	P-09.1, P-09.2
C-13	P-11.1, P-11.2, P-11.3
C-14	P-04.1, P-08.2, P-08.3
C-15	P-05.1, P-10.1, P-10.2
C-16	P-08.3, P-12.1, P-12.2
C-17	P-11.1, P-12.2
C-18	P-06.3, P-11.3, P-12.3
C-19	P-01.2, P-02.1, P-10.2, P-12.3
C-20	P-12.1, and every process as a producer

Two findings fall out at Cape Mutual. **C-08 is exercised by four processes and supported by no application** — the exposure gap Chapter 11 names. And **C-20 is exercised by everything and owned by one person**, which is the normal state of data governance in an insurer.

Process to role, as RACI

One accountable role per process: that is the whole discipline. Where I could not honestly name one, the row says so, because that is the finding rather than a gap in the table.

Process	Accountable	Responsible	Consulted	Informed
P-01	R-01	R-01, R-02	R-11, R-24, R-29, R-31	R-03, R-04, R-07
P-02	R-07	R-07	R-11, R-29	R-28, R-32
P-03	R-07	R-03, R-04, R-08, R-10	R-09, R-12	R-02
P-04	R-11	R-09, R-10, R-13, R-14	R-12, R-24	R-01, R-02
P-05	R-33	R-03, R-08, R-15	R-09, R-10	R-16, R-28
P-06	R-33	R-16	R-28	R-07, R-17
P-07	R-07	R-03, R-15	R-02, R-10	R-01
P-08	R-18	R-17, R-19, R-20	R-10, R-21, R-23	R-25, R-28
P-09	R-18 nominally; nobody in practice	R-22	R-23, R-31	R-25, R-28
P-10	R-29	R-08, R-30	R-15, R-18, R-31	R-01, R-33
P-11	R-23	R-23, R-24	R-25, R-28	R-26, R-27
P-12	R-25 and R-28 jointly; contested	R-25, R-28	R-27, R-32	R-26, R-29

Three rows are the honest ones. **P-09 has no accountable role**, which is why recoveries leak. **P-12 has two**, which is why actuarial-to-finance reconciliation is manual. And **P-05 is accountable to operations but performed largely by R-03**, a party outside the company — the delegated-authority problem stated as a RACI.

Process to data, as CRUD

Which entities each process touches, and how. The complementary question — which *application* masters each entity — is answered below, and the two together are what you need before a migration.

Process	Entities touched
P-01	D-04 **CU**, D-06 CU, D-15 CU
P-02	D-03 **CU**, D-01 CU, D-02 CU, D-15 C
P-03	D-19 **CU**, D-07 CU, D-01 CRU, D-02 C, D-04 R, D-05 C, D-06 C, D-08 C, D-15 C
P-04	D-17 **C**, D-01 CRU, D-02 C, D-05 CU, D-06 CU, D-08 CU, D-16 R, D-18 C, D-15 C
P-05	D-05 **CU**, D-06 CU, D-07 CRUD, D-08 CU, D-02 CU, D-16 RU, D-18 U, D-15 C
P-06	D-16 **CRUD**, D-08 RU, D-05 RU, D-15 C
P-07	D-05 CU, D-06 U, D-08 CU, D-19 C, D-15 C
P-08	D-09 **C**, D-10 **CU**, D-11 **CU**, D-12 C, D-01 CRU, D-02 C, D-05 R, D-06 R, D-15 C
P-09	D-13 **CU**, D-10 U, D-11 U, D-07 U, D-01 CR
P-10	D-20 **CU**, D-15 CR, D-17 R, D-10 R, D-05 R
P-11	D-18 **CU**, D-14 RU, D-13 CU, D-08 R, D-09 R, D-10 R
P-12	D-08 R, D-10 R, D-11 R, D-12 R, D-13 R, D-18 R. Creates nothing

Bold marks the process that is the natural *creator* of an entity, and that column repays reading alone. One process creates a loss event. One creates a mandate. One creates a policy version legitimately, and if two do, the second is the problem. Note that **P-12 creates nothing**: the reporting cycle is pure consumption at this grain, which is why its data problems are always upstream and never fixable inside it.

Process to application

Process	Applications
P-01	A-05, A-06, A-20, A-24
P-02	A-03, A-04, A-18, A-25
P-03	A-01, A-02, A-06, A-07, A-08, A-09, A-10, A-20, A-25
P-04	A-07, A-08, A-09, A-10, A-13, A-15, A-17, A-18, A-20
P-05	A-02, A-06, A-08, A-15, A-19, A-20, A-25
P-06	A-08, A-14, A-15, A-19
P-07	A-02, A-06, A-08, A-19, A-20
P-08	A-01, A-08, A-12, A-13, A-16, A-18, A-19, A-20
P-09	A-12, A-16, A-21 — and, in most insurers, a spreadsheet
P-10	A-08, A-12, A-19, A-20, A-24
P-11	A-08, A-11, A-12, A-17, A-21
P-12	A-17, A-21, A-22, A-23, A-24

Count the applications per process and the procurement lesson appears. **Quote and bind touches nine; new business touches nine.** Anyone proposing to replace one of them with one product is proposing to replace nine things, and that conversation belongs before the shortlist.

Application to data, with the system of record

The most useful table here. **Bold means system of record**: the application creates the entity, owns its lifecycle, and everything else holds a copy.

Application	System of record for	Also reads
A-01	nothing	D-04, D-05, D-10, D-15
A-02	nothing	D-03, D-05, D-08, D-19
A-03	**D-03**	D-01, D-02, D-05
A-04	nothing; remuneration is a component of D-08	D-03, D-05, D-08
A-05	**D-04**	D-06 as structure
A-06	nothing; rate versions sit inside D-04	D-02, D-04, D-07
A-07	**D-19, D-17**	D-01, D-04, D-07
A-08	**D-05, D-06, D-07, D-08**	D-01, D-02, D-04, D-16, D-18
A-09	nothing	D-01, D-05, D-06, D-07
A-10	the decision, which Chapter 9 holds inside D-05	D-05, D-06, D-14
A-11	nothing conceptual; masters modelled aggregates	D-06, D-07, D-09, D-14
A-12	**D-09, D-10, D-11, D-12**	D-01, D-02, D-05, D-06, D-18
A-13	nothing conceptual; masters scores and outcome labels	D-01, D-07, D-10, D-12
A-14	nothing; premium is D-08, mastered by A-08	D-05, D-08, D-16
A-15	**D-16**	D-05, D-08
A-16	nothing conceptual; masters payee bank detail	D-02, D-12, D-13
A-17	**D-14, D-18**, and D-13 on the reinsurance route	D-05, D-08, D-09, D-10
A-18	**D-01, D-02**	D-03, D-05, D-10
A-19	**D-20**	D-01, D-05, D-10, D-15
A-20	**D-15**	D-01, D-04, D-05, D-08, D-10
A-21	nothing conceptual; postings sit below this grain	D-08, D-12, D-13, D-18
A-22	nothing; consumes D-11, produces provisions	D-05, D-08, D-10, D-11, D-18
A-23	nothing conceptual; groups sit below this grain	D-05, D-08, D-10, D-11
A-24	history of everything; authoritative for nothing current	all
A-25	nothing conceptual; masters entitlement and the audit trail	D-01, D-02, D-03

Three rows deserve argument, and they are the right arguments. **A-08 masters four of the twenty entities**, which is why replacing it is a programme rather than a project. **A-11, A-13 and A-16 master nothing conceptual** yet master things that matter enormously — modelled aggregates, outcome labels, verified bank details — which shows that a twenty-entity model does not cover everything an application must own. And **A-04 masters nothing at all** here, yet nobody trusts its numbers: the problem with commission is not where the data lives but that it is derived five ways.

Application to role

Application	Roles
A-01	policyholders, R-08
A-02	R-03, R-04, R-05, R-06
A-03	R-07, R-29
A-04	R-03, R-04, R-07, R-28
A-05	R-01, R-02, R-31
A-06	R-02; consumed by every channel
A-07	R-03, R-04, R-08, R-14
A-08	R-06, R-09, R-10, R-15
A-09	R-11 as rule owner, R-12 on the exits
A-10	R-10, R-12, R-13
A-11	R-10, R-11, R-24, R-27
A-12	R-17, R-18, R-19, R-20, R-22
A-13	R-17, R-21
A-14	R-16, R-28
A-15	R-16
A-16	R-18, R-28, R-29
A-17	R-23, R-24, R-25, R-28
A-18	R-32, and every role indirectly
A-19	R-08, R-29, R-30
A-20	R-15, R-29, R-31
A-21	R-28
A-22	R-25, R-26, R-27
A-23	R-25, R-28
A-24	R-02, R-25, R-29, R-32
A-25	R-29, R-33

Read the role column instead and look for roles appearing once. **R-22, the recoveries clerk, appears in one application** and in practice works in a spreadsheet. **R-30, the ombud liaison, appears in one** and reconstructs files from email. A role with one application and a regulated output is a gap with a deadline attached.

Capability to application — the coverage map

The map that goes in the board pack. Coverage is my assessment against the invented Cape Mutual estate: **full** where one application serves the capability and masters its data, **partial** where it is served but not mastered, or served by several, and **gap** where nothing serves it.

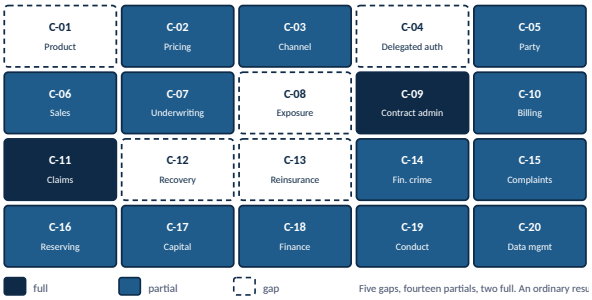


Figure 19.2 The coverage map as one picture, against the invented Cape Mutual estate. The dashed cells are the board paper.

Capability	Applications	Cape Mutual coverage
C-01	A-05, A-20	**Gap in practice.** Spread across four systems and a file share
C-02	A-06, A-24	Partial. Three rating implementations
C-03	A-01, A-02, A-19	Partial. Three channels, three customer identities
C-04	A-03, A-04, A-25	**Gap.** Identity in the commission system, mandates in email
C-05	A-18, A-19	Partial. Analytical, not operational
C-06	A-01, A-02, A-06, A-07	Partial. Quotes discarded after thirty days
C-07	A-07, A-09, A-10	Partial. Referrals by email, no decision audit
C-08	A-11	**Gap.** Nothing
C-09	A-08, A-20	Full, three times over
C-10	A-14, A-15	Partial. One collections module per platform
C-11	A-12, A-13, A-16	Full for short-term, manual for funeral
C-12	A-12, A-21	**Gap.** Reconstructed quarterly from an extract
C-13	A-11, A-17	**Gap.** One spreadsheet, one person
C-14	A-13	Partial. Scoring exists, the pre-payment hold does not
C-15	A-19, A-20	Partial. Complaints recorded, root cause not
C-16	A-22, A-24	Partial. Runs on extracts nobody can reproduce
C-17	A-11, A-22	Partial. No exposure input
C-18	A-21, A-23, A-24	Partial. No contract-level traceability
C-19	A-19, A-24, A-25	Partial. Returns from a parallel manual process
C-20	A-24	Partial. No lineage from return line to source field

Five gaps and fourteen partials is an ordinary result and should not be presented as a crisis. The useful output is the ordering. Two gaps — exposure and accumulation, and reinsurance administration — carry quantified downside and no political resistance, which makes them the first two to fund. Two more — delegated authority and product management — are organisational fights before they are system builds. Recovery management is the cheapest and the one nobody will sponsor, because its beneficiary is the loss ratio rather than a department.

Keeping these alive

One owner per dimension, named. Processes to an operations leader, capabilities to strategy, data to the head of data, applications to the lead architect, and roles to human resources — who will be surprised to be asked and then produce a better list than yours.

Change the hub last. Renumbering processes invalidates every map. Add `P-13` rather than renumbering, and retire IDs rather than reusing them, as you would product versions.

Review on a trigger, not a date. Quarterly will be skipped. Before each architecture board, before each package evaluation and after each reorganisation are triggers, and triggers survive.

Publish the gaps separately. Nobody reads reference material. The five gaps and the three honest RACI rows are a one-page paper, and that page is what the cross-maps were built to produce.

THE CHAPTER IN TEN LINES

- Hinge every cross-map on business process; it is the only dimension the business will contest, and therefore the only one that gets corrected.
- Five dimensions, five ID series: processes P, capabilities C, roles R, data entities D, applications A. Keep them local, versioned, and never renumbered.
- One accountable role per process. Where you cannot honestly name one — recoveries, usually — write that in the table rather than inventing an owner.
- Mark the system of record explicitly in the application-to-data map. Four of twenty entities mastered by one application is what makes a core replacement a programme.
- The gaps are the output. A capability with no application, a process with no accountable role and an entity with no system of record are the three findings worth a steering committee.
- These are a worked starting set for a fictional insurer. Rebuild them in workshops with the named role in the room, and expect a third of the role assignments to change in the first twenty minutes.

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APPENDIX B

Glossary

Most insurance vocabulary is ordinary English used with a precise and different meaning. These are the words, in plain language, with the dangerous ones marked.

Insurance vocabulary is treacherous in a specific way: most of it is ordinary English used with a precise and different meaning. An excess is not a surplus. A schedule is not a timetable. A reserve is not money set aside. An architect who models those words as they sound builds something that looks right and is wrong, and the error surfaces three years later in a reconciliation nobody can close.

So this glossary does two things. It defines the terms in plain language, one or two sentences each. And it marks with a dagger — † — every term that means something materially different in insurance than in everyday use. Those are the ones to be careful with in a workshop, because everyone in the room will think they already understand them.

Three habits are worth more than any definition here. Whenever somebody says "premium", ask *gross or net* (before or after reinsurance), *written or earned*, and *on what basis* (statutory, IFRS 17, or management); those three qualifiers decide whether two numbers are comparable. Establish whether you are discussing life or short-term business, because each uses different words for the same structure. And note the South African entries: local statutory vocabulary diverges from Anglo-American usage more than newcomers expect.

The accounting, prudential and conduct instruments behind several of these entries — IFRS 17, SAM, the Policyholder Protection Rules, POPIA — are registered with owners, access terms and links in Appendix B.

Term	Meaning
Accident year †	Claims grouped by the year the loss occurred, whatever year it was reported or paid in. The usual reserving basis for non-life business; underwriting year groups the same claims by when the policy was written, and the two produce different triangles from identical data.
ACORD	Insurance's standards body, founded 1970. Publishes messaging standards and a seven-facet Reference Architecture, nearly all of it subscription-gated.
Administrator (third-party administrator, TPA)	A firm that administers policies or a scheme — issuing, collecting premium, paying claims — without underwriting authority. In medical schemes an accredited role under the Medical Schemes Act.
Adviser †	A licensed person who gives financial advice and must record it. Under FAIS a regulated role with fit-and-proper requirements and a record of advice, not simply anyone with an opinion about cover.
Aggregation (aggregate exposure)	The total at risk from one event across many policies. What catastrophe reinsurance exists to cover, and a geocoded query on risk objects.
Aggregator	A comparison site that takes one set of answers and returns several insurers' prices. Appearing on one forces sub-second bindable quoting and somebody else's question set on you.
Agricultural insurance	Cover for crops, livestock, hail damage and farm assets. Weather-driven, sharply seasonal and heavily reinsured.
Anniversary †	The yearly date on which a policy re-rates, escalates premium and benefit, and resets annual limits. A dated contractual event that must be derivable for every policy, not a sentimental one.
Annuity	A contract turning a capital sum into income, usually for life. A life annuity leaves longevity risk with the insurer; a South African living annuity leaves investment and drawdown risk with the retiree.
Anti-selection	Higher-risk people buying more cover than the pricing assumed, so the pool deteriorates and the rate stops being adequate.
Assessor †	The person who inspects damaged property and quantifies the repair, most often motor. Neither a tax assessor nor the same role as a loss adjuster.
Attachment point	The loss level at which a non-proportional reinsurance layer starts to pay.
Average (condition of average) †	Insure below full value and the insurer pays the same *proportion* of the loss. Nothing to do with arithmetic means.
Bancassurance	Insurance sold through a bank, often at the point of credit. Means shared identity, policies triggered by loan origination, and commission moving between group entities.
Beneficiary	Receives the proceeds when the insured event happens. Has a share, a class and dates, so never a single foreign key.
Benefit	The amount or service payable on an insured event; life insurance's counterpart of a short-term cover.
Binder †	South Africa: a written mandate letting an intermediary perform insurer functions — issue, service, settle — within stated limits.
Binder fee / outsourcing fee	Payment for performing insurer functions under a binder. Must be separable from commission; the regulator asks separately.
Binder holder	The intermediary holding that mandate. Approval under the Insurance Act is required; the insurer stays responsible.
Bitemporal	Two time axes: when a fact was true in the world, and when the insurer knew it. Auditors ask about the second.
Bonus (life) †	A declared addition to the value of a with-profits or smoothed-bonus policy. Not a reward for performance, and usually guaranteed once declared.
Bordereau (pl. bordereaux)	A periodic itemised schedule of premiums ceded or losses paid, from cedant to reinsurer or from a delegated-authority partner to an insurer.
Broker	An intermediary who represents the client rather than the insurer and places business with several of them. In South Africa a broker may also issue, service and settle under a binder.
Business interruption †	Cover for lost gross profit and increased cost of working after insured damage, measured over an agreed indemnity period. Nothing to do with a systems outage.
Cancellation †	Either party deliberately ending the contract, with notice and usually a pro-rata refund. A decision, where a lapse is a consequence; the two must not share a status code.
Captive †	An insurer owned by a non-insurance group to carry that group's own risk. A cell captive is the version where the group rents somebody else's licence instead of obtaining one.
Case reserve	A handler's estimate of what one known claim will cost.
Catastrophe (cat) †	An accumulation of losses from one event, bounded by an hours clause and a geography rather than by how bad it felt. The unit on which excess-of-loss reinsurance pays.
Causal event charge	The South African cap on what a life insurer may deduct when an older policy lapses, is surrendered, transferred or reduced. The caps step down over time, so the rule is a dated table rather than a constant.
CBR	The Conduct of Business Return, the periodic South African conduct return carrying indicators by product, channel and outcome. It pushes reporting granularity down to cover and intermediary level.
Cedant (cedent) †	An insurer when it is buying reinsurance; the reinsurer's "customer".
Ceding commission	The reinsurer's contribution towards the cedant's acquisition costs on a proportional treaty.
Cell captive	A South African structure where a brand buys shares in a licensed insurer and takes the underwriting result of its own ring-fenced book.
Cession †	Passing a share of a risk to a reinsurer, and the record of it. In South African retail it also means pledging a policy to a lender.
Chain ladder †	The standard reserving method: project how claims developed in past periods onto periods not yet settled. Nothing to do with chains or ladders.

Claim	A demand for payment under a policy.
Claimant	Whoever asserts a claim. May be a complete stranger to the insurer.
Claims leakage	Money paid out that was not contractually owed: weak cover validation, over-settlement, missed recoveries, fraud.
Co-insurance †	Several insurers sharing one risk. In United States health cover, the percentage of a bill the patient pays after the deductible.
COFI	The Conduct of Financial Institutions Bill, intended to consolidate conduct regulation across financial sectors and absorb much of FAIS. Still in motion, so check its status before relying on a date.
Combined ratio †	Claims plus expenses over earned premium. Below 100% the underwriting made money, so lower is better.
Commercial lines	Short-term insurance sold to businesses: fire, business interruption, liability, engineering, marine, agricultural, trade credit and guarantees. Large, unlike risks underwritten case by case.
Commission	The intermediary's share of the premium, usually a percentage and often capped.
Commission ratio	Commission over premium, usually earned. The quickest read on what a channel costs, and the number most often shifted between premium and expenses to flatter a combined ratio.
Contractual Service Margin (CSM)	Under IFRS 17, unearned profit on a group of contracts, released to profit as service is delivered.
Contribution †	The sharing of one loss between two policies that both cover it.
Cooling-off period	A statutory window in which a new policyholder may cancel and get the premium back.
Cover / coverage	A promise to pay in defined circumstances, with its own limit and conditions.
Coverage units †	Under IFRS 17, the measure of how much service a group of contracts provides in a period, used to release the CSM. Not units of cover sold.
Coverholder	The Lloyd's and London-market name for a UMA or MGA holding delegated authority.
Credit life †	Cover that settles or services a debt if the borrower dies, is disabled or is retrenched. Not life cover bought on credit: it is sold at loan origination and driven by the lender's systems.
Date of loss	When the insured event occurred. It decides which policy version responds and which treaty year attaches.
DebiCheck	South Africa's authenticated debit order. The debtor confirms the mandate once with their own bank, which then checks every collection.
Debit order	A collection instruction against a bank account, now authenticated through DebiCheck. The dominant premium collection method in South Africa, and the reason collection failure rather than cancellation drives lapse.
Deductible	The first slice of every claim the insured pays themselves. United States and health usage.
Delegated authority	Any arrangement where somebody other than the insurer binds risk or settles claims on the insurer's paper.
Demarcation †	The South African regulations drawing the statutory line between health insurance and medical scheme business. A health product is prohibited as insurance unless the regulations name it.
Development †	How an estimate changes across successive valuation dates. Analysed in triangles.
Earned premium †	The part of premium relating to cover already provided. Calculated from exposure dates, not from receipts.
Embedded insurance	Cover sold inside somebody else's purchase — a handset, a loan, a retail account. In South Africa it usually reaches the market through a cell captive.
Embedded value (EV)	Net asset value plus the present value of future profits on the existing book, less the cost of capital. The life valuation measure.
Employee benefits †	The group risk and retirement business sold to employers for their staff. A line of business with its own systems and actuaries, not a list of workplace perks.
Endorsement †	A formal mid-term change to a policy, and the document effecting it. It produces a new policy version.
Engineering insurance †	Cover for plant, machinery breakdown, contract works and erection risk. Nothing to do with professional indemnity for engineers.
Ex gratia payment	A goodwill payment made without admitting the claim was covered. Flag it; regulators and reinsurers both ask.
Excess †	The South African and British word for a deductible: compulsory, voluntary, fixed or percentage. Not consideration for a supply, so no VAT.
Excess of loss (XoL)	Non-proportional reinsurance paying above an attachment point up to a limit, written in layers.
Exclusion	A circumstance the policy explicitly does not cover.
Expense ratio	Underwriting expenses over earned premium: the cost of running and distributing the book.
Facultative	Reinsurance negotiated risk by risk, which the reinsurer may decline.
FAIS	The Act licensing South African advisers, brokers and UMAs, with fit-and-proper and advice-record requirements.
FICA	The Financial Intelligence Centre Act: customer due diligence, record-keeping and suspicious transaction reporting. Risk-based, so identification requirements differ by product and client.
Financial Sector Conduct Authority (FSCA)	South Africa's market conduct regulator. Wants process and outcome over a period: who was advised, what was sold, how long it took, what the customer got, who complained.
First notification of loss (FNOL)	The moment a loss is reported: the earliest point for triage, fraud scoring and reserving.
Float †	Premium held between receipt and payout, invested for the insurer's own account. Not a share issue.

Free cover limit †	In group life, the sum assured a scheme member gets without individual medical underwriting. Not free cover: above the limit underwriting applies, and the excess benefit may not be in force until it is done.
FSP	Financial Services Provider: a firm or person licensed under FAIS to give advice or render intermediary services. Every intermediary party needs a licence number, a category and a dated authorisation status.
Fund risk	The South African statutory class for risk benefits provided through a retirement fund rather than sold to an individual. The fund is the policyholder and the members are the lives assured.
Funeral cover	Low-premium life cover paying a small lump sum fast, usually within days, frequently over several extended-family lives on one policy. The highest-volume product in the South African market.
Gap cover	Short-term insurance paying the shortfall between what a medical scheme covered and what the provider charged. One of the few health products the demarcation regulations permit as insurance.
Gross earned premium	Earned premium before reinsurance. The two pairings are independent: gross or net is about reinsurance, written or earned is about elapsed time.
Gross written premium (GWP)	Premium contracted in the period, before reinsurance and before earning. The industry top line.
Group life	Death cover for the members of a scheme under one policy held by the employer or fund. The policyholder is not the life assured, and the membership changes every month.
Group risk †	The death, disability and income protection cover sold to a scheme as a package. Not risk shared between a group of insurers.
Group scheme †	The arrangement under which an employer or association insures a defined membership. A dated membership list rather than a market segment, and reconciling it against premium received is a permanent operational cost.
Head of Actuarial Function	The statutory actuarial role every South African insurer must appoint under the Insurance Act, reporting on technical provisions, pricing and the adequacy of reinsurance.
Homeowners †	Cover for the building: structure, walls, roof, fixtures, usually required by the mortgage lender. A different product from householders, and conflating the two is a standard data error.
Hospital cash plan	Cover paying a fixed amount for each day in hospital, towards incidental costs and lost income rather than medical bills. One of the health products the demarcation regulations permit as insurance.
Hours clause †	The window — commonly 48 or 72 hours — inside which losses count as one event for reinsurance. Nothing to do with business hours, and without it no catastrophe recovery can be computed.
Householders †	Cover for the contents of a home, plus the occupier's liability. The companion to homeowners, and in most South African wordings a separate section with its own sum insured and excess.
IBNER	Incurred but not enough reported: expected further development on claims already known.
IBNR	Incurred but not reported: an actuarial estimate of claims that have happened but the insurer does not yet know about.
IFRS 17	The insurance contracts accounting standard, effective 2023. For an architect, a mandatory change to the unit of account and so to the grain of every finance number.
Inception	The date and time cover starts.
Inclusive insurance	Cover deliberately designed to reach low-income or previously unserved customers. The policy objective that microinsurance licensing exists to serve.
Indemnity †	A payout restores the pre-loss position and no better. There is no profit from a claim.
Instalment †	A scheduled collection against a premium, not premium itself. Model it as premium and endorsements become unhandleable.
Insurable interest	You must stand to suffer real loss, or the contract is an unenforceable wager.
Insurance Act 18 of 2017	South Africa's prudential insurance statute: licensing by class and sub-class, governance requirements, and the legal base of the SAM regime.
Insured	The party whose interest the policy protects. Frequently not the policyholder.
Intermediary	A licensed middleman paid to sell or service policies.
Inuring †	The order in which reinsurance contracts apply to one another. Often documented only in somebody's head.
Inwards reinsurance †	Reinsurance an insurer *accepts* from other insurers, as opposed to the cover it buys. Most primary insurers write some, and it needs its own treaty, reserving and reporting treatment.
Joint Standard	A prudential standard issued by the Prudential Authority and the FSCA together, where a matter touches both soundness and conduct. Joint Standard 1 of 2024 governs outsourcing.
Lapse †	A policy ending because premiums stopped. Not an expiry and not a cancellation.
Liability for incurred claims (LIC)	Under IFRS 17, the provision for claims already incurred: present value of expected payments plus a risk adjustment.
Liability for remaining coverage (LRC)	Under IFRS 17, the provision for cover still to be given; successor to unearned premium.
Liability insurance	Cover for what the insured may owe somebody else: public, products, employers', professional indemnity, directors' and officers'. Long-tail, so claims arrive years after the policy ended.
Life assured †	The person on whose death, disability or illness a life policy pays. Not necessarily the person who owns it.
Limit	The ceiling on what the insurer pays: per claim, per event or per year.
Loading †	An increase to the standard premium for assessed extra risk.

Long-tail	Business where claims emerge years later: liability, professional indemnity, asbestos.
Loss adjuster	An independent specialist appointed on larger or contested claims to investigate cover, liability and quantum and to recommend a settlement.
Loss adjustment expense (LAE)	The cost of investigating and settling claims: adjusters, assessors, lawyers, handlers. Allocated or unallocated, and whether it sits in the loss ratio or the expense ratio is a choice somebody must pin down.
Loss development triangle	Claims by accident period down the rows, development period across the columns. The raw material of reserving.
Loss event / occurrence	The thing that happened. One event can generate thousands of claims.
Loss ratio	Claims cost over earned premium. A read on risk selection and rate adequacy.
Mandate †	The debtor's standing authority to collect premium. Under DebiCheck a long-lived object held at the debtor's bank, not a form in your cabinet.
Marine insurance †	Cover for goods and vessels in transit, including inland goods in transit by road. Not confined to the sea.
Material non-disclosure	Withholding something that would have changed the insurer's decision to accept or price the risk.
Medical scheme †	In South Africa a not-for-profit fund regulated under the Medical Schemes Act. Neither an insurer nor a seller of insurance, however much it looks like both to a member.
Member †	On a group scheme or a medical scheme, the covered individual. Usually neither the policyholder nor a contracting party, and the membership arrives as a monthly schedule rather than as transactions.
Microinsurance	A lighter South African licence class for financial inclusion, with simpler products and proportionate requirements.
Monthly-renewable	A short-term policy that renews every month rather than annually, standard in South African personal lines. It removes the annual renewal event most imported system designs assume exists.
Net earned premium (NEP)	Earned premium after reinsurance ceded. The denominator of the loss, expense and combined ratios, and a number that moves independently of gross written premium.
Net written premium	Written premium after deducting reinsurance ceded. What the insurer actually retains.
No-claim bonus	A premium reduction or cash-back earned by not claiming. A policy-level accrual with its own history, so it survives renewals and has to be recalculated when a claim is later withdrawn.
Non-indemnity / contingency policy	Pays an agreed sum on a stated event rather than a measured loss.
Ombud	The statutory dispute-resolution office a policyholder may approach once the insurer's own complaints process is exhausted. Determinations bind the insurer and are reportable.
Onerous contract	A group of contracts expected to lose money. IFRS 17 recognises the loss immediately, not over the term.
Operator †	POPIA's word for a processor: anyone processing personal data for the insurer — UMAs, assessors, call centres, cloud providers.
ORSA	The Own Risk and Solvency Assessment: the insurer's own forward-looking view of its risks and the capital they need. An annual cycle with a standing data dependency, not a document.
P&C (property and casualty)	The United States and general-insurance name for what South Africa calls short-term and Britain calls general insurance. The same business under three names, which matters when reading vendor material.
Paid-up †	A life policy where premiums have stopped by agreement and reduced cover continues to the original term. Neither a policy paid in full nor a lapse.
Party †	Any legal person the insurer deals with. The supertype that lets one record play many dated roles.
PayShap	South Africa's real-time, low-value push payment service, addressable by mobile-number proxy, built on ISO 20022.
Peril †	The *cause* of loss — fire, flood, hail, theft — not the promise that responds to it.
Persistency	The proportion of policies still in force after a stated number of months. It decides whether acquisition cost is ever recovered.
Personal lines	Short-term insurance sold to individuals and households: motor, homeowners, householders, all-risks. Millions of small similar risks, priced by algorithm with nobody in the decision.
Policy	One sold, in-force instance of an insurance contract.
Policy administration system (PAS)	System of record for the contract: cover, premium, status and every policy event. Usually the oldest system present.
Policy loan	A loan the insurer advances against a life policy's cash value, with interest accruing against the benefit. One of several functions a short-term system has no concept of.
Policy version	The state of a policy between two changes. A policy number identifies the contract, not its state.
Policyholder	Holds the contract, owes the premium, and can change or cancel it.
Policyholder Protection Rules (PPRs)	South African conduct rules on premiums, claims management, complaints, minimum data governance and binder remuneration.
POPIA	The Protection of Personal Information Act, South Africa's data protection law, enforced by the Information Regulator.
Premium	The price of cover.
Premium allocation approach (PAA)	The simplified IFRS 17 measurement for short-duration contracts. Eligibility must still be tested against the full model, so it is no free pass.

Premium income †	Premium recognised as revenue in a period. The name misleads: premium received is largely a liability until the exposure elapses, so cash, written premium and premium income are three different numbers.
Primary healthcare insurance	Cover for GP visits, medication, dentistry and optometry. Not permitted as insurance in South Africa: the demarcation regulations require it to be provided under the Medical Schemes Act.
Producer †	United States term for whoever sold the policy. Nothing to do with manufacturing.
Product †	The reusable *definition* from which policies are made: the template, not the sale.
Product factory	The capability that makes launching a product a configuration change, not a software release.
Profit / contingency commission	Extra intermediary pay depending on how profitably their book performed. Calculated over a book, which breaks policy-centric models.
Prudential Authority (PA)	South Africa's prudential regulator, inside the Reserve Bank. Wants money at a valuation date: balances, provisions, capital and exposures.
Quantitative Reporting Template (QRT)	The prudential return South African insurers file with the Prudential Authority, amended periodically by notice.
Quota share	A proportional treaty ceding a fixed percentage of every risk.
Rating factor	Data used to calculate the price. A pricing input, not a contract term.
Recovery	Money flowing back in, reducing a claim's net cost. Not a negative payment: different counterparty, timeline and accounting.
Reinstatement †	Three things at once: repairing instead of paying cash, restoring an exhausted reinsurance limit, and reviving a lapsed policy.
Reinsurance	Insurance bought by an insurer, to transfer loss or relieve regulatory capital. The two reasons buy different structures.
Renewal †	The point at which a short-term contract is re-offered and re-priced. Not necessarily annual — South African monthly-renewable personal lines has no single renewal event at all.
Repudiation †	The insurer declining a claim. A technical outcome with a reason code, and now conduct-reporting data.
Reserve †	An *estimate* of money the insurer will have to pay. An accounting liability, not cash set aside.
Responsible party †	POPIA's word for a controller: the insurer, as the party deciding the purpose and means of processing.
Retention †	What an insurer keeps after reinsurance. Of an insured, a layer of loss they carry themselves.
Retrocession	Reinsurance bought by a reinsurer.
Rider	A life or health optional add-on benefit attached to a base policy.
Risk adjustment †	Under IFRS 17, the compensation an insurer requires for bearing uncertainty about the amount and timing of cash flows. An explicit margin inside the liability, not an adjustment to anyone's risk score.
Risk object	The thing exposed to loss: a vehicle, a building, a shipment, a fleet. It exists before and after any policy, so it cannot be a child of one.
Run-off †	A book no longer being written whose claims are still being settled.
Salvage †	What the insurer recovers by selling the wreck or goods it took over after a total loss.
SAM	Solvency Assessment and Management: South Africa's risk-based prudential regime, aligned to Solvency II, implemented through the Insurance Act 18 of 2017.
Sasria	South Africa's state-owned special-risks insurer, covering strike, riot and public-disorder loss. Other insurers and brokers attach it alongside the underlying policy, so every non-life system needs a construct for it and a separate remittance flow.
Schedule †	The personalised pages stating who and what is covered, for how much. Not a timetable.
SCR and MCR	The solvency capital requirement and the lower minimum capital requirement. Cover ratios for both go to the Prudential Authority.
Shadow accounting †	A second, unofficial set of records kept to check or challenge the official numbers. Usually commission, and always present.
Short-term and long-term insurance †	South African statutory terms for non-life and life business. Nothing to do with how long a policy has run.
Solvency cover †	Own funds divided by the capital requirement, shown as a ratio or a multiple, and reported against both the SCR and the MCR. "Cover" here means capital coverage, not insurance cover.
Special personal information	POPIA's protected categories: health, biometrics, race, criminal behaviour and more. Insurance cannot operate without them, so a separate legal gate applies.
Specialty insurance	Large, unusual or hard-to-model risk: energy, political violence, cyber, satellites, fine art, kidnap and ransom. Priced on expertise and scarce capacity rather than from a rate table.
Straight-through processing	A quote, acceptance or claim handled end to end with no human touch.
Sub-limit	A smaller ceiling inside a larger limit, such as jewellery within contents.
Subrogation	The insurer's right, after paying, to pursue whoever caused the loss. A new matter with its own parties and timeline.
Subscription market †	A market where several carriers each take a share of one risk, as at Lloyd's. Nothing to do with subscriptions: it means one risk, many insurers, and data arriving as bordereaux.
Sum assured †	The agreed amount a life policy pays, regardless of measurable loss.

Sum insured	The declared value of the insured thing, and the maximum payable for it.
Surplus treaty	A proportional treaty where the cedant keeps a "line" and cedes the surplus above it, so the ceded share varies with risk size.
Surrender †	Cashing in a savings-type life policy for its surrender value.
Suspense †	Cash received but not yet allocated to a policy. It grows, and clearing it by hand becomes a standing cost.
TCF	Treating Customers Fairly: the six conduct outcomes the FSCA supervises against, from culture and product design to post-sale barriers.
Third party	Somebody harmed by the insured, claiming against the insured's liability cover. They consented to nothing, which changes their data-protection position.
Tied agent	A salesperson representing one insurer only. Easier to control and to integrate than a broker, and the data belongs to the insurer.
Total loss (write-off) †	A claim where repair is uneconomic, so the insurer pays the value and takes the wreck. Not total financial ruin, and the threshold is a business rule that moves.
Transaction time	When the insurer recorded a fact, as distinct from when it was true. The axis auditors and reinsurers ask about.
Treaty †	A standing reinsurance agreement under which cessions happen automatically for all business within defined terms.
Twin Peaks †	South Africa's split supervision: the Prudential Authority for soundness, the FSCA for conduct. Both want data.
UMA / MGA	An intermediary with delegated authority to underwrite, price, issue and often settle claims. The insurer remains liable.
Umbrella fund †	A single retirement or group risk vehicle shared by many unrelated employers, each a participating sub-scheme. Nothing to do with umbrella liability cover.
Underwriter †	Whoever decides to accept a risk and at what price. Not an administrator.
Unearned premium	Money held for cover not yet given, and therefore a liability.
Unearned premium reserve (UPR)	The liability for premium already written that relates to cover not yet given. Under IFRS 17 its successor is the liability for remaining coverage.
Unit-linked	A life policy whose value is the number of units held times a published unit price. Needs unit allocation, unit pricing and a dated price history, none of which a short-term system has.
Utmost good faith	A heightened mutual duty to disclose everything material to the contract.
Valid time	When a fact was true in the world, as distinct from when the insurer learned it. The axis the policyholder and the date of loss care about.
Value of new business (VNB)	The present value of expected future profit on business written in the period; over premium it gives the new-business margin.
Variable Fee Approach (VFA)	The IFRS 17 measurement model for contracts with direct participation features, where the insurer's profit is effectively a fee on the underlying items. The counterpart to the PAA and the general model.
Waiting period	Time after inception before a benefit can be claimed.

A closing warning. No glossary settles a definitional dispute inside an insurer, because the arguments that matter are not about English. "How many policies do we have?" has three defensible answers, depending on whether a cancelled-and-rewritten contract counts once or twice, whether multi-life funeral cover counts lives or contracts, and whether the delegated-authority book is counted at bordereau or policy grain. Finance, actuarial and operations will each have a number and each be right. Somebody must arbitrate, the ruling must be written down, and the data layer must expose the derivation rather than the answer [1][2].

THE CHAPTER IN TEN LINES

- Insurance's hardest vocabulary is the ordinary words: excess, schedule, reserve, lapse, treaty, retention.
- A dagger marks every term whose insurance meaning differs materially from everyday use — those are the ones that cause silent modelling errors.
- Always qualify "premium" three ways: gross or net, written or earned, and on what reporting basis.
- Life and short-term insurance use different words for the same structure, so establish which business is being discussed before agreeing anything.
- South African statutory vocabulary — short-term, long-term, binder, cell captive, microinsurance — diverges from Anglo-American usage.
- A product is a definition and a policy is an instance; conflating them is why launches take quarters.
- A glossary cannot settle a definitional dispute: somebody must arbitrate, the ruling must be written down, and the data layer must expose the derivation.

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APPENDIX C

Standards, bodies and where to find them

A working register of the standards, instruments and institutions an insurance architect has to cite: what each one is, who owns it, whether you can read it without paying, and where it lives.

An architect in insurance spends a surprising amount of time hunting for the authoritative text of something. This appendix is the register: what each item is, who owns it, whether you can read it without paying, and where it lives. Checked October 2026; every row is a starting point.

Two warnings. **Almost nothing substantive in the industry-model space is free.** ACORD's marketing pages and one fact sheet are public; everything you would build from needs a subscription. **Almost everything South African is free**, because it is gazetted — so the difficulty is establishing which version is operative, since standards, communications and templates are amended by notice rather than reissued.

One distinction is worth holding onto. *Messaging standards* (ACORD AL3 and the XML families, NGDS, ISO 20022) are interchange formats: translate at the boundary, never restructure internal data to match a message schema. *Reference models* (ACORD's Reference Architecture, IBM IIW, Oracle OIDF, openIDS) are conceptual or logical assets: a vocabulary and a checklist, not a physical schema. *Regulatory instruments* are obligations with dates: they set the grain of your data, not its shape.

Industry models and messaging standards

Item	What it is	Owner	Access	Where
ACORD Reference Architecture	Seven interlocking facets: Business Glossary (6,000+ terms, interactive since January 2026), Information Model, Data Model, Capability Model, Component Model, Process Model (also published as a Process Framework) and Product Model. The traceability between them is the value.	ACORD, non-profit, founded 1970	Subscription, whole or by component; no facet is free	acord.org/standards-architecture/reference-architecture
ACORD Data Standards	Messaging by family: Property & Casualty (XML and the fixed-length, positional **AL3**), Life & Annuity (XML plus DTCC EDI), Global Reinsurance & Large Commercial (XML *and* JSON), Asia-Pacific, Southern Africa, and Multi-Functional — which holds the **Delegated Authority** standards covering risk capture through binding, premium and claims reporting, the standard answer to the bordereau problem.	ACORD	Members and participants only; licensing terms unpublished	acord.org/standards-architecture/acord-data-standards
ACORD NGDS	Object resource definitions, schemas and templates for REST APIs, microservices and IoT. Whether these are JSON Schema or OpenAPI is not stated publicly — get the artefacts first.	ACORD	Members with NGDS access	acord.org/standards-architecture/acord-data-standards/next-generation-digital-standards
ACORD Southern Africa Standards	Short-Term Insurance standards: XSDs, XML samples and a policy-life-cycle process guide, plus an Exposure Reporting standard. Current version and local adoption are unpublished — ask directly.	ACORD	Members; training materials free within membership	acord.org/standards-architecture/acord-data-standards/southern-africa_Data_Standards
IBM Insurance Information Warehouse	A warehouse blueprint: conceptual business model, enterprise entity-relationship model, and Business Solution Templates shipping the analytical *questions* as well as the schema. Its regulatory content predates IFRS 17.	IBM	Commercial licence	ibm.com/support/pages/node/626821
Oracle Insurance Data Foundation	A data foundation inside the OFSAA stack rather than a published model: staging fed by table-to-table mappings, a results area for analytical applications, conformed slowly-changing dimensions.	Oracle	Commercial licence; documentation sits behind support entitlement, so the model cannot be evaluated before purchase	docs.oracle.com/cd/E92918_01/technicalguides_8.1.x.htm
openIDL	An insurance data-sharing network under Linux Foundation governance, with AAIS as principal industry sponsor.	Linux Foundation / AAIS	Free	openidl.org
openIDS	Community data-model work hosted by openIDL. Its Homeowners Standard v1.0 (20 November 2025) is described as the first free, open, production-ready insurance data standard: one line of business, one jurisdiction, and the first credible open competitor to ACORD.	openIDL / Linux Foundation	Free; licence identifier not named in the release	openidl.org/openids
EIOPA Solvency II DPM and XBRL taxonomies	A *reporting* model: every template data point with its dimensional breakdown. The definitive open statement of what a prudential supervisor means by each number.	EIOPA	Free	register.eiopa.europa.eu/Pages/Supervision/Insurance/Reporting-formats-(previous-versions).aspx

Accounting and payment standards

Item	What it is	Owner	Access	Where
IFRS 17 *Insurance Contracts*	Effective for periods from 1 January 2023, replacing IFRS 4. Measures *groups* of contracts, holds unearned profit as a Contractual Service Margin, permits the Premium Allocation Approach for short contracts and requires the Variable Fee Approach for participating business.	IFRS Foundation / IASB	Issued standard free; bound editions sold	ifrs.org/content/dam/ifrs/publications/html-standards/english/2025/issued/ifrs17.html
ISO 20022	The message scheme under modern payment rails, PayShap among them. It matters because disbursement and collection increasingly carry richer remittance data than a batch file can.	ISO, with a Registration Authority	Registry free	iso20022.org/iso-20022-message-definitions
DebiCheck (Authenticated Collections)	The authenticated debit order: the debtor confirms the mandate once with their own bank, which then validates each collection. Types TT1 (real time), TT2 (two-day window), TT3 (card and PIN, face to face). AEDO and NAEDO, which it replaced, were switched off on 31 October 2021.	Clearing arrangements now managed by PayInc, under SARB oversight	Participation via a sponsoring bank or mandate provider	payinc.co.za/about-us/what-we-do and pasa.org.za/notice-of-sunsetting-of-aedo-and-naedo
PayShap	Real-time, low-value push payments launched March 2023 under the SARB's Rapid Payments Programme, proxy-addressable and ISO 20022 compliant. Launch clearing applied up to ZAR 3,000 — funeral and small personal-lines claims fall inside that, motor and property do not. Check current scheme rules.	PayInc, with the banks and the SARB	Scheme participation	payshap.co.za

South African instruments

Item	What it is	Owner	Access	Where
Insurance Act 18 of 2017	The primary prudential statute. Commenced 1 July 2018; restructured licensing into classes and sub-classes of life and non-life business and created a microinsurance framework.	Parliament; administered by the Prudential Authority	Free, gazetted	gov.za/documents/acts/insurance-act-18-2017-english-afrikaans-18-jan-2018
Prudential Standards	Forty-two standards on governance, risk management, internal controls and financial soundness, with separate sets for insurers, microinsurers, branches and groups, applied proportionally to scale and complexity.	Prudential Authority	Free	resbank.co.za/en/home/what-we-do/Prudentialregulation
SAM	Solvency Assessment and Management: the risk-based, Solvency II-aligned regime delivered through the Act and the Prudential Standards — SCR and MCR cover ratios, an ORSA, and statutory control functions including a Head of Actuarial Function.	Prudential Authority	Free; development-phase position papers online	fsca.co.za/Regulated-Entities
Quantitative Reporting Templates	The prudential returns, solo and group, amended by notice. Prudential Communication 7 of 2023 is required architectural reading: built under IFRS 4, so premium terms keep their pre-IFRS 17 meaning and gross technical provisions report as one total.	Prudential Authority	Free	resbank.co.za/content/dam/sarb/publications/prudential-authority/pa-public-awareness/covid-19-response/2023/Prudential%20Communication%207%20of%202023%20-%20IFRS%2017%20QRT.pdf
Insurance conduct regulations	Effective 1 January 2018: premium-	National Treasury;	Free	treasury.gov.za/comm_media/press/2017/2017121501%20Media%20statement%20release%20of%20final%20insurance%2

and Policyholder Protection Rules	review principles, **minimum data governance requirements**, claims-management minimums, complaints reporting, limits on who may hold a binder and what may be paid for it, and the principle that an insurer outsourcing critical functions stays responsible.	FSCA supervises		
Joint Standard 1 of 2024 (Outsourcing by Insurers)	Effective 1 December 2024, replacing Prudential Standard GOI 5. Due diligence before outsourcing; concentration risk where a provider serves several insurers; notification to both regulators thirty days before a material outsourcing and seven days after termination; existing arrangements compliant within twenty-four months. Whether binder, UMA and cloud arrangements are "material" is unsettled.	PA and FSCA jointly	Free	resbank.co.za/content/dam/sarb/publications/prudential-authority/pa-public-awareness/covid-19-response/2024/joint-co 2024/Joint%20Standard%201%20of%202024%20Outsourcing%20by%20Insurers.pdf
FAIS	The Financial Advisory and Intermediary Services Act 37 of 2002: licensing of providers and representatives, fit and proper requirements, advice records, disclosure. UMAs and brokers are licensed under it.	Parliament; administered by the FSCA	Free, gazetted	gov.za/documents/financial-advisory-and-intermediary-services-act
FIC Act	Accountable institutions, long-term insurers among them, must perform risk-based customer due diligence, maintain a Risk Management and Compliance Programme, keep records,	Financial Intelligence Centre	Free, with guidance notes	fic.gov.za/compliance

	and report suspicious and cash-threshold transactions.			
POPIA	The Protection of Personal Information Act 4 of 2013: lawful processing conditions, purpose limitation, minimality, data-subject rights, security safeguards, breach notification, cross-border constraints and operator obligations. A hard case here, because underwriting *is* health-data processing.	Parliament; enforced by the Information Regulator	Free, gazetted	treasury.gov.za/POPIA/POPI%20Act%204%20of%202013.pdf
POPIA health information regulations, 2026	In force 6 March 2026, applying to a named list that expressly includes insurance companies: technical and organisational measures, confidentiality duties, secure disposal, cross-border transfer only under section 72.	Made under POPIA s112(2)(c); the Information Regulator enforces, and also receives breach notifications and prior-authorisation applications for shared identifiers	Free	inforegulator.org.za
COFI Bill	The intended successor conduct framework: from fragmented rules to a harmonised, outcomes-based regime, consolidating FAIS and the old insurance acts. Notice of introduction gazetted 17 April 2026; still pending tabling as at the FSCA's July 2026 Regulation Plan.	National Treasury, with the FSCA	Free	moonstone.co.za/cofi-at-the-heart-of-fscas-2026-regulation-plan
Omni-CBR	The FSCA's cross-sectoral Conduct of Business Return: standardised conduct indicators, quarterly in the end state, demanding segmented demographics, product holdings, claims	FSCA	Free	moonstone.co.za/fsca-publishes-roadmap-for-phasing-in-conduct-of-business-returns

	outcomes by reason, turnaround times and complaints data. The hardest obligation to retrofit, because it concerns process and outcome rather than money.

South African bodies

Body	What it is	Access	Where
Prudential Authority	The prudential supervisor, inside the South African Reserve Bank. Licenses insurers, issues Prudential Standards, communications and reporting templates, and publishes the quarterly sector data.	Free	resbank.co.za/en/home/what-we-do/Prudentialregulation
FSCA	The market conduct authority for the financial sector. Licenses under FAIS, supervises against the TCF outcomes, and is driving the COFI transition and Omni-CBR.	Free	fsca.co.za
SARB (payments) and PayInc	In 2026 the SARB withdrew PASA's recognition as the payment system management body, splitting those functions between itself and **PayInc** — formerly BankservAfrica, now half-owned by the SARB — which manages the clearing houses for EFT, Authenticated Collections, PayShap, Registered Mandates and real-time clearing. Older payment governance references are out of date.	Free; participation by membership	resbank.co.za/en/home/what-we-do/payments-and-settlements/psmb and payinc.co.za
ASISA	The body for life insurers, asset managers and collective investment schemes. Publishes the life aggregates, lapse and new-business statistics, and the insurance gap studies you can use as denominators.	Statistics free	asisa.org.za
SAIA	The representative body for non-life insurers, with a code of conduct, divisional structures and an annual review.	Free	saia.co.za
FIA	The body for brokers, advisers and UMAs. Useful for how the intermediated side reads a regulatory change, which is often not how the insurer reads it.	Free news; member services	fia.org.za
National Financial Ombud Scheme	The consolidated ombud for banking, credit, life and non-life complaints, replacing the separate insurance ombud offices. Its determinations are the cheapest education available in where policy wording and system behaviour diverge.	Free	nfosa.co.za

What will date fastest

Four are moving as you read. **Omni-CBR** has slipped at least once; its template version and submission dates are the most volatile item here. **Joint Standard 1 of 2024** has a transition deadline of 30 November 2026, and whether cloud and binder arrangements count as material outsourcing will be settled by practice, not by the text. **COFI**, once passed, will invalidate the licensing vocabulary in half the documents on your shelf. And **payment governance** changed in 2026 [18], so scheme rules and the PayShap cap need re-checking against PayInc and the SARB.

Two things will not date. The Reference Architecture's shape — seven facets, traceable to one another — has been stable for years, and that traceability [1] is what you pay for. And the difference between a messaging standard, a reference model and a regulatory instrument will still be the first thing to establish about any new acronym.

THE CHAPTER IN TEN LINES

- Almost nothing substantive in the industry-model space is free; almost everything in the South African regulatory space is, because it is gazetted.
- Separate the three kinds of instrument: messaging standards to translate at the boundary, reference models to use as a checklist, regulatory instruments that set the grain of your data.
- ACORD's value is the traceability between its seven facets, not any single one of them.
- openIDS is the first credible open competitor to ACORD, but covers one line of business in one jurisdiction.
- EIOPA's Data Point Model is a free, precise statement of what a prudential supervisor means by each number, usable far outside the European Union.
- In South Africa the access problem is not getting the text but working out which amended version is operative.
- Omni-CBR, the Joint Standard 1 of 2024 transition and the COFI Bill are the fastest-moving items in the register.
- Payment governance changed in 2026: PASA's recognition was withdrawn and its functions split between the SARB and PayInc, formerly BankservAfrica.

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- **40.** South African Insurance Association https://saia.co.za/about-us/
- **41.** Financial Intermediaries Association of Southern Africa https://fia.org.za/
- **42.** National Financial Ombud Scheme South Africa https://nfosa.co.za/

APPENDIX D

References

Every source cited in this book, de-duplicated and grouped by theme — with a note on how the research was done, when, and which parts of it will go stale first.

Everything asserted in this book either carries a source or is marked as the author's judgement. This appendix collects those sources in one place, de-duplicated across every chapter, grouped by theme and numbered continuously. Where the same source supports several chapters it appears once.

When the research was done. The underlying research was compiled and verified on 15 October 2026, across four passes: the business of insurance, the application landscape, the vendor market, and data and terminology. Results and market figures are as published at that date. Dated items say so; undated vendor and standards pages were read on that date and may have changed since.

The source hierarchy used. Primary sources first: statutes and gazetted regulations, prudential standards and communications, standard-setters' own publications, regulator statistics, and audited company results. Then near-primary: exchange filings and announcements, vendor technical documentation, and industry-body publications. Then trade press, used for facts a primary source should have carried but did not — most South African market results in this book reach you through trade reporting of company announcements rather than through the announcements themselves, which is a limitation worth knowing about. Vendor marketing is used only to establish what a vendor claims, is attributed to the vendor, and is dated.

What was deliberately left out. A figure in wide circulation with no traceable primary source is not a fact. The clearest case is the share of South African short-term premium written through underwriting management agencies: estimates circulate, none could be traced to a primary source, so the book describes the model and not its size. Likewise no claim is made about which systems integrator implemented which platform at which South African insurer, because nothing sourceable was found. Where the practitioner consensus is strong but the public record is thin — the persistence of COBOL and mainframe technology in life administration is the obvious example — it is written as consensus, not as a sourced fact.

What will date fastest, in order. *Vendor ownership and product names.* Several categories here are one acquisition away from being wrong; the register already records Sapiens going private, Duck Creek taken private, and Arius moving from Milliman to Akur8. Re-verify before quoting any of it. *Regulatory timetables.* The Omni-CBR conduct return, the COFI Bill and the 30 November 2026 transition deadline under Joint Standard 1 of 2024 are all in motion. *Payments governance.* South Africa's arrangements changed materially during 2026, so anything written earlier about who runs the collection schemes is out of date. *Market results.* Every premium, ratio and margin quoted is a point in time, and the point has passed. *Standards versions.* ACORD publishes facet-level versions that the overview pages do not list, so a version number from this book should be checked against the marketplace.

What will not date. The mechanisms. That a policy is an instance of a product and both need versions. That cover must be established as at the date of loss. That insurance data is bitemporal because somebody will ask both what was true and what was known. That parties play dated roles and there is no such thing as a customer. That the bordereau boundary is where risk aggregation, reserving granularity and the single customer view all break. And that the overnight batch window is a hard constraint which only ever shrinks. Those outlast any vendor, statute or reporting template in the list below.

Sources cited by 12 chapters and appendices of this book (2, 3, 4, 5, 7, 8, 9, 10, 11, 12, 13, 14), 208 distinct sources in all.

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THE CHAPTER IN TEN LINES

- The research behind this book was compiled and verified on 15 October 2026; every market figure is a point in time.
- Primary sources first — statutes, prudential standards, standard-setters and audited results — then near-primary, then trade press, with vendor marketing used only to record what a vendor claims.
- Figures in circulation with no traceable primary source were left out, the UMA share of South African short-term premium among them.
- Strong practitioner consensus with a thin public record is written as consensus, not as fact.
- Vendor ownership and product names will date fastest, then regulatory timetables, then payments governance, then market results, then standards versions.
- The mechanisms will not date: product versus policy, cover at date of loss, bitemporality, parties in roles, the bordereau boundary and the batch window.

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